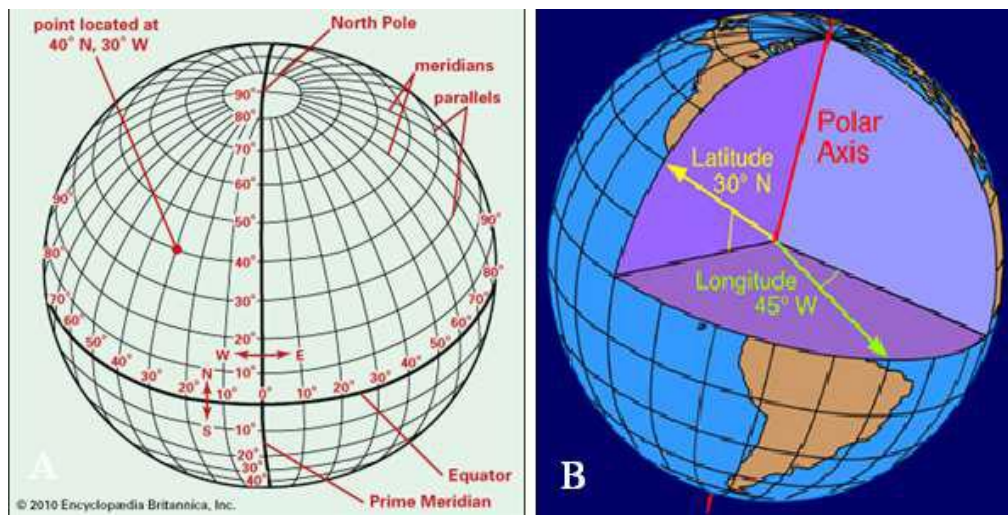


**PRE-UNIVERSITY REMEDIAL PROGRAM
FOR THE 2014 E.C. ESSLCE EXAMINEES**

GEOGRAPHY MODULE



WACHEMO UNIVERSITY

Department of Geography & Environmental Studies

Teaching material for:

Pre-University Remedial Program for the 2014 E.C. ESSLCE Examinees

Geography

Credit Hours: 4 Hours

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2015

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UNIT ONE INTRODUCTION

1.1. The Science of Geography

1.1.1. Definitions and Concepts

❖ Ancient Greeks defined geography for the first time. They combined two words:

-**Geo** – which means earth

-**Graphic** – which means writing

From the ancient Greeks to modern-day geographers, geography has been defined differently. **However, the various definitions share some common ideas.**

Here are some of the most important definitions that geographers have proposed:

- **Eratosthenes (276-196 BC)** – Geography is the description of the earth.
- **Concise Oxford Dictionary (1964)** – Geography is the science of the earth's surfaces.
- **Hartshorne, R. (1899-1992)** – Geography is a branch of knowledge that is concerned with the provision of an accurate, orderly and rational description of distributions on the surface of the earth.
- **Yeates, M. (1968)** – Geography is a science that is concerned with the rational development and testing of theories that explain and predict the spatial distribution and locations of (things and) phenomena on the surface of the earth.
- ✓ As you can see, each of these definitions includes the idea **that geography studies the earth**. Most of them specify the **surface of the earth**; emphasized the fact that geography is a spatial science.
- ✓ In the mid-20th century, it became a spatial science dealing with the arrangement and distribution of things and phenomena over the surface of the earth.
- ✓ Geography has now become **a science that examines “place and space” on the Earth's surface in relation to location, distribution, arrangement, interaction, causes and effects of (things and) phenomena.**
- ✓ Modern geography is a spatial temporal-areal science.

Accordingly, geography can be defined as *“the study of the spatial distribution of both physical and human-made things and phenomena on the earth's surface and the two-way interactions and interdependences between natural and human environments.”*

While making spatial investigations, geographers ask at least the following five pertinent questions about the phenomena they study. These are:

“**WHERE** are things located?”

“**WHY** are they located where they are?”

“**WHEN** did the things form?”

“**WHAT** things are found where?” and

“**HOW** are they arranged?”

What makes geography a science?

Science is a system of acquiring knowledge through **scientific methods**. These methods involve: **observation, identification, description, experimental investigation, and theoretical explanation of phenomena.**

Geography investigates facts and relationship related to physical and social phenomena, and examines their distribution across the world and changes over time.

The **main tools geography uses** to gather and analyze information are **observation, systematic description, systematic recording and mapping**. These facts make geography a spatial science

1.1.2 Branches of Geography

Geography has two main branches of study - -

A. Physical geography and

B. Human geography

A. Physical Geography

This branch of geography studies the distribution of the natural features of the world, such as climate, landforms, soil, vegetation, and drainage systems. Physical geography also considers causes, effects and interactions of these features.

Physical geography includes:

- **Climatology:** studies factors that create climate and examines the variation and distribution of climate and related causes and effects.
- **Geomorphology:** studies the distribution of landforms (such as mountains and plains) and the forces that change them.
- **Soil geography:** studies the distribution of soils and their characteristics.

- **Biogeography:** studies the distribution of plants and animals in relation to the environments that they inhabit.
- **Oceanography:** studies the location, causes and effects of ocean currents, waves and tides.
- **Hydro Geography:** deals with the water resources, the distribution and use of water across the planet
- **Cryosphere Geography:** Cryosphere geography explores the ice of the earth, especially glaciers and ice sheets.

B. Human Geography

This branch of geography studies the distribution and influence of human aspects of our world, including cultures, population settlement, economic activities and political systems.

Human geography includes:

- ✚ **Cultural geography:** studies the distribution and interactions of cultures, including peoples' beliefs and customs. It also examines the movement, expansion and interaction of cultures on the surface of the earth.
- ✚ **Population geography:** studies the distribution, growth and structure of population.
- ✚ **Economic geography:** studies production, consumption and exchange and the spatial distribution of goods and services and factors affecting them.
- ✚ **Political geography:** studies the distribution of political systems and the ways people use them to exercise power and make decisions.
- ✚ **Urban geography:** studies the development and characteristics of towns, cities and other urban centers.
- ✚ **Historical geography:** is the study of the geography of the past and how places or regions change over time.
- ✚ **Transportation Geography:** deals with the History of transport, modes and needs of transport systems, networks and models of transportation, cost of transport and accessibility and connectivity.
- ✚ **Medical Geography:** Medical geographers study the geographic distribution of disease (including epidemics and pandemics), illness, death and health care.
- ✚ **Recreation, Tourism, and Sport Geography:** The study of leisure-time activities and their impact on local environments, among others.

1.1.3 Scope/Coverage of Geography

The scope of Geography is the surface of the Earth. The geo-sphere is considered as geography's scope. The geo-sphere itself is made up of the following sub spheres:

- ❖ Atmosphere(air),
- ❖ Lithosphere(land),
- ❖ Hydrosphere(water) and
- ❖ Biosphere(living things),
- ❖ Anthroposphere (earth's cultural landscape) which provides the habitable zone in which humans are able to live.

The geospheres of the earth

Geospheres	Description	Geography's related area of study
Lithosphere	The solid part- i.e., the rock layers of the earth	Geomorphology, soil geography
Troposphere	The lower part of the atmosphere where weather changes occur	Climatology
Hydrosphere	The water surfaces of the earth including oceans, seas and lakes	Oceanography
Biosphere	The part of the earth that supports all sorts of life	Biogeography
Anthroposphere	The earth's cultural landscape	Cultural geography, population geography

1.1.4. Themes and approaches of Geography

- ❖ Basic themes in Geography
 - Geography has five basic themes namely:
 - Location,
 - Place,
 - Human-environment interaction,
 - Movement, and
 - Region

1. **Location**:- is defined as a particular place or position.

Location can be of two types:-

❖ **Absolute location and Relative location**

Absolute location (also known as astronomical location or mathematical location).Is defined by its exact address or it can be defined by:-

Latitude (is the distance north or south of the Equator and is expressed as a number between 0 and 90 degrees north or south, and **Longitude** (is the distance east or west of the Prime Meridian and is expressed as a number between 0 and 180 degrees east or west.

2. **Place**

Place refers to the physical and human aspects of a location.

Place is associated with:-

- ✚ **Toponym** (the name of a place),
- ✚ **Site** (the description of the features of the place), and
- ✚ **Situation** (the environmental conditions of the place).

Each place in the world has its unique characteristics expressed in terms of physical aspects like landforms, hydrology, biogeography, pedology; human aspects like characteristics and size of its human population, and the distinct human cultures. The concept of “place” aids geographers to compare and contrast two places on Earth.

3. **Human-Environment Interaction**

Humans have always been on ceaseless interaction with their natural environment. No other species that has lived on our planet has a profound effect on the environment as humans. Human-environment interaction involves three distinct aspects: **dependency, adaptation, and modification.**

- **Dependency** refers to the ways in which humans are dependent on nature for a living.
- **Adaptation** relates to how humans modify themselves, their lifestyles and their behavior to live in a new environment with new challenges.
- **Modification** allowed humans to “conquer” the world for their comfortable living.

4. **Movement**

Movement entails to the translocation of human beings, their goods, and their ideas from one end of the planet to another. So, the three dimensions of movement include: **physical movement of people; the transport of goods and the flow of ideas**

5. Region

A region is a geographic area having distinctive characteristics that distinguishes itself from adjacent unit(s) of space. Region could be:-

- **A formal region** (natural regions) that is characterized by homogeneity in terms of a certain phenomenon like soil, temperature, rainfall, physical conditions, etc.
- **A functional region** or nodal region characterized by functional interrelationships in a spatial system. Ex: Town-region which is interdependent with surrounding rural areas

1.1.4 APPROACHES IN GEOGRAPHY

The most frequently adopted approaches are:

- A. Topical or Systematic Approach
- B. Regional approach

A. Topical or Systematic Approach: The topical or systematic approach applies a specific geographical element or phenomenon over a defined geographical unit. For example, it takes a phenomenon such as climate, land forms or culture, and treats the distribution of the selected element over a country, continent or the world at large.

Examples of topical approaches to study geography:

- ✓ The geography of hunger
- ✓ The geography of climate
- ✓ The geography of agriculture
- ✓ The geography of population

B. Regional Approach

A geographic study that uses the regional approach focuses on a region – a defined geographic unit or locality. Within the region, the study examines a variety of geographic features.

The region studied could be a subcontinent, continent or a number of countries that share a common geographic factor. Examples of regional studies:

- ✓ The geography of Africa, Asia, or Oceania, etc
- ✓ The geography of sub-Saharan.
- ✓ The geography of the Middle East.
- ✓ The geography of the Balkans.

1.1.5 Major School of Thoughts in Geography

Various schools of thought have emerged with different views regarding the relationship between humans and their environment as well as the interpretation of social problems by human.

A. School of Environmental Determinism

- The idea of environmental determinism was laid down by Greek and Roman scholars.
- Is a philosophy that bases its view on the idea that the natural environment is an influencing factor on humans' mode of living.
- It believes that human activities are controlled by the environment.
- It was the dominant idea up to World War I.
- It advocated that the physical environment directs the day-to-day activity of people.
- It advocated that the physical environment is the master in determining the day-to-day activity of people.
- The physical qualities of geographical conditions are the causes:-
 - for people's physical differences;
 - for differences from place to place in people's economic activities, cultural practices and social structure.
- It tends to focus on the impact of the physical environment on people, rather than the reverse the influence of people on the environment.

Determinists consider human beings as passive agents where the physical factors determine their attitude and process of decision making.

However, this outlook was strongly criticized by geographers who favored a new school of thought known as environmental possibilism.

The prominent scholars who supported the school of determinism were: Charles Darwin, Demolins, F. Rutzel, etc.

Q??? With your friends, discuss why the philosophy of environmental determinism was severely criticized by the proponents of environmental possibilism.

B. Environmental Possibilism

- Is a philosophy that bases its view on the idea that the human beings are an influencing factor on the natural environment.
- The school of possibilism was postulated by Febvre.

- Supporters argue that human beings are masters of the environment and they can judge their benefits. They argued that there are no necessities but only possibilities.
- Proponents of this view emphasize that two-way relationships exist between humans and the environment.
- They state that people can influence the environment to enhance their way of life.
- These geographers agree that the environment can potentially affect people's activities, but they believe that we can use our knowledge and skills to regulate these effects.

Nowadays, the school of possibilism is becoming widely accepted since it recognizes human's ability to change its environment using the latest or better technologies.

Example: Human beings have been using irrigation to turn barren lands of the deserts into agriculturally productive areas.

C. The Quantitative Revolution

The Quantitative Revolution has emerged as new theoretical approach, and the analytical method of inquiry to geographical research in the early 1950s.

Socio-economic, physical, and political features and processes are spatially organized and ecologically related. As a result, a more abstract, theoretical approach to geographical research has emerged, and the analytical method of inquiry evolved this new approach. The Quantitative Revolution has used rigorous mathematical formulae borrowing from the physical sciences. This movement in geography is called the “**Quantitative Revolution**”

Contributions of the quantitative revolution to the development of geography

- ✚ The quantitative revolution led to an increased use of statistical techniques.
- ✚ It emphasized multivariable analysis and the use of computers in geographical research.
- ✚ The methods adopted included various mathematical techniques that were more precise than the descriptive methods of regional geography.

The quantitative revolution was a response to the crisis in the 1950's. The crisis was the result of the challenges that geography faced during late 1940's and early 1950's.

Some of the major challenges were:

- The shutting down of many geography departments and courses. For example, the geography program at Harvard University was abolished in 1948.

- The division between Human and Physical geography was continued demanding the autonomous subject hood of Human geography.
- Geography was seen as solely descriptive and unscientific. As some argued, there was no explanation of why processes or phenomena occur in geography.
- Geography was not useful for solving problems. Hence, it was seen as exclusively educational
- Questions regarding the nature of geography persisted, for example, it was unclear to some people whether geography was a science, an art, a humanities subject or a social science.
- The quantitative revolution was driven by the development of the computer and its ability to rapidly process data.
- Quantitative geographers “went radical” and applied computers, statistics, and mathematical models to the study of geographers.

Some of the techniques that became central to geography during the quantitative revolution were:

- ✓ Descriptive statistics
- ✓ Inferential statistics
- ✓ Basic mathematical equations and models, such as gravity models

Deterministic models e.g., Von Thünen’s and Weber’s location models

- ✓ Statistical models, using concepts of probability
- ❖ The revolution introduced a rapid change in the methodologies used in geographical research: a **shift from descriptive geography to empirical law-making geography.**

D .The Emergence of Applied Geography

Geography has been used since human beings appeared on earth. However, geographic knowledge had little chance of being used to solve geographic problems.

- Applied Geography has emerged in the latter part of the 20th Century in geography:
- Geography became a science that we can **use /apply/ to solve socio-economic and political problems.**

Applied geography had its roots in the quantitative revolution. The emergence of applied geography increased the applicability of geographic knowledge in the following areas:

- ❖ Today, many geographers work as:-
 - Urban planners, and GIS analysts,
 - Environmentalists, and cartographers,

- Location analysts, and transportation planners,
- Developing-nations specialists, and public-transportation planners,
- Highway planners, and university-facility planners,
- Transportation logisticians, and demographic analysts, etc.

Thus, applied geography is the use of geographic analysis in private business, government, non-profit organizations etc. Applied geography solves problems and aids in decision making.

1.1.6. Relationship between Geography and other disciplines

- Geography is an interdisciplinary subject.
- It has strong relationships with various disciplines in both the natural and the social sciences.

For instance,

- Human geography is highly linked with social sciences, while
- Physical geography is related to the natural sciences.

Many academic disciplines are linked with geography. Among them are:

- Biology relates with biogeography
- Meteorology relates with climatology.
- Geology links with geomorphology and soil geography.
- Astronomy, Economics, Political science, History, Demography, Sociology, chemistry, and mathematics.

UNIT SUMMARY

In this unit, we defined geography, discussed its scope, discussed its approaches, examined its major schools of thought and examined its relationship with other sciences.

In summary:

Geography is a systematic study of the spatial distribution of phenomena on the surface of the earth and of the two-way interaction between the natural and human environments. The scope of geography is very wide. It attempts to study many parts of the geosphere: the lithosphere, hydrosphere, atmosphere, anthroposphere and biosphere. Geographical studies are conducted based on two basic approaches, regional geography and systematic geography. The first studies all aspects of phenomena found in a region, while the latter investigates a single phenomenon globally.

Environmental determinism and environmental possibilism are the two dominant philosophies that geographers use when they examine the relationship between humans and their environment. The philosophy of environmental determinism is related to the idea that the environment is the factor that determines peoples' mode of living. In contrast, the possibilist philosophy emphasizes the two-way relationship between humans and the environment and the possibility for humans to change the environment. The emergence of the quantitative revolution in the 1950s and 1960s contributed a lot to the development of geography. It incorporated different statistical techniques in to geographical studies. The emergence of applied geography increased the practical applicability of geography. Applied geography solve, many different socio-economic and environmental problems. Geography is highly integrated with other disciplines. It shares a wide range of information with the social and natural sciences.

REVIEW EXERCISE FOR UNIT

I .True or False

1. The fact that deserts can be turned into agriculturally productive areas through irrigation supports the philosophy of environmental possibilism.
2. Geography is less practical in today's world than it was before.
3. According to the determinist approach, human cannot change or influence the environment.
4. Systematic geography is the study of the general characteristic of a region.
5. The facts and principles of the natural sciences can be applied in geography

II .Matching: match the items given in the “A” with the statements given in the “B” column.

A

- A. Quantitative revolution
- B. Febvre
- C. Alexander Von Humboldt
- D. Topical approach
- E. Regional approach
- F. Demography
- G. Political geography
- H. Applied geography
- I. Demolins

B

- 6. An ardent supporter of environmental determinism.
- 7. An ardent supporter of environmental possibilism.
- 8. A nomethic or empirical law- making geography that occurred in the mid-20th C.
- 9. A geographic school largely concerned with the structure of the ecological system and other social problems.
- 10. A geographic approach that emphasizes various aspects of a defined spatial unit.
- 11. A geographic approach that singles out one or two elements and treats the distribution globally.
- 12. A branch of geography that deals with the dynamic and static aspects of population.
- 13. A branch of geography that deals with boundary, communication and activities between countries in relation to political power.

III .Multiple Choices

14. Which of the following opinions do you accept?

- A. The definition of geography is static.
- B. Geography is a discipline which does not have one single definition.
- C. Geography does not have any relationship with other disciplines.
- D. The definition of geography is not subject to any change.

15. Which of the following statements about geography is correct?

- A. Physical phenomena are evenly distributed over the surface of the earth.
- B. There is a one-way relationship between humans and their environments.
- C. All elements in the natural environment affect the way of life of its inhabitants.
- D. Human beings are not capable of modifying their environments.

16. The main theme of possibilism is:

- A. The two-way relationship between human and nature.
- B. Human beings are not capable of modifying their environments.
- C. Human society is a passive agent in the man-environment relationship.
- D. Human beings are always under the influence of their environment.

17. Which of the following reflects the deterministic outlook?

- A. Self-reliance
- B. People who live in cold climates are stronger and more courageous than those living in warm climates.
- C. Human is capable of modifying his environment.
- D. Analyses social structure in terms of human activities.

IV Fill in the blank space

18. A German geographer who tried to show how people adapt to and are influenced by the environment was _____.

19. The sub field of geography that studies how plants and animals are distributed is called _____.

20. Part of the exosphere that represents the cultural landscape of the earth is_____.

21. A Greek scholar who coined the word ‘Geography’ for the first time was_____.

22.The use of geography for socio-economic and political problem solving and planning is known as _____.

UNIT TWO

MAP READING AND INTERPRETATIONS

2.1 Definition and concepts

Here is a simple definition:

- ❖ A map is a simplified, diminished, plain representation of all or part of the earth's surface as viewed from vertically above.
- ❖ A map is a two-dimensional scaled representation of part or whole of the Earth surface on a flat body such as piece of paper, black board, wood or cloth.
- Map reading encompasses a systematic identification of natural features like mountains, rivers, oceans, hills, rocks, etc. and manmade features like roads, railways, buildings, dam, etc.

Here are the main features of maps:

- ✚ A map represents all or part of the earth's surface. For example, a map might show a city such as Addis Ababa, the entire world, or a section of a garden.
- ✚ A map is a two-dimensional (plane) representation. For example, a map might be printed on a piece of paper.
- ✚ Maps show the earth's surface as if it were seen from directly above. This view is called a bird's-eye view.
- ✚ All maps are smaller than the area they represent.
- ✚ Maps are drawn to scale. In other words, the features shown on a map have the same relative proportions as they do in reality. For example, if one mountain's diameter is twice as large as that of another mountain, the first mountain would be shown on a map as twice as large as the second.
- ✚ Maps are simplified representations.
- ✚ Most maps use generally accepted symbols to represent natural, artificial or cultural features of the area they represent. They also use conventional notations to provide background information such as the map's title, date and scale.

Discuss the following question and perform the following tasks in your group.

1. What does this mean: "A map is a two-dimensional plane representation"?
2. Compare and contrast maps and globes.

2.2 Uses of Maps

The map is geography's most important tool. It can present very simple information or highly detailed results from a complicated geographical investigation.

- ❖ Maps are basically used for identifying **locations, distance, area and direction.**

Location: With the help of a map it is possible to locate a place in reference to another place. For example, Ethiopia's location can be expressed in terms of its neighboring countries, external land masses and water bodies. It is also possible to locate a place using astronomical grid references – parallels and meridians.

- **For example,** the absolute location of Ethiopia is 3°N-15°N latitude and 33°E-48°E longitude.

Distance: It is possible to calculate the distance between two or more places on a map. This is done by using the scale of the map.

- For example, the air distance between *Bahir Dar and Addis Ababa* can be calculated by using a map of Ethiopia.

Measuring distances as the scroll flies are carried out by the use of a ruler or straight edge in between the two locations, and then convert the measured distance into a real world distance using the map's scale.

- **For example**, if we measured a distance of 10 cm on a map that had a scale of 1:10,000, we would multiply 10 (distance) by 10,000 (scale). Thus, the actual distance in the real world would be 100,000 cm which is equivalent to 1km distance interval.

Area: The area of a place, a country, a region, a continent, a sub-continent or the whole world can be calculated from a map. This is done by measuring the length and the width of the given place on the map and by converting them to ground distances with the help of the scale of the map.

Direction: A map can enable us to identify the direction and bearing of any place on the map. This is accomplished by referring to another place.

- For example, a map of Ethiopia can help us to find the direction and bearing of *Mekele* by referring to Addis Ababa.

In addition to these, maps can be used to:

- ✓ **Show the distribution of physical and human phenomena on the earth's surface:** distribution maps show the locations of phenomena on the earth's surface. For example, we use distribution maps for Ethiopia to show the presence of human and animal populations, minerals, and vegetation. Similarly, we use distribution maps to show patterns of settlement, temperature, and health conditions.
- ✓ **Show surface configuration:** topographic maps give information about variations in height on the earth's surface. For example, they show heights and depths of valleys, plains and mountains.
- ✓ **Offer visual comparisons:** because the earth's surface is vast, it is difficult to compare places by direct observation. For example, it would be difficult to compare the distribution of landforms in Ethiopia and Kenya, even from an airplane. However, by offering us relatively small but accurate representations of the two countries, maps solve this problem completely.
- ✓ **Support development planning:** Maps can provide planners with vital information to plan for the future. For example, maps can show current conditions and ongoing trends, and they can help us predict a nation's socioeconomic conditions. Such information is invaluable to the country's planners as they analyze possibilities and livelihoods and then prepare for the future.

2.3 Classification of maps

Maps could be classified in terms of the following features:

- ❖ **Focus and level of detail (purpose)** – the amount of information they present about their subjects, especially their ability to show small details.
- ❖ **Scale** – the size of the area for which they give information and, therefore, the scope of the information that they give about these areas.

Classification Based on Purpose:

Maps can be classified as **general-purpose** or **specific-purpose**.

A. General-Purpose Maps

- ✓ **A general-purpose map** is a map that shows the features of a place in a relatively general way.
- ✓ It provides a wide range of information about the place it represents.
- ✓ General-purpose maps **are not topical**.
- ✓ They tend to **contain a little of many kinds of information at a relatively low level of detail**. Topographic information of each kind in lesser detail. **A topographic map** is a good example of this. It can illustrate both physical and human-made features of the earth.

B. Specific-Purpose Maps

- ✓ **Specific-purpose maps** are often called thematic maps or topical maps.
- ✓ We use these terms because specific-purpose maps emphasize on a single topic.
- ✓ These maps show detailed information about their subjects.
- ✓ Thematic maps can show almost any kind of information that varies from place to place, such as population distribution, rainfall and temperature patterns, and the distribution of types of soil or vegetation.

Soil maps

Vegetation maps

Classification by Scale

Scale is a ratio that shows the degree to which the area that is mapped has been reduced. Based on scale differences, maps can be classified into:

- Large-scale maps
- Medium-scale maps
- Small-scale maps

Let us now consider each of these types of map in more detail.

➤ Large-Scale Maps $\geq 1:50,000$

What are large scale maps?

Large-scale maps present **small areas** in **detail** with great **accuracy**.

Large-scales are greater than or equal to 1:50,000. A large-scale map, such as the map of Addis Ababa, shows the city in considerable detail.

Large scale maps present a relatively small area and show its features in considerable detail. For example, at a scale of 1:5,000, a map of a city can include many features – such as buildings. The map can also include many details, such as the bends in highways.

Medium-Scale Maps 1:50,000 - 1:250,000

These are maps that are prepared with scales that range between 1:50,000 and 1:250,000. Medium-scale maps **cover wider areas than large scale maps**, but **cover smaller areas than small-scale maps**. They are also able to present more detailed information than small-scale maps but are less detailed than large-scale ones.

Small-Scale Maps $\leq 1:250,000$

Small-scale maps are those which are prepared with scales less than or equal to 1:250,000. These small–scale maps **cover wider areas** than large and medium scale maps.

2.4 Marginal and boarder information

Maps are used to convey information. To read maps effectively, map users need information about the map. Such information is presented in the map’s margins and is known as *marginal information*.

Marginal information includes:

- **Title of the map:** Gives the map’s name. For example, “Soil Map” explains that the map presents information about soil.
- **Year of publication:** identifies the year in which the map was published. Because this information tells you how old the map is, you might be able to judge whether the map’s contents are current or might be out of date.
- **Author:** identifies the copyright owner of the map and indicates who (or what organization) has prepared the map.
- **Place of publication and publisher:** tells where the map was published and identifies the organization that published the map.
- **Scale:** This information indicates the extent to which the area that is represented in the map has been reduced.
- **Legend/Key:** explains the meaning of the signs and symbols used in the map.
- **Type of projection:** tells the kind of projection used in making the map.
- **Direction or orientation (North) arrow:** Shows the north direction on the map.
- **The magnetic declination (variation):** is the difference between Magnetic North and True North. This will be explained later in this unit.

2.5 Conventional Signs and symbols

- **Conventional signs and symbols** are those signs and symbols that are used on maps through the agreement of all map-makers of the world. They are used to represent the same detail on a map in all the countries of the world.
- **Signs and symbols** help the map reader to understand maps. Therefore, the map reader has to look first at the key or legend of the map.

When you create a map, the symbols you select should satisfy the following requirements.

- ✚ They should be uniform throughout the map.
- ✚ They should be easy to read and understand.
- ✚ The space occupation, orientation and size of the symbols should be constant.

Here are some of the symbols and conventional signs that are widely used and understood worldwide:

- Cities and towns are indicated by dots or patches of shading;
- Streams and bodies of water are often printed in blue; and
- Political boundaries are shown by dot lines/solid lines.

Symbol	Description
	All-weather road
	Bridge
	Permanent lake
	Plantation
	Mine

Figure 1 Conventional signs and symbols

Map Scale

The **scale of a map** is the ratio between the measurement of distance on the map and the corresponding measurement on the earth’s surface. For instance, the ratio of the map distance between two cities on a map and the actual distance between the two corresponding cities on the earth is the scale to which that map is drawn.

- ❖ Map scale can be linear or areal.
- **Linear scale** expresses the ratio of map distance to ground distance. It is the most common scale type.
- **Areal scale** shows the relationship between map area and ground area.

Areal scale is the square of linear scale. For example, if a map has a linear scale of 1 cm to 6 km, then the areal scale of the map is (1 cm)2 to (6 km)2, which means 1 cm2 to 36 km2.

$$\text{Areal Scale} = (\text{Linear Scale})^2 = \left(\frac{\text{Distance on the map}}{\text{Distance on the Ground}} \right)^2$$

To calculate linear scale from areal scale, you put the areal scale into a square root. For instance, if the areal scale of a map is 1 cm² to 36 km², then the linear scale of the map is the square root of the given areal scale, as shown below.

$$\begin{aligned} \text{Linear Scale} &= \sqrt{\text{Areal scale}} = \sqrt{\frac{\text{Map Area}}{\text{Ground Area}}} \\ &= \sqrt{\frac{1 \text{ cm}^2}{36 \text{ km}^2}} = 1 \text{ cm to } 6 \text{ km} \end{aligned}$$

How to Find the Scale of a Map

There are two ways of finding a map scale, if it is not given:

A. By using the known distance between two points on the map

This method is used if the ground distance between two points or places shown on the map is given. Then use the following procedure to obtain the scale:

- i. Measure the distance between the two points on the map in centimeters.
- ii. Divide the obtained distance on the map by the ground distance to obtain the ratio between the two. This gives you the scale of the map. **To understand this better see the following example.** Let us say, if the straight-line distance between **Addis Ababa and Mekele** is 555 km. using the following figure, calculate the scale of the map:

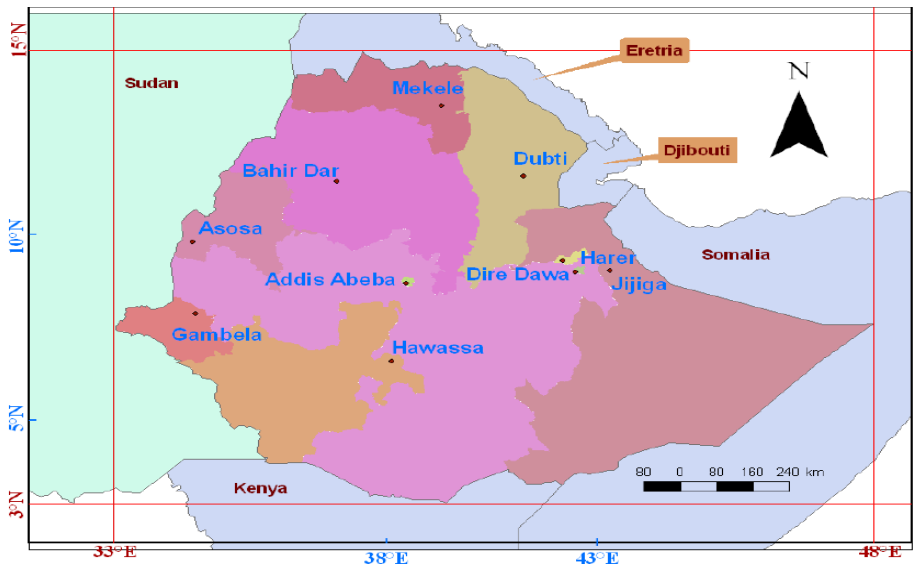


Figure 2. Map of Ethiopia

Using the distance between Addis Ababa and Mekele:

- i .Measure the distance between Addis Ababa and Mekele by using a ruler approximately 3 cm.
- The given air distance between the two places is 555 km.
- ii. Find the ratio between the distance on map and the actual distance.

This is the scale.

$$\text{Scale} = \frac{\text{Distance measured on the map}}{\text{Ground distance}}$$

3 cm to 555 km = 1 cm to 185 km = 1 : 18,500,000.

B. By using latitudes

In this method, the scale of the map can be obtained by using the values of latitudes.

For example, let us calculate the scale of the above Figure by using the 5° and 10° N latitude lines:

- 1. The degree difference between the two latitudes is 5o (10° – 5°).
- 2. The distance that 5° represents is 555 km (111 km × 5).
- 3. The distance between the two latitudes on the map is approximately 3 cm.
- 4. The scale of the map is, therefore:

$$\text{Scale} = \frac{\text{Distance on the map}}{\text{Ground distance}} = \frac{3\text{cm}}{555\text{km}} = 1\text{cm to } 185 = 1 : 18,500,000$$

- 1° latitude is about 111 km
- This relationship is derived from the circumference of the earth – 40,000 km.
- If 360° = 40,000 km
- 1° = ?
- $$= \frac{1 \times 40000\text{ms}}{360^\circ} = 111\text{km}$$

Measurements in maps [location, distance, area, direction]

Some areas have regular or geometric shapes, such as rectangles, triangles, circles and squares. Others have irregular or non-geometric shapes.

Measurements in maps area: This procedure is important for measuring areas on maps:

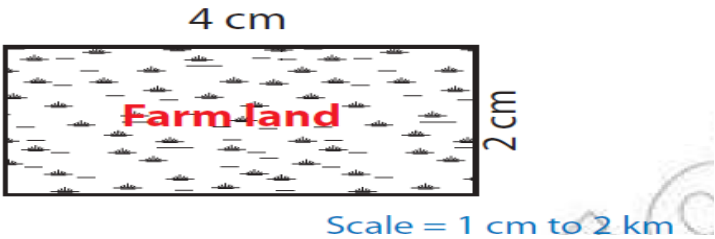
1. Measure the area of the feature on the map;
2. If the map scale is linear, convert it to areal;
3. Using the areal scale, convert the area on the map to actual ground area by using cross multiplication.

Measuring Regular-Shaped Areas

A regular shape is a geometrical shape such as a **circle, triangle or square**. If you are measuring an area with a regular shape, use the mathematical formula for its geometric shape. Then calculate the ground area by using the map’s areal scale. The following table gives you some of geometric formulae for calculating the areas of regular shapes.

Table: Geometrical formula

Regular-shaped areas	Mathematical formula
Square	$A = S^2$; where A = area, and S = side
Rectangle	$A = l \times w$; where A = area, l = length, and w = width
Right angle triangle	$A = \frac{1}{2}bh$; where A = area, b = base, and h = height
Circle	$A = \pi r^2$; where A = area, and r = radius. ($\pi = 3.14$)



What is the ground area of the farmland?

In the above sketch, the farm has a regular shape.

- Measure the length and width of the rectangle on the map. Obtain its area on the map by multiplying the length by the width (area of rectangle = $l \times w$). This gives you the area of the farm on the map.
- Change the linear scale, which is 1 cm to 2 km, into areal scale by squaring it. This gives you the areal scale : 1 cm^2 to 4 km^2 .
- Convert the map area into ground area by using cross multiplication.

In other words, if 1 cm^2 is to 4 km^2 , then what will 8 cm^2 be?

This gives you the ground area of the farm **32 km^2** .

Without changing the given linear scale into areal scale, it is possible to calculate the area.

$$L = \frac{2cm \times 2km}{1cm} = 4km \text{ and } W = \frac{4 cm \times 2km}{1cm} = 8km$$

Area = L x W; Area = 4km x 8km = 32 km^2 ; Therefore, the **area of the farm land is 32 km^2** .

Measuring Irregular-Shaped Areas

If the feature to be measured has an irregular shape, its area cannot be directly calculated by using mathematical formulae. In such cases, we can use the grid square method to measure the area of the lake on the map.

Measurement of Direction

Directions from one point to another or the bearing of one point from another can be given using two different sets of units. The traditional system uses the **cardinal compass points north, east, and south, west** and subdivisions of them.

A modern, and more accurate, method gives the directions **in degrees and fractions of degrees clockwise from north**. The relationship between the two ways of giving directions is shown in following Figure.

The following figure shows you how to determine directions and bearings on maps. The procedure involves the following steps. Example:

To find the direction from point A to point B on the map (the following figure);

- 1. Draw a line with a pencil joining points A and B on the map.
- 2. Through the point from which the bearing is required draw a pencil line parallel to true north as indicated by the meridians or the arrow indicating true north.
- 3. Using these two lines, set your protractor so that its centre is in point A and measure the angle between the true north line and the line A – B reading clock wise from north = 0°.
- 4. State the bearing either in compass directions or degrees clockwise from north.

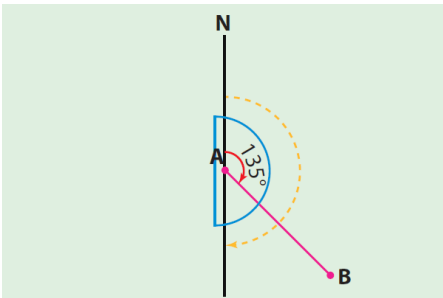


Figure: The measurement of direction

Answer:

- i .The direction from A to B is 135° or point B bears 135° from A.
- ii. Point B is roughly South East of A or point B bears South East from A.

Distance measurement

Distance measurement on a straight course is quite simple compared to measurement of irregular features like river course or road. Measuring distances as the scroll flies are carried out by the use of a ruler or straight edge in between the two locations, and then convert the measured distance into a real world distance using the map's scale.

For example, if we measured a distance of 10 cm on a map that had a scale of 1:10,000, we would multiply 10 (distance) by 10,000 (scale). Thus, the actual distance in the real world would be 100,000 cm which is equivalent to 1km distance interval.

2.6. Relief representation on maps

How can we represent relief on maps?

In order to read relief features from maps, you should first know how map –makers represent the uneven surface of the earth on a plane sheet of paper, i.e., on a map.

There are different ways of showing relief on maps. These include:

- Physiographic diagrams

- Hachures
- Layer coloring
- Hill shading
- Form lines
- Contour

Traditional Methods

A. Physiographic Diagrams

What is a physiographic diagram?

Early map makers used to represent relief features by diagrammatic pictures known as **physiographic diagrams**. They show three-dimensional pictures of landscapes as viewed from the side or oblique direction.

This method of showing relief is simple and easy to understand. However, it has the **following disadvantages**:

- It shows the side and oblique view of the landscape, unlike the modern relief map that gives you an overhead view of an area.
- Some geographic details of an area would be hidden from view behind the “backs” of the pictures of hills or mountains.
- Exact heights and slopes of the land forms are not indicated
- It lacks accuracy because it is drawn without scale.

B. Hachures

What are hachures?

Hachures are short disconnected lines that represent slopes. They are drawn in the direction in which water flows. Originally they were used to represent mountains and valleys on simple sketch maps.

Basically, hachures show the **steepness of slopes**. When **slopes are steep, hachures are put close together. For gentle slopes, the hachures are spaced wide apart**. In addition, hachures representing steep slopes are shorter than those representing gentle slopes.



Figure: Hachured map

This approach has significant limitations, such as:

- ✚ Flat areas are unshaded. Therefore, plateaus and plains can be confused.
- ✚ Hachures do not indicate height and exact gradients. They give only qualitative information.
- ✚ Hachures are laborious to draw and can be difficult to read and interpret.

Nowadays, hachures are not used alone. Instead, they are used in combination with contour lines to show landforms like escarpments, depressions and craters. (contour lines are described in a later section of this unit).

C. Hill shading

What is hill shading? What are some of the limitations of hill shading?

Hill shading is also known as oblique illumination. It is a method of showing relief on a map, assuming an oblique light that illuminates the landscape from the northwest corner of the map. Hence the northwest-facing slopes are shaded lighter than are the east-facing and south-facing slopes. The steeper the slope is, the darker it is shaded.

Hill shading offers a quick general impression of the land configuration that it represents. But still it has some limitations such as:

- It does not give absolute altitude.
- It fails to indicate clearly whether the ground is sloping upward or downward.
- It fails to indicate whether the unshaded areas are low or high-level areas. Hence, plateaus and plains can be confused.
- Detailed map information can be obscured by shading.

In general, hill shading is now used in combination with spot heights and contours to overcome some of its drawbacks.

D. Layer Coloring (Layer Tinting)

What is layer coloring?

It is a method of showing relief by using colors. The series of colors for showing different altitudes starts from sea level (see Figure).

Identify the types of colors used to represent the different elevation zones in Figure.

Elevation Zone	
	Above 3000m
	2501-3000m
	1001-2500m
	501-1000m
	0-500m
	Water body

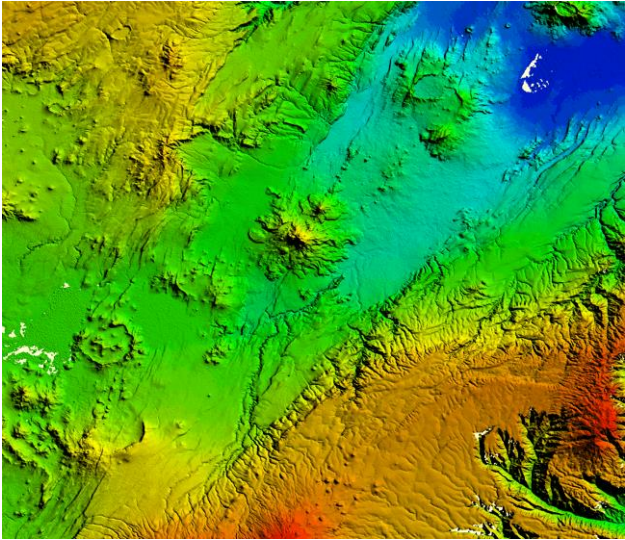


Figure: Map with layer coloring

Layer coloring has the following disadvantages:

- ✓ Color shading does not indicate gradual changes in slopes.
- ✓ The edges of the areas of different colors can suggest nonexistent physical boundaries.
- ✓ Dark colors can obscure details in the areas that they overlie.
- ✓ Some colors can create false impressions in the map reader’s mind. For example, **green might suggest vegetation or a fertile area.**

E. Formlines

What are formlines?

A formline is imaginary pecked or broken line joining points with the same approximate height on a map. Usually they are drawn on topographic maps to show where survey work is incomplete or poorly accomplished. Also, these lines are useful for showing sea depths.

Formlines have the following limitations:

- They are not drawn on a map at a fixed interval of altitude.
- Although they represent the relief of an area, they provide little or no reference to sea level.
- In many cases they are unnumbered.
- They are usually drawn with broken lines.

Modern Methods

F. Contour Lines or Isohyps

What are contour lines? How do contour lines differ from traditional methods of showing relief on maps?

Contour lines are the most common and accurate way of showing relief on modern maps. A shoreline is a good example of a contour line.

Properties of Contour Lines

What are the main properties of contour lines? Why are contour lines more accurate than the traditional methods of showing relief on maps?

General properties of contour lines

Here are some important points about contour lines:

- i. **Contour lines are imaginary lines used on a map to represent relief.** Unlike the lines that represent rivers, boundaries or coast lines, **contours do not really exist on the earth’s surface.** The only contour line that exists both on the map and in the field is the sea level.
- ii. A set of **contour lines is drawn at a fixed height interval.** For example in the following figure, contour lines are drawn at 50-meter intervals. The difference in altitude between two successive contour lines is known as vertical interval (V. I.) or contour interval (C. I.). The V. I. helps us to find out the heights of unnumbered contour lines.
- iii. **Contour lines cannot merge or cross one another** on maps except at vertical cliffs, waterfalls or over hanging cliffs.

For example, two or more contour lines run together and then separate to represent the cliff shown in the figure below.



Figure: Contour lines showing a cliff

The cliff in the preceding diagram is a vertical mountain wall. It rises from 100 meters to 150 meters.

The crossing of contours occurs only in the case of an **overhanging cliff**.

Usually contours representing a cave under an **overhanging cliff** are shown with pecked lines.

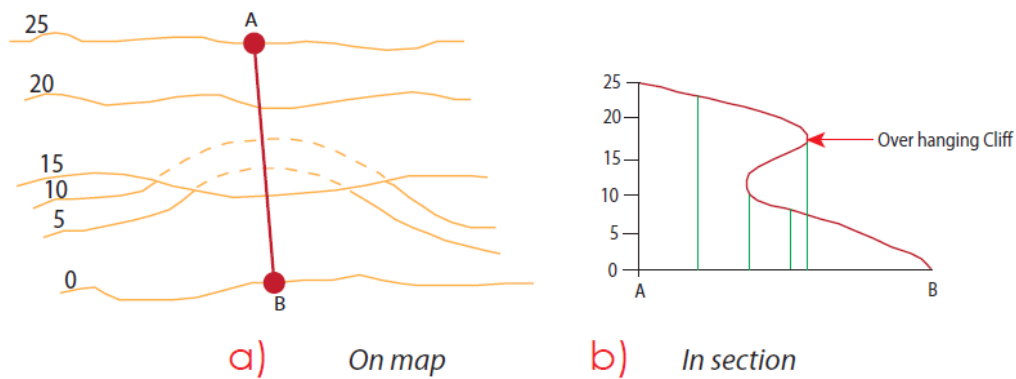


Figure: An overhanging cliff

iv. Contour lines never branch. If you see branching lines on a map, they represent features such as rivers, roads, boundaries, etc.

v. A contour line joins all points of the same altitude. For example, an altitude of 250 m will be on the 250 m contour line. The altitude of any point outside this line will be either greater or less than 250 meters.

vi. Contour lines are always numbered in the direction towards which altitude increases. These numbers can be shown with or without breaking contour lines.

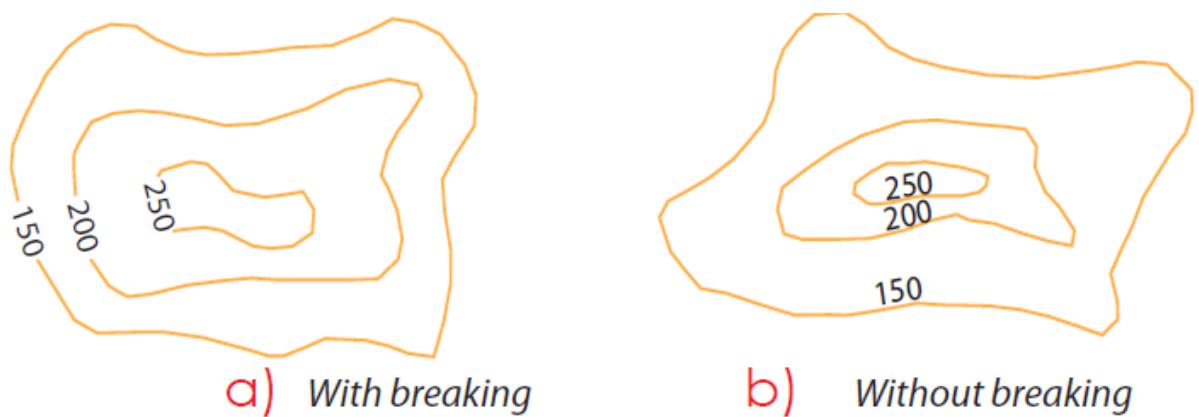


Figure: Numbering of contours

vii. Contour lines indicate the nature of slopes. When contour lines are far apart, they show gentle slopes. But when contour lines are close together, they show steep slopes.

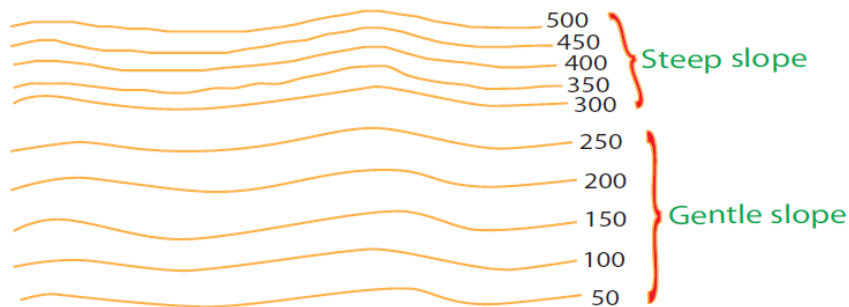


Figure: Contour-line spacing indicating slope steepness

viii. Contour lines can be printed with different thicknesses on a map. This is especially helpful in mountainous areas where altitudes may vary considerably from summits to valley floors. In order to make the reading of contour maps easier, every fifth or tenth contour line is printed thicker than the rest. Such contour lines are called index contour lines, while the rest are called regular contour lines.

ix. Contour lines can show different types of landforms, such as mountains, hills, plateaus, depressions, valleys, spurs, ridges, gorges, passes, plains, etc. Many of these relief features are readily recognized from the shapes of their contour lines.

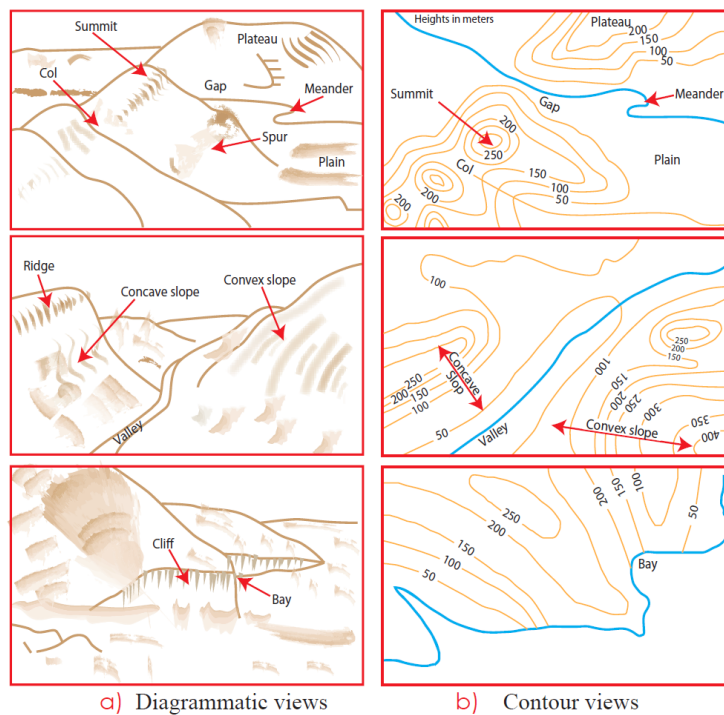


Figure: Landforms represented both diagrammatically and by contour lines

G. Different Methods of Showing Altitudes on Contour Maps

What are the shortcomings of contour lines?

How do you indicate the specific heights of hilltops, roads, railways, and towns?

Contour lines show altitude and relief on modern maps. However, they do not show the specific heights of individual features such as mountain peaks, hilltops, valley floors, towers, towns, roads or railways. Such heights are indicated on maps, using the following methods:

a) Spot heights

- ✓ They are marked on the map with a dot followed by an altitude number:
- ✓ They provide accurate altitudes for individual points, such as those along a road, on a mountain top, or between contour lines.
- ✓ Unlike contour lines, spot heights do not give a good visual impression of the general relief.
- ✓ They exist only on maps.

b) Trigonometrical points

- ✓ They exist both on maps and in the field.
- ✓ They mostly mark features such as hilltops and mountain peaks.
 - On the ground, the relevant feature is permanently marked with a pillar (concrete).
 - On maps, they are shown with a small triangle enclosing a dot, followed by the exact altitude in meters.

c) Benchmarks

- ✓ They indicate precise heights along highways or railways.
- ✓ They are shown on stones, bricks or bronze plates on walls of buildings and other convenient places
- ✓ They are useful for road construction engineers and others who wish to know the precise altitude of a main transport network.

d. Calculating Altitude:

When the altitude of a point on a contour map is not shown by any of the above methods, it can be obtained by measurement and calculation, using the interpolation method. This can be done only if the given point is located between two contour lines. In order to find the altitude of point A in **the figure**, follow the procedures given below.

- i. Draw the shortest possible straight line that passes through point (A) and join the two contour lines adjacent to it.
- ii. Measure the length of this line: = 11 mm.

- iii. Measure the distance on the map between the lower and upper contours up to point (A). They are 6 mm and 5 mm respectively.
- iv. Find the vertical interval between the two contour lines: = 100 m.
- v. Then determine the altitude of the point using the following formula:

$$\text{Altitude (A)} = \text{LC} + \left(\frac{d_1 \times \text{VI}}{D} \right) \text{ or } \text{HC} - \left(\frac{d_2 \times \text{VI}}{D} \right)$$

Where: **d1** is distance of point A from the lower contour,
d2 is distance of point A from the upper contour,
D is distance between the upper and lower contours,
VI is vertical interval,
LC is the lower contour and,
HC is the higher contour.

$$\begin{aligned} \text{Altitude of point A} &= 700 + \left(\frac{6 \times 100}{11} \right) = 754.55 \text{ meters or,} \\ \text{Altitude of point A} &= 800 - \left(\frac{5 \times 100}{11} \right) = 754.55 \text{ meters} \end{aligned}$$

Therefore, the altitude of point A is 754.55 meters.

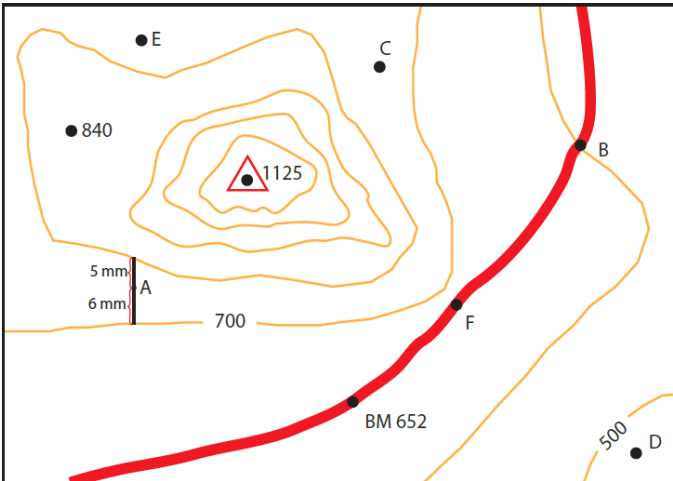


Figure: Altitudes shown on a contour map in different ways Scale 1:50, 000

Slopes and Gradients

A. Slopes on Contour Maps

What is slope? How can we determine the steepness of slopes on contour maps?

What kinds of relations exist between V.I. and slope?

Slope is the upward or downward inclination of a natural or artificial surface. It is a deviation of the surface from the horizontal.

On a map, steepness of a slope depends on:

- The distance between the contours drawn on the map. The closer the contours are, the steeper is the slope representation and vice versa (see Figure 1.24).
- The vertical interval (V.I.) between two successive contours. The bigger the V.I, the steeper is the slope representation and vice versa.

Types of Slopes

How many types of slope do you know? Mention some of them and describe how you can identify them from contour maps.

There are different types of slopes, which include:

i) Even slope

An even slope has a constant gradient from the bottom to the top. Gradient is the degree or rate of a slope. You will learn more about gradient later in this unit.

On a map of an even slope, the contour lines are evenly spaced throughout. For example, study the slope represented in the following Figure.

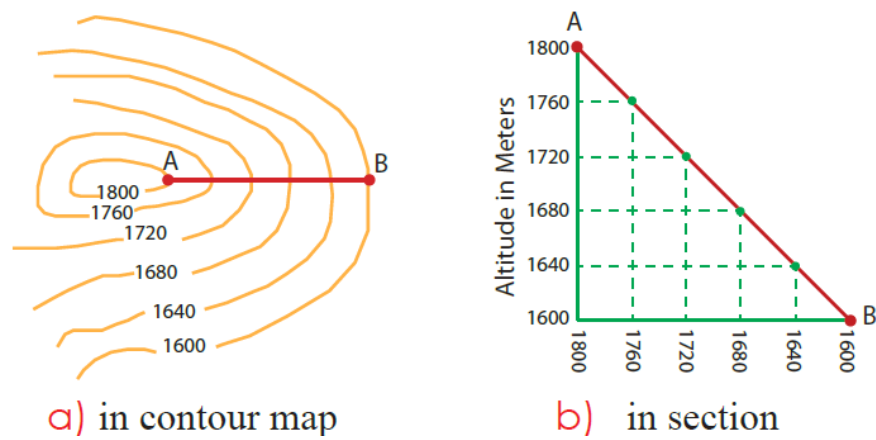


Figure: Even slope

ii. Concave Slope

In a concave slope; the contour lines are widely spaced at the base and are close together at the top. In other words, a concave slope has a steep gradient at the top.

The gradient becomes gentler towards the bottom (see the following Figure).

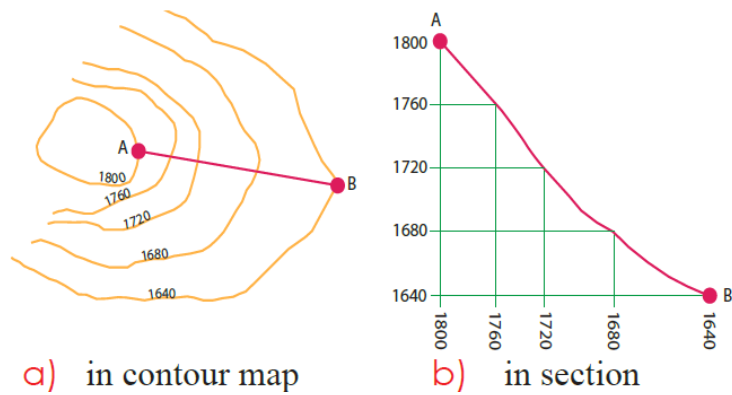


Figure: Concave slope

iii Convex Slope

In a convex slope, the contour lines are close together at the base and widely spaced at the top. The slope has a steep gradient at the bottom that becomes gentler towards the top. (See the following Figure).

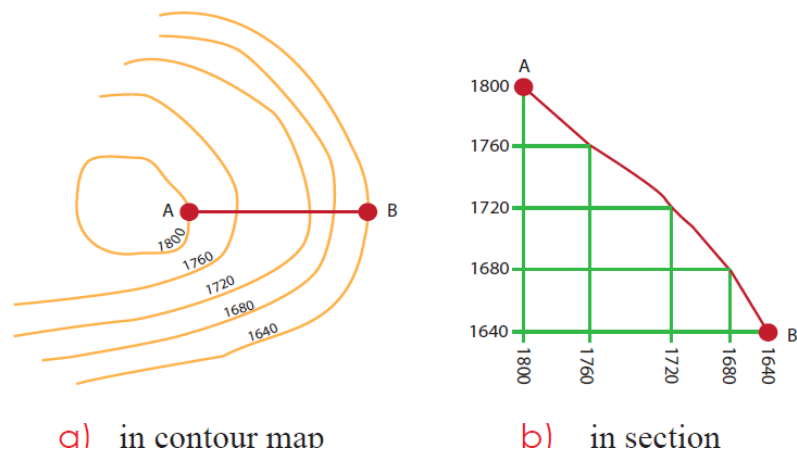


Figure: Convex slope

iv Terraced or Stepped Slope

In a terraced or stepped slope, the contour lines are alternatively close together and far apart in a regular pattern. This means the gradient changes several times between the bottom and the top of the slope (see the following Figure).

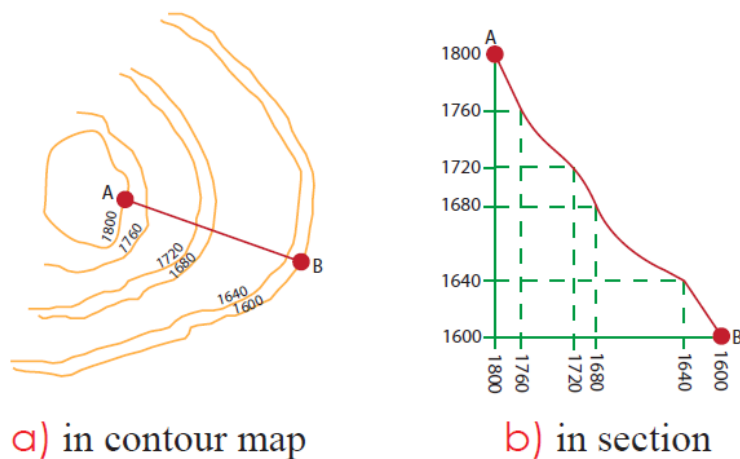


Figure: Terraced slope

v Escarpment

An escarpment is the steep slope of a plateau, especially one where the plateau ends and the lowland starts. You can also identify other two more slopes on either side of a mountain ridge. One slope is steep and the other is gentle. The steep slope is called the scarp slope. The gentler slope is called the dip slope (see the following Figure).

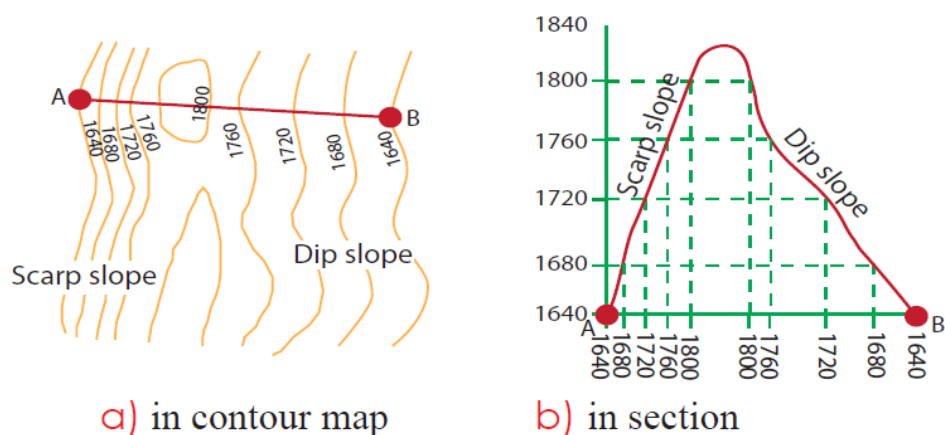


Figure: Dip and scarp slopes

B. Gradient on Contour Maps

What is gradient? What are the three common ways of expressing gradient?

How do you determine the rate of change of slope between two points?

Gradient (GR) is the degree or rate of change of slope or elevation between two points.

It is calculated using altitude difference (vertical distance) and map distance (horizontal distance) between two points. Both AD and MD must be in the same unit of measurement.

It can be expressed in any of these three different ways:

1. As a simple ratio: $GR = \frac{AD}{MD}$
- 2 As a percent: $GR = \frac{AD}{MD} \times 100$
- 3 In degrees: $GR = \frac{AD}{MD} \times 60^\circ$

Where: GR = gradient, AD = altitude difference,
MD = map distance.

Usually we express gradient as a percentage. This expression is the simplest to use, and it is relatively easy to calculate.

REVIEW EXERCISE FOR UNIT

Activity

Calculate the following:

The distance on the map between Addis Ababa and Adama is about 10 cm with the scale of 1:1,000,000 on a certain map of Ethiopia. The average elevations of the two are about 2400 and 1700 meters, respectively. Determine the gradient in ratio, in percent and in degree.

Short Answers: Give short answers to the following questions.

- 1.What is the basic difference between a contour line and a formline?
- 2.Indicate three shortcoming of hachures.
- 3.Define the term “cliff”.
- 4.Explain briefly the difference between spot heights and trigonometrical stations or points.
- 5. Things to Do:

Answer the following questions by referring to Figure 1.37.

6. Find the approximate R.F. of the map _____.

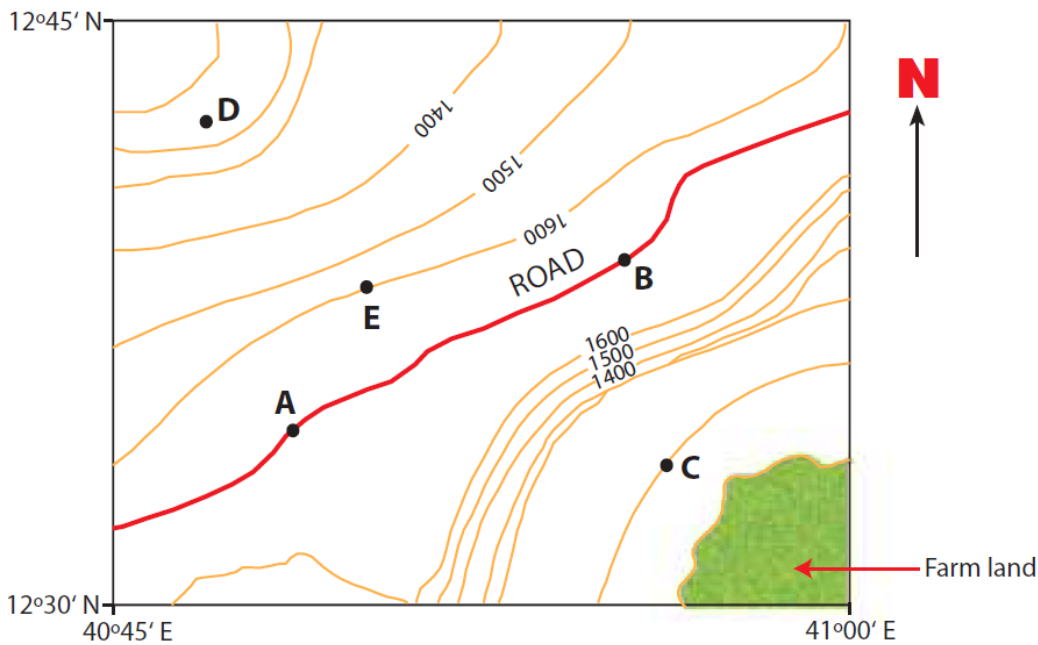
7. Measure the direction (bearing) of:

- i. A from B
- ii. B from D
- iii. D from A

8. Between points C and E, calculate

- a .The field distance
- b .The gradient in;

- i) ratio
- ii) percent
- iii) degree



UNIT THREE
THE PHYSICAL ENVIRONMENT OF THE EARTH (30 Hours)

3.1 Physical Environment of the World

At the end of this section, you will be able to:

- ✓ discuss the concept of universe
- ✓ identify the position of the earth in the solar system
- ✓ explain the origin of the earth
- ✓ demonstrate the structure of the earth
- ✓ describe the geological time scale and major geological events
- ✓ realize the major geological events of Ethiopia
- ✓ describe the concept of continental drift theory.
- ✓ describe the characteristics of each type of rocks.
- ✓ demonstrate major rock distribution in Ethiopia
- ✓ state causes and impacts of soil degradation in Ethiopia.

3.1.1 The Earth in the Universe

The universe is the totality of space and cosmos, in which everything is found. All heavenly bodies, including **all stars**, together **with the sun, comets, meteors, planets** and their **satellites** are found in the very vast space called **the universe**.

A **galaxy** is a large group of stars. The universe contains many galaxies. **The Milky Way** is our galaxy. Within the Milky Way is our solar system.

A **solar system** is a smaller group of heavenly bodies, which includes the sun at the center and the nine planets and their satellites and asteroids.

3.1.2 Forces That Change the Surface of the Earth

A. INTERNAL FORCES

Those forces that drive energy from the interior part of the earth are called internal forces. Internal forces form the ups and downs on the earth's crust by breaking and bending (faulting and folding) it. Forces inside the crust cause **folding, faulting (cracking), volcanism and earthquakes**.

- **Folding**

Folding is one of the internal processes which occur when two forces act towards each other from **opposing sides**. Due to this force, rock layers are **bent into folds**. The process by which folds are formed due to **compression is known as folding**. There are large scale and small scale folds. Large-scale folds are found mainly along destructive plate boundaries.

Types of folding: different types of folds are created, based on the nature of the forces applied on bedrock.

- If the fold is upward and convex, it is called **anticline**.
- If the fold is downward, it is called **syncline**.

Fold Mountains

- Fold Mountains are formed by crust which has been uplifted, and folded by compressional forces. They are formed when two plates move towards each other. The compressional force which is created as a result of this movement pushes sedimentary rocks upwards into a series of folds.
- Fold Mountains are usually formed from sedimentary rocks and are usually found along the edges of continents. This is because the thickest deposits of sedimentary rock generally accumulated along the edges of continents.
- **There are two types of Fold Mountains:-**
Young fold mountains (10 to 25 million years of age, example, the Atlas, Rockies and the Himalayas) and **Old fold mountains** (over 200 million years of age, example, the Cape Range, the Urals in Russia and the Appalachians of the USA).

Many ranges of mountains have been formed by folding. **The Andes, the Rocky mountains, the Alps, the Himalayas and the Australian Alps are some examples.** The Atlas mountains in north west Africa and the Cape Range in South Africa were formed by **folding**. This process of mountain building is called ***orogeny***.

- **Faulting**

Movements in the crust of the earth sometimes make cracks. These cracks are called faults. Faulting can be caused by either lateral or vertical forces, which can be either tensional or compressional. **Tension causes a *normal fault*, and compression causes a *reverse fault*.**

Major features formed by faulting include rift valleys and block/Horst Mountains.

- **Rift Valleys**

What is rift valley? How are rift valleys and Block Mountains formed?

A **rift valley** is a linear shaped lowland area between highlands or mountain ranges created by geologic rifts or faults.

A **rift valley** is a valley formed by faulting. When two parallel faults occur on the surface of the earth, and when the land between the two faults sinks down, a rift valley is formed.

The largest rift valley in the world is the **East African Rift Valley**. It extends from Syria to Mozambique, passing through the Red Sea, Eritrea, Ethiopia, Kenya, Tanzania, DR Congo, Rwanda and Burundi.

The total length of the East African Rift Valley is about 7,200 km, of which 5,600 km is in Africa.

The Ethiopian Rift Valley is a part of the East African Rift Valley. It extends from northeast to south west. Features found in the Rift Valley include active volcanoes, lakes, hot springs and fumaroles.

Block (Horst) Mountains

What is Block Mountain? How do block mountains form?

Block Mountains are formed when land between two parallel faults is pushed upward due to pressure from inside the earth. If there are two parallel faults, the crustal block between them may either rise to produce a Horst (block) mountain, or fall, to produce a rift valley.

Examples:

- ✓ The Sierra Nevada Mountains in North America.
- ✓ The Harz Mountains in Germany.
- ✓ The Afar block mountain in Ethiopia.
- ✓ The Ruwenzori in Africa.
- **Volcanism**

What is volcanic activity? *What landforms are associated with volcanic activity?*

Volcanic activity is another internal force which changes the surface of the earth. It is caused by internal movements within the earth.

Volcanism is the process by which **magma**; gases, water vapour, ashes and other solid materials are forced out to the surface. Inside the earth the temperature is very hot. This high temperature changes rocks into molten magma. When this magma reaches the surface, volcanic activity takes place. When the magma emerges on to the surface, it cools and hardens. It is then **called lava**.

Magma reaches the earth’s surface through two kinds of holes. They are **vents and fissures**. Magma may force its way violently through a small hole called a vent. If lava emerges via a vent, it builds up into a volcano (cone-shaped mound), and if it emerges via a fissure, it builds up to form a lava plateau or lava flow. Magma may pour quietly through long cracks (fissures) onto the earth’s surface.

If the magma flows to the surface through a vent, a crater is formed. Sometimes a volcano erupts very forcefully. When this happens, the top part of the volcano is blown away. This forms a large crater called a ***caldera***. Water collects in the crater or in the caldera and forms a lake. We call this a **crater lake or caldera lake**. In Ethiopia there are many crater lakes such as Zuquala, Wonchi and Dendi.

If an eruption begins again in a caldera, a new small cone-shaped volcano is formed inside the caldera. These are known as *caldera cones*. Part of the magma may not reach the earth’s surface, and when this magma cools, solidifies and forms rocks inside the crust, features such as **batholiths, lacoliths, sills and dikes**, are formed.

- A batholith is a very large mass of magma which accumulates in the crust. It is the largest structure.
- A **laccolith** is a mushroom shaped body of intrusive igneous rock.
Smaller than a batholith.
- A **dike** is formed when magma solidifies in a vertical or near-vertical crack.
- A **sill** is formed when magma solidifies horizontally or nearly horizontally along a bedding plane.

Types of volcanoes

Active volcano	Erupts from time to time	Erta’li, Fentale Dubbi and
Dormant volcano	Has not erupted for a long time but may erupt again in the future	Tatali and Dabbahu
Extinct volcano	Has not erupted within historic time	Mt. Zuquala, Ras Dashen and Batu

Importance of volcanic eruptions:

- Give us some ideas about the interior of the earth
- Provide fertile soil.
- Provide hot springs (with medical value).
- Generate geothermal energy.
- Help in the formation and concentration of minerals.
- Help in the creation of new land.

- Earthquake

What is an earthquake? Why do earthquakes occur?

Earthquakes are sudden movements in the earth’s crust. They are caused by internal movements deep down inside the earth. Earthquakes are frequently associated with faults. They take place along fault lines where the earth’s crust is weak. When an earthquake occurs, vibrations from the centre spread out in the form of waves in all directions.

The point at which an earthquake originates is called the **focus**. The point on the earth’s surface immediately above the focus is called the **epicentre**. As the vibrations spin out from the centre, the damage they cause becomes less and less.

The intensity of an earthquake is measured by an instrument called a **seismometer**, and is recorded on a seismograph. It records the vibrations produced by an earthquake. The scale which gives the magnitude is called the **Richter scale**. It ranges from 0 to 9. Each number of this scale indicates a tremor that is ten times stronger than the next lower number. An earthquake with a magnitude of 4.0 is ten times stronger than one that measures 3.0. Readings of 7.0 or higher indicate a strong or major earthquake. The strongest ever recorded earthquake was the Valdivia earthquake in Chile that occurred on May 22, 1960 (9.5 on the Richter scale).

Table 2.2: Richter scale values and the corresponding magnitude of earthquakes

The Richter Scale	Effects
< 3.5	only by instruments (seismometers)
3.5 – 4.8	Feels like a lorry passing
4.9 – 5.4	Loose things fall
5.5 – 6.1	Walls crack
6.2 – 6.9	Chimneys fall, some buildings collapse
7.0 – 7.3	Many buildings fall, landslides
7.4 – 8.1	Most buildings and bridges are destroyed
> 8.1	Total destruction

In addition to destruction of life and property, an earthquake causes:

- displacement of parts of the earth's crust vertically or laterally.
- landslides and deep cracks in surface rocks.
- the devastation of cities, fires and diseases.
- the rise or lowering of the sea floor.

About 80% of all earthquakes occur in three regions. They are

- ❖ Around the Pacific Ocean zone. The largest earthquake and volcano zone lies along the edges of the Pacific Ocean. This zone is known as the Pacific Ring of Fire.
- ❖ Across Southern Europe and Southern Asia.
- ❖ The west-coast areas of North and South America.

The two most recent earthquakes in Ethiopia measuring more than 5 on the Richter Scale occurred:

- July 14, 1960, near Lake Shalla with a magnitude of 6 on the Richter Scale.
- June 2, 1961, in Karakore.

B. EXTERNAL FORCES

❖ Weathering

What is weathering? What is the effect of weathering on landforms?

External forces can lower the level of the land by washing it away, and this process is called **denudation**. They also can raise the level of the land by deposition.

Denudation consists of weathering and erosion.

Weathering includes **disintegration** (physical weathering), which breaks rocks into smaller pieces and **decomposition** (chemical weathering), which forms new substances.

○ Physical (Mechanical) Weathering

What is physical weathering? What are the main agents of physical weathering?

Physical weathering breaks the rocks into smaller pieces. **Its main agents (causes) are temperature changes, frost action and the action of plants and animals.**

The effects of temperature changes: The temperature variation between day and night causes rock to expand and to contract. This process causes cracks to develop. In time, the cracked layer peels off and falls to the ground. This process is called **exfoliation**.

The effects of frost action: Due to frost action, rock breaks up into pieces and these fragments accumulate around the lower slopes of the rock. This material is called **scree**. Frost action is very common in the winter season in the temperate zone and in some high mountains all year round.

The effects of plant and animal action: Plants and animals also cause weathering. For example, seeds may fall in cracks of rocks. If water collects there, it forms suitable conditions for the seeds to germinate and grow. As plants develop their roots may push the rock apart. Some animals burrow, and this also helps to break up rocks.

Chemical Weathering (Decomposition)

What process is important in chemical weathering? What are main agents of chemical weathering?

Chemical weathering is a process that forms new substances, and it is affected by the minerals in the rock. Its main agents are rain action and plant and animal actions.

As rain water passes through the atmosphere, it takes in carbon dioxide (CO₂) and forms a weak carbonic acid. When this acid water comes into contact with rock, it begins to dissolve minerals in the rock. The rate at which rock dissolves depends on the type of rock. Limestone, for example, dissolves very quickly. This process is known as **carbonation**.

$H_2O + CO_2 \Rightarrow \text{carbonic acid} \Rightarrow \text{dissolves and erodes limestone and forms caves.}$

Example:
Sofomer Cave along the River Weiyb in Bale.

In underground rivers, seeping rain water continues to dissolve the limestone beneath the surface, gradually forming passages and caves. These caves contain features such as stalactites, stalagmites and pillars.

- ✓ **A stalactite** is a limestone column that hangs down from the ceiling of the cave.
- ✓ **A stalagmite** is a limestone column that builds upwards from the floor of the cave.
- ✓ **A pillar** is formed when a stalactite and a stalagmite join together.

When rain water dissolves oxygen and reacts with iron in rocks, the rocks become rusty.

Pollution in towns and cities increases chemical weathering.

How do plants and animals act as agents of chemical weathering?

Plants absorb minerals, and decaying vegetation produces organic acid, which causes a further breakdown of minerals.

Bacteria in the presence of water break down certain minerals in the soil.

Leaching is a major soil-forming process. It occurs when substances are dissolved in water that percolates through soil. Such substances include soluble chemicals that move out of biological tissues into soil - for example, rainfall causes potassium and other ions to be lost by foliage.

- **Erosion**

What is erosion? What are the major agents of erosion? What are the major types of erosion? What are the characteristics of the agents of erosion?

Erosion is the transporting of weathered material by various natural forces such as moving water, wind and moving ice.

Erosion occurs when particles of rock or soil are:

- washed away by a river
- removed by waves of the sea
- crushed under a glacier
- blown away by the wind

Erosion by Running Water

How does running water cause erosion? What processes are included in erosion?

Rivers are the most important of all natural agents which help in shaping the earth’s surface. The work of running water includes **eroding, transporting and depositing** eroded material.

There are three types of running water erosion:

1. **Sheet erosion:** occurs when surface water moves in a wide flow.
2. **Rill erosion:** occurs when surface water cuts relatively small channels.
3. **Gully erosion:** occurs when floods cut deep wide gorges.

The course of a river, from its source to its mouth, can be divided into 3 stages. The action of the river is different in these three parts. The stages are upper course, middle course and lower course.

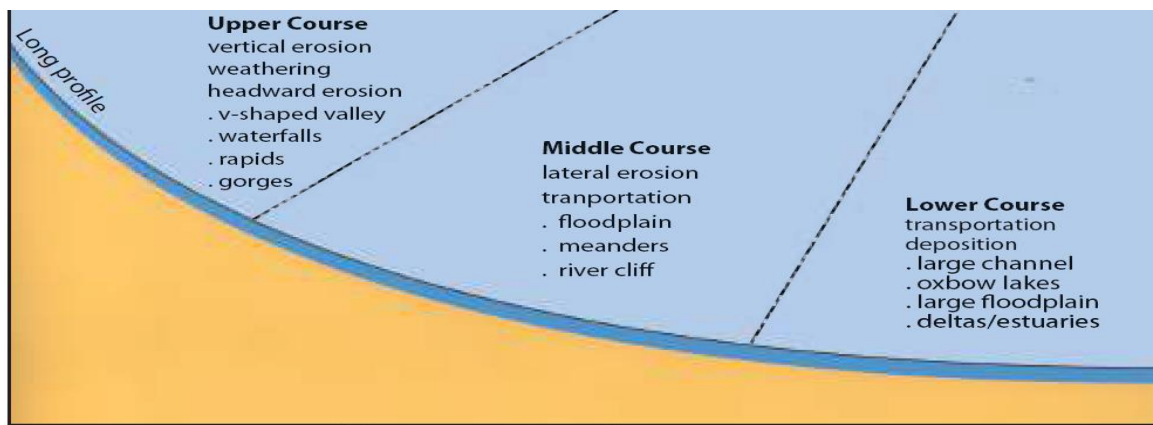


Fig: The three stages of a river

Upper Course

In this stage the river water is usually small in volume. As the river flows very fast down steep-slopes, a **V-shape valley, waterfalls and deep gorges** are formed.

The fast flow of the river causes vertical erosion and destruction. The **V-shape valley** has steep sides and a narrow floor. The fast flowing river cuts down deeply into the land.

Waterfalls are caused by sudden drops in the level of rivers. Waterfalls are formed when water flows over hard rock which cannot be eroded easily, while soft rocks are easily eroded. The hard rock produces an overhang, and the water flows over it as a waterfall.

The Middle Course

During a river's middle course, the river valley becomes wider and larger. The river may receive waters of many tributaries, which increase the volume of water. Wide-floored valleys with gentle slopping sides are the main features of the middle course of the river. Instead of taking the most direct course possible, the river begins to meander. **Meanders** are pronounced curves in the course of a river.

The Lower Course

The river flows fast, meandering over wide plains, and makes widespread deposition. The load is so large that deposition occurs. **Flat floodplains, big meanders, levees, ox-bow lakes and deltas** are the main features of this course.

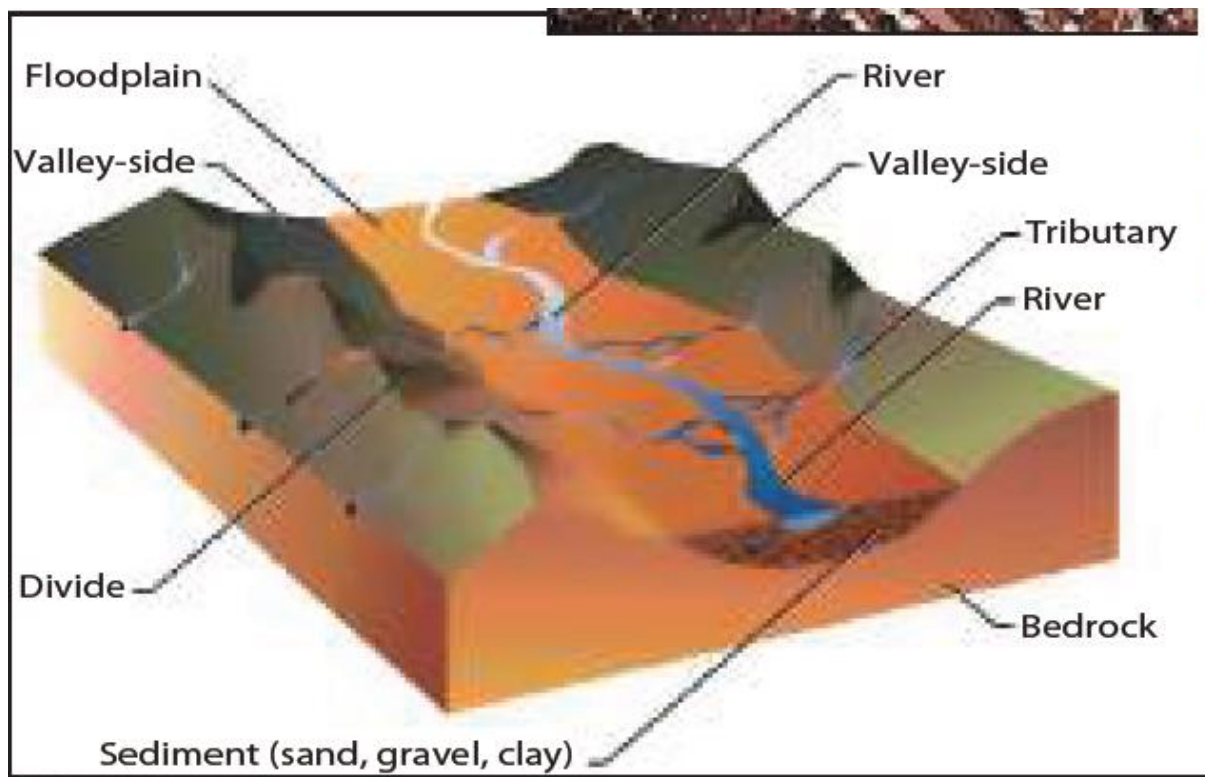


Fig: Features of the lower course of a river

- ❖ **Floodplains** are broad flat areas which border with the lower course of a river and are sometimes flooded by the river. They are covered with fertile alluvial soils which are deposited by the river when flooding.
- ❖ **Levees** are narrow ridges of alluvial deposits found along the bank of a river.
- ❖ **Ox-bow lakes** are crescent-moon shaped lakes created due to meanders that have been abandoned. They are formed when meanders are cut off from the main river channel.
- ❖ **Deltas** are usually triangular areas of land which are usually formed at the mouth of rivers.

Erosion and Deposition by Sea Waves

What are coastal landform features produced by wave erosion?

Waves are formed when wind moves over the surface of the sea. This causes the particles of water to move in a circular motion, which forms a wave. This movement of water in the sea clashes against coastal lands and picks up rock particles and throws them into the sea as sediments. The work of the sea along the coast includes erosion, transportation and deposition. Some of these features formed along the shoreline are **beaches, spits and lagoons**.

- ✓ **Beach** is a strip of land along the sea coast covered with various types of sediment.
- ✓ **A Spit** is a narrow ridge of sand or shingle. It projects into the sea but is attached to the land at one end.
- ✓ **Lagoon** is an area of saltwater separated from the sea by loose sandbanks.

Wind Erosion and Deposition

What is the most active agent of erosion in desert regions? What is the most common type of wind deposit?

Wind erosion is common in desert and semi-desert areas. Wind erosion and deposition form different landforms such as **sand dunes, barchans and loess deposits**.

- ✓ **Sand dune** is a small hill of sand formed by the action of the wind.
- ✓ **Barchan** is a sand hill that has a crescent-moon shape.
- ✓ **Loess deposit** is a deposition of fertile soil in the desert by wind.

3.1.3. Weather and Climate

At the end of this section, you will be able to:

- ✚ explain the meaning of atmosphere;
- ✚ discuss the composition and layers of the earth’s atmosphere;
- ✚ explain weather and climate;
- ✚ express the concept of temperature;
- ✚ appraise the variation of temperature;
- ✚ demonstrate how to measure and record temperature data;
- ✚ compute normal temperature lapse rate;
- ✚ interpret temperature data;
- ✚ explain the formation of rain; discuss the types of rainfall;
- ✚ relate the various roof slopes of houses in various climatic regions to the respective types of rainfall;
- ✚ practice measuring and recording rainfall data;
- ✚ differentiate types of winds (local, monsoon and planetary winds, including cyclones and anticyclones); relate direction and deflection of winds to the earth’s rotation;
- ✚ interpret wind speed and direction from wind gradient map;
- ✚ explain how conditions of wind affect structures of buildings and crop production;
- ✚ Identify types of atmospheric pressure;
- ✚ relate atmospheric pressure with temperature and altitude;
- ✚ demonstrate the pressure belts of the world;
- ✚ develop the skills of measuring and recording atmospheric pressure;
- ✚ analyze the position of the sun at various latitudes at noon time of Dec. 22/June 21;
- ✚ examine the impact of latitude on temperature;
- ✚ justify the effect of altitude on the characteristics of temperature, rainfall and air pressure;
- ✚ compare and contrast the condition of rainfall and temperature between places of coastal and interior areas;
- ✚ express the meaning and types of ocean current; identify the impacts of ocean currents;
- ✚ recognize the effects of ocean currents on temperature and rainfall on land surfaces;

Earth and Atmosphere

What is atmosphere? How do you explain the importance of atmosphere for human beings or for all life forms? The air that surrounds the earth is called the atmosphere. It is an envelope of transparent colorless, tasteless and odorless gases found above the earth’s surface. Composition of the Atmosphere: The earth’s atmosphere is a mixture of gases, suspended dust particles and condensed moisture droplets which are collectively known as aerosols. The gases are different in their volume

Table: Gases of Earth’s atmosphere

Major gases		Minor gases		Rare gases
Nitrogen	78%	Argon	0.93%	Hydrogen, ozone, methane, neon, helium, krypton , xenon, carbon monoxide
Oxygen	20.95%	Carbon dioxide	0.03%	

Table 2.4: Atmosphere layers and their characters

Name of the layer	Average height	Major characteristics
Troposphere	Extends from sea level to 8/16 kms	⇒ Contains 75% of the atmospheric mass. ⇒ Uniformly, temperature decreases with an increase of altitude. ⇒ The top boundary is known as the tropopause, which is characterized by jet streams (high velocity winds).
Stratosphere	Extends up ward to 50 kilometres	⇒ Has constant temperature. ⇒ High concentration of ozone gases. ⇒ Its upper limit is called the stratopause.
Mesosphere	Extends from 50 to 80/85 kilometres	⇒ It is the coldest part of the atmosphere. ⇒ Its upper surface is known as the mesopause. ⇒ Temperature decreases to nearly -100°C at the top of the mesosphere. ⇒ Meteorites burn and disintegrate because of friction here.
Thermosphere ⇒ Ionospheres ⇒ exosphere	Extends from 80/85 kilometres upward into space	⇒ Have extremely low density ⇒ Very little heat can be absorbed, held or conducted. ⇒ Temperature rises as high as 1200°C. ⇒ The ionosphere is a layer of electrically charged particles. These electrons and ions are useful for communication because they reflect radio waves. ⇒ The exosphere begins at an altitude of about 500 to 700 kilometres above the earth's surface and extends to interplanetary space.

Meaning of Weather and Climate

What is the condition of the atmosphere today? What is weather? What is climate? How is climate different from weather?

❖ **Weather** is the condition of the atmosphere over a short period of time. **Weather** includes daily changes in precipitation, air pressure, temperature, wind, etc. **Weather** refers to atmospheric conditions in a given location.

What is the weather like in your locality today?

❖ **Climate** is the average of all weather conditions of an area over a long period of time. These conditions include average temperature, air pressure, humidity, and days of sunshine for a period of 30 years. Climate tells us what it is usually like in the place where we live.

Major Elements of Weather and Climate

The major elements of weather and climate are **temperature, rainfall, winds, air pressure, clouds**, etc.

- **Temperature**

What is temperature?

Temperature is the amount of hotness or coldness of an object. The sun is the primary heat source for the earth and its atmosphere. The sun’s energy is called **insolation or solar radiation**, and this turns into heat energy at the earth’s surface.

How is energy transferred in the atmosphere?

Not all the energy that originates from the sun reaches the earth’s surface

- ❖ Heat transfer takes place in **three ways**. These are
 - Radiation
 - Conduction
 - Convection

Radiation is the transfer of energy from one body to another by means of electromagnetic waves. Energy transmitted from the sun reaches the earth’s surface through the process of radiation. Electromagnetic waves usually travel through empty space. When these electromagnetic waves come in contact with an object, they transfer the heat to that object. The sun warms the earth through radiation of electromagnetic waves.

Conduction refers to the transfer of heat through molecular contacts within and between bodies. Molecules are always in motion. The process of conduction is more important in solids. Air and water are poor conductors of heat. Convection is the transfer of heat due to differences in density. As gas or liquid either warms and rises or cools and falls, it creates convection currents. Convection is the method by which heat moves through gases or liquids. As gas or liquid is heated, it warms, expands and rises because it becomes less dense. When the gas or liquid cools it becomes dense and falls. Heat gained through radiation or conduction usually transfers by convection.

Measuring and Recording Air Temperature

What is the instrument that is used to measure temperatures? Explain how air temperature is measured and recorded?

We measure temperature with thermometer. **There are two types of thermometers:** maximum and minimum thermometers.

A maximum thermometer is a mercury-in-glass thermometer that has a constriction near the bulb end. When the temperature of air rises, the mercury in the thermometer expands and forces its way into the stem past this constriction. But when the bulb cools, none of the mercury above the constriction moves back into the bulb. Therefore, the length of the mercury in the stem remains the same. The end of the mercury thread, which is the farthest from the bulb, registers the highest temperature reached in a day.

- The freezing point of mercury is -38.83°C , and the boiling point is 356.73°C
- Alcohol freezes at a temperature of negative one hundred thirty degree Celsius (-130°C)

A minimum thermometer has alcohol as its liquid, and it sets a metal index.

When the temperature falls, the alcohol column drags the index towards the bulb end. When the temperature rises, the alcohol column expands and runs past the index without disturbing it. Thus, the end of the index, moves the farthest from the bulb and gives the lowest temperature attained in a day. Alcohol thermometers may be used to measure temperatures from -130°C (freezing point of alcohol) to 785°C (boiling point of alcohol). The standard thermometer for environmental measurements needs only to cover the range between -30°C to 50°C . **Maximum and minimum thermometers** are kept in a box-like shelter which is known as a **Stevenson screen**.

The temperature of the air changes from time to time. Typically we measure the daily and annual variations. The change between the highest and the lowest temperatures during 24 hours of a day is known as the **daily march of temperature or the diurnal range**.

The change of temperature from month to month within a year is known as the **annual march of temperature**. To describe this temperature variation, we have to use records for a long period of time. We use words like **average and range** to indicate the variations.

Daily average (mean) temperature: is obtained by **adding the maximum and minimum temperatures of a day and dividing the sum by two**.

Example 1:

If the maximum daily temperature is 25°C, and the minimum daily temperature is 5°C, daily average temperature = $\frac{25^{\circ}C + 5^{\circ}C}{2} = 15^{\circ}C$

- Monthly average (mean) temperature is calculated by adding all daily averages and dividing the sum by the number of days of the month.
- Annual average is obtained by adding the average monthly temperatures and dividing the sum by 12.
- Daily (diurnal) range is the difference between the maximum and minimum temperature in a day.

Example 2:

If the maximum temperature is 25°C, and the minimum temperature is 5°C,

- ✓ Daily range = maximum – minimum = 25°C – 5°C = 20°C.
- ✓ Annual range is the difference between the temperatures of the hottest and coldest months in a year.

Example 3:

If the hottest month is 40°C, and the coldest moth is – 10°C,

Annual range = 40°C – (–10°C) = 50°C

Table 2.5: Average annual temperature for Addis Ababa

Months	J	F	M	A	M	J	J	A	S	O	N	D
Annual Max.Temp (°C)	23.3	24.3	24.8	24.2	24.4	22.8	20.6	20.6	21.3	22.3	22.6	22.8
Annual Min.Temp (°C)	9	10.8	11.6	12	12.3	11.2	11.3	11.2	11	9.9	8.7	8.1

Activity

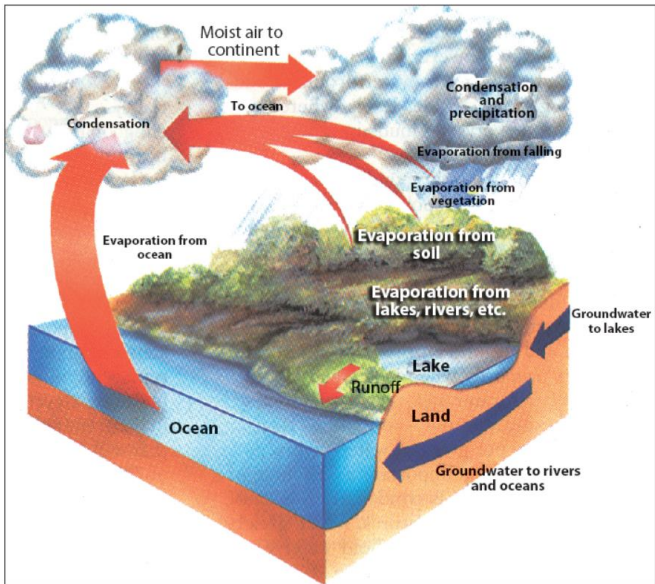
- By referring to the above Table ,
 - Calculate annual range of temperature.
 - Calculate annual average (mean) temperature.
 - Convert the data into graphs
- When do the maximum and minimum temperatures of the month occur in Ethiopia? Why?

Rainfall

What is rainfall?

Rainfall is liquid precipitation. Any moisture that falls from the clouds towards the earth’s surface is called precipitation. Precipitation may occur in the form of rain, snow, hail, sleet and drizzle. Precipitation is part of the water

cycle or hydrological cycle. The water cycle begins as water is changed from liquid to vapour by evaporation and transpiration of water vapour. Once water vapour is formed, it expands and cools. Then, **condensation occurs**, forming clouds, and the water falls as snow, sleet or rainfall. The whole process is powered by solar energy and is repeated continuously. This whole process is called the hydrological cycle.



Evaporation is the process by which liquid water is converted into gases.

- *Transpiration is the transfer and change of water from plants to water vapour in the air.*
- *Evapotranspiration is the combined loss of water through the process of evaporation and transpiration.*
- *Condensation is the process by which vapour becomes liquid.*
- *Sublimation is the process in which ice changes into water vapour without first becoming a liquid, and vice versa.*

Water is a unique substance, because it can exist in three states as liquid, solid and gas) in the atmosphere. Water either absorbs or releases heat when changes from one state to another.

Types of rainfall

What are the types of rainfall? Explain their formation.

Rain is given three different names according to the different ways in which moisture is forced to rise. They are:

- ✓ Convectonal rainfall
- ✓ Orographic or relief rainfall
- ✓ Cyclonic rainfall

- 1. Convectonal rainfall:** When the ground surface is heated by the sun, the air above it is warmed up. At high altitudes, the water vapour cools, condenses to form clouds and falls as rain. This type of rainfall is common in humid areas where temperature is high throughout the year.
- 2. Orographic (relief) rainfall:** occurs when moist air is forced to rise over mountains. As it rises, it cools, then condenses and falls as rain. Almost all orographic rainfall falls on the windward side of mountains.
- 3. Cyclonic or Frontal rainfall** when two air masses (warm and cold) meet, they do not mix freely with each other. They remain separated with a boundary surface between them. The warmer and less dense air is forced to rise over the colder and heavier air. As the warmer air rises, it cools and condenses. Then clouds form and rain falls. The place where warm air and cold air meet is called a **front**. Frontal rainfall is very common in the middle and high latitudes (60° north and south from the equator).

Measuring and Recording Rainfall

Rainfall is measured using an instrument called **rain gauge**. A rain gauge consists of a wide-mouthed funnel placed over a cylindrical container. Rain water passes through the funnel into the container below. The water in the container is poured into a measuring cylinder, and then the amount of rainfall is measured in millimeters and is recorded.

Table: Rainfall data for Debre Markos

Months	J	F	M	A	M	J	J	A	S	O	N	D
Rainfall (mm)	18.3	12.1	57.5	55.1	173.1	113	256.5	293.8	210.8	12	91	9.4

Activity

Using the preceding rainfall data for the Debre Markos station, perform the following tasks.

1. Calculate the total annual rainfall
2. Identify the season of heaviest rainfall.
3. Draw a line graph to illustrate each monthly total rainfall.

Air Pressure

What is air pressure? Explain how to record and measure pressure?

The air around us has weight. This weight exerts pressure on the surface of the earth. We call this atmospheric pressure. Atmospheric pressure is not the same all over the earth, and it is not the same even in one place all the time.

Pressure is measured by a mercury barometer. Normal pressure, at sea level, is about 760 mm/1013 mb.

The distribution of pressure over the earth’s surface **depends on**

- (1). The altitude of places above sea level and,
- (2). Most importantly, on temperature.

Pressure decreases with an increase in altitude. This explains why air pressure is highest at sea level and decreases with increasing altitude. High temperature makes air expand, so that it has a lower density and pressure.

Low temperature makes the air to contract, resulting in a higher density and creating an area of high pressure.

The distribution of air pressure over the globe is known as the ***horizontal distribution of pressure***. Pressure distribution can be shown on a map. Lines connecting all places that have the same pressure are called **isobars**.

❖ **Global Pressure Belts:**

1. Equatorial low pressure belt (Doldrums):

- Located from 5° North to 5° South.
- There is tremendous heat, and thus warm air, creating low pressure. Also, the centrifugal force is very high at the equator, because the earth’s velocity of rotation is high. Hence, the air masses tend to be thrown outwards, resulting in low pressure.
- Wind speed is low, that is why this pressure belt is called the doldrums (Belt of Calm).

2 Tropical high pressure belt (Horse Latitude):

- Located from 30° to 35° North and South.

- Except for two months, temperature is usually high.
- Here the pressure is high, because pressure depends on the rotation and movement of air (as winds from the Doldrums rise up and accumulate here. Also winds from the sub-polar low pressure belt accumulate here).

3 Sub-polar low pressure belt:

- Located from 60° to 65°, North and South.
- Here the low pressure is created because of intense high pressure at the poles.

4. Polar Highs

- Located near the north and south poles.
- The polar zones have permanent centers of high pressure known as polar highs.

The pressure distribution over the earth's surface is not a continuous belt. Except in the higher latitude of the southern hemisphere, they form belts due to the small land areas which do not affect the free flow of the atmosphere. However the position of the pressure belts and cells does not remain fixed in one position.

They move north or south with the apparent movement of the sun. During the northern hemisphere's summer, the sun is overhead north of the equator. The pressure belts then shift northward by a few degrees from their average position. During the southern hemisphere's summer, the sun is overhead south of the equator. As a result, the pressure belts move southward by a few degrees from their average position.

Wind

What is wind?

Wind is air in horizontal motion. Winds have speed and direction. Wind force (speed) and wind direction are affected by:-

- Pressure gradient
- Frictional force
- Coriolis force

On weather maps, pressure is indicated by drawing **isolines** of pressure, called isobars. The difference in distance between Isobars is called the **pressure gradient**.

If the isobars are closely spaced, we can expect the pressure gradient force to be great, and wind speed to be high. In areas where the isobars are spaced widely apart the pressure gradient is low and light winds normally exist. High speed winds develop in areas where isobars are closer.

Wind speed is often measured with **anemometer**.

Winds are named according to the compass direction of their source. Wind direction is measured as the direction from where wind comes.

For example, a southerly wind comes from the south and blows to the north. Direction is measured by an instrument called **wind vane**.

Types of Winds

There are three types of surface winds. They are:

- ✓ **Planetary**

- ✓ **Monsoon**
- ✓ **Local winds.**

Planetary winds and their relationship with pressure belts: Planetary winds blow over large areas of the earth’s surface. They are closely associated with the world pressure belts. Winds blow from areas of high pressure to areas of low pressure.

- ❖ The most common planetary winds are **trade winds**, **westerlies** and **polar easterlies**. They have wide influence over the earth’s surface.

Monsoon winds: They are seasonal winds whose movements are controlled by pressure that differs during different seasons.

Monsoon winds are very common in South and South East Asia. Seasonal changes in the direction of these winds are caused by the unequal heating of land and water surfaces. The direction of monsoon winds changes between summer and winter.

Local winds: They affect only limited areas and blow for a short period of time.

They affect climate conditions on a small scale. Local winds are caused by the nature of the physical features of the area. The main local winds are:

- Land and sea breezes
- Mountain and valley breezes

1. **Land and sea breezes:** These winds are common along coastal areas. These winds change their directions daily and affect very small areas. During the daytime, temperature on the land is higher than on the water/sea. So low pressure is formed on the land, while it is relatively high on the sea. Wind blows from the sea towards the land. This is known as a **sea breeze**. At night the land is colder than the sea. So low pressure develops over the sea. The wind blows from land towards the sea. This is known as a **land breeze**.

2. **Mountain and valley breezes:** Mountain and valley breezes arise from contrasts in temperature between a valley floor and mountain slopes. During the day the air at the bottom of the valley becomes warmer. As a result it expands and rises along the mountain slopes. This is known as a **valley breeze**.

At night the wind over the slope of the mountain becomes cool. Then this cooler and heavier mountain air slides down slopes towards the valley. This is called a **mountain breeze**.

Clouds

What are clouds? How are clouds formed? What are the major types of clouds?

A cloud is a dense concentration of very fine invisible water droplets, sleet or ice crystals. Clouds are formed by the condensation of water vapour below the dew point in the atmosphere. There are varieties of clouds, based on their height, appearance and shape.

Table: Types of clouds

Group	Cloud type	Description
High clouds > 6000 m	Cirrostratus	Thin, wispy, appears in sheets.
	Cirrus	Thin, wispy, filamentous or curly, mostly composed of ice crystals.
	Cirrocumulus	Small, puffy, patchy and/or with a wave-like appearance.
Middle clouds 2000 – 6000 m	Alto cumulus	Medium-sized, puffy, patchy, scattered clouds – often in linear bands.
	Alto stratus	Thin and uniform.
Low clouds < 2000 m	Stratocumulus	Broad and flat on the bottom, puffy on the top.
	Stratus	Uniform, flat thick to thin layered clouds. Mostly composed of liquid droplets.
	Nimbostratus	Uniform, dark, flat, low clouds that produce precipitation. Mostly composed of liquid droplets.
Vertical clouds < 500 and > 18000 m	Cumulus	Puffy and piled up.
	Cumulonimbus	Can cause lightning, hail, strong rains, strong winds and tornados.

Controls of Weather and Climate

What are the major controls of weather and climate?

The energy that the earth receives from the sun is not distributed evenly. Many factors affect the distribution. These include latitude, altitude, distance from the sea, cloud cover, ocean current, planetary winds and pressure.

Latitude

What is latitude? What is the effect of latitude on temperature?

On a global scale, latitude is the most important factor determining the strength of heat reaching the earth’s surface. When the sun’s rays are vertical (at a right angle) to the surface, the amount of heat received is the greatest. But when sun’s rays are slanting (oblique) the heat’s strength decreases.

At the equator, the overhead sun is high and of high intensity insolation is received. At the poles, the overhead sun is low, so the amount of insolation is low. **The sun is overhead at noon for six months between the equator and the Tropic of Cancer**, and it is **overhead for another six months between the equator and the Tropic of Capricorn**.

The sun is directly overhead at the **Tropic of Cancer on June 21 (the June solstice)**. The sun is directly overhead at the **Tropic of Capricorn on December 21** (the **December solstice**). At March and September equinoxes, the sun is directly overhead at the equator. At times between solstices and equinoxes, the sun is overhead in tropical areas in the zone between the Tropic of Cancer and the Tropic of Capricorn. The sun passes directly over every tropical place twice a year: once as the sun moves from being overhead at the equator to being overhead at the tropic and then again on the return from the tropic to the equator. The sun is never overhead outside of the tropics. The sun is closer to overhead in the middle and high latitudes on the day of their hemisphere’s summer solstice.

Altitude

Do you know how to compute temperature change with altitude?

Air temperature decreases with increasing altitude. The **normal decrease of temperature with height is 6.4°C per 1000 m**. This is known as the **normal lapse rate**. The normal lapse rate is the average rate or proportion at which

temperature changes per unit of altitudinal change. The highland areas of Ethiopia are good examples of places that demonstrate the effect of altitude on temperature.

Example: If the altitude of the given mountain is 4070 m above sea level and the temperature at sea level is 20°C. What will be the expected temperature at the top of the mountain?

Solution: At normal lapse rate temperature decreases 6.4°C per 1000 m

Altitude of mountain = 4070 m

$$\begin{array}{l} 1000 \text{ m} = 6.4^{\circ}\text{C} \\ 4070 \text{ m} = ? \end{array} \qquad \frac{4070 \text{ m} \times 6.4^{\circ}\text{C}}{1000 \text{ m}} = 26.048^{\circ}\text{C}$$

Therefore, the temperature at the top of the mountain = 20°C – 26.05°C = - 6.05°C

Distance from the Sea

How distance from the sea affects the distribution of temperature over the surface of the earth?

Land heats more quickly than water. But it loses its heat quicker than water. This is because of the transparency of water reflections from water surfaces, evaporation, mixing and currents in water. All these conditions result in greater and more rapid temperature changes on land than in the seas. Therefore coastal regions are cooler than inland regions. In winter in mid-latitudes, the ocean surface is much warmer than land areas. So, onshore wind brings heat to coastal lands. During the summer, coastal areas remain much cooler than inland sites.

Cloud Cover

What is cloud? How cloud affects temperature? Cloud reduces the amount of solar radiation reaching the earth’s surface and the amount of radiation reflected from the earth’s surface. When there are no clouds both types of radiation will be at a maximum level.

Ocean Current

What is ocean current? How ocean current affects the distribution of temperature?

Ocean current is the horizontal movement of ocean waters caused by winds and differences in temperature. The effect of ocean currents on temperatures depends upon whether the current is cold or warm. The water at the equator is warmer and less dense than that in polar areas.

Convection currents in the oceans result in a pole ward flow of warm, light, surface water. Compensating heavy cold water flows through the ocean depths towards the equator. Ocean currents are the source of temperature regulation on the earth. Ocean currents change the temperature of a particular region. A warm ocean current makes the weather of the nearby regions to go up in temperature, whereas cold ocean currents reduce the surrounding temperature. For instance, the European coasts get warmth from the ocean current of the Mexican Gulf stream, and southwestern Africa’s Namib Desert is cooled by the Benguela current off the continent’s western coast.

Winds

What is wind? What is the effect of wind on temperature?

Winds indirectly change the temperature of places near the sea according to the areas from which they blow. There are two ways by which distribution of temperature is influenced by winds.

- 1. Winds carry the temperature from one place to another.
- 2. Winds blow the surface layers of a body of water in the direction of their flow.

In this way, the winds have the effect of raising the temperature of the shore towards which the wind is blowing and lowering the temperature of the opposite shore.

Q??? What would be the temperature be at the top of mountain Ras Dashen assuming that the temperature at sea level is 30°C?

Climatic classification

At the end of this section, you will be able to:

- Realize the criteria used for classifying climate;
- Compare the climate classifications of the Greeks’ and that of Köppen;Express causes of climate change;

Explain major consequences of climate change

How does climate differ from weather? What are the elements of climate? What are climate controls? Why is climate classification needed? What are the major types of climate and climatic regions of the world? In which climaticregion do you live?

Classification of the Climates of the World

How is climate classified?

There are many ways to classify climate, each with its own advantages and disadvantages, depending on the purposes for which it is used. In the following sections, we consider two systems of climate classification with their climatic regions:

- *the system of the ancient Greeks – based on temperature and sunshine within latitudinal boundaries*
- *the Koppen system – based on temperature and precipitation as reflected in vegetation zones*

Classification by Ancient Greeks

Which zone is warm year - round?

After considering worldwide temperature and sunshine distribution, the ancient Greeks divided the world into three climatic regions. Note that they are bounded by specific latitudes:

- 1 *The low-latitude tropical (or torrid) zone:*** a winterless tropical region. It is bounded by the northern and southern limit of the sun's vertical rays (23½°N and 23½°S). In this climatic region, the noon sun is always high, day and night are of nearly equal length, and it is warm year-round.
- 2 *A middle-latitude temperate zone:*** sandwiched between the other two zones i.e., (23½° – 66½°N), and 23½° – 66½°S this climatic region has distinct summer and winter seasons and exhibits characteristics of both extremes, in terms of seasonal temperatures.
- 3 *The high-latitude polar (or frigid) zone:*** bounded by (1) the Arctic Circle, at 66½°N-90°N, and (2) the Antarctic Circle, at 66½°S-90°S. Places with this climate are considered summerless, because they are cold all year round due to long periods of winter darkness and a low summer sun.

The ancient Greeks’ system is some what simplistic method of climatic classification because it does not consider precipitation, and therefore it does not differentiate between wet and dry regions.

The Köppen System of Climate Classification

What makes Koppen classification more acceptable than the others?

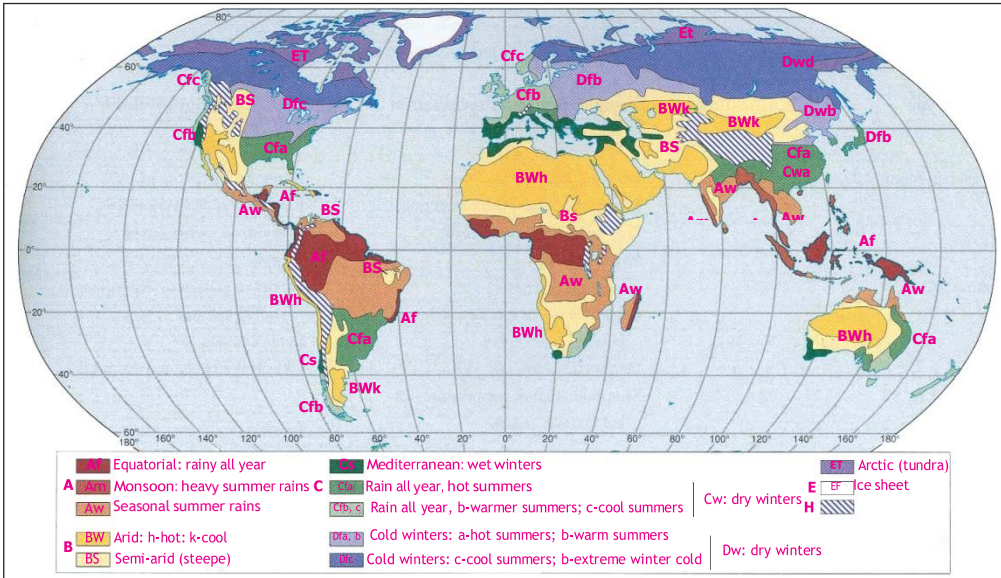
This widely used classification of world climates is based on seasonal variations in the annual and monthly averages of temperature and precipitation. It was devised by the famous German scientist Waldmir Köppen (1846 – 1940). Faced with the lack of adequate observing stations throughout the world, Köppen related the various climates to the distributions and types of

native vegetation in the world. In this way, where no climatological data were available, climatic boundaries could be approximated with vegetation zones.

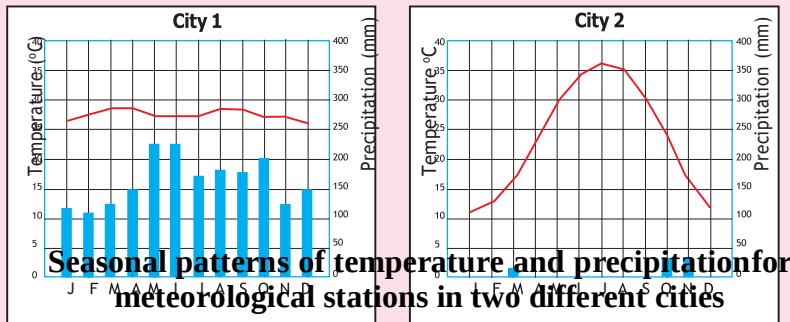
The Köppen climate classification, is now used by climatologists throughout the world. It defines five principal climatic groups. Each type is designated by a capital letter. Note that a single climatic region can exist in multiple locations on the earth. A Köppen *climatic region* is not a spatial or areal region.

- A Tropical moist climate:** All months have an average temperature above 18°C (64°F). There is enough moisture to support abundant plant communities. Since all months are warm, there is no real winter season.
- B Dry climates:** precipitation is deficient most of the year. Potential evaporation and transpiration exceeds precipitation.
- C Moist mid-latitude climates with mild winters:** C climates have warm-to-hot summers with mild winters. The average temperature of the coldest month is below 18°C (64°F) and above -3°C (27°F).
- D Moist mid-latitude climates with severe winters:** D climates are similar to C climates, but have distinct summer and winter seasons. They have enough moisture to support abundant plant communities. The average temperature of the warmest month exceeds 10°C (50°F), and the coldest month average drops below -3°C (27°F).
- E Polar climates:** They have extremely cold winters and summers. The average temperature of the warmest month is below 10°C (50°F). Winters are extremely cold, and even the summers are cool. Since all months are cold, there is no real summer season.

H Highland climate: At low latitudes the effect of altitude can produce tundra and polar conditions. Glaciers on tropical mountain sum its attest to the cooling effects of altitude highland climates follow the pattern of Earth’s mountain ranges.



General climate classification



- 1 Study the monthly temperatures of City 1 and City 2.
- 2 Compare and contrast. Next study the seasonal patterns, in each city, for each of the two climate elements – temperature and precipitation. Then perform the following activities and answer the following questions:
 - a Describe the pattern of rainfall and temperature distribution for each city.
 - b Which city has the warmest climate?
 - c Which city has the driest climate?
 - d Which city is probably situated in a desert?
 - e Which city is probably located in a tropical zone?
 - f Which city has uniform annual precipitation distribution?

Part 2

With your teacher, visit the meteorological station found nearest to your school. Ask the experts there to give you the most recent available 12 months of data for your locality's monthly temperature and precipitation.

- 3 When you return to your classroom, use the data to perform these activities,
- 4 Draw two graphs, similar to those in Figure 2.24, that show seasonal distributions of
 - a temperature
 - b rainfall;
- 5 Identify
 - a the warmest three months
 - b the coldest three months
 - c the driest three months
 - d the wettest three months

Part 3

Work with your group to choose a group representative to present your idea to the class.

Climate Change

What is climate change? Is climate change a reality? What are some of the manifestations and consequences of climate change?

Climate change is a long-term change in the statistical distribution of weather patterns over periods of time that range from decades to millions of years. It may be a change in the average weather conditions or a change in the distribution of weather events with respect to an average, for example, greater or fewer extreme weather events. Climate change may be limited to a specific region, or may occur across the whole Earth.

Causes of Climate Change

What causes climate change?

The earth's climate is dynamic. It is always changing through a natural cycle. What the world is more worried about is that the changes that are occurring today have been speeded up because of man's activities. These changes are being studied by scientists all over the world and tried to identify the causes. The causes of climate change can be divided into two categories - those that are due to natural causes and those that are caused by human factors.

A Natural causes

There are a number of natural factors responsible for climate change. Some of the more prominent ones are continental drift, volcanoes, ocean currents etc. Let's look at them in a little detail.

Continental drift

The continents that we are familiar with today were formed when the landmass began gradually drifting apart, millions of years back. This drift also had an impact on the climate because it changed the physical features of the landmass, their position and the position of water bodies. The separation of the landmasses changed the flow of ocean currents and winds, which affected the climate. This drift of the continents continues even today; the Himalayan range is rising by about 1 mm (millimeter) every year because the Indian land mass is moving towards the Asian land mass, slowly but steadily.

Volcanoes

When a volcano erupts it throws out large volumes of sulphur dioxide (SO₂), water vapour, dust, and ash into the atmosphere. Although the volcanic activity may last only a few days, the large volumes of gases and ash can influence climatic patterns for years. Millions of tones of sulphur dioxide gas can reach the upper levels of the atmosphere (called the stratosphere) from a major eruption. The gases and dust particles partially block the incoming rays of the sun, leading to cooling.

Ocean currents

What causes ocean currents?

The oceans are a major component of the climate system. Ocean currents move vast amounts of heat across the planet - roughly the same amount as the atmosphere does. But the oceans are surrounded by land masses, so heat transport through the water is in channels.

Ocean currents have known to change direction or slow down. Much of the heat that escapes from the oceans is in the form of water vapour, the most abundant greenhouse gas on earth. Yet, water vapour also contributes to the formation of clouds, which shade the surface and have a net cooling effect.

B Human Causes

How do human activities affect climate?

The industrial revolution in the 19th century saw the large - scale use of fossil fuels for industrial activities. These industries created jobs and over the years, people moved from rural areas to the cities. This trend is continuing even today. More and more land that was covered with vegetation has been cleared to make way for houses. Natural resources are being used extensively for construction, industries, transport, and consumption. Consumerism (our increasing want for material things) has increased by leaps and bounds, creating mountains of waste. Also, population growth has increased to an incredible extent. All this has contributed to a rise in greenhouse gases in the atmosphere.

Greenhouse Gases and Their Sources

What are the sources of greenhouse gases?

Carbon dioxide is undoubtedly, the most important greenhouse gas in the atmosphere. Changes in land use pattern, deforestation, land clearing, agriculture, and other activities have all led to a rise in the emission of carbon dioxide.

Methane is another important greenhouse gas in the atmosphere. About 1/4 of all methane emissions are said to come from domesticated animals. These animals produce methane during the cud - chewing process. Methane is also released from rice or paddy fields that are flooded during the sowing and maturing periods.

A large amount of nitrous oxide emission has been attributed to fertilizer application.

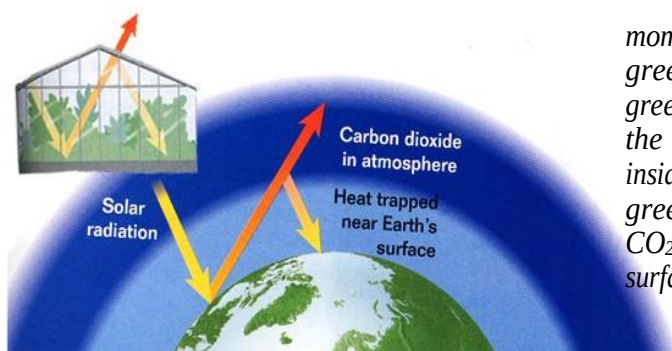
Greenhouse effect

What aggravates greenhouse effect?

The greenhouse effect is a process by which thermal radiation from a planetary surface is absorbed by atmospheric greenhouse gases, and is re-radiated in all directions. Since part of this re-radiation is back towards the surface, energy is transferred to the surface and the lower atmosphere. As a result, the temperature there is higher than it would be if direct heating by solar radiation were the only warming mechanism.

This mechanism is fundamentally different from that of an actual greenhouse, which works by isolating warm air inside the structure so that heat is not lost by convection.

Global warming, a recent warming of the earth's surface and lower atmosphere, is believed to be the result of a strengthening of the greenhouse effect mostly due to human - produced increases in concentration of atmospheric greenhouse gases, CO₂ in particular.



The greenhouse effect the moment you step into a greenhouse, you feel the greenhouse effect. Heat trapped by the glass walls warms the air inside. Similarly, atmospheric greenhouse gases, particularly CO₂, trap heat close to Earth's surface.

Since the beginning of the industrial revolution, the burning of fossil fuels has increased the levels of carbon dioxide in the atmosphere from 280 ppm to 390 ppm (parts per million). The emissions of carbon are directly proportional to energy

Consequences of climate change

- *What are the possible consequences of climate change resulting from global warming?*

Global warming

Global warming is the increase in the average temperature of earth's near - surface air and oceans since the mid - 20th century and its projected continuation. According to the 2007 fourth assessment report by the Intergovernmental Panel on Climate Change (IPCC), global surface temperature increased by about $0.74 \pm 0.18^{\circ}\text{C}$ ($1.33 \pm 0.32^{\circ}\text{F}$) during the 20th century. Most of the observed temperature increase since the middle of the 20th century has been caused by increasing concentrations of greenhouse gases, which result to human activity such as the burning of fossil fuel and deforestation. Global dimming, a result of increasing concentrations of atmospheric aerosols that block sunlight from reaching the surface, has partially countered the effects of warming induced by greenhouse gases.

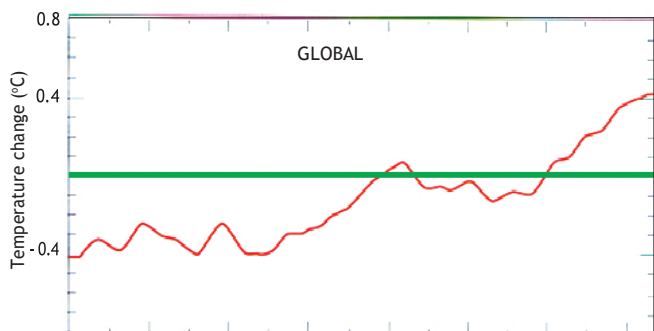


Figure 2.25:

Average global temperature variations from 1860 through 2000
The zero line represents the average surface temperature from 1961 to 1990 (Source: P. D. Jones, et al., and Climate Change 2001: the Scientific Basis, 2001, by J.T. Houghton, et al.)

Desertification

Desert - like conditions may spread over human habitats, crop lands and wet lands. Grass land and forest areas might turn into desertified environments as a result of global warming and hence climate change.

Drought

Because of climate change which results in extensive damage to plants, crops and animals, droughts are likely to occur more frequently. Up to three billion people could suffer from increased water shortages by 2080. In Ethiopia, for example,

drought started to reoccur more frequently, at an interval of 3 to 8 years in the most drought prone regions.

A rise in the sea - level

Globally, the average sea - level could rise by 18 to 59 cm, or more, by the end of the century. Rising sea levels could swamp some small, low - lying island states and put millions of people living in low - lying areas at greater risk of flooding.

It is also possible that salt from rising sea levels may contaminate underground fresh water supplies in coastal areas. The sea expected to encroach coastal areas and cover land surfaces.

Shift of the direction of global winds

If the average global temperature increases the jet stream will weaken and global winds will shift from their “normal” position. This in turn may disturb the “normal” amount of seasonal pattern and distribution of precipitation.

Shift of Tropical Zone

The tropical zone expected to extend north and southwards by not less than one degree in the northern and southern hemisphere respectively.

Expansion of Tropical Diseases

It is widely recognized that zone of mosquitos breeding and hence malaria spread farther into upper grounds, lower Woina Dega agroclimatic zones in case of Ethiopia.

Loss of Biodiversity

A global temperature rise could make some species extinct. There are already changes to the way plants and animals live.

Further changes in rainfall and temperature will affect many animal and plant species around the world. Some species might be unable to adapt quickly enough and habitats might not be available for them to move into. If global temperatures rise by two degrees celsius, 30 percent of all land-living species could be threatened by an increased risk of extinction.

The atmospheric greenhouse effect blocks the escape of some radiation to upper space. Therefore it heats the earth's atmosphere and surface. This process is due to the presence of blanketing “greenhouse gases” such as CO₂, CH₄, N₂O, CFC, and water vapor in the atmosphere. Their presence acts as a barrier to radiation and heat that would normally escape through the atmosphere. The atmosphere warms as it absorbs and then emits

infrared radiation downwards, while allowing shortwave radiation to pass out, away from the earth. The most noticeable result of the greenhouse effect is global warming, which is characterized by global temperature rises ranging between 0.5°C and 0.6°C.

Activity

In your geography work group, perform the following activities.

- 1 Discuss climate change: what it is and how it differs from climate variability. List some of its most noticeable effects.
- 2 What changes do you observe in your locality in relation to climate change? Do you think any of Ethiopia's endangered plant and animal species still exist in your woreda and region? Have patterns of temperature and rainfall changed during your lifetime or that of your parents or of the elders in your family?

Ask your parents and elderly relatives about temperature and rainfall changes in your locality? Have such changes been significant within their lifetimes? Within your lifetime, so far? Have such changes affected any of your lives? If so, how?
- 3 Consider any other changes in your locality that you think have resulted from climate change.

Create a report of all of your findings and opinions in your science journal.

- 4 Have you heard about any rivers, lakes or swampy areas in your locality that have shrunk in volume or even disappeared due to climate change? If so, write about it and report your ideas in class.
 - 5 Travel, either with the group or alone, to nearby rural area and talk with the peasants about climate change. Have they encountered the term climate change? Are they aware of climate changes in their area? If so, what are the changes, and how have they affected their work and earning capacity? For example, have climate changes led them to change their farming activities, and have those changes affected their ability to earn a living?
- Work with your group to prepare a report on your activities findings and opinions that you developed during in Tasks 1–5 of this Activity. Then work together to choose a group representative to present the report to the class

3.1.4 Natural Regions of the Earth

The Concept of Region and Regional Studies

Major Natural Regions of the Earth

What is a natural region? Are all environments similar? What makes them different?

Different environments result in differences in human activities, population distribution and economic development. Mostly, natural regions are named after the dominant vegetation found in them. But their **division is based on climate**.

The earth can be divided into **three major temperature zones**. These are:

- Tropical (hot) zone.
- Temperate zone.
- Frigid (cold) zone.

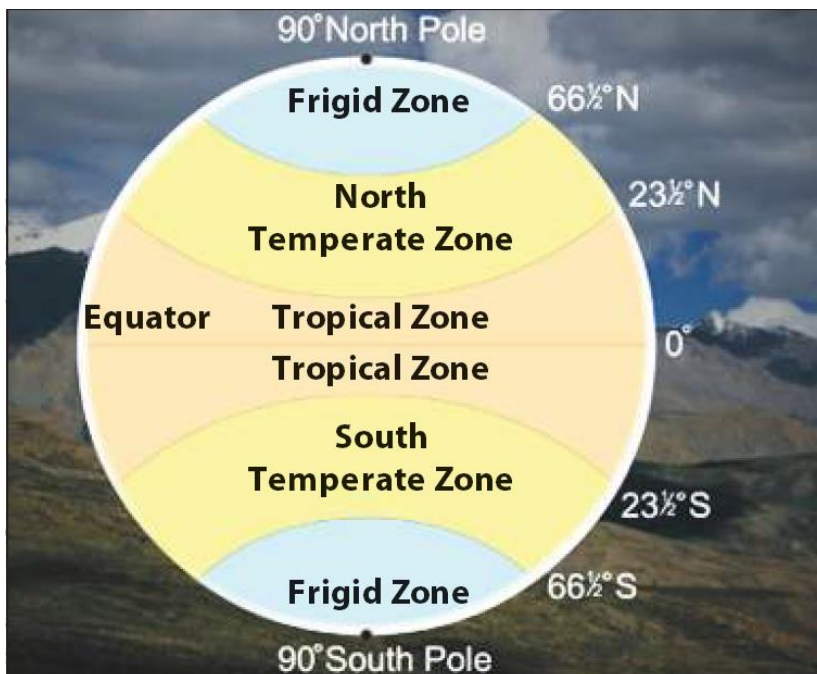


Figure: Temperature zones of the earth

Within each zone, there is a great deal of variation. Each zone can be divided into a number of sub-regions.

I. Tropical (Hot) Zone

What is the location of Tropical zone? Explain the general characteristics of temperature?

The tropical or hot zone is the zone between the Tropic of Cancer ($23\frac{1}{2}^{\circ}\text{N}$) and Tropic of Capricorn ($23\frac{1}{2}^{\circ}\text{S}$).

The tropical zone is characterized by high temperature throughout the year. The annual average temperature is above 20°C , which decreases polewards from the equator. Rainfall decreases and becomes seasonal as one moves away from the equator. Along the equator there is dense forest which gradually changes with distance from the equator, to savanna grassland, semi-desert vegetation and true desert vegetation. Agriculture and mining are the major economic activities of the zone.

The tropical zone is divided into four sub-regions. These are:

- the equatorial rainforest region
- the savanna grassland region
- the desert region
- the tropical monsoon land.

A. The Equatorial Rainforest Region

Location

They are found in the lowland areas within **6° north and south of the equator**. The main areas in the world include.

- The Amazon Basin
- The Congo Basin
- Southern Malaysia and some Indonesian islands

Climate

Equatorial rainforest region has uniformly high temperature and heavy rainfall throughout the year. The mean annual temperature is about 27°C , and the annual range of temperature is less than 3°C . But the daily range is greater than the annual range of temperature. Rainfall in this region is mainly of a convectional type which falls every day, usually in the afternoon. The total annual rainfall is very high, usually between 1500 mm and 2500 mm. All months have rainfall with a small variation in amount.

Natural Vegetation

The region has dense, tall, broad-leafed and evergreen trees. The high temperature and abundant moisture of this region makes plants growth continuous. The ground is covered with tropical rainforest, the most luxuriant type of natural vegetation in the world. Most of the trees are hardwood, such as rosewood, ebony, and mahogany, which are valuable for making durable furniture.

Wild Animals

The Equatorial rainforest is inhabited by varieties of animals, such as insects, birds, reptiles, mammals. Most mammals are arboreal. They are small in size and are tree climbing, example apes, monkeys, bats, squirrels, etc. Water body animals, such as crocodiles, alligators, hippopotamuses, fish, etc, are also available.

Human Activities

The human activities found in equatorial rainforests include

- Primitive hunting and gathering,
- Shifting cultivation,
- Plantation agriculture, and
- Mining

B. Tropical Deserts

What is a desert? Where are the major tropical hot deserts? How were tropical deserts formed? What are the major characteristics of hot deserts?

Location

Deserts are almost barren lands. The tropical hot deserts are located between 15° to 30° north and south latitudes of the equator. The largest hot desert is the Sahara Desert. There are also other important deserts such as the Kalahari, Atacama, Colorado, Arabia, Thar and Australian deserts. Deserts are formed due to their location

- in the trade-wind belts
- on the leeward side of high mountains
- in the interior of continents
- along cool ocean currents

Most deserts arise due to atmospheric wind conditions. Warm air masses create two belts of desert, one along the Tropic of Cancer and the other along the Tropic of Capricorn. Other deserts result from the effects of ocean currents on land masses, where cool air masses carry fog and moist, but little rain, along coastal regions.

Deserts are located on the western margins of continents, except for the Sahara Desert, which is in North Africa and extends to the northeastern coast of the African continent.

Climate

The main characteristics of desert are that the climate is hot throughout the year, and there is very low and unreliable rainfall (not more than 25⁰ mm per year). Temperature ranges from 25°C to 40°C or above during the daytime, and they can fall even below 0°C during the night, due to the absence of clouds. The winds are warm and dry.

Vegetation

The climate of the hot desert is not favourable for plant growth. However, there are some plants that have a special way of behaving or special features (adaptations) which enable them to survive. Plants in deserts have long roots, few or no leaves, hard bark and an oily leaf surface. These features help the plants to extract every drop of moisture from the soil and reduce the loss of water by evapotranspiration.

These are called xerophytes. Xerophytes are woody and thorny plants, grasses and herbs, for example cacti.

Animal life

Animals such as gazelle, hare, fox, snakes, lizards, etc. are commonly available in the desert region. For the desert people camel is their best companion. It has special adaptations, which helps it to survive the desert.

Human Activities

The major economic activities found in the hot deserts are pastoralism, crop cultivation and mining.

Pastoralism is the grazing of animals. Many of the desert people practice a nomadic way of life. Nomads do not live in one area. They move with their animals from place to place looking for water and pasture. They keep animals like camels, sheep, goats, cattle, etc. Small-scale crop cultivation is possible along wadis (small streams) and in the oases; dates, vegetables, fruits and cereals are grown. In areas where the deserts are crossed by rivers, irrigation on a large scale is possible. The waters of the Nile, Tigris-Euphrates and Indus have made possible the cultivation of crops in desert areas.

Most deserts are important sources of minerals. The most important mineral is oil. The Middle East alone accounts for about 60% of the world's total oil reserve. Other minerals produced in hot deserts are nitrates and copper in the Atacama Desert, gold in the Australia Desert, diamond in the Kalahari, natural gas in the Sahara Desert and the Middle East, phosphate and iron ore in the northwest Sahara Desert. Therefore, mining is an important occupation in deserts.

Case Study 1

Vertical Distribution of Climate in Ethiopia

Ethiopia lies within the tropics (3°N to 15°N latitude). It lies in a zone of maximum insolation where every place has overhead sun twice a year. However, as it is a highland country, tropical temperature conditions are not experienced everywhere. They are limited to the lowlands in the peripheries.

The traditional classification of climatic zones of Ethiopia is divided into five:

1. **Bereha (hot arid):** found below 500 m a.s.l., where the average annual temperature range is between 30°C and 40°C or higher.

2. **Kola (warm to hot semiarid):** It is a climate of hot lands with altitudinal ranges of 500 - 1,500 m a.s.l. Average temperature is between 20°C – 30°C.

3. **Weina dega (warm to cool semi-humid):** This is a zone which covers the temperate highlands that fall in altitudinal ranges of 1,500 – 2,300 m a.s.l. Its average annual temperatures are between 15°C and 20°C.

4. **Dega (cool to cold humid):** This describes the cool temperate highlands with an altitude ranging from 2,300 – 3,300 m a.s.l. and average temperatures between 10°C to 15°C.

5. **Wurch (cold moist temperature):** It coincides with the Afro-alpine areas on the highest areas of the plateaus. The lower limit of the wurch zone, generally, coincides with the transition from coniferous forest to the vegetation dominated by Asta. This happens above 3,300 m.a.s.l. where Average temperature is below 10°C.

➤ *Though Ethiopia is a tropical country, its climate types range from desert to hot steppe, and from tropical savanna and rainforest to warm temperate cool highlands. Ethiopia is a multifaceted tropical country, because of the influence of altitude.*

II Temperate Zone

What does temperate zone mean? Where is the temperate zone located?

What are the climatic characteristics of the temperate zone?

Location

The temperate zone is located between 23½°, and 66½°, both north and south of the equator. The temperate zone is the transitional zone between the tropical zone and the frigid zone. It is also known as a mid-latitude region.

Climate

Temperate regions usually have four distinct seasons. They are summer, winter, autumn and spring. Weather changes are observed in each season. Precipitation ranges from 750 mm to 1500 mm. Temperature ranges from below freezing point during the winter, to 30°C or more during the warmest days of summer. There are important factors that bring about frequent weather changes in the temperate zone. These are:

- *Convergence of tropical and polar air masses: This results in cyclonic storms and fronts.*
- *Angle of the sun: The sun is never overhead in the temperate zones. Therefore it does not heat the earth very much.*
- *Length of day: There is a big difference in the length of day between summer and winter. Days are longer in summer and shorter in winter.*
- *Effects of continents: This is more important in the northern hemisphere, where there are very large areas of land. In winter, the centres of the continents become very cold because they are far away from the sea.*

In this season, the angle of the sun is very oblique and it does not give much heat. During summer, the interiors of the continents become very hot. The annual range of temperature is therefore very great.

The temperate region is, therefore, the most important producer of both agricultural and industrial productions in the world.

Sub-regions: The temperate zone is a very extensive zone. It contains many regions with different characteristics. But the two main divisions are the warm temperate and the cool temperate regions.

The warm temperate region includes:

- *The Mediterranean region areas.*
- *The temperate monsoon lands.*
- *The warm temperate grasslands of the southern hemisphere.*
- *The temperate deserts (in the interiors of the continents).*

The cool temperate region includes:

- *The temperate grasslands of the northern hemisphere.*
- *The cool temperate western margins.*
- *The cool temperate eastern margins.*
- *The coniferous forest belts.*

III. Frigid (Cold) Zone

Where is the location of frigid zone? Why is this region cold?

The frigid zone is located between 66½° to 90°, both N and S of the equator. In some areas natural conditions of this zone are not limited by latitudinal location.

For example, in Canada, this zone bends southwards to about 55°N. The boundary between the cold temperate and the frigid zone is marked by 10°C isotherm of the warmest month. Common vegetations of this zone are lichens, mosses, grasses, cushion plants, and low shrubs which spring to life during the short summer season. They remain dormant for about ten months of the year. However on the ice bound areas, such as Antarctica, there is no vegetation because the temperature is not above the freezing point and there is no sufficient sunlight that serves as a source of energy for life. The most widespread people in the region are the Inuit (Eskimos), Indians, Lapps, Chukchi, Koyaki, and Yakuts. They are found in Alaska, Canada, Greenland, Finland, Norway, Sweden and Siberia. They follow their traditional ways of life as hunters, reindeer herders, fishing, fur trapping, and they are being trained for more settled occupations, such as Farming and mining.

All of the native Arctic people have developed a unique ability to survive in their harsh environment by skillfully using the few materials available to them. From snow, ice and animal skins and bones they have fashioned a simple technology that enables them to build shelters, weapons, and forms of transportation such as sleds and kayaks (small, skin-covered boats). Land and sea animals and fish provide their main source of food.

The frigid zone has two sub-regions. These are **tundra and the polar ice caps**.

Tundra Region

What is tundra? Where are the areas of tundra located? What are the climatic characteristics of the tundra sub-region?

The word tundra refers to an area where the growth of trees is prevented due to low temperatures and short seasons that restrict growth.

The tundra is a transitional zone between the polar ice caps and the coniferous forest belt. Tundra is found only in the northern hemisphere. It extends along the northern Arctic shores of the continental land mass.

Climate

Tundra has two distinct seasons. These are the summer and winter seasons. During the summer season, there is long daily duration of sunshine, but the temperature is very low. Only three or four months have average temperatures above the freezing point. This is because the sun's rays are very slant, and therefore lack heat. Winter is cold and long. During the daytime, the sky is dark, and stars can be seen shining. There is twilight in the middle of the day. The rainfall level in the tundra is low. The amount of annual total rainfall is below 250 mm.

Natural Vegetation

Is the type of climate experienced in the tundra suitable for plant growth? Why? What kind of plants grow in the tundra?

As plants cannot grow in temperatures below 6°C, there is little or no vegetation in tundra areas. In places where the snow is not thick, low forms of plants such as lichens, mosses and sedges grow during

the summer season. On the southern margins of the tundra lands, and in lower areas, there are short trees and flowering plants.

Human Activities

What are the main human activities in tundra?

The tundra is a sparsely populated region. There are a few scattered tribes such as Eskimos (Inuit), Lapps, Finns, Aleuts and Tunguses. These are primitive peoples who have adapted themselves to the cold and difficult environment of the tundra. The people lead primitive ways of life and depend on hunting, fishing, and reindeer herding. In North America, the Eskimos settle along the seashores in igloos (houses made up of blocks of ice) in the cold winter. During the summer, the people move inland and live in tents which are made of animal skins.

Polar Ice Cap Region

What is polar ice cap? Where is the location of polar ice cap?

A polar icecap is a high latitude region of the earth that is covered with ice. The two polar ice caps are located near the North and South poles. These natural regions include Greenland and the Arctic islands and Arctic Ocean in the northern hemisphere and the continent of Antarctica and the South Sea in the southern hemisphere.

Climate

Ice caps are generally characterized by negative temperatures. The average temperature for each month of the year is less than 0°C. The precipitation is very low and falls in the form of snow. In Greenland the total annual precipitation is about 80 to 100 mm. The whole of Antarctica receives an average precipitation of less than 150 mm per year.

Natural Vegetation

Plant growth is impossible as the regions are permanently covered by ice and snow. The soil of the region is permanently frozen. Except few areas the surface is covered with ice. In those areas, a little vegetation, like mosses, lichens, and algae may grow with the help of the light of the very short summer.

Human Activities

There are no people living permanently in Antarctica. Only scientists and hunters visit this cold continent. At present, the only resource of Antarctica is whale oil.

But it is believed to contain rich deposits of coal. The Arctic Ocean is rich in animals that can be used as sources of food and material for clothing. The Arctic region is also important for military strategies. For example, the USA has an air base at Thule in Greenland.

3.1.5 Ecosystem

What is an ecosystem? What are the components of an ecosystem?

An ecosystem is a community of living things and their non-living environment. It is a basic functional unit of biosphere, consisting of organisms (plants and animals) and their environment (air, water, soil and rock). The non-living features of the environment are the abiotic factors, and the organisms in the environment are the biotic factors. Populations and communities make up an ecosystem. An ecosystem may be small organisms in a drop of water, or even the whole earth as one system. There are many types of ecosystems, even within a relatively small area. Some are on land, others are aquatic. All components of an ecosystem function together as a closed unit of biological community or association.

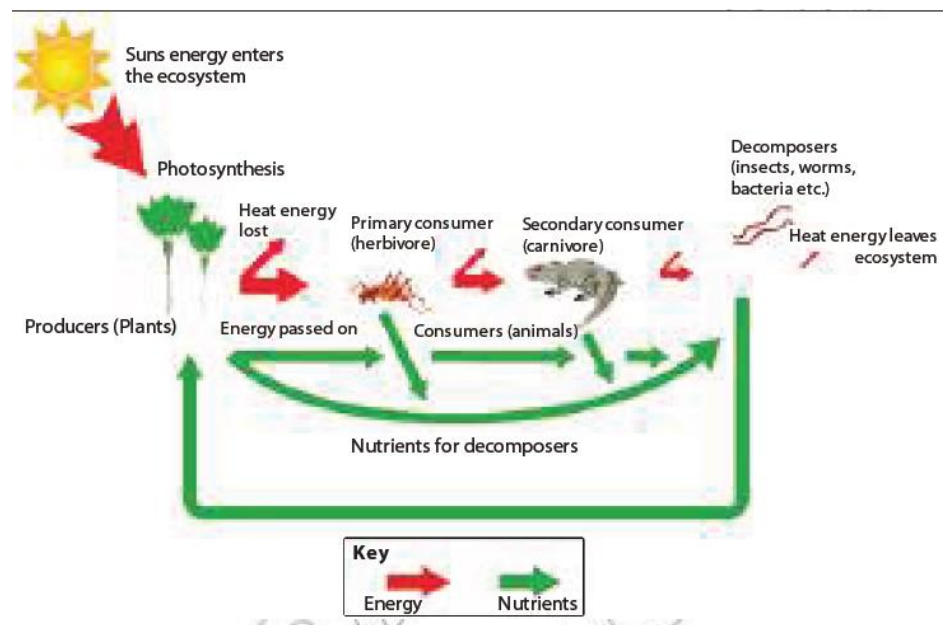


Figure 2.65 All plants and animals are linked with the natural environment. This is an ecosystem

All ecosystems require energy in order to exist. This is provided by sunlight, with only minor contributions from other sources. Energy from the sun reaches the earth's surface in two forms. These

are heat energy and light energy. Heat energy cannot be used directly by plants and animals. Light energy can be captured only by green plants during the process of photosynthesis. Only about one percent of light energy falling on leaves is converted into food energy and stored as carbohydrate molecules.

Ecosystems have lots of different living organisms that interact with each other. The living organisms in an ecosystem can be divided into three categories: producers, consumers and decomposers. They are all important parts of the ecosystem.

- **Producers** are the green plants. They make their own food by means of photosynthesis. Green plants are the primary passage from one organism to another along the food chain. A food chain is a simple way of showing how energy in the form of food passes from one organism to another.
- **Consumers** are animals, which get their energy from the producers or from organisms that eat producers.
- ❖ **There are three types of consumers:**
 - ✚ **Herbivores** (primary consumers) are animals that eat plants.
 - ✚ **Carnivores** (secondary consumers) are animals that eat herbivores and sometimes other carnivores.
 - ✚ **Omnivores** are animals that eat both plants and other animals.
- **Decomposers** are plants and animals that break down dead plants and animals into organic materials that go back into the soil. When organisms die, their bodies decompose and form a source of energy and nutrients for other organisms. Similarly, waste matter passed from the bodies of living organisms are also sources of energy and nutrients. These materials are not wasted by their ecosystems. They form the food for many other organisms, which are referred to as decomposers. Decomposers are microorganisms, mainly fungi and bacteria, which live on dead organic matter.

3.2 Physical Environment of Africa

3.2.1 Position, Size, and Shape of Africa

At the end of this section, you will be able to:

- *demonstrate the relative and absolute location of Africa using world map;*
- *compare the size of Africa with other continents;*
- *discuss the impacts of the coastal and continental shape of Africa on its development;*
- *describe the regional division of Africa;*
- *show the geographical location of each region;*
- *relate the sub regions in terms of size, access to the sea and major relief structure; and*
- *analyze the socio-economic and geo-political similarities of each region.*

3.2.1.1 Position of Africa

Africa's location can be expressed in two ways, namely, absolutely and relatively.

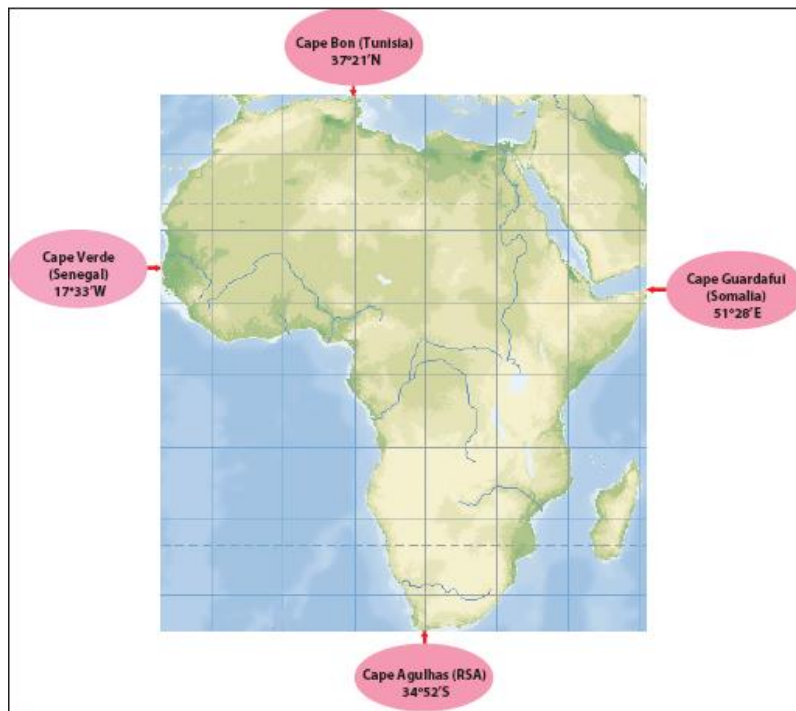


Figure: Africa's Astronomical Location

As you can see from the above Figure, Africa lies between 37°21'N and 34°52'S latitudes and between 17°33'W and 51°28'E longitudes. This means that the continent extends for about 37° to the north of the equator and 35° south of it. Similarly, Africa extends for about 17° west and 51° east of the Greenwich Meridian. From this absolute location, we can understand that the continent stretches in all the four hemispheres.

The four extreme points of the continent, which mark the extreme points of Africa are the following.

- Extreme North - Cape Bon (Tunisia) - 37°21'N
- Extreme South - Cape Agulhas (Republic of South Africa (RSA)) - 34°52'S
- Extreme East - Cape Guardafui (Somalia) - 51°28'E
- Extreme West - Cape Verde (Senegal) - 17°33'W

From the map that shows the absolute location of Africa, we can learn that:

- when we measure Africa's greatest north-south and east-west extents, we find that they are almost equal:
 - north-south – approximately 8000 kilometers
 - east-west – approximately 7,600 kilometers
- the equator crosses Africa almost at its north-south center. The north-south extents above and below it are almost equal, although the northern area is greater than the southern one. The land area north of the equator is about twice that of the south.

- Africa is the only continent crossed by all of the following: the Tropic of Cancer and Tropic of Capricorn, the Equator and the Prime Meridian.
- almost $\frac{3}{4}$ of the continent's total area is found within the tropics, and therefore much of the continent experiences tropical climates.

When Africa's position is expressed in relation to the major landmasses and water bodies that are close to it, it can be described as follows. Africa is found to the:

- South of Europe
- Southwest of Asia
- South of the Mediterranean Sea
- West of the Indian Ocean
- East of the Atlantic Ocean
- North of the Southern Ocean



Figure: Africa's Relative Position in the world

As the map shows, Africa gets closest to Europe across the **Strait of Gibraltar**, which is about 22 kms wide between Morocco and Spain. With relation to Asia, the continent comes closest across the **Strait of Bab-el Mandab**, which is about 40 kms wide. A narrow stretch of land called the **Isthmus of Suez**, which is cut into two by an artificial canal called the **Suez Canal**, connects Africa with Asia.

The relative position that Africa has, in relation to the rest of the world, gives the continent many advantages. The following are the most important ones.

- The continent occupies a central location in the world. This makes the continent close to the rest of the world, providing it geographical accessibility.
- The geographical proximity that the continent has with Europe and Asia has resulted in socio-cultural contacts. This has encouraged socio-economic and cultural exchange and integration among the three continents.

3.2.1.2 Size of Africa

Africa is the second largest continent in the world, following Asia. The continent's total area is about 30,335,000 km². This constitutes 20.2% of the earth's total land surface. With this area, the continent is about two-thirds of the size of Asia and 3.36 times larger than Europe. The following Table indicates the sizes of the world's continents.

Table 3.1: Area of the Continents

Continent	Area (000' km ²)	% of World Area
Africa	30,335	20.2
Asia	45,000	30.0
Europe	10,000	6.67
Australia and Oceania	8,700	5.8
N. America	24,200	16.14
South America	17,000	11.34
Antarctica	14,000	9.3
World total	149,900	100

Africa's large area gives the continent several advantages, including the following:

- A large area of land that can be used for settlements, agriculture, and other economic activities.
- Huge resource potentials in terms of resources like soil, water, minerals, flora and fauna and the like, which are vital for its development.

However, Africa's large size also has disadvantages. For example, it makes geographical connectivity difficult among the people of the region and of the world. As a result, there are problems related to integration among peoples of the continent and others outside, making both integrated development and international trade a challenge.

3.2.1.3 Shape of Africa

What is shape? What is its significance?

How do you describe Africa's shape?

Shape is defined as the geographical form of an area. In other words, it is the external geographical appearance of a place. It has great impact on the socioeconomic integration and flow of goods and services within each region.

The shapes of places can be described in different ways for example, as moderately compact, elongated, fragmented, perforated, and the like. A moderately compact shape is close to that of a circle. An elongated shape is a shape with one side longer than the other. A place with a fragmented shape is made up of multiple disconnected areas. For example, some countries are composed of islands. Africa has a relatively compact shape. This means that the continent's longest east-west and north-south distances are almost the same. The following evidence supports the above statement.

- *The east-west and north-south extensions are almost equal, with a minor difference.*
- *Africa has a fairly unbroken coastline. The continent's coastline does not have many indentations, inlets, bays or gulfs. This condition creates a relatively smooth coastline, compared to those of other continents, and therefore a relatively short one. For example, if we compare Africa and Europe, the coastline of the latter is, by far, longer than the former.*

The unbroken coastline of Africa makes the continent poor in natural harbors.

- Many places in Africa are not very far from the coast. Almost all places are at most located 1500 km from the coast. Therefore, many countries of the continent are not very far from the sea.
- The unbroken coast line discourages external communication.

Regional Division of Africa

The mainland of Africa has five major regions. They are defined in terms of socio-economic and cultural similarities. Each region consists of a number of countries with their own regional variations. The regions are

- Eastern Africa
- Central Africa
- Northern Africa
- Southern Africa
- Western Africa

The geographical location of these regions and their political units are given in the following figures. A brief description of the socio-economic conditions of each region is also made below the figures. Study them carefully and try to identify each region's relative location as well as the countries that constitute the region.

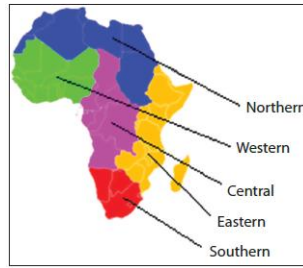


Figure: Regional division of Africa

3.2.2 Geological and Relief Structure of Africa

The present relief structure of Africa is the result of long, complex and continuous processes that have taken place for several millennia. In this lesson, we will consider the geological history of Africa. Throughout the **4.5 billion year history of the planet**, several geologic processes acted on the planet and formed its relief. In this process, Africa underwent several major geological events and changes. The continent of Africa was part of the old continent that we call **Pangea**. In the **Jurassic period** of the **Mesozoic era**, some 200 million years ago **Pangea broke into two, forming Laurasia and Gondwanaland**. **Laurasia** was the northern continent, and **Gondwanaland** was the southern. Africa was part of Gondwanaland. As time passed, each of these continents was further broken down, forming the seven present-day continents. In this process of change, the planet experienced several geologic events that resulted in the formation of the world's present surface structure. In Africa, major geologic events and processes acted on the continent's surface and left their scars, forming the continent's diverse relief. Now let us see the major events that took place in the African continent in each era.

Precambrian Era (4.5 billion - 600 million years before the present)

This era is the oldest and largest division of the geological time scale. It covers almost 5/6th of the geological history of the planet. Due to its remoteness in time and lack of fossil evidences, not much is known about this era. However, two processes are believed to have been dominant. These are **orogenesis** (a series of mountain-forming processes) and **metamorphism**.

For Africa, the following events are assumed to have been dominant in this era.

- Formation of the Basement Complex Rocks: the oldest rocks of the continent, which are called Precambrian or crystalline basement complex rocks, were formed during this era. These rocks cover nearly two-thirds of the continent. **Precambrian rocks are rich in metallic mineral deposits, such as gold and copper.**
- **Orogenesis:** is a mountain-forming process. In this era, many mountains that make the face of the continent very rough and undulating, were formed.

Paleozoic Era (600 - 250 million years before the present): This is the second-longest and second-oldest era in geological history. This era witnessed no major rock formation processes. As a result, it is a gap, relative to during this era. The following are the most important ones.

- **Series of denudation and peneplanation:** during this era, internal and external forces acted on the face of Africa, resulting in denudation and peneplanation of its surface. Denudation is the lowering of the earth's surface, while sinking of land and its resultant peneplanation refers to the formation of almost level surfaces as a result of lowering in altitude.
- **Heavy erosion:** the denudation and peneplanation processes were facilitated by the heavy erosion that affected many places in Africa.

The eroded materials accumulated in the Maghreb region, the Western Sahara, and the Southern Cape. From Eastern Africa, sediments were taken to Southern Africa and the Middle East. These sediments finally formed sandstones, shale and limestone. Also during this era, Fold Mountains that run parallel to the Great Karroo, the interior plateau, formed.

- Formation of coal during the Carboniferous period.

Carboniferous period: The coal Age: is that part of the Paleozoic Era when coal was extensively formed. Thick layers of partially decayed swamp vegetation, covering coastal lowlands were buried under marine deposits when the coastal lands sank. More swamps were formed when the water grew shallower and the process was repeated. As the deposits became compressed and hardened, the vegetation matter formed coal.

Mesozoic Era (250-70 million years before the present)

The Mesozoic era is the third-largest and third-oldest era in the geological history of the earth. For Africa, the era was a time of alternate sinking and rising of the land. The era is divided into three periods. These are the Triassic, Jurassic and Cretaceous. During the first period, the Triassic, there was sinking of the land in the eastern part of the continent. The land remained under the sea during the Jurassic period. Rising (uplifting) of the land began in the Cretaceous.

The following are the most important events that took place during the Mesozoic era in Africa.

- Sinking of the Horn of Africa resulted in the gradual transgression of the sea during the early years of the Triassic period;
- Formation of sedimentary rocks like those in eastern Africa, which were the results of the alternate sinking and rising of the land;
- Uplifting of the land in the Horn of Africa, resulted in the regression of the sea during the Cretaceous period;
- The flooding of the Sahara region, by water that advanced from Tethys (a sea that separated Laurasia from Gondwanaland), and the subsequent accumulation of sediments.

Cenozoic Era (70 million - present)

This era is the most recent and the shortest era in the geological history of the earth. It covers the time from 70 million years ago to the present. As a result, the era is sometimes called the living era. There are two periods in this era. These are the Tertiary and Quaternary periods. A number of geologic events that changed the face of the continent took place in this era. The major geologic events of this era are the following.

- Formation of the Mediterranean Sea, the Great East African Rift Valley, the Red Sea and the Gulf of Aden.
- Formation of many of the volcanic mountains, plateaus and young fold mountains (Atlas Folds) of Africa. The volcanic mountains that shape the landscape in eastern Africa were formed during this era.
- Climatic change that resulted in the cooling and later warming of the earth's climate. The cooling of the climate resulted in the Pluvial (Fluvial) rains in tropical Africa, which caused heavy erosion in many parts of the continent. Later, warming of the climate resulted in heavy evaporation and drying up of many water surfaces. In some instances, such a change resulted in the formation of extensive salt plains like the one in northeastern Ethiopia (Afar).

The various geological events and processes that took place in the four major eras of the geological time scale have affected the African landmass. The diverse landscapes that we see today in Africa are the results of these processes. The following Table gives a summary of the major geologic events that took place in Africa during the four eras and during their periods and epochs.

Table: Summary of the Geological History of Africa

Era	Period	Epoch	Geologic Events in Africa
Cenozoic	Quaternary	Recent	River terraces and raised beaches formed; glaciations in Eastern African mountains; down-warping of the Chad basin; heavy deposition in basins such as the Kalahari basin.
		Pleistocene	
	Tertiary	Pliocene	Mountain building (for example, the Atlas Mountains); intense volcanic eruption in East Africa.
		Miocene	
		Oligocene	East African Rift Valley developed; extensive lava flows in Ethiopia
		Eocene	
Mesozoic	Cretaceous		Marine sediments deposited in many areas, e.g., southern Nigeria.
	Jurassic		Mountain ranges uplifted, e.g., Cape Mountains.
	Triassic		
Paleozoic	Permian		Central and Southern Africa in grip of ice age .
	Carboniferous		
	Devonian		
	Silurian		
	Ordovician		
	Cambrian		Ice age in parts of Northern Africa.
Pre-Cambrian			Mountain building; oldest rocks (4 billion years) in Southern Africa.

The Relief Structure of Africa

What is relief? How do you describe the relief of Africa?

Africa's relief is made up of huge mountains, extensive plateaus, deep valleys and gorges, plains, and the like. However, much of Africa is plateau. The continent's relief consists of 71% plateaus, 25% plains and 4% mountains. Africa is the only continent that is predominantly covered by plateau lands, and that has only a small proportion of plains.

The relief of Africa ranges from 5,895 m above sea level (at Mt. Kilimanjaro in Tanzania) to 132 m below sea level (at the Qattara Depression in Egypt). This makes the maximum relief of Africa to be 6027 m. The plateau nature of Africa can be seen in the following Table.

Table: Percentage distribution of major landforms of the earth, by continent

Continent	Landform Types (%)		
	Mountains and Hills	Plateaus	Plains
Africa	4	71	25
Asia	44	24	32
Europe	25	8	67
North America	24	24	52
South America	20	24	56
Oceania	28	24	48

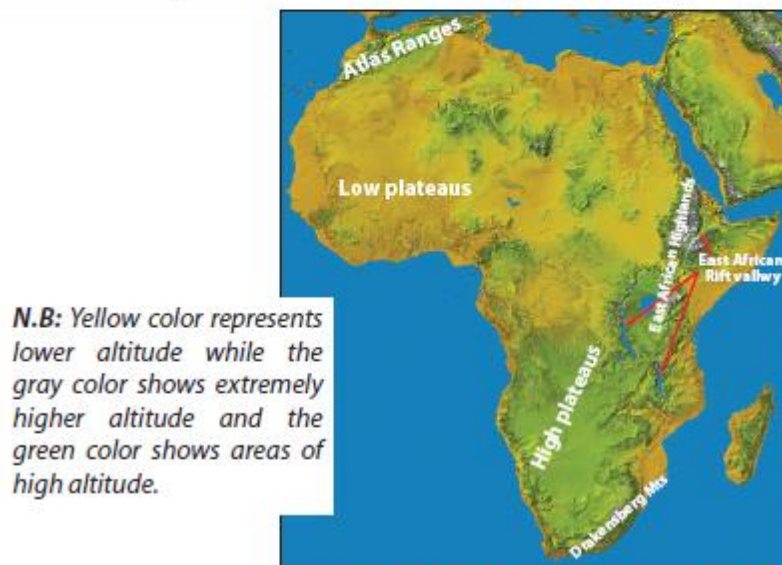


Figure 3.5: The Topography of Africa

As the above Table indicates, much of Africa is composed of plateau landscape. In contrast to the rest of the continents of the world, Africa has the greatest part of are mountains, plains and the Rift Valley. Now, we shall discuss each of these relief features.

- ❖ Maximum relief is the difference in altitude between the highest and lowest points of a place.

Mountains

What is a Mountain? Where are most of the Mountains of Africa found?

A mountain is a high land with steep slopes and a peak. Africa's landform is dominated by two major types of mountains. These are volcanic and fold mountains.

Volcanic mountains: were formed as a result of the great volcanic activities that took place in the Tertiary period of the Cenozoic era. They constitute the highest points (peaks) of the continent, with many of them being above 4,000 m above sea level. Many of the volcanic mountains of the continent are concentrated in Eastern Africa. The following Table shows you the major volcanic mountains of the continent with their heights and locations.

Table: Volcanic Mountains of Africa

Mountain	Elevation (m)	Location
Killimanjaro	5895	Tanzania
Kenya	5200	Kenya
Ras Dejene/Dashen	4620	Ethiopia
Meru	4567	Tanzania
Elgon	4321	Uganda and Kenya border
Cameroon	4070	Cameroon

N.B. Mount Ruwenzori (5119 m) is among the highest mountains in Africa the formation of which was associated with tectonic movement.

Fold Mountains: are found in the northern and southern extremes of Africa.

They are of two types: young and old. The **young fold mountains** are located in northwestern Africa, particularly in the Maghreb region, a region that covers areas in Morocco, Algeria and Tunisia. They are called the **Atlas Mountains**. These mountains are contemporary to the Alps of Europe and the Himalayas of Asia, and they were formed during the Alpine orogenesis during the Cenozoic era. Their general elevation declines from west to east. The **old fold mountains** are found in South Africa. They were formed during the Hercynian orogeny in the Mesozoic era. They are contemporary to the Australian Alps. They are called the **Cape Ranges**.

The Great East African Rift Valley

What is a rift valley? How do you describe the Great East African Rift Valley?

The East African Rift Valley is part of the world's Great Rift Valley system, which stretches from Syria, in the Middle East, to Mozambique, in Southeastern Africa, over a distance of about 7,200 km. It was formed in the Tertiary period as a result of faulting processes that acted on the crust of the earth in the region. This rift system extends a distance of about 5600 km in Africa, touching 15 countries in the continent. The Rift Valley has four trenches (branches). These are:

- Ethio-Eritrea-Djibouti-Northern Somalia Branch

- Western Branch
- Eastern (Gregory) Branch
- Malawi Rift Valley

Ethio-Eritrea-Djibouti-Northern Somalia Branch

As the name itself implies, this branch runs through the four countries of the Horn of Africa. It is the northern most part of the African Rift Valley, extending north from Lake Turkana in Kenya. It branches out into three trenches at the Afar Triangle, forming the Red Sea and the Gulf of Aden. It has many lakes, including Ethiopian Rift Valley lakes like Lakes Hawassa, Langano, Shalla, Abijata, and Ziway.

Western Branch

This part of the Rift Valley runs from Uganda in the north, southwards through the Democratic Republic of the Congo, Rwanda and Burundi to Tanzania. This part of the Rift Valley hosts the continent's highest block (horst) mountain, Mount Ruwenzori, and also lakes such as Lake Edward, Kivu and Tanganyika.

Eastern (Gregory) Branch

It runs from Lake Turkana in northern Kenya, crossing Tanzania to the east of Lake Victoria. Lakes such as Turkana, Naivasha, Norton, Manyari and Eyasi are located in it.

❖ ***The Great Rift Valley is a geologic depression that extends from Syria in southwestern Asia to Mozambique in southeastern Africa. It takes the form of a series of valleys and bodies of water that are bounded by parallel fault lines. The Great Rift Valley is widening , slowly but surely, and in the process is causing many volcanic eruptions and earthquakes in the area.***

Africa's largest lake, Lake Victoria (83,000 km²) is found trapped between the western and eastern trenches of the Rift Valley.

The following are the major characteristics of the East African Rift Valley:

- It is bounded by steep escarpments (edges);
- It has numerous active and dormant volcanoes;
- It is often affected by earth tremors like earthquakes, volcanism and landslides, making the Rift Valley very unstable;
- Has a hot and dry climate in its many parts, making the place difficult for human habitation;
- Many structural basins (lakes) occupy the floor of the Rift Valley.

3.2.3. CLIMATE OF AFRICA

Controls of Weather and Climate in Africa

Africa has varied climatic conditions. Why are there varied climates in Africa? The answer is that the variation in the climatic conditions of the different places in Africa is due to the intervention of weather

and climate controls. The most important factors that are responsible for this variation include latitude, altitude, distance from the sea, mountain barriers, ocean currents and major planetary winds and pressure belts. These factors are called climate controls because they regulate the conditions of the elements to produce different weather and climatic conditions. They are discussed below in detail.

Latitude

What is latitude? How does it affect the climate of Africa?

Latitudes indicate the distance places have from the equator. A place's latitudinal location affects the amount of incoming solar radiation the place receives, and thereby its temperature. As we discussed earlier, much of Africa (nearly 2/3rd) lies within the tropical latitudes. Hence, the continent receives high sun angles throughout the year. Due to its latitudes, Africa is the hottest of all the continents.

Altitude

What is altitude? How does it affect the conditions of temperature and rainfall in Africa?

Much of Africa is plateau. There are also high mountains in many parts of the continent, especially in Eastern Africa. These plateaus and mountains have great impacts on the continent's climate.

The plateaus and high mountains reduce maritime influence. This condition reduces temperature. For instance, Eastern Africa, despite its closeness to the equator, experiences highland climates. Similarly, the Atlas Mountains of North Africa and the Cape Ranges of South Africa experience the coldest temperatures in the continent. There are several reasons for this. One of the reasons that temperature is low in higher altitudes is that the air is very thin. This reduces the temperature-retention capacity of the air. The other reason is that, when air rises, it cools. The third factor is the fact that the atmosphere is heated from below (by the earth) directly, not by the sun. These are some of the factors why we feel colder at mountain tops than in valley bottoms.

Distance from the Sea

How do you describe the impacts of the Sea on the climate of Africa?

Water bodies that are adjacent to land masses have great impacts on the temperature and rainfall conditions of those places. Africa in its northern part is very wide. This makes much of the area far from the sea. In addition, the extensive plateaus of Africa, which almost reach the coast with steep edges, form barriers greatly reducing the influence of the sea. Furthermore, the continent's relatively straight and smooth coastline also reduces the impact of the sea in the interior areas. As a result, many parts of interior Africa experience continental climates with insignificant maritime influence.

Ocean Currents

What is Ocean Current? Which ocean currents affect the climate of Africa?

Oceanic water moves in two dimensions, vertically and horizontally. The horizontal movement of oceanic water is called ocean current. Based on their origin, ocean currents are of two types. These are

warm and cold. Warm ocean currents have high temperatures and high moisture content, and therefore they have warming effect. They also tend to bring moisture to the coastal areas. In contrast, cold ocean currents have cool temperatures and low moisture content.

Therefore, they have cooling effects on the areas that they blow over. They also bring no rain to adjacent areas, making places over which they blow very dry and desert.

Three major ocean currents affect the African continent. These are the Canary Cold Current, the Benguela Cold Current and the Mozambique Warm Current. The first two make the northwestern and southwestern parts of Africa both cool and dry. On the other hand, the Mozambique warm ocean current makes Southeastern Africa warm and wet.

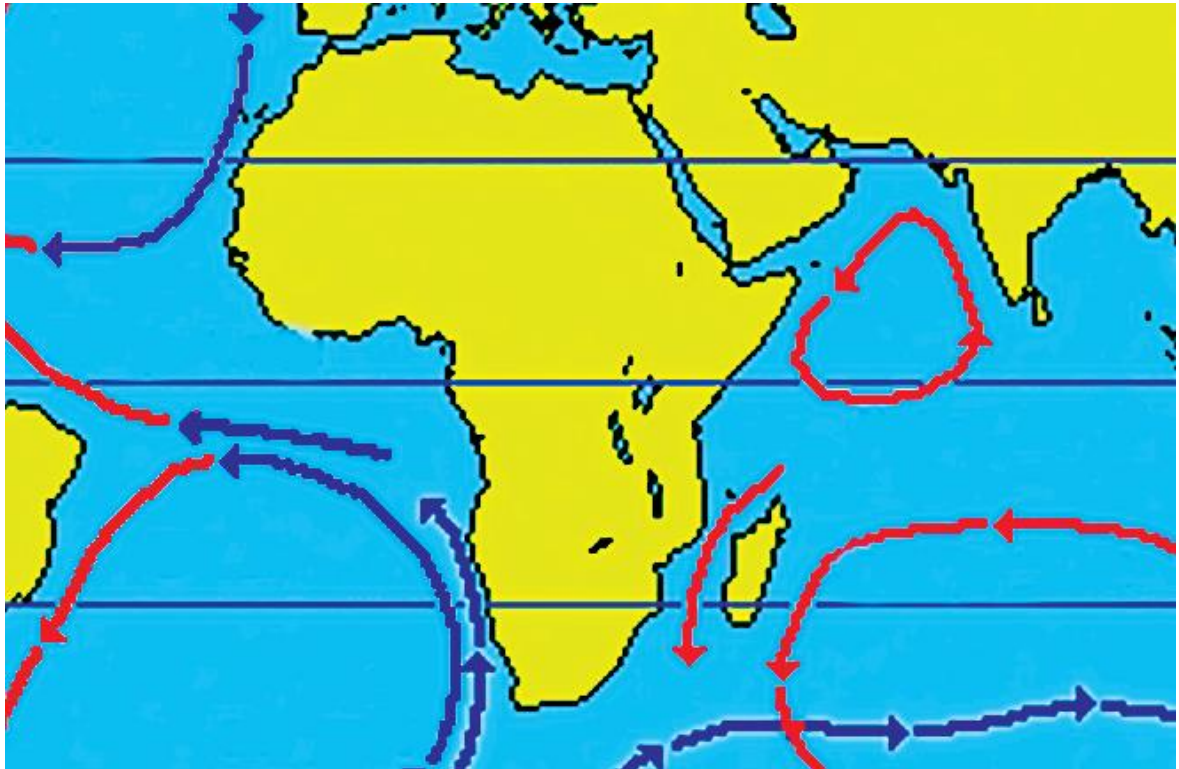


Figure: The major ocean currents affecting the climate of Africa

Major Planetary Winds and Atmospheric Pressure

What is wind? What about atmospheric pressure?

What are the major winds and pressure systems that affect Africa?

Most of Africa lies within the tropics. Hence, much of it lies within the trade wind belts. The southeast trades and northeast trades dominate the climate of tropical Africa. The westerly winds from the subpolar high-pressure belts of the world reach the southern and northern tips of the continent. The Guinea monsoon winds (equatorial westerlies) also have significant impacts on the climate of equatorial Africa. Of the global pressure belts, the subtropical highs (around 30°N and S) and equatorial lows (doldrums) (between 5°N and 5°S latitudes) are the main pressure belts that affect the climate of the continent. In addition, the Inter Tropical Convergence Zone (ITCZ), which is the zone of convergence between the trade winds, regulates the winds that blow into Africa at different seasons.

The ITCZ moves between the tropics of Cancer and Capricorn, following the overhead sun, pulling winds towards it. Hence, it controls the distribution of rainfall in Africa.

Temperature Conditions in Africa

Spatially, lowlands (coastal areas) and the desert and semi desert areas of Africa experience the highest temperatures in the continent. The Sahara, the largest desert in the world, has the highest temperature. The Kalahari Desert in the south has relatively cooler temperatures as a result of the cold Benguela ocean current. The Ethiopian and East African highlands, on the other hand, have lower temperature condition as a result of their high altitude. The Atlas of Morocco and Cape Ranges of south Africa have the lowest temperature in the continent as a result of their high altitudinal and latitudinal location.

The distribution of temperature in Africa also has seasonal variation. Such variation is the result of the apparent movement of the overhead sun between the two tropics. The months of December, January and February constitute the summer season in the southern hemisphere. During this time, the sun is overhead, south of the equator. As a result, areas of high temperature are found in Southern Africa. The Northern and northeastern parts of the continent remain relatively cool and dry at this time. Similarly, the summer season in the northern hemisphere corresponds with the months of June, July and August. This season is a season of high sun angle in Africa north of the equator. Hence, the Sahara and other parts of Northern Africa experience high temperature conditions. In contrast, the areas south of the equator are characterized by lower temperatures at this time.

Rainfall Distribution in Africa

The distribution of precipitation (rainfall) in Africa is controlled by the Inter Tropical Convergence Zone (ITCZ). The position of ITCZ is, in turn, determined by the position of the overhead sun. For instance, in July the overhead sun is located near the Tropic of Cancer. Hence, winds that carry moisture from the

Atlantic and Indian Oceans, the Guinea monsoon (equatorial westerly) and the southeast trade winds, invade parts of Africa north of the equator, causing the region to receive high rainfall. In this season, West African coastal areas, the Ethiopian highlands, and eastern Madagascar get their heavy rainfall.

In January, the sun is overhead near the Tropic of Capricorn, pulling the northeast trade winds southward over Africa. These winds are continental in origin, carrying limited or no moisture. In this season, therefore, Northern Africa remains dry, except for the Maghreb region, where the Mediterranean type of climate dominates. Southern Africa, however, gets its maximum rainfall from the southeast trades of

the Indian Ocean. The Congo Basin gets rainfall from the moist winds of the Atlantic Ocean. To better understand the points so far discussed, study the following points.

- The Guinea monsoon(equatorial westerly) winds from the Atlantic Ocean bring rainfall to the north of Africa up to the southern fringes of the Sahara in July;
- Moist winds from the Atlantic Ocean and the southeast monsoon winds from the Indian Ocean bring rainfall to Southern Africa in January;
- The westerly winds bring rainfall to the Maghreb region of North Africa and the Cape Province of South Africa in their respective winter seasons. Summer is not the wettest season in these places;
- Summer is a season of heavy rainfall in all parts of Africa, except in the Sahara desert and the Mediterranean-climate regions;
- Winter is dry in all parts of Africa, except for the equatorial and Mediterranean climate regions.

Climatic Regions of Africa

The distribution of temperature and rainfall show great spatial and seasonal variations in Africa. As a result, many different climatic regions are established by these variations. A climatic region is a geographical area with more or less similar climatic characteristics mainly of temperature and rainfall. Multiple areas can be characterized by a single climatic region,

There are seven main climatic regions in Africa. These are:

- ✓ Equatorial
- ✓ Tropical continental (savanna)
- ✓ Tropical desert and semi-desert
- ✓ Tropical maritime (monsoon)
- ✓ Warm temperate continental
- ✓ Highland (mountain)
- ✓ Mediterranean

The following map shows the locations of the major climatic regions of Africa. Study it carefully and try to identify the geographical locations of each region.

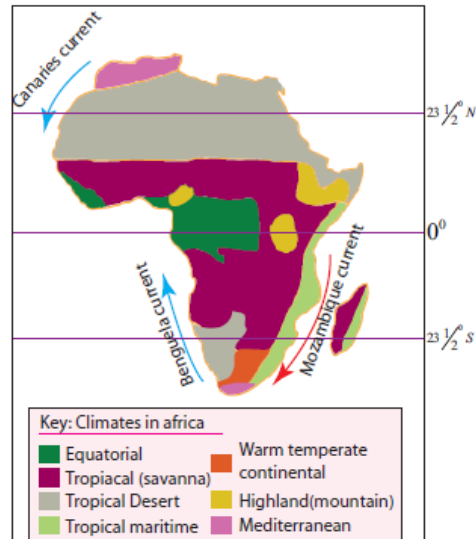


Figure: Climate Regions in Africa

The Equatorial Climate

The parts of Africa that are found around the equator have this type of climate.

The region surrounds the equator extending between 6° or 7°N and S latitudes.

The region is characterized by:

- High sun angle throughout the year;
- High mean monthly and mean annual temperatures;
- High daily and low annual range of temperature; and
- High total annual rainfall, with rain falling throughout the year.

Tropical Continental (Savanna) Climate

Where in Africa do we have the savanna climate? What climatic features characterize this type of climate?

This climatic region is found between 5° and 15° north and south latitudes. It occurs north and south of the tropical wet zone, in many parts of Western Africa and Southern Africa and in most of Madagascar. The region is situated between the wettest and driest climate zones of the continent. As a result, it is said to be the zone of transition between the equatorial and the desert climatic regions of

Africa. In this region, summer is very hot, with an average temperature of 25°C, and winter is cool, with an average temperature of slightly above 15°C.

The region is characterized by:

- a well-defined dry season of three to eight months, with annual rainfall ranging between 500 and 1,500 mm;
- a progressive decline in total annual rainfall, north and south wards;

- that part of the savanna which borders the equatorial rainfall region receives high rainfall, with that amount decreasing as distance from the equator increases.
- slightly higher ranges of temperature than the equatorial climatic zone, with the range increasing with distance from the equator; and
- high daily temperatures averaging more than 30°C in its northern section throughout the year, with relatively lower temperatures in its Southern and eastern sections, due to higher altitudes.

The Tropical Desert and Semi-desert Climate

What characterizes the desert and semi-desert climatic regions? Where in Africa are these types of climate found?

This type of climate is found bordering the tropical savanna climatic region, especially in north central and Southern Africa. There are two types of deserts in Africa. These are the coastal and continental deserts. The coastal type includes deserts that are found along the western coast of the continent, like the Namib and western Sahara.

In contrast, the Sahara, which constitutes Africa's continental type of desert, is located north of the equator.

The following characteristics distinguish the tropical-desert and semi-desert climatic region from the others.

- A short rainy season of up to three months. There are about 250 to 500 mm of rain per year in the semi-desert areas and less than 250 mm in the desert regions;
- Variable, unreliable and insufficient precipitation, which hinders plant growth;
- High daily average temperature, which ranges between 25°C and 36°C;
- Significant annual temperature variations and also extreme fluctuations in temperature over the course of a day. For instance, in the Sahara desert, daytime summer temperatures exceed 50°C, but winter night temperatures drop below freezing.

Drought in Africa

Drought is a condition of unusually dry weather within a geographic region.

Expected rainfall does not occur during drought. Therefore drought conditions differ greatly from conditions in an area that is normally, or at least seasonally, dry.

The term is usually applied to a period in which an unusual shortage of rain causes a serious hydrological imbalance a situation whereby water-supply reservoirs empty, wells dry up, and crop damage follows. The severity of a drought is measured in terms of the degree of moisture deficiency, its duration, and the size of the area affected. Droughts tend to be more severe in some areas than in others. Disastrous droughts occur mostly at latitudes of about 15°-20°, in areas bordering the permanently arid regions of the world. As most parts of Africa lie within these latitudes, the continent is one of the most drought-affected areas in the world. Repeated drought is common in many areas.

Major causes of drought in Africa include unwise use of natural resources, including deforestation, overgrazing and over cropping, expansion of farm lands and settlements, and the resultant environmental degradation.

The repeated drought that occurs in many areas of the continent causes environmental degradation, habitat destruction, shortage of water and famine (shortage of food). This, in turn, results in the displacement/migration and death of people and animals. The huge displacement of people and the high rural-urban migration that occurs in many countries of Africa is associated with this phenomenon. Drought also affects the biodiversity of a place. As drought prevails, plants and animals may fail to survive. As a result, some species of plants and animals may become extinct or be exposed to danger of extinction.

This is happening in the drought-affected areas of the continent. The Sahel region of Africa is one of the extremely drought-affected areas in the continent. The region is a transitional zone between the Sahara on the north and the wetter tropical areas to the south. Desertification of the Sahel was aggravated by extended drought between the late 1960s and early 1980s, the worst in 150 years. The stress of increasing human and livestock populations is another major contributor.

Desertification is the process whereby soil loses its ability to retain moisture, and then desert encroaches on arable land. Desertification is shrinking the size of the Sahel and causing famine in many parts of the region. This climatic condition covers many countries that are found to the south of the Sahara desert. As a result, they are among the areas repeatedly affected by drought.

The following figure shows you the drought-prone areas of Africa. Study it carefully and try to identify the countries that are found in the zone.

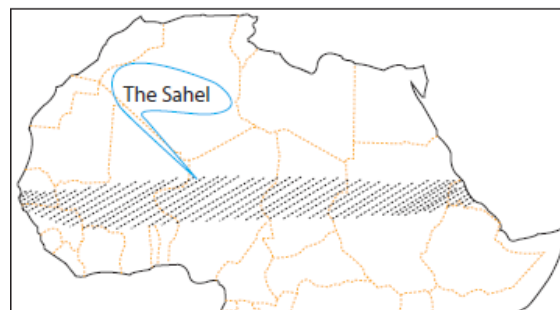


Figure: The Sahel, drought-prone areas in Africa

3.2.4 Drainage in Africa

Africa is rich in terms of water resources. Thousands of rivers that originate in African highlands drain the extensive landmass of the continent. Furthermore, large numbers of lakes are found in the continent. Swamps are also dominant drainage features in Africa. These resources have tremendous potential. However, the people of the continent are far behind the developed world in terms of their socio-economic status, and therefore also in the technology and other resources needed to develop this potential.

The Major Rivers and Drainage Systems of Africa

The entire area that a river drains is called its catchment area or drainage basin.

A group of drainage basins, which are supplied by multiple rivers, with common characteristics, such as a common destination, form a drainage system.

Nine major rivers drain the African continent. These are the Nile, Congo, Zambezi, Limpopo, Orange, Niger, Volta, Gambia and Senegal rivers. These basins are categorized into four major types of drainage systems, based on their flow direction. These major drainage systems are

- Atlantic Ocean drainage system;
- Indian Ocean drainage system;
- Mediterranean Sea drainage system; and
- Closed (Inland) drainage system.

A .The Atlantic Ocean Drainage System

What are the major rivers that constitute this system?

This system is made up of all major rivers that drain westward and southward and empty into the Atlantic Ocean. The system is the largest in terms of catchment area, annual discharge and drainage density. It accounts for about 90 percent of the continent's surface flow. The major river basins that constitute this system are the Congo, Niger, Volta, Gambia, Orange and Senegal basins. Several other, small rivers are also found in this system.

The following Table 3 shows you the major river basins and their tributaries.

Table: Major rivers in the Atlantic Drainage system

River Basin	Length (Km)	Remark
Congo	4,380	Largest river in Africa in terms of total annual discharge, 2 nd longest river in Africa and 2 nd largest in the world.
Gambia	1,130	One of the smallest major river basins of Africa
Senegal	1600	Has two major dams. the Manantali Dam in Mali, and the Maka-Diama dam on the Mauritania-Senegal border.
Niger	4000	Drains Guinea, Mali, Niger, Benin and Nigeria. 3 rd longest in Africa.
Volta	1500	Has Africa's largest artificial lake, Lake Volta; has three major parts. Black Volta, White Volta and Red Volta.

B. The Mediterranean Sea Drainage System

Which river is the most important in this system? Why?

This drainage system includes all the major rivers that flow northwards into the Mediterranean Sea. However, the system has only one major river basin, which is the Nile. The Nile is the longest river in the world. Two major tributaries, namely the White Nile and Blue Nile, form the Nile proper when they

converge in the Sudan, at Khartoum. The White Nile originates from Lake Victoria in Uganda, while the Blue Nile emerges from Lake Tana in Ethiopia, near Bahir Dar.

The Nile flows generally northwards through the Sudan and Egypt and empties into the Mediterranean forming an extensive delta in north Egypt.

C. The Indian Ocean Drainage System

Which rivers of Africa form the Indian Ocean Drainage System?

The Indian Ocean Drainage System includes all the major rivers that flow eastwards into the Indian Ocean. This system is the second largest system, in Africa in terms of drainage density, catchment area and annual discharge. The Zambezi, Wabishebele and Juba river basins are the major basins in this system.

Table: Major River Basin of the Indian Ocean drainage system

River Basin	Length (Km)	Remark
Zambezi	2750	Has Africa's largest waterfall, Victoria Falls; has two major dams: Kariba (Zambia and Zimbabwe) and Cabora Bassa (Mozambique)
Wabishebele	2000	Has High seasonal variation, and therefore usually fails to reach the Indian Ocean in winter. However, it manages to reach the ocean during the summer season.
Jubba	1600	Forms one of Somalis most fertile agricultural region in its lower valley.

D. The Closed (Inland) Drainage System

What are the rivers that form the Inland drainages system of Africa?

The rivers of Africa that do not have direct access to the sea form this drainage system. Most of the rivers in this system have multiple flow directions. This makes the system different from the rest of the drainage systems in Africa. This drainage system covers nearly 32% of the total area of the continent. It receives nearly 4% of the continent's total annual runoff. The Awash and Ghibe/Omo rivers in Ethiopia, the Okovango Swamp in Botswana, the Sudd Basin in the Sudan, the Danakil Basin in Ethiopia and Eriterea and Lake Chad Basins are among the major inland drainage basins of the continent. The Chad Basin is the largest inland basin in Africa.

Table: Catchment Areas and Lengths of Some of the major Rivers of Africa

River	Length (km)	Catchment Area ('000' km ²)	Main Tributaries
Nile	6650	2862	Blue Nile, White Nile, Sobat, Atbara
Zaire/Congo	4380	3700	Kassai, Ubangi
Niger	4000	1112	Benue
Zambezi	2700	1300	Cubango, Cuando
Ghenale/Juba	1600	168	Dawa, Weyb, Mena, Welmel
Wabeshebele	2000	205	Erer, Fafen, Ramis

General Characteristics of African Rivers

Most African rivers share some common characteristics:

a) **Steep long profile:** Most of the rivers in Africa have steep courses, as a result of the continent's relief. Furthermore, most of the rivers empty into the major water bodies by falling from the edges of interior plateaus of the continent. This affects the navigability of many of the rivers of the continent.

b) **Waterfalls and Rapids:** Many African rivers are interrupted by waterfalls and rapids, which impede navigation. The waterfalls and rapids are due to Africa's number of plateau lands, with their steep sides, and strong erosion-resistant rock, which the rivers encounter in their courses. Some examples of rivers with these features are the Congo, Nile, Niger, Zambezi, Orange and Cunnen Rivers. Study the following table, which presents Africa's major waterfalls.

Table: Major Waterfalls in Africa

Name and Location	Height of Falls (m)
Tugela, South Africa	614
Chirombo, Zambia	268
Kalambo, Tanzania, Zambia	221
Maletsungane, Lesotho	192
Victoria, Zimbabwe, Zambia	108
Tis Isat, Ehtiopia	45

c) **Seasonal Fluctuation:** Many of the rivers of Africa have their origins in areas of seasonal rainfall distribution. As a result, there are variations in their volumes between the wet and dry seasons. The Nile and Niger Rivers are good examples of this phenomenon. Both originate in wet highlands with seasonal rainfall. In contrast to the other African rivers, the Congo River does not show significant volume

variation. It is the only African river with a steady volume throughout the year. This characteristic is due to the fact that it has tributaries running from both within and south of the equator.

d) **Deltaic Mouths and Mangrove Swamps:** Many of the African rivers have low pressure force along their lower courses. This allows the rivers to branch out into distributaries, and results in the formation of deltas and mangrove swamps at the mouths of the rivers. For instance, the Nile, Niger and Zambezi rivers have extensive deltas and are swampy at their mouths. Such conditions affect the penetrability of the rivers from the coast.

e) **Exotic Nature of the Rivers:** Many of the rivers of Africa, like the Nile, Senegal and Orange, travel across different physiographic regions that range from cool to extremely hot climatic conditions. As a result, they lose much of their water through evaporation and seepage before they reach their final destinations. The Nile River faces the greatest impact in this case. It loses nearly 64% of its total run off through evaporation and seepage. The Senegal River loses 54%, and the Orange River has a total loss of 44% in its runoff.

❖ Deltaic mouths, mangrove swamps, the fluctuation regime of the rivers, and waterfalls and rapids across the major rivers of Africa hinder the navigability of their courses. However, these rivers have high HEP potential. The water falls of Africa, if regulated, could produce enormous amounts of hydro-electric energy.

Lakes and Swamps of Africa

What is a lake? What about swamps?

Where are most of the lakes of Africa found?

Africa has many lakes and swamps. Some of them are natural, and others are human-made (artificial). The lakes differ in their size and depth. For instance, lakes Tanganyika and Malawi are deep and large, while Victoria and Tana are wide and shallow, respectively.

Lakes of Africa

The lakes are divided into **two types-natural and artificial** (anthropogenic).

Natural lakes are formed by tectonic, volcanic and/or denudation processes.

Anthropogenic lakes are formed when water is accumulated at the backs of dams that are constructed across rivers for various purposes, including hydro-electric production and irrigation.

Natural Lakes

What is a natural lake? How are lakes formed naturally?

These are lakes that are formed under natural conditions. Based on location, natural African lakes are divided into two groups: Rift valley and Non-Rift Valley lakes. The first group includes all the lakes that occupy the floor of the Great East African Rift Valley. Lakes, like Turkana, Tanganyika, Kivu, Albert, Malawi, Edward and the numerous Rift Valley lakes in Ethiopia belong to this group.

The major **non-Rift Valley** natural lakes include Victoria, Chad and Tana.

The following Table presents the major natural lakes of Africa and some of their characteristics.

Lake	Area (km ²)	Maximum depth (m)	Type	Remark
Victoria	83,000	92	Non Rift valley	Largest in Africa, largest tropical lake in the world, 2 nd largest freshwater lake in the world and 3 rd longest in the world
Tanganyika	32,890	1435	Rift valley	World's longest fresh water lake, Africa's deepest lake and second in the world, Africa's second largest lake
Malawi	30,800	706	Rift Valley	The most southern lake in the Great African Rift Valley system
Chad	18,000	12	Non-Rift valley	-
Turkana	8,660	72	Rift valley	-
Albert	5,500	17	Rift valley	The northernmost of the chain of lakes in the Great Rift Valley
Meru	4,920	NA	Rift valley	-
Tana	3,600	9	Non-Rift valley	Shallowest in Africa
Edward	3,550	NA	Rift valley	-
Kivu	2,650	475	Rift valley	-

Artificial (Anthropogenic) Lakes

What are artificial lakes? How and why are artificial lakes formed?

Africa has large numbers of rivers that have enormous hydro-electric power and irrigation potential. To utilize the rivers for these and many other purposes, large dams have been constructed across their courses. The major anthropogenic lakes in Africa are Nasser, Koka, Volta, Kaindji and Kariba.

Table presents the major artificial lakes in Africa.

Lake	Dam	River	Country
Nasser	Aswan High Dam	Nile	Egypt
Koka	Koka	Awash	Ethiopia
Volta	Akosombo	Volta	Ghana
Kaindji	Kaindji	Niger	Nigeria
Kariba	Kariba	Zambezi	Zambia and Zimbabwe

Swamps of Africa

What is swamp? How are they formed?

There are many seasonal and permanent swamps (marshes) in Africa. They develop mostly in depressions and areas of seasonal flooding along the courses of the major rivers. The major swamps of Africa include:

- Sudd swamps, along the Nile river basin;
- Kamulando swamps, in the Congo Basin;
- Batorse and Kafue swamps, in the Zambezi Basin;
- Okovango swamps, in Botswana;
- Swamps adjacent to lake Chad;
- Mangrove (coastal) swamps along the deltaic mouths of the major rivers;
- Timbukto swamps, in Mali along the Niger river.

The Uses of African Rivers and Lakes

The rivers and lakes of Africa have great potential for development. They can contribute to the socio-economic development of the continent if they are properly developed and utilized. Some of their uses are discussed below.

Hydro-Electric Power (HEP)

Africa has about 40% of the world's HEP potential. The steep profile of the rivers and the waterfalls and rapids that develop along their actual and potential courses make the continent rich in this respect. Although, very little (about 5%) of this potential is actually being utilized, large dams have been constructed for this purpose. For example, see the following Table.

Table: Major Dams of Africa

Dams	River	Location
Aswan High Dam	Nile	Egypt
Owen Falls	White Nile	Uganda
Koka	Awash	Ethiopia
Akosombo	Volta	Ghana
Kaindji	Niger	Nigeria
Kariba	Zambezi	Zambia and Zimbabwe
Inga I and II	Congo	Congo, DR
Cabora Bassa	Zambezi	Mozambique
Ghibe I and II	Gilgel Ghibe	Ethiopia
Tekeze	Tekeze	Ethiopia

Irrigation

The rivers and lakes in Africa have great potential for irrigation. However, what has been utilized so far is insignificant compared to this huge potential. Lack of technology, capital and skilled human power, as well as conflicts and political unrest, are among the factors that result in low development of irrigation in Africa. Among the major irrigation schemes, some are the Gezira and Kenana irrigation in Sudan, Nile Delta and lower Nile irrigation in Egypt; Fish river, Orange and Pongola irrigation in RSA.

Fishing

What is fishing?

Africa's rivers and lakes are rich in fish resources. There are about 2,000 different species of fish in the continent.

The most widespread human use of lakes in Africa is for fishing, but this economic sector is poorly developed. Most fish production in the continent is for home consumption. As a result, very little is taken into the market. This situation is primarily due to insufficient technical skills in the continent.

Navigation (Inland Waterways)

Most of the rivers in Africa are characterized by waterfalls, rapids, steep profiles, deltaic mouths and seasonal volume fluctuation. These conditions hinder their navigability. However, the Nile, Niger, Senegal and Gambia Rivers are navigable along parts of their courses, especially in summer. The Congo River is navigable for a good part of its course throughout most of the year. Lakes like Chad, Victoria, Tana and Malawi also provide transport services for a good number of people.

Tourism and Recreation

In many countries in Africa, rivers and lakes are good tourist destinations and centers of recreation. The waterfalls along the rivers and the birds of the lakes are attractive. The Nile in Egypt, Victoria Falls on

the Zambezi River, and Lake Nakuru in Kenya and Lake Malawi in Malawi and Tis Abay on the Abbay River, and Awash Falls in Ethiopia are good examples.

Fresh-Water Supply

The rivers and lakes in Africa constitute most of the continent's fresh-water resources. The piped water supplies for urban Africa and the water supply for the rural population, are all dependent on the rivers and lakes of the continent.

Source of Minerals and Construction Materials

Rivers and lakes carry various rocks and their fragments that are good sources of minerals and construction materials. For instance, the alluvial deposits along the major rivers of Africa contain gold and diamonds, as well as other minerals, for example, tin in Ghana and Namibia. Salt and potash as well as sand and gravel can be obtained from rivers and lakes.

The Hydro Politics of the Nile River

The Nile River is one of the most politically significant rivers in Africa. In its basin countries have significant interest over its water. The countries that are found in the Nile basin are Ethiopia, Egypt, Kenya, Sudan, Uganda, Rwanda, Burundi, Democratic Republic of Congo, Eritrea and Tanzania. Based on the general alignment of the river these countries are categorized into two as upper course and lower course countries. The upper course countries are those that contribute the water for the river and are generally found at higher altitude where the major tributaries of the Nile originate. On the other hand, the lower course countries, namely Sudan and Egypt, are those that are found at lower elevation where the water of the river flows gently over vast plains.

Blue Nile is the largest contributor of water to the Nile proper. Together with Baro-Akobo and Tekeze rivers, Ethiopia accounts for about 84% of the water of the Nile. The Hydro-politics of the Nile is, therefore, related with the degree to which the Nile river is utilized in its upper and lower courses. Of the total estimated 110bn m³ of annual water resource, nearly 65% (72 bn m³) occurs in the Ethiopian portion of the Nile basin. Of this, about 52.6 bnm³ is accounted by the Abay river alone.

Historically, the two countries, Egypt and Sudan have been the most benefited of all the countries in the Nile basin. This is especially true for Egypt where the Nile River's water and alluvial soil along its flood plain have become the source of life in the desert affected Egypt. The Aswan high dam that is constructed along the Nile River in Egypt has been the most important source of water, energy, fish and recreation for the Egyptians. In Sudan, too, the river has been developed to a greater extent. Contrary to this, the upper course countries have been the least benefited. This unbalanced and unfair utilization of the river between the upper and lower course countries has been a great area of interest. However, the countries of the basin are now in a situation where by they are working together to bring about equitable utilization of the river. The Nile basin initiative is a good example in this case. Through the initiative, the countries are working together to maintain balanced utilization of the river in the upper course and lower course countries.

3.2.5 Natural Vegetation and Wild Animals of Africa

There are different types of climates in Africa. As a result, varied types of natural vegetation develop over the landmass of the continent. There are also various species of wild animals that inhabit the various physiographic regions of the continent.

Major Vegetation Zones of Africa

What is Natural Vegetation? What are the major vegetation types that develop in Africa? What causes a variation in vegetation type and distribution in Africa?

The type of natural vegetation that develops in a certain environment is a reflection of the climatic characteristics of that place. Africa has a number of different climatic regions with their own distinguishing climatic features. As a result, we have various types of natural vegetation covering the different geographic regions of the continent. In general, five different vegetation zones can be identified in Africa. These vegetation types are discussed below.

- Tropical rainforests
- Tropical grassland (savanna)
- Desert and semi-desert vegetation
- Afro-montane (Afro-alpine) vegetation
- Mediterranean vegetation

Tropical Rainforests

What is a rainforest? Where in Africa do we have rainforests?

What unique features do rainforests have?

Tropical rainforests develop in areas with equatorial climates. In Africa, they are confined to Central and Western Africa, and eastern Madagascar. In these places, the climate is typically tropical, with high rainfall and high temperatures throughout the year. Tropical rain forests are also known as equatorial broad leaf evergreen forests. Africa's most extensive rainforest is found in the Congo Basin.

The following points give you some ideas about the nature of these forests.

- Rain forests are complex, with these three distinct layers:
- Top layer – made up of tall trees (30 - 50 m) with buttress roots;
- Middle layer – made up of tree ferns, lianas (creepers), epiphytes, and trees with heights of 19 - 34 m; and
- Bottom layer – consists of ferns, herbaceous plants, saprophyte (plants which live on dead plants) and trees with heights of up to 17 m.
- They contain broad-leaved evergreen trees (green throughout the year);
- Different plants exhibit different stages of growth at the same time, due to the absence of climatic seasons. Some are in flower, some in fruit, and others in the leaf-fall stage;
- They have little undergrowth, as the canopies of the tall trees prevent light penetration;
- Most of the trees are hardwood, like mahogany, ebony, ironwood, rosewood and green heart;

- They have high species diversity and thick growth.

Study sketch given below to learn what the vegetation in tropical areas looks like.

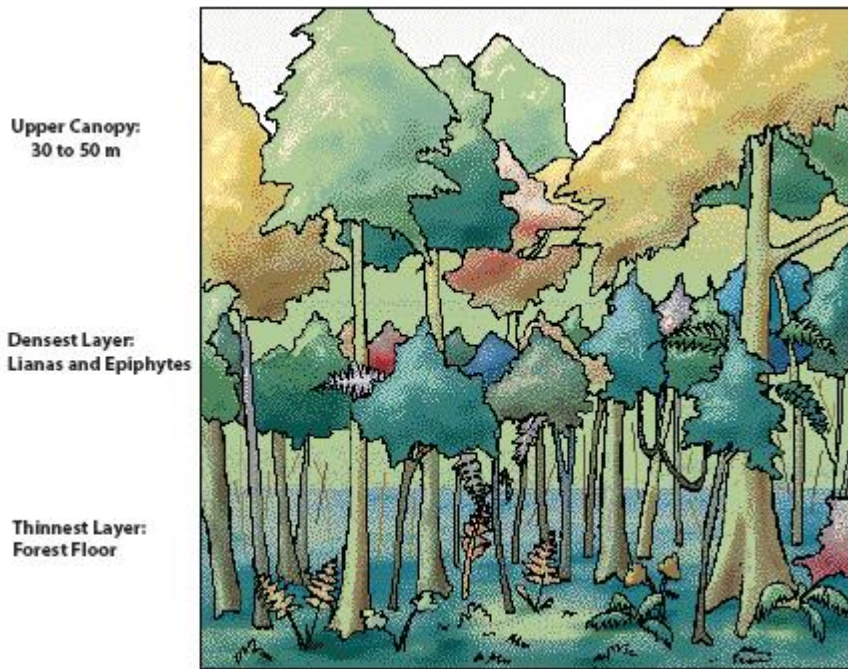


Figure: A Sketch of the Tropical Rainforest

Tropical Grassland (Savanna)

What is a Savanna? Which parts of Africa have savanna vegetation?

This kind of vegetation develops in areas of seasonal rainfall. Tropical grassland vegetations are extensively developed in areas that have wet summers and dry winters. Tropical grasslands exist in Northern and Southern Africa, and they encircle the equatorial rainforests. Rainfall in savanna areas varies, decreasing over the range from the forest margins to the edges of the desert. Savanna areas that are close to the equatorial rainforests receive more rainfall, compared to the areas at the edge of the desert. This results in the development of three Savanna zones in Africa.

These are:

- Park (Wetter) Savanna:** it is found close to the rain forests and made up of many trees and grasses. Geographically, it is found in Western Africa, northern Congo, southern Sudan and central Malawi.
- High (True) Savanna:** this type of grassland is developed between the wetter and drier savannas. It is made up of more grasses than park-savanna areas, and contains scattered trees only. It is found in Zimbabwe, Malawi, southern Kenya, eastern Tanzania and Western Africa.
- Thorn Scrub (Drier) Savanna:** this savanna zone is developed along the desert margins and is made up of short grasses with widely scattered thorny trees, thorn bushes and low scrub. It is common in the

semi-arid areas of the Sahel region that extends from Senegal to Ethiopia, Northern Kenya, Angola and Botswana.

The main features of savanna vegetation include the following.

- They have tall grasses that are often as tall as 2 m;
- Trees are more common than grasses in areas bordering the forest, and grasses are more common than trees in areas bordering the desert;
- Most of the grasses wither and turn brown in the dry season and regain in the wet season;
- The trees survive the dry season by shading their leaves, storing water, having long roots, thorny leaves, and only a small number of leaves.

Desert and Semi Desert Vegetation

What is a desert? What about a semi-desert climate? What type of vegetation develops in such climatic regions? In which parts of Africa do we have desert and semi-desert vegetation? Why?

Desert and semi-desert vegetation is developed in areas where rainfall is scant.

This climatic condition is found in the driest areas of the continent. The Sahara, the largest desert in the world, the Namib deserts and the Sahel region of Africa have such vegetation.

You may think that deserts are devoid of natural vegetation. The reality, however, is that it is different in Africa. The continent's drier areas have high species diversity. For instance, more than 3,000 species of plants (about 20% endemic) are found in the desert and semi-desert zones of Northern Africa. Even the southern deserts support plants. Cactus, thorn bushes and coarse grasses are among the most common plants in this vegetation zone.

The desert and semi-desert climatic regions are characterized by very low rainfall, extremely high evaporation and low humidity. The plants that develop in such regions are xerophytes with high drought resistance. To withstand such problems and survive in deserts, desert plants have different mechanisms of adaptation. For example,

- They have long roots that can penetrate to great depths to reach the underground water table;
- They store water in their spongy leaves, stems, roots, fruits, and the like (example, cactus);
- They have waxy or needle-shaped leaves to reduce water loss through transpiration;
- They produce seeds that lie dormant for several years during extreme dry seasons until rain falls;
- Their leaves are small in size and few in number, reducing water loss through transpiration; and
- They have thorny leaves to protect them from being eaten by herbivores, etc.

Afro-Montane (Afro-Alpine) Vegetation

What type of vegetation is Afro-Alpine vegetation? In which parts of Africa do we have this type of vegetation?

This type of vegetation develops over the tropical highlands of Africa, mainly over the Ethiopian and Eastern African highlands. In these areas, the climate is highly modified by altitude, with temperature decreasing as altitude increases.

As a result, the vegetation that would have existed in the tropical climate at lower temperatures is replaced by vegetation typical of temperate regions. The vegetation consists of highland (temperate) forests and temperate grasslands. As altitude decreases, vegetation varies. For example, alpine (cold-resistant) plants like *Aster* and *Gibber* (in Ethiopia) grow in areas with altitudes above 3000 m.

Afro-montane forests grow in altitudes up to 3000 m. Bamboo forests are found at 2000 – 2500 m. Then we find temperate evergreen (coniferous) forests of trees such as *Taxus*, mountain grassland, and heath.

Mediterranean Vegetation

Which parts of Africa have this type of vegetation? What are the common trees of the Mediterranean vegetation in Africa? As the name indicates, this type of vegetation develops in the northwestern and southwestern extremes of the continent, where Mediterranean climate is dominant. The region is rich in plant species. Evergreen and deciduous trees constitute a good part of the vegetation of this zone. Cork oak, maquis, and wild olive, are among the most common plant types in the region.

Mediterranean climatic regions have hot dry summer seasons. Thus, plants develop certain adaptation mechanisms to withstand the summer drought. Among others, the following adaptation mechanisms are common.

- Storing water in their leaves and bark and using it during the dry season;
- Having waxy thick leaves to reduce water loss through transpiration;
- Having spiny small leaves to reduce water loss through transpiration; and
- Having long roots to tap underground water.

Factors Affecting the Natural Vegetation Of Africa

Despite the richness of the continent in terms of natural vegetation, the resource is far from being properly utilized. Deforestation, overgrazing, burning (wildfire), and the expansion of settlements and farmlands are among the major problems affecting natural vegetation in the continent.

Of all the challenges of natural vegetation in Africa the most serious one is deforestation. Deforestation is indiscriminate cutting or over-harvesting of trees. The deforestation rate in the continent is both very high and escalating. In the late 1970's, for instance, the annual rate of deforestation in the continent was about 3.6 million hectares per year. Slightly less than a decade and a half later, this rate (according to 1993 FAO estimations) had reached 4.3 million hectares/year. At the present time, the figures are even greater than what we have seen above.

Clearance of tropical forests for various reasons is a common practice in many African countries where rainforests develop. People clear the forest for many reasons. The major ones, however, are the following.

- They clear the land for shifting cultivation, especially in the equatorial areas where soil leaching is a common problem;
- The need for land for permanent agriculture;
- The increased need for fuel wood - nearly 90% of African energy demands are satisfied by using fuel wood that is collected from forests;
- Extractive forest uses, such as selective forestry, to get logs for industries.

Deforestation has multifarious impacts. Forests regulate the climatic conditions of the earth and reduce soil erosion. Thus, deforestation can cause climatic change, resulting in problems like desertification and soil loss through erosion. In addition, where the forest habitats are destroyed, the animals that live in the forests are affected. As a result of habitat destruction, animals may migrate and even die. Deforestation results in the extinction of some of the wild animals and plants of the continent. There are also many endangered plants and animals in the continent due to deforestation and related impacts.

Possible Conservation Measures

The problem of deforestation can be reduced through the application of different forest-conservation measures. These measures include reforestation, afforestation, agro-forestry and social forestry.

Reforestation: is planting trees in areas where the original forest cover has been removed. It is done to replace the trees that have been cut by humans for different purposes.

Afforestation: is planting trees in areas where there was no original forest cover. For example, afforestation is appropriate for areas where the land is left empty and therefore is exposed to erosion.

Agroforestry: is forestry combined with farming. It is a practice of integrating the planting of trees into farming to provide fuel, fruit, forage, shelter for animals or crops, and other benefits. In short, it refers to associating crop production with forest development.

Social forestry: refers to planting trees in urban areas in association with human settlements.

Wild Animals of Africa

It is a well-known fact that natural vegetation serves as habitats for wild animals. Thus, the different vegetation zones of Africa, together with the varied climate and topography of the continent, create an ideal situation for wild-animal diversity.

For example, the equatorial rainforests of Africa are habitats for different species of wild animals. The region hosts different kinds of tree climbing (arboreal) animals like monkeys, apes, baboons, and gorillas, as well as birds.

The aquatic environment of this zone hosts large animals like hippopotamus and crocodiles. The region has the highest species diversity in Africa. Most of the time wild animals that live in the equatorial rainforest have small body sizes. This is because the thick and dense forest of the region limits movement in the forests. There are also large numbers of herbivorous and carnivorous animals in the Savanna lands of the continent. The herbivore animals include numerous species of antelope, zebra, giraffe, buffalo,

African elephant, and rhinoceros. The carnivore wild animals include the lion, leopard, cheetah, hyena, jackal and mongoose.

There are also various species of wild animals in the desert and semi-desert areas of the continent. The desert fox, hares, gazelles, jerboa, the wild ass and different reptiles like snakes, lizards and tortoises are among the most common species. Similarly, the rivers, lakes and swamps of the continent are inhabited by different species of aquatic animals.

There are crocodiles like the Nile crocodiles, hippopotamuses, fish (about 2,000 different species), and different species of birds like the guinea fowl (the leading game bird in Africa), pelicans, goliath herons, flamingos, storks, egrets, and the ostrich, mainly in Eastern and Southern Africa. In addition to the above wild animals, the continent also has a variety of destructive insects, notably mosquitoes, driver ants, termites, locusts, and tsetse flies.

This wild-animal resource of Africa is of great importance for the people of the continent. They serve a wide range of purposes, including the following. They:

- are used as sources of animal protein in many countries of Africa;
- help to maintain the balance of nature by feeding on each other and plants;
- serve as source of income through tourism, legal hunting and legal sale of live wild animals;
- provide scientific and educational opportunities to researches, students and the like;
- provide inputs (industrial raw materials such as skins and excreta) for various industries;
- add aesthetic value to the environment, hence serving as source of recreation for people;

Factors Affecting Wild Animals

Despite all these and many other advantages that wild animals provide to the people of Africa, the attention paid by the people to animal conservation is inadequate. Many animals face serious problems that emanate from human interference-for example, with their habitats. Such interventions threaten many species. As a result, some animals and birds are extinct and still others, like the Mountain Nyala and Walia Ibex in Ethiopia, and the Ostrich in Algeria, are endangered or threatened.

Wild animals in Africa are facing different challenges. The following are the most serious ones.

- **Illegal hunting (poaching):** In many parts of Africa, poaching, or illegal hunting, is a common practice. People hunt wild animals for many reasons. While some are economic, others are socio-cultural. For instance, elephants are hunted for their tusks, rhinos for their horns, lions, and cheetah, for their skins etc. This activity threatens some species like the African elephant and black rhinoceros. Some animals in Africa are also brutally exterminated for several reasons. For example, animals are killed because they are considered to be pests. They are also killed for food. People also kill animals to be honored by society, as there are cultural group that attach bravery/heroism to killing large dangerous wild animals. This practice exists particularly in the very traditional parts of the continent.
- **Human Encroachment:** in many areas of Africa, people encroach into the natural habitat of wild animals. For instance, many people in Ethiopia enter into the habitats of wild life in the Bale Mountains, Semien Mountains and Awash National Parks and disturb the habitat. The

encroachment is the result of the increased need of human populations for farm and grazing land, settlement areas, fuel wood and the like. Such increased needs result in: -

Deforestation – people in Africa clear forests for shifting and permanent cultivation, fuel wood, charcoal, settlement and the like. This disturbs the habitats of wild animals and causes the migration and death of wild-animals.

Burning of vegetation cover in order to obtain land for shifting or permanent agriculture, people set fire to vegetated areas.

Overgrazing – When land is grazed beyond its carrying capacity, overgrazing occurs. This, in turn, leads to environmental degradation.

Desertification – it is the expansion of desert-like climatic conditions, which occurs as a result of changes in the characteristics of the local climate.

Drought – is extreme shortage of rainfall. It occurs when expected rains fail to fall in an area.

Possible Conservation Measures

These and many more challenges are affecting Africa's wildlife resources. Animals migrate, and even die due to the disturbance of their environment as humans encroach. To curb the situation the following measures can be taken.

- **Conserving natural vegetation:** the conservation of one type of natural resource means the conservation of others. Conserving natural vegetation that serves as habitats and sources of food for wild animals can ultimately help with their conservation.
- **Establishing national parks, game reserves and sanctuaries:** these areas provide protection and conservation for wild animals. They are established to conserve wild animals and their habitats so that they have a secure environment that is safe for their survival. In this regard, Kenya, Tanzania, the Republic of South Africa and Uganda are significant. In Ethiopia, too, nine national parks and many other sanctuaries and game reserves have been established to promote the conservation of the wildlife resources of the country.
- **Controlling illegal hunting:** it can be done by developing strict legislation whereby illegal hunters are punished or penalized for their illegal acts. For instance, an international ban on ivory trade, instituted in 1989 by the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) and supported by over 120 countries, has diminished the illicit ivory trade and reduced the related wild-animal killing.
- **Raising the people's awareness:** this can be done by educating the people about the uses of wild animals so that their attitudes and activities change. This is the most important conservation measure, and it needs the closest attention, because nothing can be done without getting the support of local communities.
- **Changing the economic condition of the people** through good and applicable policies and programs.

3.2.6. SOILS OF AFRICA

Soil is defined simply as a loose and unconsolidated material that overlies the crust of the earth. Soil is Africa's most important resource. This is because the majority of the people in the continent, and almost the entire economy of the continent, depend on activities that are directly or indirectly linked with this resource.

Major Soil Types of Africa

Why is Africa endowed with varied types of soils? What are the most common soil types of Africa?

The diverse climatic conditions, natural vegetation and the geology of Africa result in the presence of different soil types. The Food and Agricultural Organization of the UN (FAO) has classified the soils of Africa into several groups, of which the following are the most important:

- Pedalfers (the largest group in Africa)
- Pedocals
- Hydromorphic soils
- Azonal soils

Based on their geographical distribution, the soils of Africa are classified into the following types. These are Ferrasols, Nitosols, Acrisols, Lixisols, Plinthosols, Luvisols, Planosols, Vertisols, Calcisols, Solonchaks, Gleysols, Fluvisols, Arenosols, Regosols and Leptosols. The geographical locations and characteristics of each of these soils is discussed in brief in the next sections.

A. Pedalfers

Pedalfers are soils with aluminum deposits. They are soils without a layer of accumulated calcium carbonate. Such soils have high content of iron and aluminum. Soils of this group include the following.

Ferrasols: are found in the central parts of Africa around the equatorial forests and savanna lands. They are red and yellow in color. Since they are found in areas of heavy rainfall, they are affected by leaching. As a result, they are characterized by high concentrations of iron, clay and aluminum.

Nitosols: These soils mainly develop in humid climatic regions. Their parent materials are usually volcanic rocks. Such soils have a deep profile and are rich in humus content. Therefore, they are the most productive soils in Africa. As a result, they are ideal for crop production.

Acrisols: These are soils that develop in hilly areas with wet tropical and monsoon climates. These soils are weathered, acidic and shallow. As a result, they are unproductive. This soil type is found in Western Africa and the Lake Region of East Africa.

Lixisols: Lixisols are found in the savanna and semi-arid areas. These soils are reddish and sometimes yellowish in color. Geographically, they are abundant in the plains of Western Africa, Eastern Africa and east-Central Africa. Lixisols are more fertile than ferrasols and acrisols.

Plinthosols: these soils exist on plains and gently-sloped areas. They are soft and laterite. Such soils also develop in rainforest areas and the savanna regions, where marked dry and wet seasons characterize the climate.

Luvisols: these soils are developed in the Mediterranean climatic regions of the continent. They have high mineral reserves and are fertile.

Planosols: these soils dominate the High Veld of South Africa, particularly the waterlogged plains of the country. They are used mostly for grazing.

Fluvisols: fluvisols develop in seasonally flooded plains, valleys and tidal marshes. They are found in the Nile and Zambezi River Deltas, and the coasts of Western Africa and Lake Chad. They have a brown color. Most of these soils are young and fertile. As a result, they are suitable for large-scale irrigation.

B. Pedocals

Vertisols: These are black basaltic soils with clay character. Due to their clay, they become sticky during the rainy season and crack during the dry season. As a result, working such soils is very laborious. They are found in the Sahel region at the southern border of the Sahara. In some parts of Africa, these soils are cultivated with the help of irrigation and rainfed agriculture. However, generally they are used for grazing.

Calcisols: They are found in the Sahara and Namib deserts of Africa. Though they are potentially fertile in terms of mineral content, they are poor in humus. These soils are used mostly for grazing.

Solonchaks: these soils are found in inland river basins, bottoms of ancient lakes, depressions and coastal areas. They are saline and not very productive.

C. Hydromorphic Soil

Gleysols: Like solonchaks, they are found in depressions and low-lying areas of shallow ground water. They are extensively found in the Niger Delta, the Congo Basin and interior parts of Angola. In Africa, these soils are used for the production of rice, sugar cane, yam and vegetables.

D. Azonal Soil

Arenosols: They are found in the humid tropical parts of Africa, the semi-arid zones of the southern Sahara, southwest Africa and Africa's coastal plains. They are used mainly for grazing and dry farming.

Regosols: They are found in arid areas extending from West Africa to Ethiopia and Somalia. They are used for pastoralist grazing.

Leptosols: These are young, shallow and stony soils that are highly susceptible to erosion and drought. They are found in the strongly dissected uplands of Northern Africa, the Sahara, and in Southern, Central and Eastern Africa. Terracing is the most important mechanism for cultivating these soils. Otherwise, they are devoted to transhumance, forestry and tourism.

3.6.7. Problems and Conservation Measures of Soils in Africa

Problems of Soils in Africa

What are the major problems of soil in Africa? What measures, do you think, should be taken to conserve Africa's soil resources?

Soil erosion and environmental degradation are among the major problems concerning soils in Africa. The major causes of soil erosion and degradation in the continent include traditional farming practices, overgrazing, deforestation, and over-exploitation of vegetation for domestic uses. Due to these and many other factors, the continent loses huge amounts of soil every year to erosion.

Also, such problems result in the deterioration of the quality and productivity of the soil.

Soil erosion is affecting Africa in many ways. Among others, the following are the major impacts. It results in the:

- deterioration and depletion of agricultural and range (pasture) lands;
- decline of productivity of the major cereal crops;
- the collapse of agriculture and thereby the migration of people;
- downstream pollution, sedimentation, floods and damage to settlements, irrigation and farmlands;
- consumption of national economic resources to control erosion. For example, Zimbabwe invests 3% of its total annual budget for applying fertilizer to replace nutrients lost through erosion.

As we have said earlier, soil erosion is serious throughout Africa. However, the problem is more intense in some areas than others. These areas are highly affected by the problem.

- Most of the Sahel Region of Western Africa, and the Sahara and Namib deserts of Africa where wind is the major agent of erosion;
- The sub-humid (savanna) regions and the tropical rainforests, where water is the main agent of erosion; and
- The tropical highlands and mountain areas of Africa such as those of Ethiopia and other East African countries.

Conservation Measures

Major soil conservation measures that could be taken to increase soil fertility in Africa include the following.

Terracing: constructing stair like structures along hillsides to reduce the speed at which water flows down the slope, thereby reducing erosion.

Agroforestry: is associating agriculture with forest development.

Afforestation: is planting trees in areas which originally were not covered by forests.

Reforestation: is planting tree seedlings to replace cut forests.

Windbreaks and shelter-belt plantations: planting trees along a line to break the speed of the blowing wind and reduce its erosivity.

Check dams: are small ditches that are prepared along sloppy areas to reduce the impact of the down slope surface flow.

Strip cultivation: is planting two or more types of crops on the same farm, using a pattern of stripes of alternating crops. This approach reduces soil erosion because different types of plants use different ways of binding soil particles to themselves.

Contour plowing: is plowing the land sideways, following contours. It is commonly used in sloped areas, forming furrows perpendicular to the angle of the slope. These furrows act as blocks, slowing the flow of downhill water.

Crop rotation: planting different crops alternately on a farm.

“Green manure”: This approach uses plants that have soil-nutrient value to enrich the soil in the same way that animal faeces are used as fertilizer. The “green manure” plants are cultivated on the land and then ploughed under to mix them with the soil.

Mulching: is covering the soil with plant residue to let the soil regain some nutrients as the residue decays.

Fallowing: is leaving the farm idle for a while until the soil regains its fertility.

3.3. PHYSICAL ENVIRONMENT OF ETHIOPIA

LOCATION, SIZE AND SHAPE OF ETHIOPIA

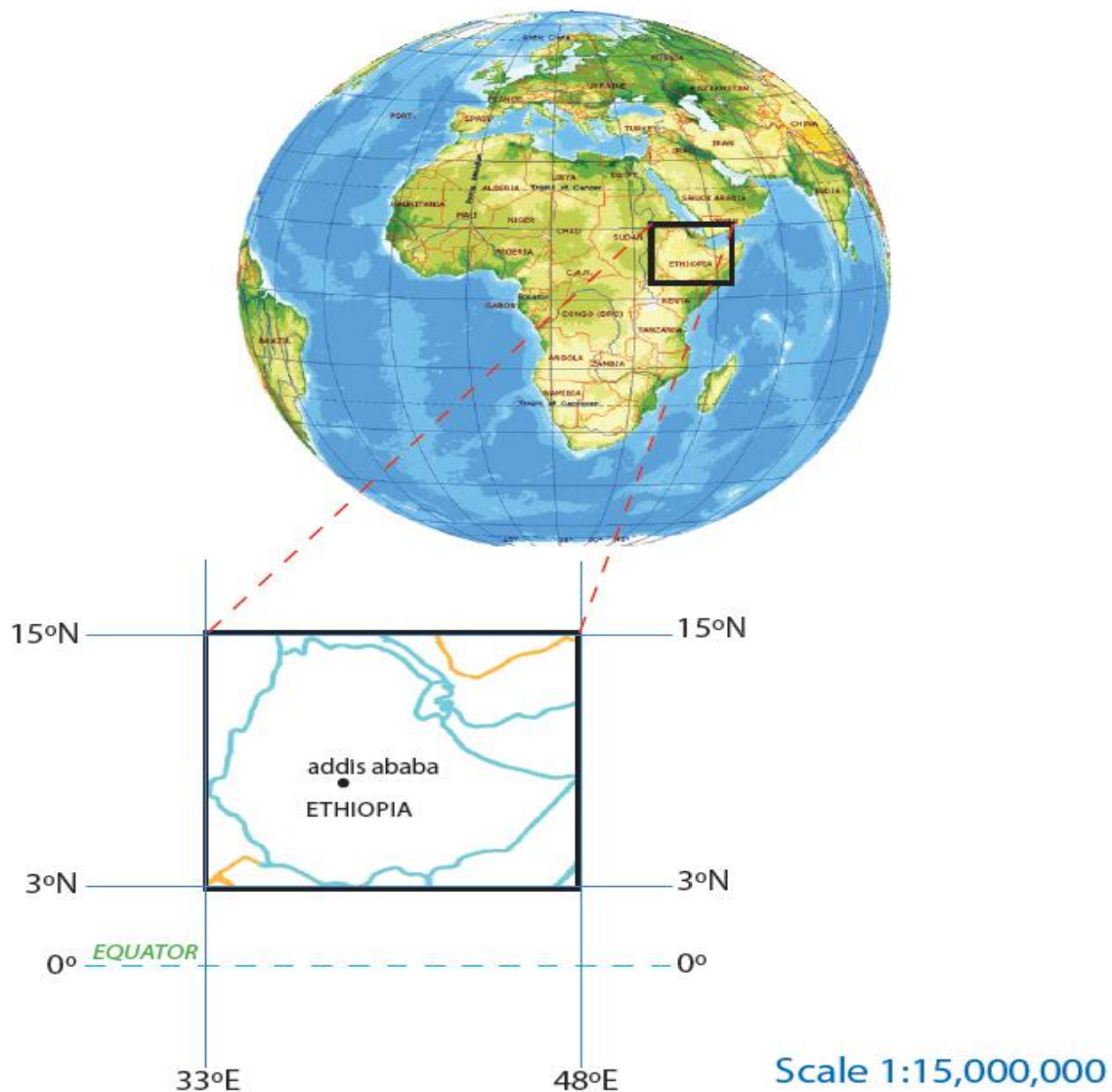


Figure: Location of Ethiopia

Source: Adapted from School Atlas of Ethiopia

Location of Ethiopia

Do you know the extent to which the location of a given place is important in world politics? What locational significance does Ethiopia have as a country that is in the Horn and near the Red sea route?

Every place has its own particular **location** in relation to its surroundings. Ethiopia's location can be expressed in two ways: **relative location** and **absolute location**.

Relative Location of Ethiopia: **Relative location** can be expressed in vicinal and geological terms.

I. Relative (Vicinal) Location of Ethiopia

Ethiopia is a landlocked country that is surrounded by five neighboring countries. Each country shares different lengths of Ethiopia's borderlines. The total length of Ethiopia's boundary line is **5260 km**.

Table: Ethiopia's boundary line length, as shared with neighboring countries

S/N	Bordering Countries	Shared boundary-line length in kms
1	Djibouti	310
2	Eritrea	840
3	Kenya	760
4	Somalia	1600
5	Sudan	1750

This Table indicates that:-

- Sudan shares the longest length of boundary line, followed by Somalia.
- The Republic of Djibouti shares the smallest boundary line length.

II. Strategic (Global, Geological) Relative Location of Ethiopia

Ethiopia's geological location can be described in the following ways. It is found:

- to the southwest of the Asian continent,
- to the south of Europe,
- to the northwest of the Indian Ocean,
- in the Nile Basin, and
- in northeastern Africa.

Absolute (Astronomical) Location of Ethiopia

Ethiopia's absolute location is expressed as follows.

Ethiopia is located between

- 3°N – 15°N latitudes and 33°E– 48°E longitudes

As a result, Ethiopia's extreme points lie at

- Badime in the north (Tigray)
- Moyalle in the south (Borena)
- Akobo in the west (Gambella) and
- The tip of Ogaden in the east (Ogaden).

Size of Ethiopia

What is size in terms of spatial distribution? Does size influence the economic strength of a country?

Ethiopia is the tenth largest country in Africa, with a total area of 1,106,000 square kilometers. It contains about 0.7 percent of the world's land area and about 3.6 percent of Africa's land mass.

Table: Comparison of Ethiopia's areal size with its neighbors'

Country	Total area extent (km ²)	Rank	Size in comparison to Ethiopia
Ethiopia	1,106,000	2	–
Djibouti	23,200	6	Almost 50 times smaller
Eritrea	118,000	5	9.42 times smaller
Kenya	580,000	4	1.91 times smaller
Somalia	637,661	3	1.73 times smaller
Sudan	2,505,813	1	2.3 times bigger

❖ **Ethiopia is the largest country in the Horn. This status in size, in combination with its status of having a large population, confers many advantages to Ethiopia in the Horn area.**

Advantages: Ethiopia's large size lets it:

- possess diverse agro-ecological zones, resulting in a wide variety of fauna and flora,
- possess a large amount of arable land,
- have a great variety of mineral resources, and
- be the home of diverse ethnic groups.

Disadvantages: Its large size compels Ethiopia to:

- require great financial power to construct infrastructural facilities,
- have a large army to protect its sovereignty,
- preserve an efficient and popular government to administer its vast territory.

Shape of Ethiopia

Countries vary not only in location and size but also in shape. Some have nearly circular (compact) shapes, others have elongated (linear) shapes, and still others have truncated (shortened) shapes. These shapes affect each country's administration, defence and economic integration, both within the country and in respect to outside areas.

Ethiopia's shape is of the compact type. Its shape is considered to be compact, or essentially circular, because the extreme north-south and east-west spans of the country cover comparable distances. You can easily see this approximate circularity in your school atlases and wall maps.

There are three theoretical indicators of the compactness of an area:

- *the boundary-circumference ratio (B/C)*
- *the area-boundary ratio (A/B)*
- *the actual area-area of the inscribing circle (A/A')*

Each of these theoretical assumptions is based on a value of 1 as indicating a perfectly compact shape. They consider 0.5 –1.5 values as deviating only slightly from circular/compact and therefore indicating approximate compactness. In contrast, smaller values indicate greater divergence from compactness, especially as they approach zero (0). These small values reflect tendencies to elongation or truncation.

For example, let's use the boundary-circumference ratio to measure Ethiopia's degree of compactness or index of compactness. In the ratio, circumference is based on an inscribing circle that touches the north, south, east and west boundaries of Ethiopia described earlier in the "Absolute (Astronomical) Location" section.

$$\text{Index of Compactness} = \frac{\text{Boundary Length}}{\text{Circumference of the inscribing circle}}$$

r – is not given, we have to find it.

The inscribing circle is the circle drawn through the extreme points on the boundary of Ethiopia. The radius is obtained by taking half the length of the distance between the astronomical extremes of west and east of Ethiopia i.e., $48^{\circ}\text{E} - 33^{\circ}\text{E} = 15^{\circ}/2 = 7^{\circ}30'$ (this is radius of the inscribing circle).

Then, change the obtained length ($7^{\circ}30'$) into kilometers:

$$1^{\circ} = 110.5 \text{ km}$$

$$7^{\circ}30' \times 110.5 \text{ km} = 828.75 \text{ km}$$

$$\text{Therefore, index of compactness} = \frac{5260}{2 \times 3.14 \times 828.75} = \mathbf{1.01}$$

The value obtained, 1.01, indicates that Ethiopia's shape to be nearly a perfect compact shape.

The value obtained indicates that Ethiopia's shape deviates by only 32% from being perfectly compact. Therefore, Ethiopia's shape is closer to the compact type than to either of the other shapes.

3.3.1. Geological Structure and Relief of the Horn of Africa

The geological history of the Horn cannot be separated from the geological history of Africa. It deals with various geographic activities that have occurred for many millions of years in the past. The geological history of the Horn shows us that four major geological eras have elapsed. Each era is divided into periods, and each period is sub-divided into epochs.

Each geological era is distinguished from the other's, based on grounds of the following three characteristics:

- the relative positions of the continents
- the character of the prevailing climate
- the predominant life form

Summary of Major Geological Events in the Horn

Let's begin by considering the different geological eras and then study the events that took place in those eras. Here are the geological eras, in chronological order.

- *the Precambrian Era – the oldest era (from 4.5 billion years to 600 million years ago;*
- *the Paleozoic Era (from 600 million years to 250 million years ago)*
- *the Mesozoic Era (from 250 million years to 70 million years ago)*
- *the Cenozoic era (from 70 million years to the recent time)*

The Precambrian Era (from 4.5 Billion to 600 Million years ago)

What do you understand by the term Precambrian?

- the Precambrian Era is the oldest and longest geological era, covering about 5/6 of the earth's geological time. The following geological events occurred in the Horn during this era.
- frequent orogenic movements
- intensive volcanic activities
- denudation during the later periods
- formation of folded mountain ranges in a NNE – SSW direction

During the Precambrian era

- ✓ The first forms of life emerged, such as amoeba, and jellyfish.
- ✓ The oldest rock formed – the old crystalline basement. This rock underlies all other rocks.

Today, in a few areas of Ethiopia, outcrops of old crystalline basement complex rocks are found on the surface, due to continuous denudation.

Example:

- *In central and northern Tigray.*
- *In Mettekel, Assossa, Illubabor and the Abbay.*
- *In central Sidama, southern Omo, southern Bale and Borena.*
- *In central, western and northern Eritrea.*

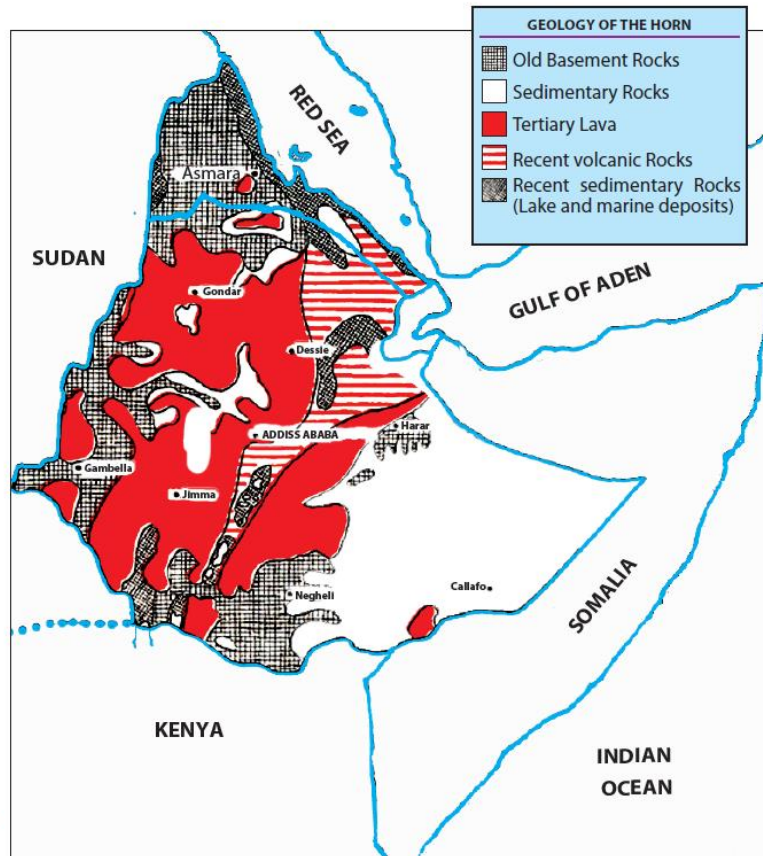


Figure: Geological map of the Horn

The Paleozoic Era (from 600 Million – 250 Million years ago)

Which life form was dominant in the Paleozoic era?

The main geological events of the Paleozoic era were **denudation and peneplanation**. No significant structural formation took place. The massive denudational activity resulted in the formation of inselbergs in some parts of Ethiopia and the Horn.

- ❖ The Paleozoic era is known for the predominance of invertebrates.

Mesozoic Era (from 250 Million – 70 Million years ago) an Era of Reptiles

Which life form was dominant in the Mesozoic era?

The most important geological occurrences of the Mesozoic era in the Horn were the sinking and uplifting of the landmass.

Periods	Geological time scale
Cretaceous	70 m
Jurassic	142 m
Triassic	250 m

The landmass sank during the Mesozoic's Triassic and Jurassic periods:

- In the Triassic Period, the landmass sank due to internal forces. This event was followed by transgression of a nearby sea into the mainland of today's Somalia, and southeastern Ethiopia. During the Triassic Period, the oldest sedimentary rock known as Adigrat sandstone was formed.
- In the Jurassic Period, the transgression of the sea continued into the mainland in the northwest direction. This event deposited another sedimentary rock known as Hintalo limestone.
- In the Cretaceous Period, the landmass began to rise and the sea started to regress towards the southeast, depositing sedimentary rock known as Upper Sandstone. Upper Sandstone is the youngest sedimentary rock, and therefore overlies the rest.

Because of the direction of the regression and deposition, Upper Sandstone is the youngest and thinnest in the southeast, and is the oldest and thickest in the northwest.

The Mesozoic Era was an era of sedimentary rock formation in Ethiopia and the Horn.

- The deposited sandstones vary in age and thickness from the northwest to the southeast direction.
- The Adigrat sandstone is older and thicker in the southeast and progressively decreases in age and thickness to the northwest.
- ❖ The transgression of the sea extended up to northwestern Ethiopia, as far as central Tigray and the western slopes of the western highlands.
- ❖ The sedimentary rocks formed in the Mesozoic Era were later buried by overlying Cenozoic igneous rocks. However, the sedimentary rocks have been exposed at the surface in some areas of Ethiopia. They are thinnest (because they are the youngest) in the southeast and thickest (because they are the oldest) in the northwest.
- ❖ One can see them exposed at the surface mostly in the south eastern lowlands of Ethiopia, central Tigray, and in the Abbay and Wabishebele gorges. (For more information, look at the geological map of Ethiopia.)

Biological Events of the Mesozoic Era

The Mesozoic is also known for the predominance of reptiles. Huge reptiles, such as dinosaurs, were dominant. However, at the end of this era, two other significant biological events occurred:

- one was the disappearance of the dinosaurs, and
- the other was the emergence of mammals, birds and flowering plants.

The Cenozoic Era (from 70 Million to Recent Years)

Do you know the era in which the human form of life appeared?

The Cenozoic is the most recent geological era. Very significant structural, climatic and biological events have occurred in the Horn.

In order to make things simple and easily comprehensible, we shall discuss only the geological events of this era into events of the Tertiary and Quaternary Periods.

Geologic Events of the Tertiary Period - (70 million - 2 million years ago)

In the Tertiary Period, the uplifting that began in the Cretaceous Period of the Mesozoic Era continued and reached its maximum height.

In Ethiopia and the Horn it formed huge blocks of dome over the greater part of the region. As the uplifting continued through time, great cracks opened in the crust and resulted in the pouring out of extensive basaltic lava (known as the Trappean lava series). The lava resulted in the formation of:

- The Eritrean Highlands
- The Northwestern Highlands
- The Southeastern Highlands
- The Somali plateaus

As the cracking continued during the period, it formed the Great East African Rift Valley System – of which the Ethiopian Rift Valley System is part.

Geologic Events of the Quaternary Period (2 million - recent years)

In the Quaternary Period, these structures were formed in Ethiopia and the Horn:

- The Afar Horst that extends into Djibouti
- The active volcano of Ertalle in Afar
- The dormant volcanic mountain of Fentalle in Eastern Oromia
- The extensive lava field of Methara.

Climatic Events in the Cenozoic Era

In addition to the geological events that happened in this period, a significant change of climate took place in the Horn and Ethiopia. A massive flood, called the pluvial period, occurred. This flood formed deep gorges, moraines and lacustrine deposits.

- ❖ One good example of the deep gorges formed as the result of the flood is the **Abbay gorge**.
- ✓ It is believed that Lake Langano, Lake Abijata and Lake Shalla were one sheet of water during the time of the pluvial period. The same is believed true of lakes Abbaya and Chamo. Today these lakes are widely separated.
- ❖ The Cenozoic Era in its Quaternary Period is assumed to be the period in which modern man evolved.

Here is a simplified presentation of the rock profile in Ethiopia and the Horn:

- The lowlands

The Highlands of the Horn

Highlands are lands with altitudes of over 1000 meters above sea level (masl).

The Horn's highlands are:

1.The Northwestern Highlands

2.The Southeastern Highlands

1 The Northwestern Highlands of the Horn

The Northwestern Highlands of the Horn stretch from Ras Kassar in Eritrea to the highlands of Gamo-Goffa in southwestern Ethiopia. They are separated from the southeastern highlands by the Rift Valley, which is part of the Great East African Rift Valley. They consist of:

- The Plateau of Eritrea
- The Plateau of Tigray
- The North Central Massifs
- The Plateau of Shewa
- The Southwestern Highlands

The Plateau of Eritrea

The plateau of Eritrea is located between the course of the Barka River and the coastal plain of Eritrea. It is bisected by the upper basin of the Mereb River. The plateau is capped by basaltic rocks.

The Plateau of Tigray

Is the plateau of Tigray contemporary to the other plateaus of Ethiopia?

The Plateau of Tigray is the most northerly plateau in Ethiopia. It is separated from the Eritrean plateau by the Mereb River. It lies to the southeast of the upper course of the Mereb/Gash River and to the northeast of Tekkeze River Gorge. It is an area composed largely of sandstones and limestones, as the overlying basalt have been eroded. As a result, the soils are poor and thin. The plateau has been exposed to severe erosion due to long periods of human inhabitation.

There are very high mountains on this plateau with elevations of over 3000 meters above sea level. Three of these are:

- Mount Tsibet – 3988 m.a.s.l
- Mount Ambalage – 3291 m.a.s.l
- Mount Assimba – 3248 m.a.s.l

The North-Central Massifs

What makes the Northern central massifs distinct from the massifs of central

Ethiopia?

These are the most rugged and dissected plateaus of Ethiopia. They are capped by basalts and surrounded by deep gorges. Within each of the plateaus are small arable lands known as ambas. The ambas are isolated from one another by gorges.

The Tekezze, together with its tributaries, is the main river that drains the region. Tekezze River has divided the North Central Massif area into western and eastern massifs, which are connected by the Yejju-Wadla-Dilanta plateau. The western massifs make up the massifs of South Gondar (Semein), while the eastern ones make up the Lasta and Wollo Massifs. South of these is found the Gojjam Massif. The Semein Massif is dominated by Ras Dashen, while the eastern forms the watershed of the Nile drainage basin.

The Gojjam Massif is the most extended tableland and is comparatively less dissected. It is formed on the core of the Amedamit-Choke mountains. It is carved by the Abbay river that effectively separated it from the Shewa plateau in the south and the Amahara Saynt Massif in the northeast.

The North-Central Massifs are made up of numerous high mountains. They are known for the production of cereals, such as teff, pulses and oil seeds.

Famous mountain peaks on these massifs are

- MT. RAS DASHEN (in Semein) – 4620 masl
- MT. LEGEDA (in Gondar) – 4532 masl
- MT. ANALU (in Gondar) – 4480 masl
- MT. TEFAW LEZER (in Gondar) – 4456 masl
- MT. KOLO (in Lasta) – 4300 masl
- MT. GUNA (in Gondar) – 4231 masl
- MT. ABUNA YOSEPH (in Lasta) – 4190 masl
- MT. HEY (in Gondar) – 4154 masl
- MT. BIRHAN (in Gojjam) – 4100 masl

The Plateau of Shewa

Have you ever had the opportunity to travel from Addis Ababa to Debre Markosor Bahir Dar? If yes, what do you recognize all the way through until you reach the Abbay Gorge?

The Shoan Plateau is a dome-shaped plateau that serves as a watershed between the Awash and Abbay River basins. It extends westwards into western Wellega through Horo Guduru and forms a crescent shape which causes the Abbay to swerve and drain northwards. The Shoan plateau is separated

- from the plateau of Gojjam by the Abbay gorge in the north
- from the southeastern highlands by the Awash River and the Rift Valley.
- from the Highlands of Keffa by the Ghibe River.

The Plateau of Shewa is drained by the tributaries of the Abbay River in the west and the Awash River in the east. Its high mountains are found on its northeastern and south eastern margins; they are:

- Mt. Abbuye Meda (on the northeastern margin) – 4000 masl
- Mt. Guraghe (on the southeastern margin) – 3721 masl

The Southwestern Highlands of the Horn

Which regional zones are found in the southwestern highlands of the Horn?

These include the highland areas of Wellega, Illubabor, Gamo Goffa and Keffa.

These Ethiopian highlands lie south of the Abbay trough which is greatly eroded due to torrential rain that pours down on in the area for almost all of the year. It is the wettest region of the country with a total average annual rainfall of above 1500 mm.

The region is drained:

- northwards, by the Dabus and Dideessa tributaries of the Abbay River
- westwards, by the headstreams of the Baro-Akobo River
- southwards, by the Omo-Ghibe River, which ends in Lake Turkana
- eastwards, by the right-bank tributaries of the Omo-Ghibe (the Gojeb-Ghibe River of Jima and Yem zone)

The general elevation of these highlands is relatively low, when compared to that of the Northern and Eastern Highlands. Only a few areas are above 2500 meters.

The highest points in the region are the:

- Gamo-Konso Highlands (in GamoGoffa)
- Maji-Korma Highlands (in Keffa Zone)
- Kulo-Konta Highlands (Keffa zone)
- Benishangul mountain (in Benshangul Gumuz)
- Tullu Wallel (in West Wellega)

Mt Gughe has the highest altitude: 4200 masl. It is found in the Gammo plateau.

The Southeastern Highlands of the Horn

The Southeastern Highlands of the Horn include the plateaus of

- Hararghe
- Arsi
- Somali highlands
- Sidama
- Bale

Their formation is similar to that of the North and Southwestern Highlands; and they are capped by basaltic rock. They are the main sources of the Wabishebele and Genalle rivers.

They are bounded:

- in the west, by the fault line of the Rift Valley
- in the east, by the Ogaden Lowlands
- in the south by the Elkerie and Borena Lowlands.

These highlands are subdivided into the plateaux of Hararghe, Arsi, Bale and Sidama. Each is discussed in turn as follows.

The Plateau of Hararghe

The Plateau of Hararghe rises sharply from the fault line of the Rift Valley and extends gently to the east up to Jijiga. After Jijiga, a fall in elevation takes place, giving way to the Ogaden Lowlands. The Plateau is drained by the left-bank tributaries of the Wabishebbelle River. The basaltic rocks have been worn away, exposing limestones and earlier sedimentary rocks.

The Hararge Plateau area and its foothills are significant producers of coffee, chat, sorghum and millet. The highest points of this plateau are Mount Gara Muleta (3381 masl) and Mount Jebel Tita (3122 masl).

The Plateau of Arsi

This plateau area consists of the Gugu and Chillalo Massifs. It is an extensively rolling plateau; it is a very suitable plateau for farming. These features are due to the fact that erosion on the Arsi plateau has been comparatively low. The Arsi plateau is known for its wheat. The highest points on the Arsi plateau are Mount Chillallo (4136 masl), Mount Bada (4139 masl) and Mount Kaka (4180 masl).

The Bale Massif

The Bale Massif is next to the Arsi Plateau but is separated from it by the headstreams of the Wabishebbelle River (popularly known as the Wabe). In the north, the massif consists of a flat form that is similar to basaltic plateaus. In the south, the massif consists of huge mountains. The Bale Massif is dominated by

- Mount Tulu Dimtu (4377 masl)
- Mount Batu (4307 masl)

The Plateau of Sidama

This plateau area is next to the Bale Massifs, but is separated from them by the Genalle River. The area constitutes the southwest extension of the southeastern highlands. The plateau slopes away gently to the south and is drained by the Dawa River and its tributaries. The Jemjem is the dominant part of the plateau.

The Somali Highlands

The highlands are the extension of the southeastern highlands of Ethiopia. Their average altitude doesn't exceed 1500 masl. They rise gently in the west and descend sharply to the Indian Ocean coasts.

The Rift Valley System

The Ethiopian Rift Valley System is part of the Horn's Rift Valley System, which is part of the Great East African Rift System. The Great East African Rift System is a set of fractures in the earth's crust that extend from the Dead Sea in the north, through the Red Sea, and then across East and Central Africa to Mozambique in the south.

Major faulting and rifting took place at the end of the Tertiary Period of the Pleistocene Epoch as a result of tectonic epeirogenic activity. These events formed the Great East African Rift System.

Let's now consider the Ethiopian Rift Valley System. It has been the scene of intense volcanic activity and minor faulting. Even today, active volcanic activity exists there. It is therefore, the most unstable physiographic division of the area. As you can see in the following map, the Rift Valley runs diagonally from northeast to southwest and divides Ethiopia east-to-west. It covers a total length of 1700 kilometers in Ethiopia and Eritrea.

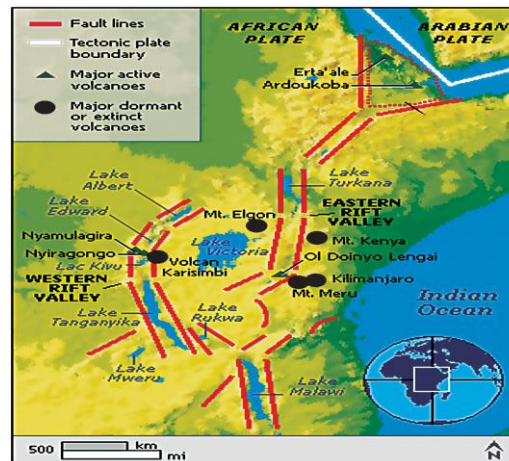


Figure: The Rift Valley System of Ethiopia and the Horn

Subdivisions of the Ethiopian Rift Valley System

Covering a length of 1700 km, the Ethiopian Rift Valley System comprises 18 percent of the country's total area. It is subdivided into three main parts:

- The Afar Triangle (northern)
- The Main Ethiopian Rift (central)
- The Chew-Bahir Rift (southern)

The Afar Triangle (Northern Subdivision)

The northern subdivision of the Ethiopian Rift Valley System, i.e., the Afar Triangle, is the largest and widest part of the system. Its altitude is generally low, ranging from 116 meters below sea level at the Kobar Sink to about 900 meters above sea level at Awash. This part of the Ethiopian Rift System is characterized by

- faulted depressions (the Dallol Depression) and grabens (also called troughs)
- volcanic mountains.

Also, a large part of the area is covered by extensive salt plains and lakes (Lake Assale and Lake Afrera).

The Afar triangle is bounded by parallel fault lines on the east and west. Its floor is made up of:

- grabens such as the Tendaho Graben

- volcanic ash and lava deposits
- Lacustrine and fluvial deposits, and
- Volcanic mountains such as Mount Fentale.

What is more, the Afar Triangle has special characteristics that do not exist in other regions of Ethiopia do not bear. It consists of:

- fossil rich sediments
- rich archeological sites which have shown us that the area was the home of the ancient ancestors of primates and hominids.

The Main Ethiopian Rift (Lake Region or Central Part)

- The central subdivision of the Ethiopian Rift system, i.e., the Main Ethiopian Rift, covers the area from the lower Awash basin up to Lake Chamo. This subdivision is the most elevated part of the Ethiopian Rift Valley. It is also the wettest, most densely vegetated, and most densely populated.
- Sedentary farming is practiced here. The area also includes numerous lakes of enormous aesthetic value.

The Chew-Bahir Rift (Southern Subdivision)

The southern subdivision of the Ethiopian Rift system, the Chew-Bahir Rift, is also known as the Omo-Ghibe trough. It is the smallest section of the Ethiopian Rift System. It consists of an extensive shallow marshy area covered by tall grasses.

In the vicinity of Arba Minch, this part of the Ethiopian Rift valley system is split into the Ganjuli and the Galena Valleys by the Amaro mountain range.

The Lowlands of the Horn

These landform divisions occupy the peripheries of Ethiopia on its eastern and western sides, running from north to south. They are generally below the 1000 meter contour line and have relatively harsh and very hot climates. The lowlands more than 35 percent of the total area of the country. They are inhabited mainly by pastoralists.

The lowlands of the Horn are subdivided into

- Western lowlands
- Southeastern lowlands

The Western Lowlands

These lowlands extend from western Eritrea in the north up to the Omo-Ghibe River in the south, bordering the Sudan. They have a general elevation of 500 – 1000 masl.

These lowlands are sub-divided into

- The Setit and Barka Lowlands (in Eritrea)

- The Tekezze and Angereb Lowlands (in Tigray and Amhara regions)
- The Abbay Dinder Lowlands (in Benishangul and the Gumuz Region)
- The Baro-Akobo Lowlands (in Gambella)
- The Omo-Ghibe Lowlands (in SNNP)

The Western Lowlands are characterized by arid and semi-arid conditions. The Baro-Akobo lowland is the wettest lowland.

Because of climatic hardship in most parts of these lowlands, the communities practice nomadic and semi-nomadic pastoralist ways of life. However, there are notable towns, such as Humera, Kurmuk, Omedla and Metema, that serve as business centers for the communities living along the Ethio-Sudanese border.

The Southeastern Lowlands

Like the Western Lowlands, the Northeastern and Southeastern Lowlands run from northwest to southeast. They begin in Djibouti and run all the way to Somalia in the southeast, bordering the Indian Ocean. They consist of

- the Red Sea coastal plains (in Eritrea)
- the Afar plains (in Afar) which are included in the Afar Triangle.
- the Ogaden plains (in Somali region)
- the Elkeri plains (in Bale)
- the Borena plains (in Oromiya)
- the Benadir plains (in Somalia)
- Djibouti

These lowlands are highly extensive lowlands. They are characterized by low annual rainfall-often below 500 mm. Most of these lowlands are covered by sandstones and recent marine deposits. The people practice a pastoralist way of life because of the harsh climate.

3.3.2. CLIMATE OF ETHIOPIA

Distribution of Major Elements of Climate in Ethiopia

Temperature Distribution

Temperature in Ethiopia is determined primarily by altitude and latitude. Ethiopia lies within the tropics, a zone of maximum insolation, where every place has overhead sun twice a year. However, considerable portions of Ethiopia are highland areas, and their altitudes give them non-tropical temperatures. Ethiopia's tropical climate occurs in lowlands at the country's peripheries.

Away from the peripheral lowlands, the land begins to rise gradually and considerably, culminating in peaks in various parts of the country. The highlands form the heartland of the country. Thus temperature in most of the center of the country is affected by altitude, and temperature essentially decreases from Ethiopia's peripheries towards the interior. Ethiopia experiences both hot and cold extremes of temperature (see Figure).

Daily Temperatures: Ethiopia's daily temperatures are more extreme than its annual averages. In terms of spatial distribution, Ethiopia's daily maximum temperature varies from a high of more than 37°C over the lowlands of the northeast and of the southeast to a low of about 10°C-15°C over the highlands of Ethiopia. In terms of temporal distribution, the months of March, April and May are generally the hottest throughout the country.

Monthly Temperatures: Ethiopia's monthly temperatures also exhibit extremes.

The lowest monthly minimum temperatures mostly occur over the highlands of the country. Most of the highlands experience mean minimum temperatures as low as 0°C between November and January (**Bega** season). The highest mean monthly minimum temperature (20°C - 30°C) is observed in the Dallol Depression. The lowest mean monthly minimum temperature (0°C or less) occurs in January in the highlands in the northwest (Gonder and Gojam), central (Shewa) and southeast (Arsi-Bale), and the highest (30°C) occurs in the lowlands of the western, southeastern and northeastern areas.

Annual temperatures: Mean annual temperature varies from a low of about 10°C, in the northwest, central and southeast highlands, to a high of about 35°C at the country's northeastern edges, which contain the Dallol (Denakil) Depression. As you can again infer, altitude is the most important temperature-controlling factor in Ethiopia.

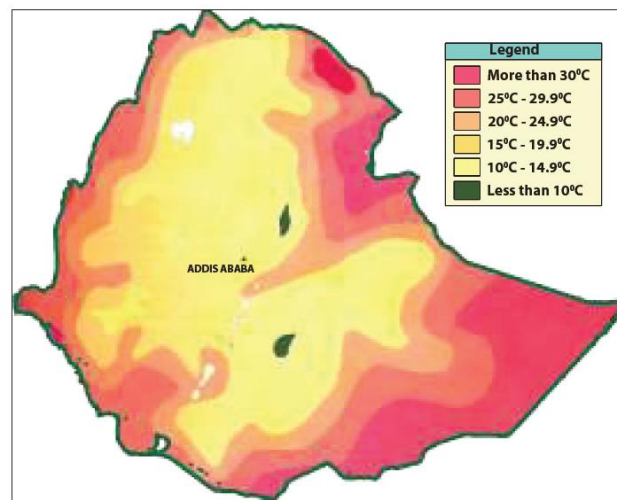


Figure: Mean annual temperature distribution Ethiopia

Temperature Ranges

Daily (diurnal) temperature range: In the northern hemisphere, during the winter months of December, January and February, the sun apparently shifts to the southern hemisphere and Ethiopia experiences its **bega** season. During this period, the sky is clear, without blanketing cloud cover to retain heat or cold on earth. Therefore, during the day the heat from the sun is intense. But at night, because of the clear sky and relatively longer nights, temperature drops very low. Reflecting such clear-sky conditions, Ethiopia's daily temperature range is greatest during **bega** in most parts of the country.

Annual temperature range: Ethiopia's annual temperature range is the highest in the lowlands, and it decreases with increasing altitude.

Distribution of Rainfall

Overall, mean annual rainfall ranges from over 2200 mm in pockets of areas in the southwestern high lands to less than 400 mm over the northeast and southeastern lowlands of the country (Figure). However, the low end of the range is much less in the southeast lowlands, the Ogaden area, standing at 200 mm. In the northeast lowlands, the Afar region, it is even less than 100 mm.

The map shows that southwestern Ethiopia is the region of heaviest rainfall. It is the wettest part of the country, with only two to four dry months in a year. The mean annual rainfall for southwestern Ethiopia is about 1500 mm, but in some areas it is much higher – reaching up to 2800 mm in Mocha, southwestern parts of Gore and Arjo. Also well over the national average, mean annual rainfall can exceed 2000 mm in parts of Gimira and Kafa, Limu, Gore, Buno Bedele, Sore and Geba and the southern extreme of Gimbi. The adjoining western lowlands of Gambella and Assossa and Benishangul, which are found on the windward sides of the western highlands, receive over 1000 mm of annual rainfall.

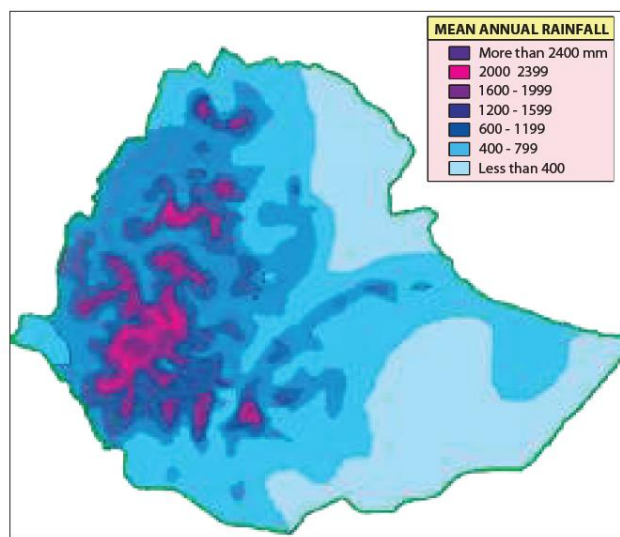


Figure: Mean annual rainfall distribution in Ethiopia

From the southwest, mean annual rainfall gradually decreases towards the northeast and east. In central and north-central Ethiopia, the annual amount is moderate, about 1100 mm. But, there are some pockets where annual rainfall reaches over 2000 mm. These include the western parts of Awi zone, and parts of Metekel and Kola Dega Damot. In parts of northern Gonder and central Wegera and central Semien, the mean annual exceeds 1600 mm. In southeastern Ethiopia, the mean annual rainfall is about 700 mm. However, this amount varies from over 2000 mm in Jemjem, and over 1200 mm in parts of Genale and Dolo in the Bale zone and parts of Webera in Harerge, to less than 400 mm in most of the Ogaden area. In northern Ethiopia, including Tigray and Wollo, the mean annual rainfall is about 500 mm, but in some areas it rises to over 1200 mm.

Seasonal Pattern of Rainfall

Summer Rainfall

As the ITCZ (Inter - Tropical convergence zone, zone of convergence of north east and south east trade winds) drifts towards the north in northern hemisphere summer (Keremt), the equatorial westerly winds from the South Atlantic Ocean invade most parts of Ethiopia, while the trade winds from the north retreat from the country. The ITCZ's southward drift marks the onset of the trade winds from the north, which causes the retreat of the equatorial monsoons. Such periodical shifts in the flow pattern of winds causes rainfall to be variable and seasonal in Ethiopia.

Rainfall in Ethiopia is seasonal, varying in amount over space and time. There is the long and heavy summer rain, which is normally referred to as "the big rain" or keremt. There are also short and moderate rains in autumn (Tebi), winter (Bega) and spring (Belg). They are collectively called "the little rains". In contrast to the rest of Ethiopia, southwestern Ethiopia gets rain for a long period that stretches usually for more than eight to ten months.

Other regions, for instance the southeastern lowlands, receive rain twice a year. These rainy seasons do not correspond with keremt or bega. In most of highland Ethiopia, the main rainy season is in summer (June to September), when the ITCZ is to the north of Ethiopia. During this season, the whole country with the exception of a few places is under the influence of the southwest equatorial westerly winds from the South Atlantic Ocean or of the south easterly winds from the Indian Ocean.

Effects of the southwest equatorial westerly winds: The southwest equatorial westerly winds originate from the South Atlantic Ocean, and they blow over the humid regions of the Gulf of Guinea, the Congo basin and Central Africa on their way to Ethiopia. When these winds approach Ethiopia they encounter highlands. When they start ascending over the highlands, they cause heavy rain in southwestern Ethiopia because they are moisture - laden. However, the amount of rainfall gradually decreases as the winds move north and northeastwards.

Highland Ethiopia receives the widest coverage of these keremt rains, though the amount that falls in different areas varies. The southwest experiences the longest Keremt rain. Keremt lasts for only two to three months in the extreme northeast highlands. The eastern escarpments of the northeastern highlands and associated lowlands (the Afar region) remain dry, because they are in what is called a rain shadow.

Effects of the southeasterly winds: The southeasterly winds that originate from the Indian Ocean blow over Ethiopia's southeastern highlands and associated lowlands. However, these winds lose their moisture over the East African highlands before they reach Ethiopia. Therefore, the country's southeastern highlands and associated lowlands that receive the winds remain relatively dry.

Winter Rainfall

In winter the ITCZ shifts farthest south. Most of Ethiopia comes under the influence of North East Trade winds, which originate from west Asian high pressure centers. These winds are cold and dry, and they carry little or no moisture, giving most of the country a dry winter. Only the Red Sea coastal plains, including parts of the Afar region, receive rain. This rain is little in amount, and is due to the area's proximity to the Red Sea.

Another exception to this lack of rain is southwestern Ethiopia. In winter this region is still under the influence of the equatorial westerly winds, although they are now weak. The moderate rain they supply to the region at this time is the area's smallest annual amount.

Spring and Autumn Rainfall

In spring the ITCZ drifts to the north and lies across Ethiopia. At this time, a strong low-pressure cell develops over the Sudan. This center attracts.

- Winds from the Gulf of Aden
- Indian Ocean highs

These moist south easterly winds blow across central and southern Ethiopia, and they produce the big rains in southeastern Ethiopia.

The same winds produce the little rain of spring for the east central part of the northwestern highlands. These rains are often called *Belq* rain.

Spring is the major rainfall season in the southeastern highlands and associated lowlands (Ogaden, Borena of Oromia and the South Omo zone). The area's second rainfall season is in autumn. In the southeast peripheral lowlands, the towns of Moyale and Kelafo also receive their rainfall in spring and autumn. Moyale's annual rain is about 1000 mm. Kelafo's is about 500 mm. For Moyale, about 50% occurs in spring, and about 37% occurs in autumn. For Kelafo, the percentage distribution are approximately 60% and 33% respectively.

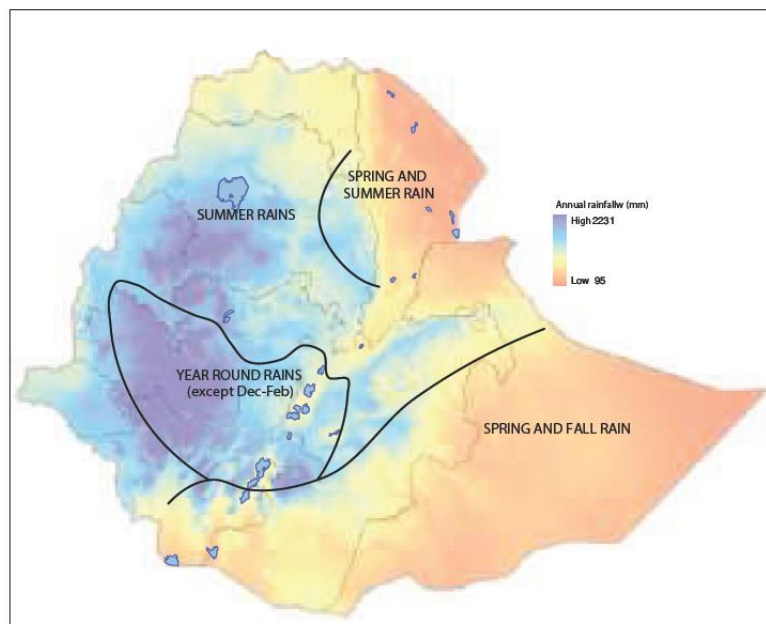


Figure: Rainfall Regimes and Climates of Ethiopia

Major Climate Controls in Ethiopia

Many physical factors influence Ethiopia's climates and their distribution. They are collectively called controls of climate.

Ethiopia's most important climate controls are

- latitude
- altitude
- cloud cover

The above noted factors, together with other less important ones, determine the distribution of climate in Ethiopia. As noted earlier, Ethiopia is located within the Tropics. Therefore, in principle, it is a tropical country and might be expected to have tropical climate throughout. However, since Ethiopia is a highland country, much of its climate is affected by altitude. It is only in lowland areas that a tropical climate (Kolla) prevails. Altitude is the most important climate control in the highland parts of the country. For instance, rainfall amount tends to increase with altitude, while, in contrast temperature decreases with increasing elevation.

Next to altitude in importance is latitude. See the following Table.

Table: Relationships between temperature, latitude and altitude

Town	Latitude	Altitude (m)	Mean annual temperature (°C)
Addis Ababa	9°	2400	17
Awash	9°	916	25
Ambo	9°	2130	18
Nekemte	9°	2005	18.3
Dire Dawa	9.6°	1160	25

In Ethiopia's highlands, cloud cover is another important climate control during the long rainy season – June to September. Although this period has high sun, comparatively low temperatures prevail due to the season's cloud cover, which absorbs and reflects away much of the incoming rays of the sun. The other period of high sun in Ethiopia is March to May. In the highlands, this season is not characterized

by cloud cover and its temperatures are relatively high. The northeast and southeast lowlands have both low altitude and clear skies, and therefore their temperature is high.

- ❖ Climate controls are factors that act on climate elements and produce different climate types. On a global level, important climate controls include latitude, altitude, distance from the sea, ocean currents, wind and air pressure, major and local relief, and sky conditions. In the Ethiopian context, altitude and latitude are the most important determinants of climate. Although two places might lie on nearly the same latitude, if their altitudes vary, they probably have large temperature differences.

Main Seasons and Climatic Zones of Ethiopia

Main Seasons of Ethiopia

Ethiopia is a tropical country. It is located completely within the tropical zone. Because of its tropical location there is little variation in the length of days and nights. The maximum difference, about 30 minutes, occurs in December and June. Many people of the country are not aware of the differences. There are four main seasons per year: Keremt, Tseday, Bega and Belg (Table 2.2).

Table: Temporal distribution of the main seasons in Ethiopia

Months*	Name of season		Location of the sun
	English	Amharic	
June, July and August	Summer	<i>Keremt</i>	Northern hemisphere
September, October and November	Autumn	<i>Tseday</i>	Around the equator
December, January and February	Winter	<i>Bega</i>	Southern hemisphere
March, April and May	Spring	<i>Belg</i>	Around the equator

Traditional Climatic Zones of Ethiopia

The traditional Ethiopian classification of climatic zones is based on altitude and temperature. This system divides the nation into the following five major climatic zones: Bereha, Kolla, Woina Dega, Dega and Wurch. Bereha: Bereha is the hot arid climate. Bereha is the climate of the desert lowlands

that are found below 500 m above mean sea level where the average annual rainfall is less than 400 mm, and average annual temperature is over 30°C.

Bereha is usually characterized by strong wind, high temperature, low relative humidity, and little cloud cover. Evapotranspiration is always in excess of rainfall in some places.

Kolla: Kolla is a (warm-to-hot semi-arid climate). Kolla is the climate of the hot lowlands with an altitudinal range of 500 to 1500 m a.s.l. Average annual temperatures are between 20°C and 30°C. Although mean annual rainfall ranges between 410 mm and 820 mm, it can be as high as 1600 mm in the wet western lowlands of Gambella. Rainfall is highly variable from year to year. This region is intermediate between the hot arid climate and the humid climates.

Woina Dega: This is subtropical warm-to-cool semi-humid zone which corresponds to roughly with what is commonly known as the warm temperate climate. Woina Dega has distinct dry and wet months in winter and summer respectively.

The average annual temperature is between 15°C and 20°C, and annual rainfall is generally around 1200 mm. Woina Dega covers the temperate highlands that fall with altitudes between 1500 m and 2300 m amsl. In the southwest, rainfall reaches 2400 mm.

Dega: Dega corresponds roughly with the temperate climate. Dega is the climate of the cool temperate highlands. It covers a region with an altitude range of 2300 m to 3300 m amsl. The coldest month is less than 10°C. The area experiences adequate rainfall. Rainfall ranges from about 1000 mm, in most areas, to 2000 mm in higher altitudes.

Wurch: Wurch is a type of Alpine climate. The annual average temperature is less than 10°C. Annual rainfall is between 800 and 2000 mm. The zone exists at altitudes equal to or more than 3300 m amsl.

This zone exists as afro-alpine areas on the highest areas of Ethiopia's plateaus. It is found in small isolated high areas such as the Senate plateaus (Bale zone), Semen mountains (north Gonder), Mount Guna (south Gonder), Amara Saint (south Wollo), and the Choke mountains (Gojam).

Table: Traditional Temperature and Altitude Zones in Ethiopia

Traditional Zones	Global	Altitude (meters) Zones	Mean Annual Temperature (°C)
<i>Wurch</i>	Alpine	Above 3300	Below 10°C
<i>Dega</i>	Temperate	2300 - 3300	10°C - 15°C
<i>Woina Dega</i>	Sub - Tropical	1500 - 2300	15°C - 20°C
<i>Kolla</i>	Tropical	500 - 1500	20°C - 30°C
<i>Bereha</i>	Desert	Below 500	>30°C

Drought in Ethiopia

Drought is a condition that happens when much less rain is received than is normally expected. It is a period of abnormally dry weather that is sufficiently long enough to cause serious problems for agriculture and other activities in the affected area. This extended period usually involves months or years during which the region receives consistently below-average rainfall. Drought is one of the world's major environmental hazards. It affects human and animal life catastrophically, and it can cause severe crop failure.

Consequences of drought

Unless drought-causing conditions are reversed by putting into practice measures of mitigation, drought can have adverse consequences both on the natural environment and on the socio-economic life of the people in drought-prone areas. Some of the general consequences include climate change, drying up of surface and subsurface water sources (ponds, streams, swamps, reservoirs, lakes, etc.), decline in underground water tables, loss of soil moisture, crop failure, starvation and famine, death of animals and human beings, lack of seeds, livestock and labor, loss of biodiversity and environmental deterioration, desertification, rural out-migration, etc.

Although famine and starvation can result from drought, they are not its necessary consequences, even though this has generally been the case in Ethiopia and other less developed countries. Drought can happen anywhere in the world. But a country's ability to resist it is a function of their developmental stage. Less developed countries like Ethiopia are highly vulnerable, and affected by post-drought effects. In addition, their political, economic and social affairs are highly influenced by drought and its product, famine. However, economically well-to-do countries can manage drought and resist its after effects with their systems and wealth. A country under the influences of drought and famine for long periods of time may develop dependency syndrome.

This may lead to loss of self esteem and national pride and end up in longing for foreign donations.

Drought-Prone Areas in Ethiopia

The three drought-probability zones of Ethiopia include:

i. High drought probability zone (zone III) covered about 25% of Ethiopia in 1988. This zone includes the Bereha climatic zones of the largest part of the Somali and Afar regions, part of the Tigray region, the eastern portion of the Amhara region, and the southern part of the Oromia region.

ii. Medium drought probability zone (Zone II) covers part of the Somali, Oromia and Afar regions, and small portions of the eastern Tigray and Amhara regions. It covers about 10% of the country.

iii. Low drought probability zone (Zone I) also covers about 16% of Ethiopia's area, including most of the semi-arid (Kolla) climatic zones. It includes part of the eastern highlands of Tigray and Amhara regions, the eastern and southern parts of Harerge, a portion of the Bale, Borena and Guji zones, and the southern portion of the south Omo zone.

Drought-Coping Mechanisms in Ethiopia

Geographers expect major droughts to reoccur in Ethiopia at about ten-year intervals and expect minor droughts at two-years intervals. Hence, the Federal and Regional Governments, the nation's farming community and other stake holders, should make themselves ready for the challenge of reoccurring drought and related problems by mobilizing the resources of rural farming and pastoralist communities and government facilities to avert the situation by implementing as many of these mitigation and adaptation measures as possible.

- carefully planned land use to increase agricultural productivity and minimize erosion
- encouraging farmers in the drier areas to plant crops that have low water-dependencies
- drilling deep water wells, installing water pumps and building simple irrigation channels.
- rainwater harvesting collecting and storing rainwater on various scales
- recycling waste water for reuse after treatment and purifying
- building reservoirs and canals and redirecting rivers as massive attempts at irrigation in drought-prone areas
- instituting outdoor water-use restriction - regulating the water use of sprinklers or buckets for watering outdoor plants, for filling pools, and for water-intensive home maintenance tasks
- planting trees in degraded areas
- ensuring wise use of forest rangeland resources
- maintaining reserves of food (for example, grains) and other facilities
- limiting household size by regulating population growth
- instituting water and soil conservation programs.
- Other measures include resettlement and rehabilitation of drought victims, the distribution of seeds, oxen, fertilizers and livestock in pastoralist regions, and construction of roads.

3.3.3. NATURAL VEGETATION AND WILD ANIMALS OF ETHIOPIA

Types of Natural Vegetation of Ethiopia

Natural vegetation refers to any original plant grown in and covering an area.

The distribution of natural vegetation is influenced by many factors. The most important ones are:

- Altitude
- Climate
- soil type, and
- drainage

The types of natural vegetation in an area are strongly determined by temperature and rainfall. That is why the natural vegetation of an area is a good indicator of the area's climatic conditions.

Natural Vegetation's Relationship to Altitude and Rainfall

In Ethiopia, the types of the natural vegetation of an area are highly correlated with altitude and rainfall, as they are with temperature. The lowlands have harsh environments due to low rainfall and are characterized by xerophytic plants, while the highlands are characterised by different types of tree stands and forests.

Based on altitude, we can classify the natural vegetation of Ethiopia into the following five types:

- Afro-alpine and sub-Afro alpine
- Forests
- Woodland savanna
- Steppe vegetation
- Desert and semidesert vegetation

Afro-Alpine and Sub Afro-Alpine Vegetation

This type of vegetation is found at very high altitudes (above 3300 meters).

Afro-alpine and sub-afro-alpine vegetation is very similar to European alpine vegetation. Sub-afro-alpine vegetation is found between 3000 – 3300 m, while Afroalpine vegetation is found at higher altitudes than these. In Ethiopia, Afro-alpine and sub-Afro-alpine vegetations are found in the:

- Highlands of Semein and
- Highlands of Bale

Afro-Alpine vegetation consists of tussock grasslands, serules, scattered mosses and lichens. SubAfro Alpines are predominantly woodland scrubs. Gibra (*Lobelia rhynchopetalum*) and Asta (*Erica arborea*) are dominant plant species in this region of natural vegetation.

Forests

In Ethiopia, forests are characterized by broad altitudinal ranges (450 – 3300 m) and large variations in mean annual rainfall (200 – 2200 mm). This wide variation in altitude and rainfall results in the formation of highland and lowland forests.

These two types of forests have very different characteristics since they are the results of altitudinal zonation.

A. Highland Forests (forests that grow between 1500 - 3300 masl altitude) Ethiopia's highland forests consist of

- Kerkha (*Arundinaria*) 2800 – 3000 masl
- Tid (*Juniperus Procera*) or Coniferous trees 2200 – 2800 masl
- Zigba (*Podocarpus*) 1800 – 2200 masl
- Woira (*Olea Africana*) and Kosso (*Hagenia Abyssinia*) (1500 – 1800)

B. Lowland Forests (forests that grow below 1500 m altitude)

These forests are known as gallery/riverine forests. In Ethiopia, they grow along the banks of the Awash, Wabishabelle, and Ghanalle Rivers where moisture is available in the soil. The predominant trees are Sholla and Warka. In areas where mean annual rainfall exceeds 500 mm, Baphia forest predominates.

Woodland Savanna

Like forests, Savanna woodlands are found in both highland and lowland areas. Their altitudinal range is 250 – 2300 m, and their mean annual rainfall range is between 200 – 1400 mm. (Example: acacia, grass etc.) However, such vegetation is dominant at lower elevations and drier climates than the forests. Savanna grasslands experience marked seasons and are characterized by scattered acacia trees.

In areas where mean annual rainfall is more than 1000 mm, these grasslands can form attractive park-like areas with acacia, wild fig, sycamore and kosso trees. Ethiopia's savanna grasslands are found in the southern half of Ziway, Langano, Abiyatta and Hawassa.

Semi-desert and Desert Vegetation

Semidesert and desert vegetation consists of short acacia, thorn bushes, succulent plants and a few rough grasses. In Ethiopia, vegetation of this kind is found in the Eastern, Northwestern and Southeastern Lowlands, i.e., in areas where annual rainfall is below 500 mm and drought persists for a long period of time.

Wild Animals of Ethiopia

The diversity in Ethiopia's topography, climate, and vegetation has given the country a wide variety of wild animals. Ethiopia's wild-animal stock is generally similar to that of East Africa as a whole, due to topographic similarity and other aspects.

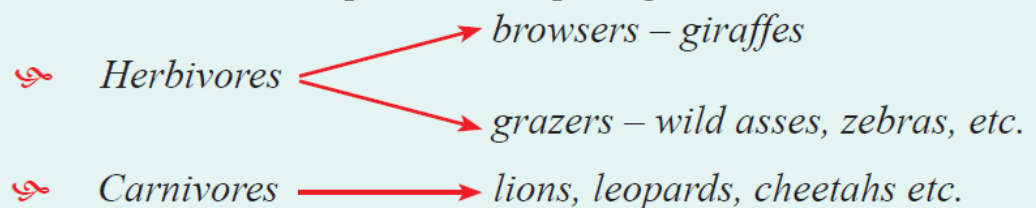
Ethiopia has about 277 species of mammals and 862 species of birds. Of these, seven species of mammals and twenty five species of birds are endemic to Ethiopia.

Types of Wild Animals in Ethiopia

The many types of wild animals found in Ethiopia can be grouped into the following broad categories.

- **Common wild animals:** These ones are animals commonly found in many places of the world. Many areas of Ethiopia have many common wild animals, including the hyena and the jackal.
- **Game animals:** These are animals that are killed for sport. Ethiopia's game animals include herbivores and carnivores. They are found in the lowlands.

Here are some examples of Ethiopia's game animals:



- **Arboreals:** These animals are animals that climb up trees. Ethiopia's arboreal animals, such as the Colobus monkey and baboons, are mostly found in the rainforest regions of Ethiopia.
- **Aquatic animals:** These creatures are animals that live in lakes and rivers.
Example: Fish, crocodiles and hippopotamus.
- **Birds:** Ethiopia has different kinds of both endemic and migratory birds; for example, Pelicans and flamingoes.
- **Rare/Endemic animals:** These are wild animals found only in Ethiopia. These days Ethiopia's endemic animals exist in only very small numbers. They inhabit highland and other areas. They are at great risk of extinction. The following are some of them.
 - Walia Ibex (wild goat), found in the Semein highlands.
 - Mountain Nyala (Dega Agazon), found in the Bale mountains.
 - ‘Gelada’ or ‘Chelad’ baboon, found in the Semein highlands.
 - Menilik's Bushbuk (‘Dikula’) in the Shoan and Bale highlands.
 - Swayne's Hartebeest (‘Korkay’), found in the Nechsar park and the Sankalle sanctuary.
 - Semein Fox (‘Key Kebero’), found in the Bale and Semein Highlands.
 - Wild Ass (Yedur Ahiya), found in the Afar and Southeast Lowlands.



Photograph: Wildlife of Ethiopia

These rare animals, especially the Walia Ibex and Semien Fox, are approaching extinction.

Causes of the Extinction of Wild-Animals

The main reasons for such conditions are the shrinking and destruction of habitats, which are mainly forest land by way of:

- rapid expansion of farmland, settlements, and industrialization
- expansion of grazing land
- wide spread practices of illegal hunting by the local people in search of meat, skin, fur, horn and ivory.
- frequent wild fires

What is more,

- migration of the wild animals to neighboring countries, due to shortages of food and water in Ethiopia adds to their perpetual disappearance.

Conservation Measures

Here are some conservation measures that have been recommended:

- establish national parks, game reserves and sanctuaries.
- Monitor and administer existing conservation areas properly.
- establish and implement strong laws that effectively prohibit illegal hunting.
- educate the public about environment protection.
- protect habitats.
- educate and encourage local communities to protect their animals' habitats and resources.

As you can see, some of these mitigation measures involve direct protection of the animals – for example by establishing and properly administering protected areas; and by training people in how to protect these areas.

Establishing Protected Parks, Reserves, and Sanctuaries.

National Parks: National parks are conservation areas for wild animals in which legal hunting is allowed, with some restrictions. The Ethiopian National Parks are

Table: Ethiopian National Parks

Name	Area in km	Location
Gambella	5,061	Gambella
Yongudirasa	4,731	Afar
Omo	4,068	SNNPR
Gerallea	3,858	Somali
Allattish	2,665	Amhara
Bale Mountains	2,471	Oromiya
Maggo	2,162	SNNPR
Chebbera churchura	1,215	SNNPR
Abiyatta – Shalla Lakes	887	Oromiya
Awash	756	Oromiya-Afar
Nech Sar	514	SNNPR
Semein Mountains	225	Amhara

Game Reserves

Game reserves are wild-animal conservation areas where tourists are allowed to practice licensed hunting. The game reserves of Ethiopia are listed in Table.

Table: Ethiopian game Reserves

Name	Area in km	Location
Borena	45,366	Oromiya
Lower Wabishebele	23,788	Somali
Arsi	10,876	Oromiya
Bale	9,663	Oromiya
Western Shoa	9,136	Oromiya
Afdim Gewane	5,932	Afar-Somali
Akobo	5,049	Gambella
Mursi	4,561	SNNPR
Mizan Tefferi	4,172	SNNPR
Jikawo	3,375	Gambella
Western Omo	3,200	SNNPR
Chercher-Arbagugu	3,045	Oromiya
Erer – Gotta	2,386	Somali
Geddeo	2,347	SNNPR
Dabus Valley	2,127	Benshangul-Gumuz
Boyyo	58	SNNPR
Segan Valley	N:A	SNNPR

Sanctuaries

Sanctuaries are wild-animal conservation areas where hunting is strictly prohibited. Example: Afar Gewane.

Table: Sanctuaries of Ethiopia

Name	Area in km²	Location
Babille	6982	Oromiya
Sankele	54	SNNP
Yabello (yavello)	2500	Oromiya

Human Intervention in Forest Lands

One intervention is in deforestation. At the beginning of the 20th C, the forest cover of Ethiopia was estimated at 40 percent of the entire country. Now it is less than 3 percent. This deforestation rate is alarming. We estimate that Ethiopia loses 100,000 – 200,000 hectares of forestland every year.

Causes of Deforestation

The main causes are:

- unwise tree cutting for supplies and materials
- overgrazing
- slash-and-burn practices
- fuel wood
- furniture
- construction etc.

Overgrazing: This is the practice of placing too MANY livestock on a given piece of land. The activities of these animals strips the land bare. In addition to overgrazing pasture areas, people are increasingly converting forest land to pasture land. This practice has expanded grazing land at the expense of forest land – in other words, through deforestation.

Slash and burn practices: People are involved in slash-and-burn practices to clear forestland in order to prepare it for farming. This practice essentially strips the forest bare by slashing (cutting down and digging up) all or most of the trees and other vegetation and then burning the piles away. Slash-and-burn is also used to periodically strip an area of farmland in order to leave it lying fallow to regain its nutrients. In Ethiopia, slashing and burning forests destroys a large area of forest annually.

This approach to gaining farmland is commonly practiced in southwestern Ethiopia. Expansion of built up areas: Built-up areas are areas occupied by factories, residence, recreational sites etc. Ethiopia's expanding human population increasingly requires more area for housing and other services. Some of the land that is converted to built-up areas is forestland.

Mitigation Measures

The following approaches have been suggested for slowing down and/or mitigating Ethiopia's rapid deforestation rate:

- conservation
- capacity building
- institutional development

These approaches are described below. However, despite their having great potential, these approaches alone are not enough. More ideas are needed, and sustained effort must be applied to enhance the mitigation effort.

Conservation of Natural Vegetation

These conservation measures have been suggested for preserving Ethiopia's forests:

- Reforestation – planting trees after every tree harvest.
- Afforestation – planting trees on bare and unproductive lands.
- Controlling burning practices (i.e., the slash-and-burn activities).
- Practicing agro-forestry.

- Reducing the use of fuel wood by adopting alternative sources of energy for household consumption.
- Controlling overgrazing.

Capacity Building and Institutional Development

Here are some approaches to preserve Ethiopia's forests through capacity building and institutional development:

- Providing environmental education to enhance public awareness about the use and management of natural vegetation.
- Developing forest-related curricula for schools, colleges, universities, forestry institutions, and forestry-management institutions. Then implement those curricula.
- Supporting and protecting community forests by applying strict legal measures.
- Moderating the existing rapid rate of population growth
- Facilitate community participation in combatting deforestation.
- Changing the life style of the people in terms of ongoing deforestation activities.

These goals must be vigorously implemented if we are to attain the country's ongoing goal of defeating deforestation. For example, the anti-deforestation rules and laws must be vigorously executed by relevant agencies.

3.3.4. SOILS OF ETHIOPIA

Formation of Soils in Ethiopia

Soils are the uppermost loose or unconsolidated material overlying the earth's crustal rocks. Its major components are water, air, organic and inorganic minerals.

It is a dynamic, natural and complex substance which can support animals and plants.

Soils of Ethiopia owe their origin to:

- parent rock material, which has been broken into small particles by way of weathering and natural decomposition
- climatic conditions, which largely determine the speed and nature of the processes that form the soil; for example, extreme heat, or cold, could stop the work of bacteria, and the amount of moisture influences on several aspects of soil formation.
- vegetation cover, which adds humus to the soils and renders support to the soil making animals and bacteria

On the plateaus that make up extensive areas in north, north western, south western, and central and south eastern Ethiopia the parent rocks are volcanic origin and experience sufficient rainfall. The soils formed in these areas are red basaltic and black basaltic soils. Unlike the red basaltic soils, the black soils have high clay content that makes the soil difficult for farming.

Types of Soil in Ethiopia

The soils of Ethiopia are basically derived from crystalline, volcanic and Mesozoic sedimentary rocks. According to the latest classification made by the FAO, there are eighteen classes of soil in Ethiopia.

Here are the main soil types. They cover more than 85 percent of the country.

Nithosols (red basaltic soils): These soil types:

- cover about 12 percent of the country.
- are basically associated with high rainfall and are found in areas that were previously covered with forest.
- are predominant in the Western Highlands of Wellega, Keffa, Illubabor, the Southern Highlands of Sidama, the Central and Western Highlands of Shoa, the Highlands of Gojjam and the Eastern Highlands of Hararghe.
- are matured soils with deep profiles. They are highly leached and lack soluble minerals like Sodium, Calcium, etc., but they are rich in iron and aluminum.
- are potentially good for farming and other agricultural practices since they are friable, and have a stable structure; as a result,
- are the most widely cultivated soil type. They are the best soils for coffee, inset and cereals.

Vertisols (black Basaltic soils): soil types of these sort:

- cover about 10 percent of the total land of Ethiopia.
- have high clay content; so, are sticky. For this reason and for poor drainage qualities, such soils are difficult to be used for farming purposes.
- have excellent nutrients that could provide support for agriculture, but their poor drainage qualities limit their use for grazing purposes.
- are largely found in Arsi, Bale and central Hararghe, where there are pronounced wet and dry seasons.

Acrisols: These soil types:

- are found associated with Nithosols.
- cover about 4.5 percent of the country.
- are widely found in the Southwestern Highlands of Ethiopia, where there is high rainfall.
- FF are extremely leached; and therefore, have low productivity capacity.

Cambisoils: Such soils:

- are soils that developed from the recent lava deposits of the Quaternary Period.
- are young and shallow.
- are found on the rugged and sloping terrain of the Plateau of Shoa (eastern escarpment) and Chercher Highlands.

Regosols: These types:

- like the cambisols, are shallow and young; but they are coarse-textured.
- have low agricultural value.

- are found in the Danakil and Ogaden plains.

Xerosols: These soils:

- These soils are generally young and shallow, and are found in arid and semi-arid regions. They have a weakly developed profile.
- are found extensively in the Northeastern escarpment, Northwestern and Southeastern Lowlands.
- are characterized by high salt content and humus deficiencies.
- have little significance for agriculture except places where they could be irrigated.

Yermosols: Soils of this sort:

- are found in desert and semi-desert areas, as are xerosols.
- are salty, acidic and have a weakly-developed profile.
- are not suitable for cultivation, even when irrigated, due to their salty and acidic nature.

Luvisols: These soils:

- are well-developed in areas where there are clearly marked wet and dry seasons and when leaching is not very high.
- are among the best soils, since they have good chemical nutrients.
- are intensively cultivated, except in areas that are steeply sloped or water-logged.
- are found around Lake Tana, and in the eastern part of the Northern Central Highlands and in the Southern Lowlands.

Lithosols: Such soils:

- are similar to cambisols and regosols in their poor maturity and their location on steep slopes.
are found in areas of low precipitation.
cover the escarpments of the Northeastern and Chercher Highlands.

Fluvisols: This type of soils:

- are soils that rivers have transported from highlands to lowlands.
- cover about 10 percent of the country's total area.
- are associated with river, sea and lake deposits.
- have very good agricultural potential.
- are found extensively in the lower regions of the Omo, Awash, Abbay and Baro-Akobo Rivers.

Soil Problems and Conservation in Ethiopia

Ethiopia's Soil Problem – Erosion

For countries like Ethiopia, where the mainstay of the peoples' livelihood is agriculture, issues related to soil are extremely important. Therefore, whatever setback happens must be carefully handled. One such setback is erosion.

Soil Erosion by Running Water

Every year enormous quantities of soil are carried away by Ethiopia's rivers to neighboring countries. As a matter of fact, the main problem related to soil in Ethiopia is erosion by running water.

Examples:

- River Abbay alone carries away 3000 – 4000 million cubic metric tons of soil annually.
- The Ethiopian highlands experience a loss of about 2000 tons of soil per square kilometer per year.

Factors that Accelerate Soil Erosion in Ethiopia

It is known that natural and human-made factors accelerate soil erosion. Both natural and human-made factors play a role in accelerating erosion.

Natural Factors

Two main natural factors contribute to soil erosion in Ethiopia.

- **Topography:** Many Ethiopian rivers set in motion from high places and have steep profiles as they progress to lower altitudes. These conditions are highly conducive for erosion to take place at ease.
- **The heavy summer rains:** Although most of Ethiopia's rivers have very low flows during the dry season, the heavy summer rains turn them into rushing torrents. The cracks that have appeared during the dry seasons turn rapidly into deep gullies. These conditions greatly facilitate the mechanical action of raindrops to wear away the surface soil, eroding it deeply.

Human-Made Factors

Human activities accelerate the soil erosion that is caused by the natural factors that we have just considered. Here are some of the many ways in which people have contributed to the erosion of the land that they depend on.

Deforestation: Most of the highlands of Ethiopia were once covered by forests. These forests provided cover for the land and protected the soil from the physical action of rainfall. Deforestation is progressing at a rapid rate and has become one of the main causes for Ethiopia's hastened erosion rate.

Bad cultivation practices: Bad cultivation practices also speed up erosion by making the soil vulnerable to the natural forces that we have just described a few lines above. For example,

Overcropping – This is one bad cultivation practice. It is the act of planting an area too densely. This approach to farming uses up the soil's nutrients faster than natural processes can replenish them and destroys the land's fertility.

Overcultivation – This one entails tilling land every year. For example, in order to keep up with increasing food requirements, people over-cultivate their lands. Too frequent tilling can remove nutrients from the soil faster than natural processes can replenish them.

Slash and burn – With such a practice, people slash (cut down and dig up) all or most of an area's vegetation and then burn the results. In farming, this practice is sometimes used periodically to strip an area before leaving it to lie fallow to regain its nutrients. However, since the soil is now bare and therefore unprotected, the land is vulnerable to the forces of erosion. Slash-and-burn techniques are also used to clear forestland in order to prepare it for farming, but unwittingly exposing the soil for erosion.

Keeping too many livestock (overgrazing): Most areas used for pastoral activities are overgrazed because too many animals have been kept there relative to their grass. The livestock destroys the vegetation faster than it can replenish itself. When the vegetation is destroyed, the land is laid bare and is therefore vulnerable to erosion. Goats are particularly damaging. They destroy all kinds and parts of an area's vegetation, including the roots.

Unscientific tilling: Most peasants do not use (being unaware) scientific farming techniques, such as terracing and contour ploughing, which serve as checks against soil erosion.

Using inadequately maintained tracks and trails: Without adequate maintenance, frequently used routes across the countryside can gradually change into deeply eroded gullies and eventually make them impassable, for they are partly eroded. Most of this destruction is caused by the action of rain and other running water on the soil, which has been laid bare by continual foot and vehicle traffic.

Conservation Measures to Combat Soil Erosion

Here are some conservation measures that have been recommended for minimizing soil loss by erosion in Ethiopia.

- ✓ **Reforestation** – replanting trees immediately after tree harvesting;
- ✓ **Afforestation** of areas that are not used for cultivation, are along steep slopes that are exposed to erosion.
- ✓ **Proper terracing of slopes and are set a side for cultivation.**
- ✓ **Intercropping** – this is the practice of growing two or more crops on the same field so that the land is not exposed to erosion.
- ✓ **Controlling livestock populations.**
- ✓ **Developing improved grass types that can feed more livestock.**
- ✓ **Installing fences to control the movement of grazing animals,** thereby restricting their activities and protecting the areas outside of the fences.
- ✓ **Proper construction** of tracks and routes so that drainage could not wash them easily.
- ✓ Constructing check dams.
- ✓ Installing shelter-belts and windbreaks in arid and semi-arid areas.
- ✓ Extending irrigation schemes along the major river basins and their tributaries so that water will be available in proper quantities at proper times. This measure would alleviate both drought and

flood conditions. Stored water would be used during dry periods, and flood waters would be controlled and therefore would no longer be destructive or wasteful.

UNIT SUMMARY

- Ethiopia and the Horn countries occupy a tropical location in Eastern Africa within the 1°S – 18°N latitudes and the 33°E – 51°24' E longitudes.
- The countries of the Horn vary in size. Ethiopia is the biggest while Djibouti is the smallest. This variation of size among the countries has resulted in advantages and disadvantages.
- Regarding shape, Ethiopia and Djibouti have more or less compact shapes, while Somalia and Eritrea have elongated shapes.
- Ethiopia and the other Horn countries have experienced four geological eras. The geological structure of the region is the result of the geological events that occurred during these eras.
- The landforms of Ethiopia and the Horn are largely the results of the geological forces that took place in the Cenozoic era's, Tertiary and Quaternary periods. Altitudes increase from the peripheries to the centers.
The highest peak is Ras Dashen (4620 meters) in Ethiopia.
- The altitude plays a significant role in determining the climate, natural vegetation, drainage systems as well human activities in the region.
- Due to the altitude effect we see spatial variation of :Temperature; Rainfall; Vegetation and Population density
- The drainage system itself is the altitudinal effects, rivers as flow out from the central highlands to the peripheral lowlands. This has made the rivers flow swiftly.
- The highlands are homes of different wild animals of which some are endemic. These endemic animals are at the risk of extinction.
- Although the forests are confined to the highlands, human intervention for both economic and other purposes is destroying it at an alarming rate. The forest resource stock that had covered 40 percent of the country has come to less than 3 percent these days.
- Ethiopia has different types of soils. The volcanic soil, especially nithosol, is the most widely utilized soil. The country's cereals, coffee, inset, etc. are grown on this soil type. However, the topography, climate and traditional practices have become major accelerants of its removal. Attempts have been underway to check erosion. Awareness is being created at all levels through school syllabi and extension services to mitigate the problem.
- Soils are the upper most loose or unconsolidated material overlying the earth's crustal rocks. Soils of Ethiopia owe their origin from their parentmaterials. They are basically derived from crystalline, volcanic and mesozoic sedimentary rocks. The main types of soil in Ethiopia are nithosols, vertisols, acrisols, cambisols, regosols, xerosols, yermisols, luvisols, fluvisols and lithosols.
- The major problems of Ethiopia's soils is erosion by running water. Both human-made and natural factros are responsible for erosion.

REVIEW EXERCISE FOR UNIT

I. Short Answers

1. Describe the major geological events that occurred during the Cenozoic Tertiary Period in Ethiopia and the Horn.
2. Identify the dominant factors that determine the drainage systems and change patterns in Ethiopia and the Horn.
3. Name two areas in Ethiopia with a high rainfall variability b very low rainfall variability
4. Which soil classes in Ethiopia are very unproductive? What are the reasons for this unproductivity?
5. Which regional administration is involved in the protection of wild animals and conservation of areas for them? Name at least four of Ethiopia's protected wild animals.

II Completing Sentences: Complete the sentences below by filling in the spaces left blank.

6. The two non-anthropogenic factors that accelerate soil erosion in Ethiopiaare _____ and _____.
7. One of the attainable measures for reducing soil erosion is afforestation. It is the _____.
8. The year-round rainfall region corresponds to _____.
- 9 The current economic potential of the Ethiopian Rift Valley is assumed to be _____.

UNIT FOUR

HUMAN POPULATION (27 Hours)

4.1 Concept and Facts about Human Population

Human population is the number of people living in a definite area. It is all of the people living in a specified area-such as a city, region, country, or continent-at a given time. Population is one of the most critical factors that determine the socio-economic and environmental conditions of every country in the world.

The study of human population is also necessary for development and socio-economic activities. For example, population is the major source of the labour force for the productive and non-productive economic sectors, such as agriculture, manufacturing, teaching, health services, etc. Human population is also the main productive force and creator of material wealth. This makes the study of population extremely important for the overall socio-economic development of a country.

In geography, population geography is :-

- ✓ A branch of human geography that usually focuses on the spatial patterns and variations of the various characteristics of human populations.
- ✓ Particularly emphasizes the spatial dimensions of population size, structure, composition, distribution, settlement, density, migration, growth, and other demographic processes and facts.
- ✓ Considers demographic facts in terms of their present contexts and causes, characteristics and possible consequences to the geographical environments (i.e., both the physical and cultural environment).

Uses of Population Data

- ❖ **Having accurate data of the population is essential for effective socioeconomic development planning and administration:**

Population studies in geography and other fields yield knowledge essential for planning, especially by governments, in fields such as health, education, housing, social security, employment, food security, and environmental preservation. Moreover, the studies provide information needed in the formulation of governments' population policies, which seek to modify the trends of population size, composition, and distribution in order to achieve economic, social, and environmental objectives.

4.1.2. Components of Population Change

A change in the overall size of a population is the result of the collective effects of changes in **fertility, mortality and migration**.

The three factors are collectively known as **population-change dynamics or determinants or components**. Fertility and mortality are **biological factors**, while migration is purely **non-biological**. The combined effect of the three factors controls the changes in population size and composition.

❖ **Fertility:**

- It is the actual occurrence of live births in a given population; it is the beginning of life. Specifically, fertility refers to the actual reproductive performance of a population.
- It refers to the number of live births women have.
- ❖ The study of human fertility occupies a central position in the study of population because it is responsible for biological replacement and maintenance of the population of the human race. The growth of world population depends largely on human fertility. This is so because any society replenishes itself through the process of human fertility. Thus, in population dynamics, fertility is a force of expansion of population by counteracting the force of attrition caused by mortality.
- ❖ Fertility is affected by cultural, social, economic, and health factors. Most of these factors operate through four other factors: (1) the proportion of women in sexual unions; (2) the percentage of women using contraception; (3) the proportion of women who are currently not fecund (primarily because of breastfeeding); (4) the level of induced abortion.

❖ **Mortality:**

Mortality refers the occurrence of death. Mortality rates, though decreasing worldwide, are higher in the developing countries than in the developed ones. This difference is caused by variations in standards of living, nutrition, medical services, personal hygiene and environmental sanitation.

While we all eventually die, the probability of dying during a given time period is linked to many factors, such as age, sex, race, occupation, and social class. The incidence of death can reveal much about a population's standard of living and health care.

- ❖ **Migration:** is the geographic movement of people across a specified boundary for the purpose of establishing a new permanent or semi-permanent residence. Along with fertility and mortality, migration is a component of population change. The terms “immigration” and “emigration” are used to refer to moves between countries (international migration). The parallel terms “in-

migration” and “outmigration” are used for movement between areas within a country (internal migration).

4.1.3. Spatial Distribution of Human Population

Population distribution is the way that population is spread out over a given area. That area can be anything from a small region to the earth as a whole. For several reasons, human population is spread very unevenly over the earth’s surface. This distribution of population forms a pattern. While some areas are heavily populated, others are sparsely populated.

- The availability of natural resources, the nature of topography, the productivity of the soil, and many other factors has resulted in an uneven distribution of people. There are densely, moderately and sparsely populated areas in the world. The densely and sparsely populated regions of the world are outlined below.

4.1.3.1. Densely Populated Regions:

There are three main belts of high population density and concentration in the world. These areas are also called “Population Clusters” or “Population Nodes”.

The **three belts** alone account for well over two-thirds of the total world’s population. These regions are:

Asiatic population belt:

This belt is confined to the river basins of Monsoon Asia. The region accounts for about 60 percent of the world’s total population. The belt includes:

East Asia - Japan and China are the most populous in the sub-region.

South Asia - India, Pakistan and Bangladesh are densely populated.

Peninsular Europe:

Europe makes up 10.8 percent of the world’s total population. It is the most developed cluster, in both economic and demographic terms globally.

Northeastern North America

This part of North America (mainly the USA) is highly urbanized and industrialized. It accounts for about 5 percent of the world’s total population.

4.1.3.2. Sparsely Populated Regions

In contrast to the above discussed population clusters, there are areas of thin population cover. These areas include:

1 **Tundra**:- These areas support only a very few nomadic peoples.

2 **Hot-dry lands**:- Include many of the tropical desert lands. Shortage of water and high temperature do not encourage permanent settlement. However, in desert areas where water is available, such as oasis and river basins, and mining sites, there are settlements located right in the heart of hot deserts. Example: the Sahara Desert.

3 **Hot-wet lands**:- Because of the unfavourable climate, there are few people living in these areas.

4 **High relief**:- Includes regions of mountainous areas with high altitude, rugged topography and cold temperature. These areas do not encourage large settlements because the terrain makes movement and interaction difficult.

4 **Areas with poor soil**:- Include very large areas of the world which are covered by infertile soil. Such areas do not support agricultural practices and hence are sparsely populated.

4.1.4 Population characteristics

The most commonly used characteristics of population include **size, density, composition, and distribution**.

Size: deals with aspects like how many people live in a given population group, what changes are taking place in the size of the group and how many people live in a given place at a given point of time.

4.1.5. Sources of Population Data

Population data refers to population information, such as number, age, marital status, births and deaths, occupation, religion, educational status, and other characteristics of the human population. Such information is vital for studying the different aspects of the human population. That is why we say that studying human population is dependent on reliable population data. Statistical information about human population is obtained through a variety of ways.

- ❖ Population information is important for many reasons. The data acquired through various methods is used for many purposes that include the following.
 - ✓ It provides statistical information for making decisions related to social and economic affairs.
 - ✓ It indicates future requirements of the population in terms of social, health, education, employment and other socio-economic needs.
 - ✓ It helps policy makers to forecast the needs of the population and to understand how to meet them.

The following are among the most commonly used methods of gathering information about human populations.

- i. Population census**
- ii. Sample survey**
- iii. Vital registration**

I. Population Census

Censuses are the main source of population data in many countries including Ethiopia. The history of census goes back to ancient times in places such as Rome, Greece, Palestine, India, etc.

Census is usually defined as “The total process of collecting, analyzing, compiling and publishing demographic, economic and social data of all persons in a country or delineated territory at a specified time.” In other words, it is the official counting of all the people of a certain geographical area with a well-defined boundary so as to collect socioeconomic and demographic information about the population.

Census provides a wealth of demographic and socio-economic data. The size, growth rate, fertility and mortality characteristics, marital status, employment situations, religion, ethnicity, income, educational status, housing conditions, sex and age structures and much more information about the population of a given area can be obtained through census.

In Ethiopia, the census is one of the most important ways of collecting population data. So far, three censuses have been taken. The first was in 1984, the second in 1994 and the last one in 2007. Census in Ethiopia is conducted every ten years.

The Main Features of Census

The following are the main features that distinguish census from the rest of the sources of population data.

- **Government Sponsorship:** It is the responsibility of the national government to provide or seek the resources necessary to conduct census. Obtaining them requires adequate legislative support or legal authority.
- **Defined territory:** census refers to specified geographical unit such as state, country, or province. The unit needs to have its own demarcation or boundaries.
- **Universality:** Enumeration must include every person within the scope of the territory without omission or duplication. Census of population usually try to count every one in the country as of a fixed day, often known as Census Day.
- **Simultaneity:** The census should have a specific well-defined time, and it should be conducted simultaneously throughout the country.
- **Individual Units:** in censuses, data is collected on an individual basis. The information that is obtained normally relates to individuals rather than groups. It is compiled and published.
- **Periodicity:** Ideally, census should be taken at certain specified time interval, usually every ten or five years. For example, in the United States and Ethiopia, census is conducted every ten years, while Canada and the UK conduct census every five years.
- **Expensiveness:** Census is the most expensive method of data gathering. This is because of its universality which necessitates complete coverage of the entire territory and population of the place where the census is taken.

II. Sample Surveys

A sample survey is the second most important and widely used source of population data. It collects information only from a part of the whole population.

They are usually undertaken to generate socio-economic and demographic data in greater detail than conventional censuses can provide.

Surveys are usually undertaken during a period of transition between two consecutive censuses. They are employed to arrive at estimates of demographic characteristics (age, sex, marital status, etc), population size, population distribution, mortality, fertility and migration. Sample surveys are also an important source of population statistics in areas where census is not conducted.

In addition, they are important for gathering data to fill population information gaps. As there are many countries that do not conduct census regularly, the main source of information about world population comes from the United Nations Organization through sample surveys.

- ❖ As compared to census, sample survey is:
 - Less expensive because it does not involve the entire population;
 - Capable of providing very detailed information about the population;
 - If the sample takers are well-trained and strict supervision is in place, it can provide data of a better quality than census; and
 - Able to estimate margins of error, or the accuracy of the data.

III. Vital Registration

What is vital registration?

Vital registration is another source of population data. It refers to the continuous registration of such vital events as births, deaths, marriages and divorces. These events are recorded as they occur. Data on vital events are gathered from birth and death certificates, marriage licenses, divorce records and other official registers. The building up of a reliable vital registration system is an expensive and extensive process.

The completeness of registration and the reliability of the data gathered through vital registration can be affected by several factors including the following:

- ✓ Level of literacy and awareness of the importance of population data in the population;
- ✓ Social custom may lead to non-registration as a result of superstitions or taboos;
- ✓ The rural population may be widely dispersed or inaccessible.

Factors Affecting the Quality of Data

As we have seen so far, population data can be acquired through different methods.

Though a variety of information can be obtained by using such systems, the gathered information may not necessarily be accurate. This is because there are a number of socio-economic and culture related problems that affect the quality of the data.

Some of these problems include:

- i. Inadequate financing.
- ii. The use of different methods of census enumeration by different countries.
- iii. Inadequate awareness about the importance of population data among the general population.
- iv. Double entry (counting of a person twice).
- v. Neglecting or ignoring certain groups of people.

- vi. False information about the age and the occupation of people.
- vii. Lack of equal acceptance of the importance of vital registration by all countries.

4.1.6. Population theories

The main purpose of population theory is to seek appropriate answers to the following population-related questions.

- ✓ *How large can world population ultimately become, and more particularly how many human beings can the planet feed and the environment sustain?*
- ✓ *How does population growth determine the socio-economic issues of countries?*
- ✓ *How do biological, economic, social, and political factors determine population growth?*

The theories of population developed by different scholars, in response to the above questions, can be grouped into two broad categories: Malthusian/neo-Malthusian and anti-Malthusian. The two groups of theories have opposing views about the relationship between population growth and socio-economic, political, and environmental factors. Malthusian and neo-Malthusian theories are pessimistic, and the anti-Malthusian theories are optimistic.

The next figure indicates the characteristics and differences between these two types of population theory.

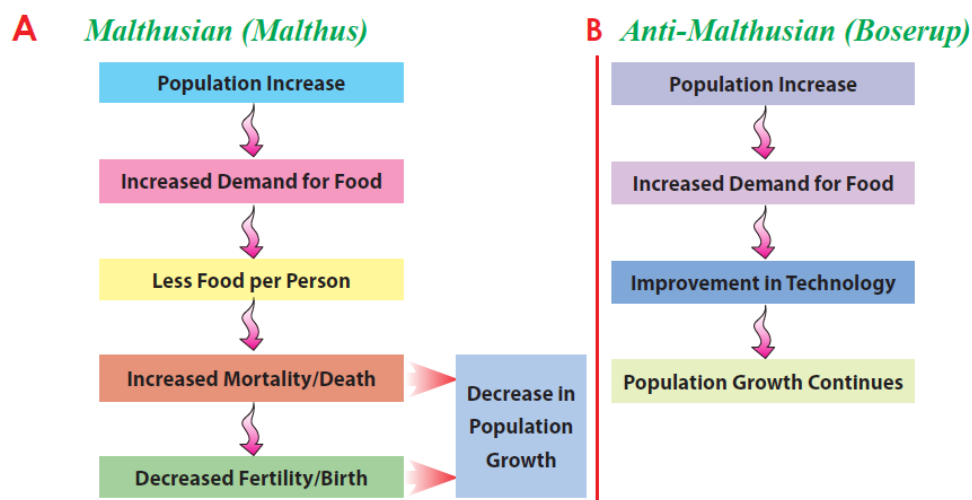


Figure: The opposing views of Malthusian and anti-Malthusian population theories

4.1.5.1. Malthusian Population Theory

In his work, Malthus warned of a constant tendency for human population growth to exceed food production. He classified the various ways that such growth would, in consequence, be slowed.

Malthus took an exceedingly pessimistic view, arguing that human populations are inescapably caught in a conflict between their “need for food” and the “passion between the sexes”.

In his essay, **Malthus put forward the following three basic propositions.**

- + Passion between sexes is inevitable and universal;
- + Food production is limited, and population cannot increase beyond the means of subsistence; and
- + If population growth outstrips the means of subsistence, “positive checks” will apply.

Malthus was concerned that **population would grow faster than the supply of food**. He believed that the supply of food can only increase by a constant amount, in **arithmetical progression** (1 – 2 – 3 – 4 – 5), but that the human population has a tendency to multiply in **geometric progression**, (1 – 2 – 4 – 8 – 16), multiplying itself by a constant amount each time. Therefore, eventually population would outstrip food supply until a catastrophe occurred.

This would be in the form of famine, diseases or war. Such catastrophes would occur as human beings fought over increasingly scarce resources. Malthus referred to the catastrophes as **positive checks** that control fast population growth and in the long term maintain a balance between population and resources.

Malthus considered that, in order to avoid the inevitable occurrence of the positive checks, human beings should adopt **preventive checks**. By preventive checks, Malthus meant “**moral restraint**”, which includes late marriage, avoiding sexual conduct before marriage and having fewer children. However, Malthus was against family planning methods; he was totally against all artificial methods of conception or birth control. Malthus, therefore, appealed to people to control their natural sexual urges in order to control the fast growth of population and to, therefore, avoid the occurrences of the disastrous **positive checks**.

❖ In general, Malthus believed that the “hot passion or sexual urge” of people could lead to overpopulation, which, in turn, would result in poverty and other catastrophes (positive checks). He concluded that “the poor are to be blamed for their own poverty. Neither wages nor providence, nor society is to be blamed.”

Today, supporters of Malthus’s pessimistic view, who fear that population growth will outstrip food supply and other resources, leading to the catastrophic consequences (the positive checks) predicted by Malthus, are called Malthusians or **neo-Malthusians**.

4.1.5.2 Anti-Malthusian Population Theories

Anti-Malthusians are optimists who argue against Malthus. They believe that either population growth will slow down well before it is limited by scarcity of food and other resources or the ingenuity/cleverness of humankind will solve the problems, overcoming potential scarcities of food and other resources. Let’s consider two anti-Malthusian theories – one was developed by Karl Marx, and the other by Ester Boserup.

A. Marxian Population Theory

In his theory, Marx stated that there could be no population problem under a socialist mode of production and of ownership of resources. He believed that population becomes a **problem of surplus people** only under capitalist modes of production. His view was that the capitalist system not only views some of the population as **surplus** in an economic sense, but also causes this phenomenon:

When existing food supplies are inadequate to support some of the population, capitalism calls the unsupported people surplus population. However, Marx stated, such food scarcities are purely the result of unequal distribution of resources by capitalism.

Marx believed that the capitalist system can produce food and other necessities for an indefinitely expanding population, and that it is only capitalism’s unequal distribution of social wealth that makes it seem as though these resources were limited and, therefore, that population growth must have a natural limit. Moreover, in Marx’s view, the system of capitalist production is not targeted to meet the needs of poor people. Instead, its aim is to increase the accumulation of capital for the wealthy. Along the same lines, Marx stated that capitalism benefits from the conditions that produce a so-called surplus population. Those conditions create competition for jobs, thus driving down wages and therefore maximizing profits for the wealthy. Marx’s solution to the problem of overpopulation was socialism with a new economic structure of society within a new social order.

- ❖ While Malthus focused on individual actions and considered “moral restraint” (preventive checks) to be the solution to population problems, Marx focused on the economic structure of society and believed the solution would be found in a new social order.

B. Boserupian Population Theory

Boserup formulated an optimistic theory about the influence of population growth on agricultural development in support of her theory; she elaborated facts that showed population growth would lead to agricultural development. For Boserup, population is a variable that contributes positively to agricultural development, because it leads human-kind to innovation. In order to explain the positive effect of population growth on agriculture, Boserup presented the following series of historical agricultural transformations that were caused by population growth, in which humans progressed from “a” through “e” in their approach to agriculture between the nth and nth centuries. For each transformation, Boserup

- described the approach to agriculture of the specified time and explained how it used land
- examined the labor requirements of that approach between the nth and nth centuries.

Boserup noted that, as the human population grew, the amount of land available per-capita shrank but that, in response, people developed more effective ways to use the land. With each development in land use, labor requirements grew, and the increasing population provided the labor – and, therefore, the means to implement these developments. In addition, the increased labor requirements provided occupation for the growing number of people.

a)Forest-Fallow Cultivation

- Long periods of forest growth alternate with short periods of cultivation.
- Per-capita, the population owns and must cultivate large areas of land. However, because the work required - and therefore the labor requirement - of the forest-fallow approach to cultivation is low, food supplies are adequate.

b) Bush-Fallow Cultivation

- Relatively long periods of bush growth alternate with short periods of cultivation. The bush-growth periods are not longer than the forest growth periods.
- The labor requirement is larger than for the first (forest-fallow) approach.

c). Short-Fallow Cultivation

- The land is cultivated with intervals of only a few years of abandonment. These intervals last only for only a few years. They allow the land to regenerate and are used to enhance its productivity.
- The labor requirement is larger than for the first two approaches.

d) Annual Cropping

- The land is cultivated annually.
- Fertilizers are required.
- Labor requirement is larger than for all of the earlier approaches.

e) Multiple Cropping

- Different crops are grown at the same time, on the same land, throughout the year. The land is used throughout the year.
- The labor requirement is larger than for all of the earlier approaches.

Boserup concluded that population growth could not be controlled by scarcity of food and other necessities, as Malthus believed. Instead, population would continue to grow because the increased demand for food and other necessities would lead to new innovations and technological advancement which, in turn, would allow production to keep pace with the needs of the increasing population.

❖ Boserup thought that “As the size of population increases, it results in technological innovations and advancement.” For her, population is an independent variable/factor, and agriculture is a dependent variable/factor. She believed that population growth could not be controlled by scarcity of food supply. Rather, the increasing demand for food caused by population growth would increase agricultural productivity by stimulating innovation in agricultural systems and technologies. In this regard, she said that “Necessity is the mother of invention”.

4.2 Human Population of the World

4.2.1 Size and Trend of World Population Growth

For most of our history, human populations have been small, compared to those of other species. Studies of hunting and gathering societies (before 5000 BC) suggest that total world population was probably

only a few million people. A major change occurred about 10,000 years ago, when humans began to domesticate animals. This advance led to increased food supplies, which allowed the human population to grow, reaching perhaps 50 million people by 5000 B.C.

However, for thousands of years, the human population increased very slowly, and it took more than 1,500 years to reach the 500 million mark. Growth was not steady, but was marked by great fluctuations dictated by climate, food supply, disease and war.

Table: World population growth and doubling time

Date	Population (millions)	Doubling time (years)
5000 B.C	50	-
800 B.C	100	4,200
200 B.C	200	600
1200 A.D	400	1,400
1700 A.D	800	500
1900 A.D	1,600	200
1965 A.D	3,200	65
1990 A.D	5,300	38
2020 A.D (estimate)	8,230	55

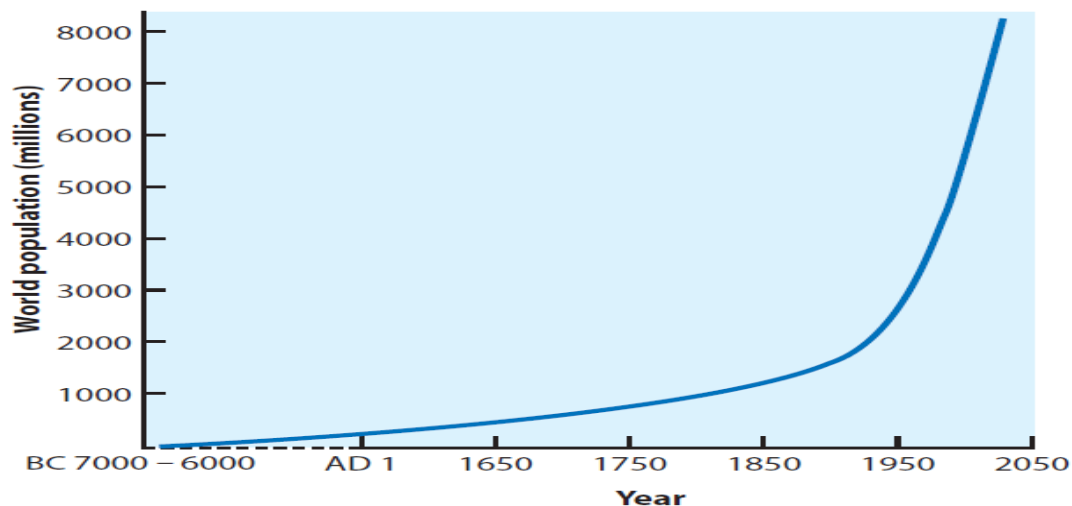


Figure: World population growth

Both above Table and Figure show that, by 1900, world population had reached 1.60 billion, and by 1960 it stood at 3.04 billion.

The United Nations estimated that world population reached 6 billion with an annual growth rate of 1.5% by 2000 A.D. The number added to world population per year would increase from 80 million to 90 million if the growth rate of 1.5% continues, and world population would double in a period of 40-

50 years from about 2000 A.D. Such rapid and dramatic growth rate of the world population is known as **population explosion**.

- ❖ Population doubling time is the time taken, in years, for a population of a given region or country to double at a given rate of population growth. It can be calculated using the following formula:

$$\text{Doubling Time} = \frac{70}{\text{Growth Rate}}$$

Population growth rate determines the time required for a population to double. The lower the growth rate, the longer is the doubling time, and the higher the growth rate, the shorter is the doubling time (See the following Table).

Table: World population size growth rate and doubling time

Region	Population (millions) 2010	Annual growth rate in % 1975 - 2009	Doubling time (year) 1975 - 2010
World	6,909	1.53	46
Asia	4167	1.62	43
Africa	1033	2.59	27
Europe	733	0.23	304
L.America	589	1.73	41
N.America	352	1.07	65
Oceania	36	1.49	47

As it is shown in the above Table, Europe and Africa have shown very contrasting doubling times. The doubling times of the former and the later are 304 years and 27 years, respectively. This implied that even among the developing regions, Africa has the fastest population growth rate in the world.

Generally, nations are considered to be less developed if they have a lower standard of living than the developed nations. A large share of the population in these less developing countries live at subsistence levels, and medical resources are limited. However, population growth in less developed nations occurs at a much faster rate than in the developed nations.

As of 2000, 1.2 billion people lived in the developed nations of the world, and 4.9 billion people lived in the less developed countries. By region, over half the world's population was in East and South Asia: China, with 1.3 billion inhabitants, India with 1.1 billion, and Indonesia with 0.23 billion were the dominant contributors. Europe and the countries of the former USSR contained 14% of the world population, North and South America made up 14%, Africa had 13%, Asia 58% and the Pacific islands had about 1% (see the following Table).

Table: Percentage distribution of population, over years, by continents

Continents	1750	1800	1850	1900	1950	2000	2050
Africa	13.4	10.9	8.8	8.1	8.9	13.1	20.3
Asia	63.5	64.9	64.1	57.4	56.2	60.7	59.0
Europe	20.6	20.8	21.9	24.7	21.4	12.0	7.1
L. America	2.0	2.5	3.0	4.5	6.5	8.6	5.3
N. America	0.3	0.7	2.1	5.0	8.6	7.9	7.9
Oceania	0.3	0.2	0.2	0.4	0.5	0.5	0.5

The nations currently defined as developed represented 20% of the world population in 2000. This percent is expected to fall to 15% by 2050. Nine out of every ten persons who are now being added to the world's population are living in the less developed countries.

More specifically, when we compare the population trends of Africa and Europe (see the above Table), Africa's population has increased since 1900. It had added about 5.0% of the world population between 1950 and 2000. This is expected to increase to 20.3% by 2050. Whereas, the population of Europe has shown a trend of decline by 12.7% in the same period.

4.2.2 Spatial Distribution of World Population

The most important characteristic feature of current world-population distribution is its extreme unevenness. It was estimated, some years ago, that one-half of the world's people was contained within about 5% of the earth's land area while, in contrast, about 57% of the land area contained less than 5% of the population.

World population distribution, and hence concentration, varies considerably across the earth, among different continents, between individual countries, between developed countries and developing countries, between rim lands and hinterlands, between highlands and lowlands, and between the Northern Hemisphere and the Southern Hemisphere (see the Figure below).

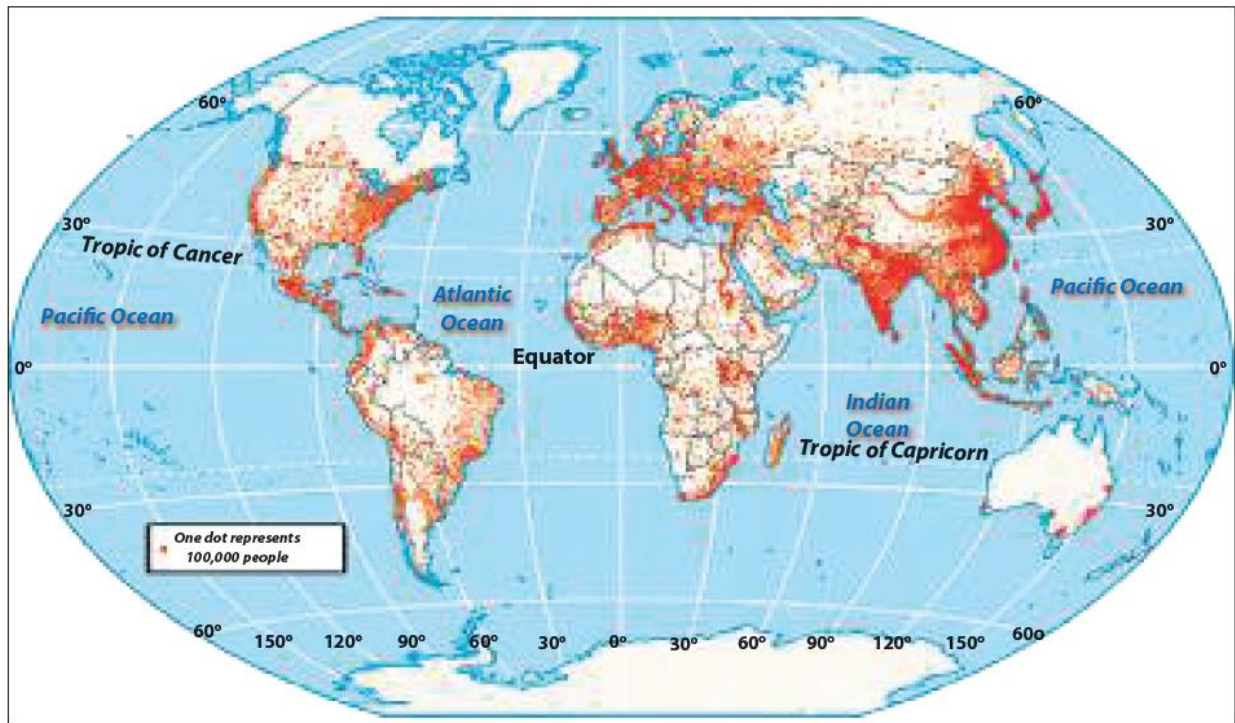


Figure: Dot map showing world-population distribution, 2000

Factors of Population Distribution

Numerous factors have positively and negatively influenced population distribution. The extreme current unevenness in population distribution is due to these factors:

- Ecological (physical) factors
- Socio-economic factors
- Political factors
- Demographic factors

Some factors, for example, favourable climate, fertile soil, adequate water supply, and industrial development positively influence population concentration. Others, such as unfavourable climate (for example, extremely hot or cold and dry climates), mountainous regions of very rugged topography with poor soils, and inhospitable areas where communication and trade are difficult, discourage large human settlements.

The four population clusters and the sparsely populated regions are found in varying geographical locations, have different population sizes and socioeconomic characteristics. The following Table gives

an overview of the various centers of high and low population concentrations, their locations and determining factors.

Table: Overview of the major densely and sparsely populated areas of the world

Population distribution	Regions or areas	Distribution factors	
		Ecological	Socio-economic
Areas of very high population concentration	Asiatic population belt (supports more than 60% of world population). ➡ East Asia-China, Japan and Korean Peninsula (supports 25% of world population).	Favourable climatic condition for agriculture.	Birthplace of one of the principal human races. Ancient history of habitation; high industrial development (Japan).
	➡ South Asia - India, Pakistan, Bangladesh, Myanmar (Burma) and Sri Lanka (supports) accounts for about 25% of world population.	Favourable monsoon climate,	Pre-modern civilization of small-scale agriculture, which essentially is traditional in nature and subsistence levels. Irrigation technology (for rice cultivation).
	Europe (more than 10% of world population).	Enjoys a very favourable climate.	Largest portion of its land area is devoted to agriculture. High industrial development.
	American cluster: Northeastern part of the USA and Canada (supports about 5% of world population).	Favourable climate, Short growing season, Rich natural resources.	During its formative period, large influx of immigrants and fast natural increase of the population. Efficient agricultural management. High industrial development.
Sparsely populated areas	Tundra (part of N. America and Eurasia).	Very cold climate (no cultivation).	
	Hot and dry lands (tropical deserts).	High temperature. Shortage of water.	
	Hot and humid regions: Amazon, Congo and Indonesia basins.	Very high humidity, rainfall and temperature.	
	High relief areas.	High mountains with rugged topography, Cold temperature.	
	Areas with poor soil.	Poor soil, badly degraded lands.	

Population Density

It was noted earlier that population distribution over the earth's surface lacks uniformity. Some areas are overcrowded, while others are sparsely settled or uninhabited.

Table: Population density of the world and its regions in 2000

Continent	Population, in million	Area, in km ²	Density (p/km ²)
World	6,892	150,000,000	46
Africa	1030	30,251,000	34
Asia	4157	45,066,000	92.2
Europe	739	10,530,750	70.2
Latin America	585	17,825,479	33
North America	344	24,346,000	14.1
Oceania	37	7,772,021	4.7

Population density, also termed as the spread of population over space, is measurable in various ways. For example, we can relate numbers of inhabitants to area of lands inhabited:

a) Crude density or Arithmetic density

b) Agricultural density or Rural density

a) Crude density or Arithmetic Density: is a measure of the number of inhabitants per unit area. This type of measurement is also known as **human-land ratio**.

$$\text{Crude density} = \frac{\text{Total population of an area}}{\text{Total area, in sq. km, of the area}}$$

b) Agricultural density or Rural density: It is the ratio of agricultural (rural) population to cultivated land. Agricultural density considers only agricultural population, the segment of an area's population whose livelihood depends on agricultural activities. The area under consideration can be a single contiguous piece of land, or it can be multiple unjoined areas – such as the land occupied by all rural populations in all developing countries. In countries like Ethiopia, almost all rural areas are occupied by people who are engaged in agricultural activities and whose livelihoods depend mainly on agricultural income. This may not be the case in the developed regions.

Activity

Answer the following questions, based on the table below:

Country	Area (km ²)	Population (thousands)	% of urban population	% of cultivated land
A	95,000	18,000	25	30
B	90,000	22,000	80	10
C	150,000	30,000	45	35

5 What are the crude and agricultural population densities of each country?

6 For which country does agricultural density appear to be a better measure than crude population density? Why?

Agricultural population density is a more meaningful measure than crude population density for developing countries where agriculture is the dominant economic activity. It also gives a better indication of population pressure on natural resources.

4.3 Population of Africa

4.3.1 Aspects of Population, Economy and Natural Resources of Africa

Population Size, Growth and Distribution

A. Size

How large is Africa's Population?

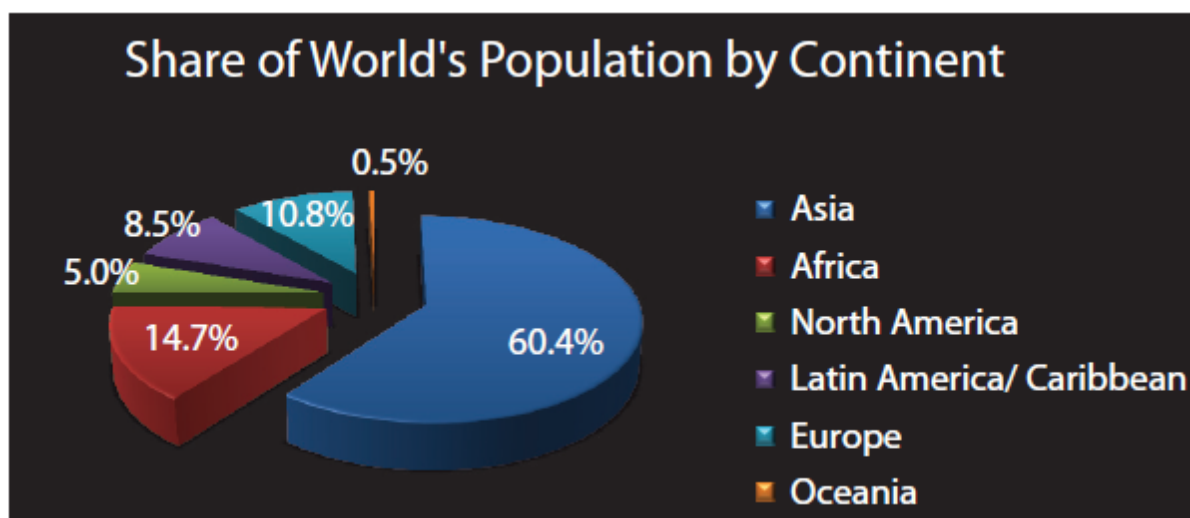
What percent of the world's population is constituted by Africa's population?

Africa has one of the world's largest populations. The continent is the world's second most populous next to Asia. According to the World Population Data sheet, by the mid of 2009, the continent had a total population of 999 million (nearly 1 billion). This makes up nearly 14.7 % of the world's total population.

As the table below depicts, Africa's population is nearly 1/4th of that of Asia, the most populous, and 28 times that of Oceania, the least populous.

Table: World population distribution

No	Region	Population (in millions)	Percentage of world total
1	Asia	4117	60.4
2	Africa	999	14.7
3	North America	341	5.0
4	Latin America/ Caribbean	580	8.5
5	Europe	738	10.8
6	Oceania	36	0.5
7	World Total	6811	99.9



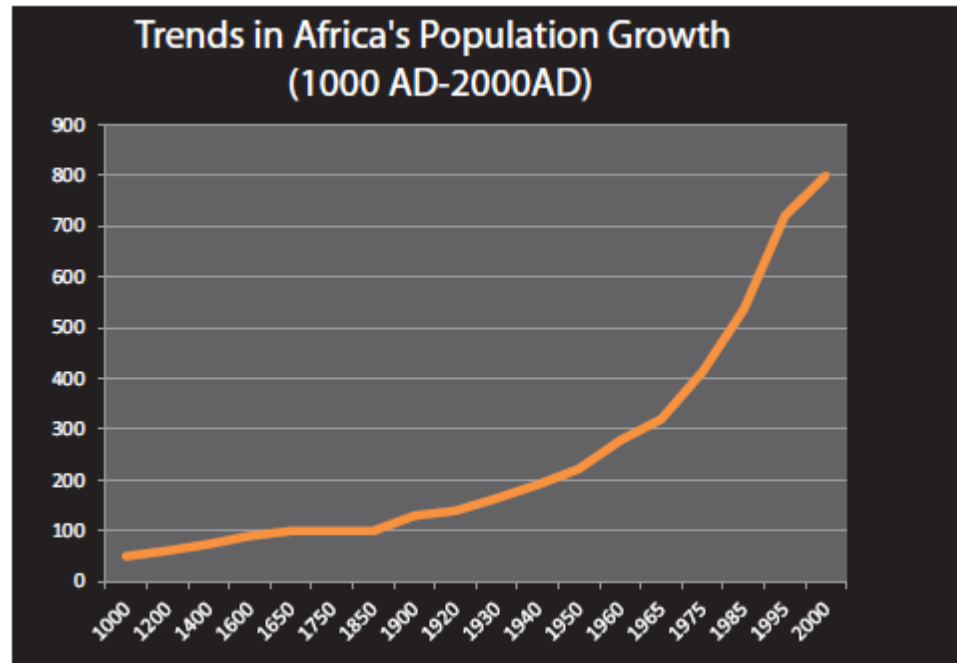
B. Growth

- What is population growth? What are the components of population change?
- How fast is Africa's population growing?

Africa has one of the world's fastest growing populations. The continent has been experiencing rapid changes in its population size as a result of many factors including the interactions between fertility and mortality. Despite the gradual decline in the birth and death rates, the continent continues to experience high rate of population growth. As some historical sources indicate, Africa's population had been increasing very slowly until recent times. However, since the 1950, the continent's population began to increase very rapidly. As a result, the continent's population had almost been quadrupled between 1950 and 2000. The following Table shows you the trend in the population size of Africa between the years 1000 A.D and 2000 A.D.

Table: Population growth in Africa from 1000 to 2000 A.D

Year (AD)	Population in million	Year (AD)	Population in million
1000	50	1930	164
1200	61	1940	191
1400	74	1950	222
1600	90	1960	278
1650	100	1965	320
1750	100	1975	413
1850	100	1985	536
1900	130	1995	720
1920	140	2000	800



As it can clearly be seen from the above table, Africa's population had been increasing very slowly until the early 1950s. The 1000 A.D. population of Africa had taken more than 650 years to double itself. Another nearly 300 years were passed until the 1650 population doubles again. Since the 1950's the continent's population began to grow faster and it took only about 25 years to double itself.

Another population doubling has still been experienced in the last 25 years of the 20th century between 1975 and 2000. Since 2000 till 2009, another 200 million people were added making Africa's population

nearly a billion. The population growth rate of Africa is the fastest in the world. According to the 2009 world population data sheet, the continent had a rate of natural increase of 2.4%. The general growth rate, on the other hand, was 2.3%. This appears being the highest rate in the world and twice as fast as the world's average growth rate. In terms of the period that Africa's population needs to double itself, the continent has the world's shortest doubling time of about 29 years. Such a rapid population growth rate and short doubling time is the result of the very high fertility rate that the continent has. Although birth rate is declining in the continent, it is still high as compared to the rest of the world. The declining mortality in the continent is also the highest in the world. The difference between the two has produced a very high rate of population growth and short period of doubling time in Africa.

Rate of Natural Increase (RNI) is the difference between Birth Rate and Death Rate expressed in percentage. On the other hand, General Growth Rate is the difference between RNI and Net-migration Rate expressed in percentage. The doubling time of a population is calculated by dividing 70 by the growth rate of the population.

Table: Demographic characteristics of the world by continents

No	Region	Birth Rate/1000 population	Death Rate/1000 population	RNI (%)	Doubling Time in years
1	Asia	19	7	1.2	58.3
2	Africa	36	12	2.4	29.2
3	North America	14	8	0.6	116.7
4	Latin America/ Caribbean	20	6	1.4	50
5	Europe	11	11	0	-
	Oceania	18	7	1.1	63.6
7	World	20	8	1.2	58.3

Table: Demographic characteristics of Africa by major regions

No	Region	CBR	CDR	RNI	TFR	IMR	NMR
1	Sub-Saharan Africa	39	13	2.6	5.3	80	-1
2	Northern Africa	25	7	1.8	3.0	38	-1
3	Western Africa	40	14	2.6	5.5	80	-1
4	Eastern Africa	40	13	2.7	5.4	76	-1
5	Central Africa	42	14	2.8	6.1	95	0
6	Southern Africa	24	15	0.9	2.8	48	3
7	Africa	36	12	2.4	4.8	74	-1

C Distribution

What is population distribution? What are the factors that affect the distribution of population?

Which areas in Africa are most populous? Why?

Population distribution refers to the way people inhabit and occupy the earth's surface. As elsewhere in the rest of the world, Africa's population distribution is uneven. As a result, there are areas of high and low population density. Some parts of the continent, particularly the vast Sahara, have few permanent residents.

While, others rank among the world's most densely populated areas. The Nile Valley of Egypt, the Atlantic coastal area that stretches from Côte d'Ivoire to Cameroon, Rwanda, Burundi, and South Africa's province of KwaZulu-Natal are among the most densely populated areas in the continent. The continent's average crude population density during the mid of 2009 was about 33 people/km².

Certain human and physical factors are responsible for such spatial variation in the distribution of population in the continent. Among the physical factors some are climate, relief, availability of water, fertility of soil, distribution of minerals and the like. On the other hand, the human factors that determine the extent to which humans inhabit a certain place include economic conditions, political situations and other social factors.



Figure: Africa density of population, 2008

However, there exists a significant regional variation in population density. As it can be seen from the following table, Eastern Africa has the continent's highest crude density of 49 p/km² followed by Western Africa with 48 p/km². Contrary to this, central Africa has the continent's lowest crude density of 19 p/km² followed by Southern Africa (22 p/km²) and Northern Africa (24 p/km²).

Table: Population density by major regions of Africa

No	Region	Crude Population Density/km ²
1	Sub-Saharan Africa	34
2	Northern Africa	24
3	Western Africa	48
4	Eastern Africa	49
5	Central Africa	19
6	Southern Africa	22
7	Africa	33

On country level, the most populous countries are Nigeria, Egypt, Ethiopia, the Democratic Republic of the Congo (DRC) and the Republic of South Africa (RSA). These countries all together account for about 43% of the continent's total population. In terms of population density, while some are heavily populated, others have very low population density. The following table gives you the ten most densely and most sparsely populated countries of Africa.

Table: The ten-top densely and sparsely populated countries of Africa

Ten Top Densely Populated Countries		Ten Top Sparsely Populated Countries	
Country	Density/km ²	Country	Density/km ²
Mauritius	625	Western Sahara	2
Mayotte	503	Mauritania	3
Seychelles	375	Botswana	3
Reunion	324	Namibia	3
Comoros	302	Libya	4
Burundi	298	Gabon	6
Rwanda	191	Central African Republic	7
Sao tome and Principe	169	Chad	8
Gambia	165	Mali	10
Nigeria	142	Congo	11

Determinants of Population Change in Africa

The three components that determine population change are birth rate, death rate and migration.

Population change in a given country is affected by:

- The difference between deaths and births also known as natural change
- The balance between immigration (coming in) and emigration (going out) also known as net migration.



In Africa, the current population change (growth) is a function of fertility, mortality and migration. There is frequent migration of people within a country.

However, these days there are not significant migrant flows from country to country or from region to region due to lack of pulling factors. Since most African countries are in their lower stage of development, there are little or no opportunities to accommodate a huge influx of people from another country. Africa's migration pattern is highly related to political instabilities and natural catastrophes.

Fertility Patterns in Africa

Fertility refers to the occurrence of birth in the human population. It is a natural positive factor that tends to increase the human population size. Africa's fertility rate is the highest in the world. Birth rates are high in Africa because:

- people have less access to contraceptives;
- people are backward and poverty stricken;
- women are of low status;
- many children are needed to work on the land;
- children are considered as symbol of virility;
- people are of low educational background; and
- Many religions encourage large families.

According to the UN World population prospects of the year 2009, the average birth rate for Africa is estimated at 36/1000. This is the highest even for the standard of developing countries, which is (22/1000).

Table: Patterns of rates and life expectancy by major regions in Africa (2009)

Region	Crude Birth Rate per 1000 population	Crude Death Rate Per 1000 Population	Rate of Natural Increase (%)	Infant mortality rate per 1000 live Births	Life expectancy at birth (years)
World	20	8	1.2	46	67
Developed	12	10	0.2	6	74
Developing	22	8	1.4	50	65
Africa	36	12	2.4	74	53
Northern Africa	25	7	1.8	38	67
Western Africa	40	14	2.6	80	50
Eastern Africa	40	13	2.7	76	50
Central Africa	42	14	2.8	95	49
Southern Africa	24	15	0.9	48	50

Although fertility is high in Africa, it still varies from country to country or region to region. Currently African countries having birth rates below 30/1000 include:

⇒ Mauritius 14/1000	⇒ Seychelles 18/1000
⇒ Tunisia 17/1000	⇒ Reunion 18/1000
⇒ Morocco 21/1000	⇒ Libya 24/1000
⇒ South Africa 23/1000	⇒ Namibia 25/1000
⇒ Botswana 24/1000	⇒ Algeria 23/1000
⇒ Egypt 25/1000	⇒ Lesotho 27/1000

Other African countries comprise rates ranging from 31/1000 in Swaziland to 50/1000 in Guinea Bissau and Nigeria.

N.B. Ethiopia's CBR was about 39/1000 in 2009.

- ❖ Birth rate is defined as the number of live births per 1000 population in a year. For example, when we say the birth rate of Africa is 36/1000, we mean that 36 births will occur for every 1000 people, on the average, every other year.
- ❖ Death rate is the number of deaths per 1000 population in a year. For example, Africa's death rate is 12/1000. This is to say that 12 deaths will occur for every 1000 people on the average every other year.

The difference between birth rate and death rate is conceived as a difference influenced by natural change. The natural change for Africa can, therefore, be computed as:

36/1000 – 12/1000 or (36/1000 (birth rate) – 12/1000 (death rate))

$$= \frac{24}{1000} \times 100 = 2.4\%$$

2.4% is the natural increase for Africa

There is an inverse relationship between the level of economic development and the fertility rate. In Africa, the regions that have a relatively better economic progress are likely to have low rate of fertility.

Mortality Patterns in Africa

What is mortality? How does it affect the population of Africa? What are the factors that affect mortality in Africa?

Simply defined, mortality refers to the occurrence of death in the human population. Africa's mortality rate is the highest in regions. The continent had a crude death rate of 12 deaths per 1000 population.

When regional comparison is made, death rate ranges from 7/1000 in Northern Africa to 15/1000 in Southern Africa. In terms of infant mortality rate, which is the death of infants under one year per 1000 live births, the continent still has the world's largest rate of 74 deaths per 1000 live births. Infant mortality rates also vary from region to region. Accordingly, Central Africa has the highest rate of 95/1000 followed by Western Africa 80/1000 and Eastern Africa 76/1000 live births. The lowest Infant Mortality rate is in Northern Africa which only 38/1000.

Africa's high death rate is attributed to the following major factors:

- ✓ Low standard of living
- ✓ Low access to health facilities
- ✓ Poor sanitary practices
- ✓ Civil war and political instability
- ✓ Wide spread of famine caused by recurrent drought
- ✓ Poor nutrition, and
- ✓ High incidence of disease and infections

Africa's death rate is declining since world war II due to development in medical technology, and sanitary practices, and the discovery of medicines and vaccines for tropical diseases. However, the continent's crude death rate remains being the highest in the world.

- ❖ Infant mortality rate is generally regarded as a prime indicator of socioeconomic development. Infant mortality has declined from 138/1000 in 1950 to 88/1000 in 1975 – 80; and now, it is down to 74/1000.

There is disparity in life expectancy between rich and poor countries. But this has shown a significant converge over the last 50 years despite widening wealth gap between them.

- ❖ Life expectancy at birth: is the average number of years a newborn infant can expect to live under current mortality levels.

Table: Level of life expectancy for selected developed and developing countries

Country	Years	Country	Years
Japan	83	Swaziland	46
Australia	81	Ethiopia	53
France	81	Mozambique	43
Italy	82	Angola	46
Sweden	81	Malawi	46
Switzerland	82	Zambia	43

The current birth and death rates of Africa reflect a very young (and economically dependent) population of low life expectancy. Almost all sub-Saharan countries have a population of under 15 years of age rated in percent at more than 40. This has resulted in high dependency ratio in the continent. The average life expectancy for the whole continent is 54 years. (Table 4.9)

Table: Life expectancy at the regional level

Region	Life expectancy
World	68
Developed	77
Developing	55
Africa	54
Sub-Saharan Africa	50
Northern Africa	69
Western Africa	51
Eastern Africa	49
Central Africa	51
Southern Africa	49

Africa's life expectancies are the lowest in the world. There is though, a wide disparity of this fact from regions to regions within Africa ranging from 69 years for Northern Africa to 49 years for Eastern and Western Africa.

Migration

What is migration? Why do people migrate?

Migration is a socio-cultural components of population change. It has both negative and positive impacts based on its net effect. A negative net migration rate tends to reduce the population of region. On the other hand, a positive net migration rate tends to increase population size.

The third component of population change is migration. It is an important control of population change next to fertility and mortality. Migration is the movement of people from their home place to another one. The most important aspects in the study of migration are its motives. The motives leading to migration in Africa include:

- ✓ prevalence of diverse and wide range of climatic zones
- ✓ presence of diverse and multitude natural resources
- ✓ coincidence of the continent's large size
- ✓ pervasiveness of varying economic activities ranging from agriculture to trade: and existence of divergent level of economic and social development.

Characteristics of African Population

1 What does a characteristic of population mean?

2 What are the major characteristics of African population?

The structure or composition of African population falls under two categories classified thus on grounds of **biological and cultural characteristics**. **Biological characteristics** include race, ethnicity, sex and age. The **cultural characteristics** comprise education, health, nutrition, religion, occupation, and the like.

Biological Characteristics

Age structure

How do you describe the age composition of Africa's population?

Africa is a continent which one can deduce, could be characterized by young population. This can be easily identified from the median age. If the median age is low, it implies that the population is young; for example, Africa's median age is 18 years. In contrast, if the median age is high, the population is likely to be adult and old age group; for example the median age for Europe is 37 years. Low median age reflects high fertility and high population growth rates whereas high median age prevails in a population where population growth rate is low and slow.

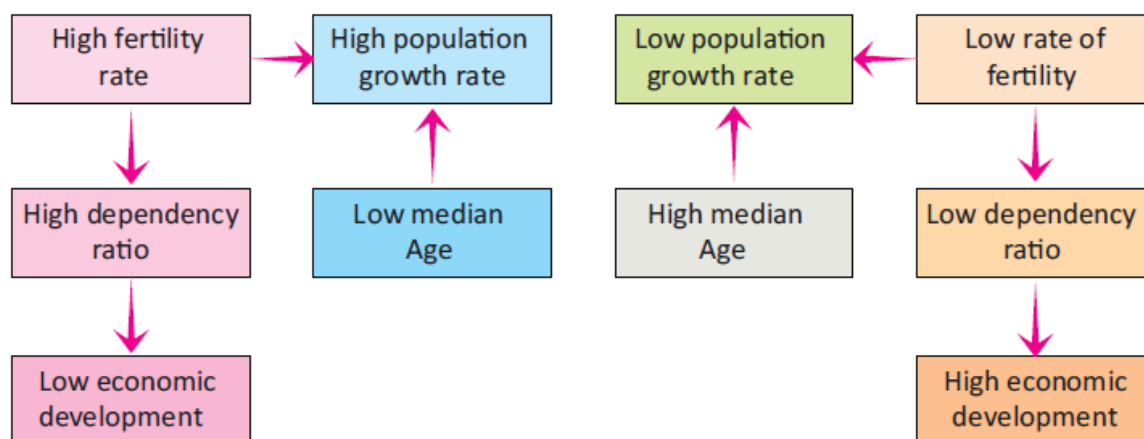


Figure: The adverse consequence of low median age on economic development

Table: Africa's Population by age and major regions

No	Region	0-14	15-64	65+	ADR
1	Sub-Saharan Africa	43	54	3	85.19
2	Northern Africa	33	62	5	61.29
3	Western Africa	44	53	3	88.68
4	Eastern Africa	44	53	3	88.68
5	Central Africa	45	52	3	92.31
6	Southern Africa	33	62	5	61.29
7	Africa	41	56	3	78.57

As it is shown in Table, Africa is the only continent which comprises high young population, but very low old age population. In developing continents like Africa, the level of fertility is very high; and so the rate of population growth is high, too. This is readily attributed to their predominantly youthful age structure.

As high fertility persists, the pressure on scarce resources for development increases. The demand for food, education, health facilities, employment opportunities, housing and other services also increases. In order to maximize these needs so as to meet the demand of the growing population, physical and institutional infrastructures have to be increased abundantly. Persisting high rate of population growth constantly dwarfs whatever is achieved in the economic sector. Resources which could be used for development purposes, could be shifted to the provision of basic supply for the additional members.

The Demographic structure of a given country (region) is best illustrated by the use of population pyramids (like the one in Figure 4.4). Population pyramids are used to portray the proportion of males

and females in different age groups. In Figure below, each bar represents a five year age group apart from the upper most bar which illustrates the population above 85 years. The male population is shown on the left of the vertical axis; and that of the females, on the right.

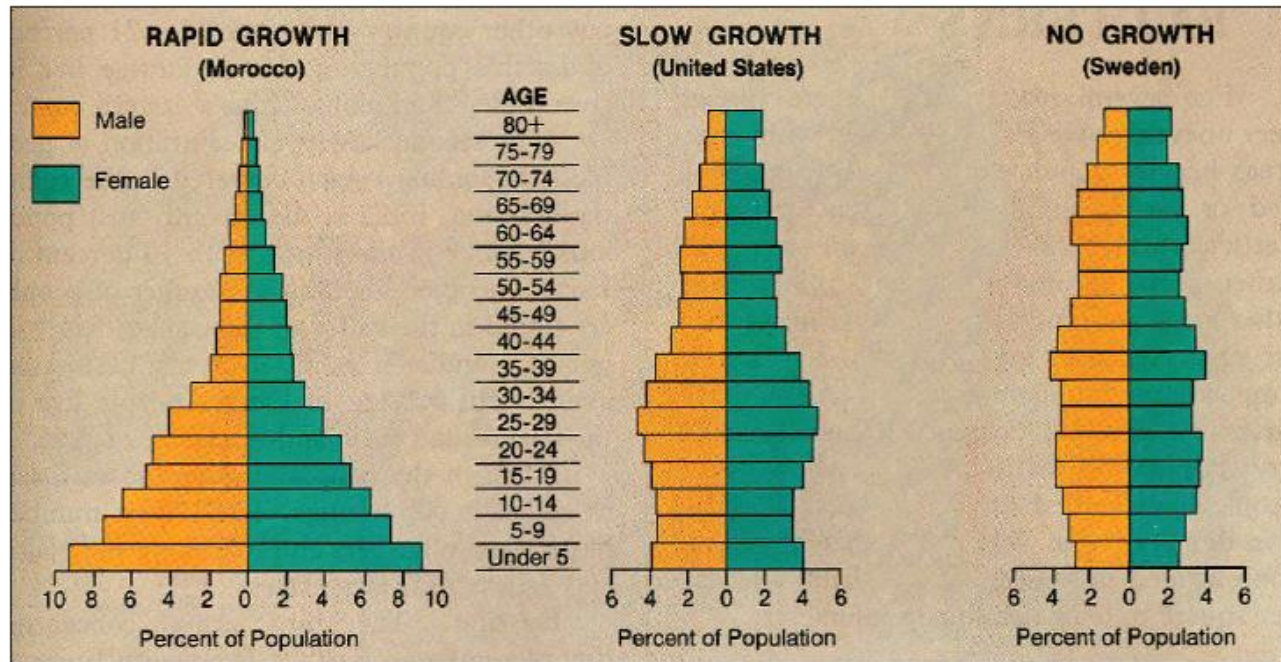


Figure: Population Pyramid

As it is portrayed in the figure above, the population pyramid for Sweden:

- ✓ is a narrow pyramid indicating equal numbers in each age group.
- ✓ reflects a low birth rate and a low death rate indicating a steady and static population growth.
- ✓ illustrates that more females live over 75 years than males do.
- ✓ gives a picture of the fact that a large proportion of the population is in the pre- and post-reproductive age groups and that a small number is in the 15 – 64 age group, which is the one that produces children and most of the national wealth.
- ✓ This can be shown in the dependency ratio expressed as:

Economically dependent	Children (0 – 14) and elderly (65+)
Economically active	Working age (15 – 64)

Let us take hypothetical countries called “X” and “Y” to calculate dependency ratio.

Country	Non-economically active (0 -14)	Economically active (15 – 64)	Old age populations
X	13,387,000	31,616,000	7,307,000
Y	15,431,000	20,200,000	2,145,000

i The dependency ratio for country x is:

The number of dependent population under-15 years = 13,387,000

Number of old age (dependent) population > 64 = 7,307,000

Number of independents 15 – 64 = 31,616,000

$$\text{Age dependency ratio} = \frac{13,387,000 + 7,307,000}{31,616,000} \times 100$$

Therefore, the age dependency ratio of country “X” is 65.45%

This simply means, for every 100 people of working age, there are 65.45 people who depend on them.

ii Country Y’s dependency ratio is:

$$\frac{15,431,000 + 2,145,000}{20,200,000} \times 100$$

The dependency ratio is, therefore, 87%.

The above dependency ratio is explained as for every 100 economically active people. The case country “Y”, for instance, can be stated as: there are 87 persons that depend up on 100 persons. This dependency ratio is the highest in the world.

When we see the age dependency ratio, of Africa by its major region, central Africa has the highest ADR of 92.31% followed by western and eastern Africa each of which having (88.68%) and northern and southern Africa having 61.29% each. The overall ADR of Africa was 78.57% and that of sub Saharan Africa was 85.19% in 2009.

The following are among the major characteristics of the African Population

- ✓ High fertility and mortality rates;
- ✓ Generally young population;
- ✓ Triangular population pyramid indicating the prevalence of high fertility and mortality rates;
- ✓ High age dependency ratio and youth dependency ratio and low old dependency ratio;

- ✓ Low life expectancy; and
- ✓ High population growth rate.

Sex Structure

Sex ratio is an important demographic characteristics. It refers to the proportion of males to females in the overall population of a given area. Sex ratio is expressed in terms of the number of males for every 100 females.

$$\text{Sex Ratio} = \frac{\text{Male Population}}{\text{Female Population}} \times 100$$

Sex ratio is an important indicator of the economy of a given country and is useful for regional analysis. The proportion of the two sexes in the population of a region has impact on other demographic elements such as marriage rate, occupational structure, growth, etc.

With regard to sex ratio, Sub-Saharan Africa sex ratio is below the world's average which is less than 98%. This means there are 98 males per 100 females, which is equitably balanced. Most Sub-Saharan African countries, however, have sex ratio of far less than 100. That is to say, there are greater number of women than men. Such an event is associated with a huge loss of men population due to war and migration. In some countries of the region, though, male births consistently exceed female births due to combined biological and social reasons. Some among such countries are equatorial Guinea, Uganda, Gambia and Kenya.

The context of high sex ratio in these countries is attributed to the prevalence of a large number of immigrants. The fact that these countries exceptionally comprise high sex ratio for reasons of immigration indicates that migration is age and sex selective. If there is a large influx of population to a given country, then that country will have high sex ratio. High sex ratio means a large proportion of active population age group will be added to that country. In countries where there is a strong rural to urban migration, the population structure of the areas affected can be different.

4.4 Population of Ethiopia

4.4.1 Population Size and Growth Rate

The population of Ethiopia was about 12 million toward the beginning of the 20th century (1900). It had an annual growth rate of 0.2%. Had this growth continued, the population would have doubled in 346 years. The rate of growth of the population was relatively very slow until 1920. But, after 1920, the population of Ethiopia started growing very fast, and it took only 60 years to double. The 1950's was another remarkable period. By 1960 the growth rate quadrupled, doubling the size of the population to 23.5 million. Then, between 1960 and 1990, i.e., within thirty years, the population once again doubled. During that period, its growth rate tripled, compared with the rate of 1920. The smallest doubling period recorded was in 1996, after which it started increasing again.

Ethiopia conducted its first census in May 1984. So far, the governments have conducted three successive population and housing censuses. The first census report showed that the population of Ethiopia (together with Eritrea) was 42.2 million. When we compare the 2007 census results with those of 1984 and 1994, we see that the population of the country increased by more than 31 and 21 million over 23 and 13 years, respectively. The population growth rate increased from 2.9% in 1984-1985 to 3.0% in 1994-1995.

Table: Estimates of population growth in Ethiopia

Year	Population (million)	Annual Growth Rate (%)	Population Doubling Time (year)
1900	11.8	0.2	346
1910	12.1	0.5	139
1920	12.9	1.0	69
1930	14.4	1.2	58
1940	16.2	1.5	46
1950	19.2	2.0	38
1960	23.5	2.2	32
1970	29.5	2.3	30
1980	37.7	2.8	25
1990	51.2	3.0	24
1993	53.4	3.1	23
1996	60.0	3.2	22
1999	61.6	2.9	24
2000	63.4	2.8	25
2010	79.0	2.6	27

Since 1999, the growth rate has decreased at an average rate of about 0.2% per year (see the above Table). By 2010, the rate had dropped to about 2.6% per year. With the current growth rate of 2.6%, the country's population is projected to grow by 1.9 million annually, and to attain a size of about 79

million by 2010, 94.5 million in 2015, about 118 million by 2025. This trend would make Ethiopia the most populous country in Africa, next to Nigeria. This rapid growth seriously strains socio-economic development.

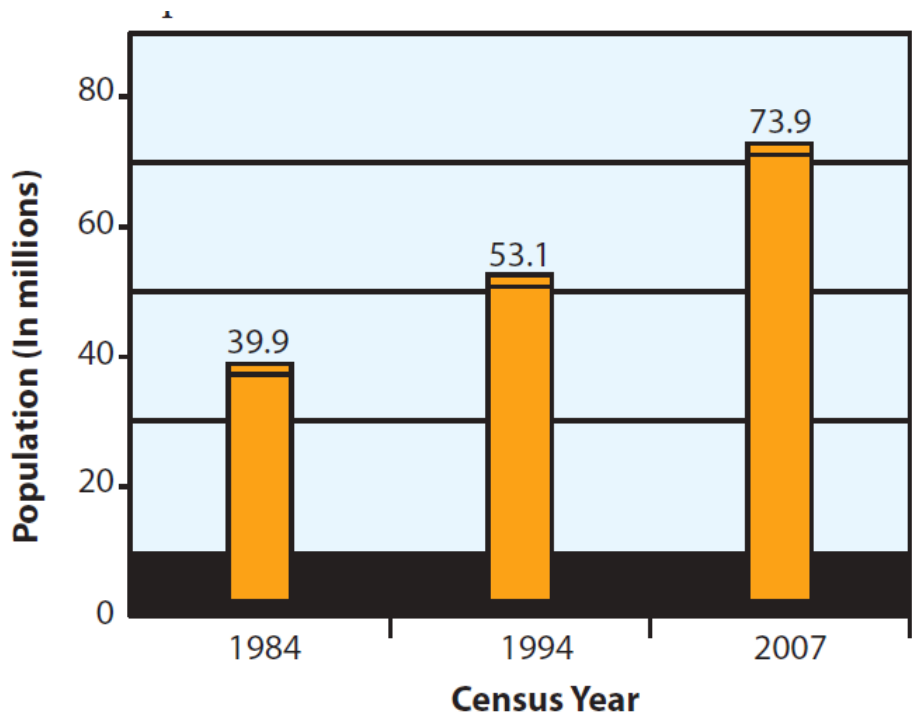


Figure: Population size of Ethiopia (in million) 1984 – 2007 (CSA, 2007)

Spatial Distribution of Population

Spatial distribution of population is the pattern created as a result of human occupation of land surface for settlement. The pattern of Ethiopian population distribution is markedly uneven. For example, we have very high population densities in the highlands. On the other hand, the peripheral lowlands of the country, which account for more than 50% of the total area of the country, are very sparsely populated. There are variations in the distribution at the national level as well as at lower levels for example, between and within regions, zones, weredas and kebele administration units. The following Table shows the country’s population distribution at the regional level.

Table : Ethiopia's population distribution, by region: 1994 and 2007

Region	1994		2007	
	Number	%	Number	%
Tigray	3,136,267	5.9	4,314,456	5.8
Afar	1,060,573	2.0	1,411,092	1.9
Amhara	13,834,297	25.0	17,214,056	23.3
Oromiya	18,732,525	35.0	27,158,471	36.7
Somalia	3,198,514	6.0	4,439,147	6.0
Benishangul Gumuz	460,459	0.9	670,847	0.9
SNNP	10,377,028	19.4	15,042,531	20.4
Gambella	181,862	0.3	306,916	0.4
Harari	131,139	0.2	83,344	0.2
Addis Ababa	2,112,737	4.0	2,738,248	3.7
Dire Dawa	251,864	0.5	342,827	0.5
Special Enumeration	Not available	Not available	96,570	0.1
Country total	53,477,265	100.0	73,918,505	100.0

More than 80% of the population lived in three regions – Oromiya, Amhara and Southern Nations, Nationalities and Peoples Region (SNNPR). Their populations accounted for 36.7%, 23.3% and 20.4% of Ethiopia's total population, respectively. According to the 1984 Ethiopian census, the nation's crude population density was 34 persons per square kilometer. By 1994, 10 years later, it had increased to 48.3. By 2007, the crude population density was almost double that of 1984 – about 67 persons per square kilometer.

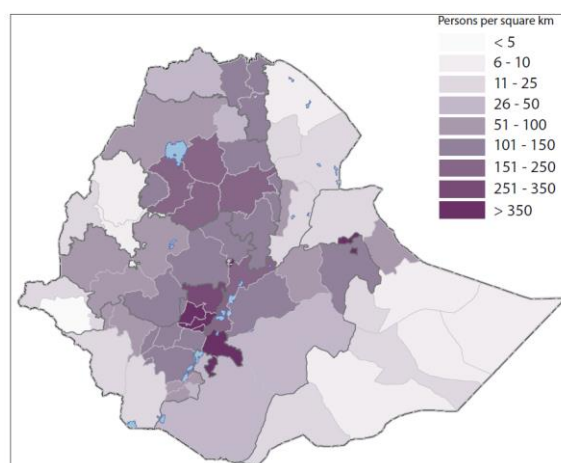


Figure: Population density of Ethiopia

The above Table presents Ethiopia's ten most densely and sparsely populated zones, as of 2009. Population densities exceeded 150 persons per square kilometre. All of the zones listed in the first half of the table, except for the last four shown in the table, are in the SNNPR and are *enset*-growing areas. Of the SNNPR zones, the most densely populated are Gedeo Kembata and Tembaro, and Sidama. Their densities exceed 400 persons per square kilometre. In contrast, there are places with very low population densities, below 25 persons per square kilometre. Almost all very low-population areas are found in the peripheral lowlands of Benishangul Gumuz, Gambella, Afar and Somali regions. Some of zones of very low-population densities include Agnewak, Metekel, Kemashi, Zone one (Afar), Etang, Zone two (Afar), Asossa zones, Warder zone, Korahe zone and Fik zone.

The following Table: Ethiopia's ten most densely and most sparsely populated areas at the zone level, 2009.

No.	The ten most densely populated areas		No.	The ten sparsely populated areas	
	Zone	Density (persons/km ²)		Zone	Density (persons/km ²)
1	Gedeo	772.6	1	Agnewak	5.7
2	Kembata Tembaro	535.8	2	Metekel	10.1
3	Sidama	482.5	3	Kemashi	11.5
4	Wolayita	386.0	4	Zone 1 (Afar)	14.8
5	Hadiya	368.1	5	Etang Special Zone	18.8
6	Gurage	231.0	6	Zone 2 (Afar)	20.6
7	West Arsi	183.7	7	Asossa	22.5
8	East Shewa	177.6	8	Warder	6 - 25
9	Jimma	175.5	9	Korahe	
10	East Hararge	167.2	10	Fik	

The spatial distribution of Ethiopia's population is uneven, characterized by high population density on the highlands while lowlands have very low population density. Such uneven distribution of population has been the result of ecological (physical) factors and human factors. More important physical factors include climate (mainly rainfall and temperature), soil, vegetation and relief (altitude). The human factors primarily consist of historical factors, for example peopling of the country and migration, development of infrastructure, types of economic activities and political factors related to government policies such as resettlement programs, land tenure system, villagization process, etc. All these factors individually or collectively have influenced human settlements favorably or adversely. They have either encouraged or discouraged settlement.

Table: Population distribution, related to altitude

Altitude (metre)	% of total area of Ethiopia	% of total population of Ethiopia (1984)	Density (persons/ km ²) (1984)
> 2600	5.8	10.4	85.0
1800 – 2600	31.8	67.1	157.5
1400 – 1800	28.1	11.5	68.14
1000 – 1400	13.4	8.5	69.14
< 1000	21.5	2.8	29.0
Total	100.0	100.0	6.2

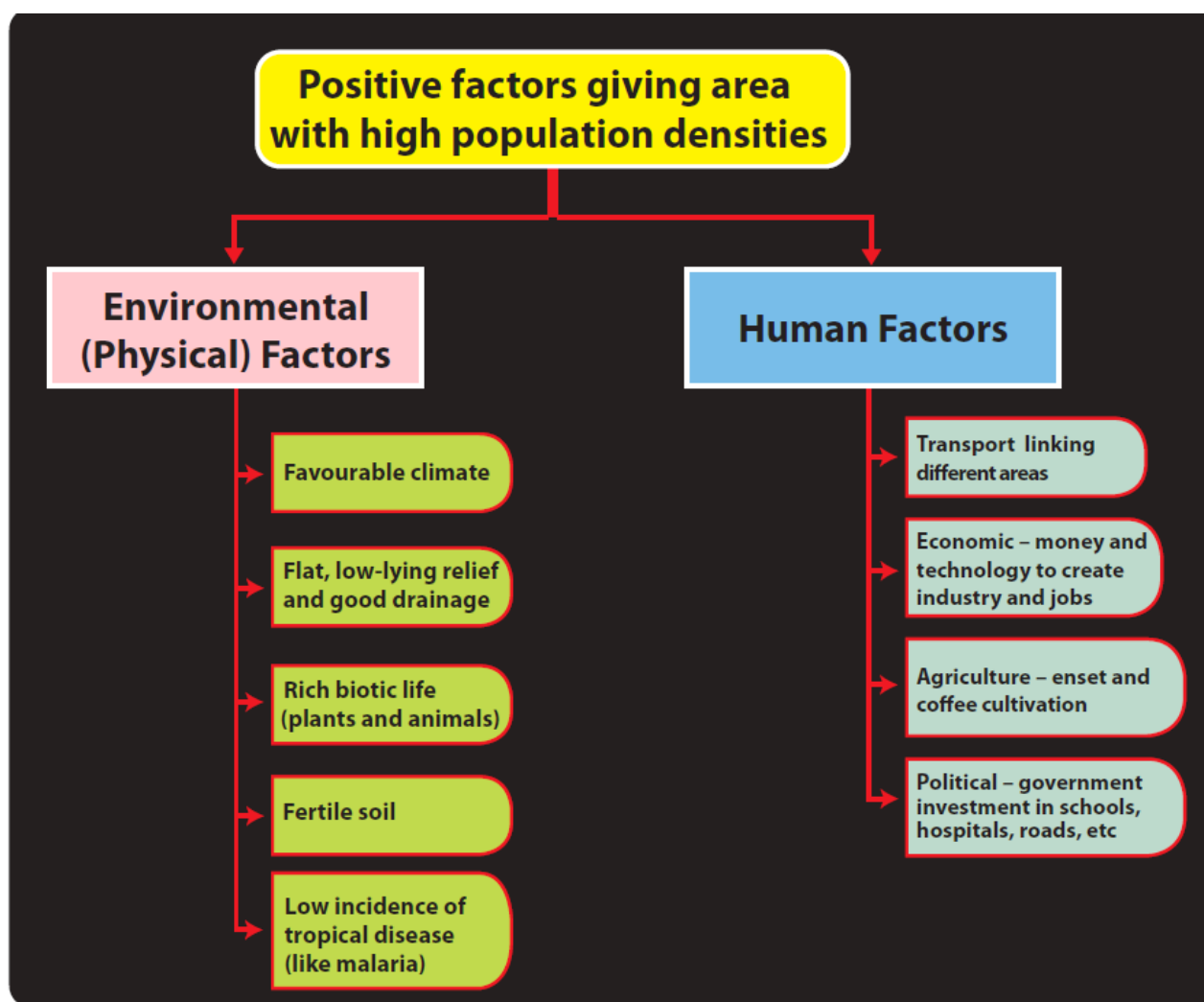


Figure: Factors that contributed to high population density in Ethiopia

4.4.2. Components of Population Change in Ethiopia

A. Fertility in Ethiopia

Fertility is one of the three principal components of population change that determine the size and structure of the population of a country. In its general sense, fertility refers to the occurrence of birth in a given country or region.

Before studying the fertility characteristics of the population of Ethiopia, become familiar with the following measures of fertility.

- **Crude Birth Rate (CBR):** this is the total number of births occurring in a given year, per 1000 population.
- **General Fertility Rate (GFR):** this one is the number of births occurring in a given year per 1000 women in the reproductive ages (i.e., women aged 15-49).
- **Age Specific Fertility Rate (ASFR):** this is the number of births that occur in a given year per woman in the reproductive ages, presented in five-year age groups.
- **Total Fertility Rate (TFR):** this is the number of children a woman may have produced by the end of her reproductive period, given the current ASFR.

Ethiopia's total fertility rate is one of the highest in the world. In 1984, TFR was 7.52 per woman. By 1994 it dropped to 6.74, and then to 5.4 in 2010. There is significant fertility variation between urban and rural areas. For example, in 1984, TFR in urban areas was 6.33, while it was 8.08 in rural areas. In 2010 these rates, respectively, dropped to 4.5 and 7.19. Fertility also varies from region to region. In 1994, Oromiya and the SNNPR recorded the highest TFRs, reaching as high as 7.26 and 7.16, respectively. In contrast to this, Addis Ababa, Dire Dawa, Harari and Gambella recorded TFRs of less than 4.3.

- ❖ The total fertility rate (TFR) for Ethiopia is 5.4 children per woman. This means that the total number of children an Ethiopian woman would have by the end of her child-bearing period (reproductive age) is about 5.4 children.

In Ethiopia, there are also substantial differentials in fertility among regions, ranging from a low TFR of 1.4 in Addis Ababa to a high of 6.2 in Oromiya. With the exception of Somali and SNNP, fertility levels in the other regions are lower than the national average (5.4).

Table: Measures of fertility in Ethiopia, by major regions, 1994

Region	Total		Urban		Rural	
	TFR	CBR	TFR	CBR	TFR	CBR
Ethiopia	6.74	44.34	4.50	34.88	7.19	45.80
Tigray	6.95	43.89	5.38	39.33	7.28	45.11
Afar	6.39	36.21	4.47	39.51	6.61	36.05
Amhara	6.76	43.86	5.40	43.40	6.94	43.95
Oromiya	7.26	45.63	6.10	44.14	7.42	45.83
Somali	6.73	36.38	5.84	35.04	6.89	36.67
Benshangul and Gumuz	6.46	45.63	5.46	44.70	6.56	44.61
SNNPR	7.16	49.95	6.42	48.06	7.22	49.72
Gambella	4.72	36.24	4.19	38.14	4.82	35.10
Harari	4.68	36.50	3.40	26.63	6.98	51.48
Addis Ababa	2.14	18.78	2.09	16.37	7.40	41.10
Dire Dawa	4.95	37.82	4.28	35.21	6.89	43.37

Study the following figure, which shows the level of fertility (TFR and CBR) for Ethiopia as a whole, and for its urban and rural areas.



Figure: Levels of Fertility in Ethiopia (TFR and CBR) (2005)

- ❖ In general, fertility rates are high in Ethiopia due to the following major factors.
- ✚ Low levels of family planning practices, due to lack of awareness and religious beliefs;
- ✚ Early marriage, particularly of females;
- ✚ Perception of high social and economic value of children;
- ✚ Low social status of women; and

- ✚ Relatively high infant and child mortality (death) rates, which leads parents to produce larger number of offspring, in compensation.

B. Mortality in Ethiopia:

Mortality is the measure of frequency of deaths in a population. Ethiopia has one of the highest levels of mortality in the world (see the following Table). The main causes of such high mortality are the effects of poverty, low living standards and poor access to health services.

Before studying the mortality characteristics of Ethiopia's population, become familiar with the following measures of mortality.

- **Crude Death Rate (CDR):** this is the total number of deaths occurring in a given year, per 1000 people.
- **Infant Mortality Rate (IMR):** this is the number of deaths in a year among infants under one year of age, per 1000 live births.
- **Child Mortality Rate (CMR):** this is the number of deaths in a year among children between one and five years of age, per 1000 children between the same ages.
- **Under-Five Mortality Rate:** this is the number of deaths in a year among infants and children between birth and five years of age, per 1000 live births.
- **Maternal Mortality rate (MMR):** this is the annual number of maternal deaths occurring during pregnancy, child birth, or within two months after the birth or termination of pregnancy, per 1000 women between 15 and 49 years of age (reproductive age).
- **Adult Mortality Rate (AMR):** this is the number of deaths in a year among adults between 15 and 49 years of age, per 1000 adults in the same age group.
- **Age-Specific Death Rate (ASDR):** this is the number of deaths that occur in a given year per 1000 of the population in five-year age group.
- **Life Expectancy:** this is the average number of years a newborn baby is expected to live if he/she is exposed throughout life to the prevailing pattern of age-specific death rates.

Infant mortality rate (IMR) is the number of deaths among infants between the moment of birth and the first birthday. Child mortality rate (CMR), on the other hand, is the number of deaths among children between the exact ages of one and five, (i.e. From the first birthday, up to but not including the sixth birthday). The under-five mortality rate, however, includes both IMR and CMR, as it refers to the number of deaths of infants and children between birth and the fifth birthday. One of the targets of

Ethiopia's millennium development goal (MDG) is a two-thirds reduction in infant and child mortality by 2015, to be achieved by way of:

- upgrading the proportion of births attended by skilled health personnel.
- increasing immunization against the six vaccine-preventable diseases.
- upgrading the status of women through education and enhancing their participation in the labor force.

Other significant factors include civil war, and recurrent drought and famine. Deaths of children under the age five accounts for over half of all deaths in the country. The main causes of children death are measles, malaria, malnutrition, and acute respiratory infections such as pneumonia and influenza. However, despite all of the current factors that cause deaths in Ethiopia, mortality rates have decreased because of improved medical technology and the population's improved access to health services. During the last two decades, crude death rate dropped from 20 per 1000 population in 1970 to 16.4 per 1000 in 1990 and then to 12 per 1000 in 2010. Similarly, the infant mortality rate declined from 153 deaths per 1000 live births in 1970 to 110 in 1990 and then to 77 in 2010 (see the following Table).

Unfortunately, however, many people now fear that such decreasing trend (and that of increasing life expectancy) will soon be reversed due to the increasing prevalence of HIV/AIDS in Ethiopia's population. For 1999, estimates put the number of infected people in the country at 3 million. Now, ten years later, some experts are predicting increases in mortality rates and decreases in life expectancy.

Table: Crude death rate, infant mortality rate and life expectancy in Ethiopia, 1970 – 2010

Year of Survey	Crude death rate			Infant mortality rate			Life expectancy	
	Rural	Urban	Total	Rural	Urban	Total	Rural	Urban
1970	20.0	16.9	20.0	155	134	153	43.8	45.0
1981	18.1	15.7	17.9	141	117	139	46.5	46.8
1984	17.2	14.17	17.1	112	94	110	46.0	58.1
1990	–	–	–	111.6	94.2	110.1	47.4	49.4
1998	17.8	14.0	17.0	109.9	92.0	109.0	48.0	50.9
2010*			12			77	–	51

4.4.3. Population Structure of Ethiopia

Age Structure: Age structure is the distribution of a given population into age groups. This structure becomes clear after we group all of the people in that population by age. In other words, age structure is the pattern that results from the distribution of members of a population into different age categories.

Two important statistical tools for understanding age structure are

- ✓ Age groups and population pyramids

Age Groups: Although we can use different sets of numbers to define age groups, the most widely used age groups are the five-year age groups 0-4, 5-9, 10-14, ..., 60-64, 65+ and broad age groups 0-14, 15-64, 65+. In the broad age groups, age groups **0-14**, **15-64** and **65+** are known, respectively, as **young age** (the young dependent population), **working age** (the economically active population) and **old age** (the elderly dependent population). The following Table presents Ethiopia's population age distribution. As you can see from the recent population pyramid of Ethiopia shown in Figure 3.6a and from the data in the following Table, Ethiopia's population is predominantly young. It has persistently remained so, at least since 1970. On average, about 45% of the population is young and economically dependent. Elderly people account for about 3% of the total. This would mean only 52% of the population is economically active and supports both itself and the rest of the population. Ethiopia has one of the largest non-productive populations in the world. This is a direct result of the country's high fertility rate.

Table: Age structure (composition) of the Ethiopian population

Year	Population in millions	Percentage distribution of population by age		
		Under 15	15 – 64	65 and over
1970	-	45.3	51.5	3.2
1981	-	45.5	51.0	3.5
1984	41.1	48.1	47.1	4.8
1990	-	50.0	43.8	6.2
1994	53.4	45.4	51.4	3.2
1999	-	43.8	53.1	3.1
2007	73.9	45.0	51.8	3.2
2010*	79.0	44.0	53.0	3.0

With such a large dependent young population, government expenditure for education, health, shelter, food and other basic social services is very high. The only way that Ethiopia can release this large amount of budgetary resources for more economically productive investments is by reducing its dependency ratio.

- ❖ Fast population growth, like that of Ethiopia and many other developing countries, produces a youthful population dominated by children who are economically dependent. This age-structure has the potential to increase the momentum of population growth as those numerous children grow up and enter their reproductive years and have multiple children of their own. This demographic instability of large, highly dependent and consuming, rather than effectively producing, populations cripples the socio-economic development process and can impede sectorial planning. Moreover, unemployment and under employment tend to prevail due to national economic issues.

Population Pyramid: It is the graphic representation of the age distribution of a given population by sex. In countries where birth rates are high and death rates are also high, the population pyramid has the form of a triangle. This pattern is typical of the population of developing countries like Ethiopia, in which many children are born, but few reach old age. In contrast, in the developed countries, with their lower birth rates and fewer people dying young, the population pyramid is more rectangular, narrowing only near its top. In these countries, both birth and death rates decline, with a result showing the number of people in each group at equivalent state.

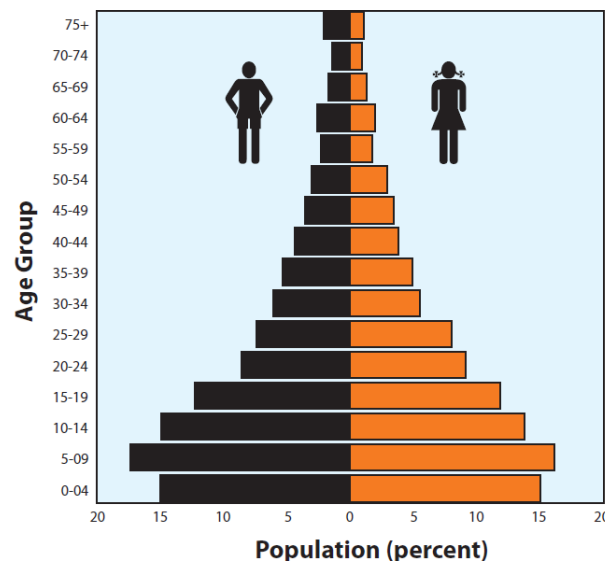


Figure: Population Pyramid of Ethiopia

In general, population pyramids of developing countries like Ethiopia have very broad bases, showing the dominance of the young-age population. These pyramids become increasingly narrower towards the top, advancing through the age groups, showing that the percentage of the population becomes less and less in the upper age groups (65-69, 70-74, etc.).

- ✓ Age structure/composition is one of the most important demographic characteristics of a population.
- ✓ Age information is often used to understand the sizes of school-age, labor-force, elderly, and other populations.
- ✓ A population pyramid is usually employed to show the age distribution of a given population by sex.
- ✓ The population pyramid of Ethiopia has a broad base that narrows towards the top as age increases. This shape is typical of a population with high fertility rate.

As shown in the above Table, Ethiopia's young-age (0-14 years) dependent population declined from 49.8 percent in 1984 to 45.0 percent in 2007 but was still very large. In contrast, Ethiopia's old-age (65+ years) dependent population was very small - only 3.2% in the 1994 and 2007 Census. Although the proportion of the population of working-age group (15-64) has shown modest increases over the course of time of the three census, it accounts for only a little more than half of the total population. The population pyramid for Ethiopia demonstrates that the country's population as a whole has a high preponderance of young population, with a median age of not more than 18 years. This is a typical feature of a rapidly growing population. The high percentage for the young-age group in Ethiopia is the result of a high birth rate, while the small percentage of the old-age group reflects a high mortality rate, which accounts for the low life expectancy of the population of the country.

The Age Dependency Ratio (ADR) is the relationship between the working or economically active population and the non-working population. The Age Dependency Ratio (ADR) is used to show the magnitude of this dependency in a given population. This means that the dependency burden, represented by the non-working population in the young and old age groups, on the working-age population can be shown by the age dependency ratio. The formula for calculating the age dependency ratio (ADR) is:

$$ADR = \frac{(\% \text{ of population aged } 0 - 14) + (\% \text{ of population aged } 65+)}{\% \text{ of population aged } 15 - 64} \times 100$$

The negative implications of heavy youth dependency in Ethiopia can be summarized as follows.

- Even higher levels of fertility are likely to be reached when the youth group reaches reproductive age.
- The capacity to save is highly constrained at both the household and the national level.
- Limited national resources must be diverted from investment and other developmental activities in order to use those resources to provide services and meet the young peoples' basic needs, such as food, housing, education, health care, etc.
- The demand for the nation's employment opportunities will increase as the youth group reaches working age; this results in a further high levels of unemployment.

In general, knowing about the age structure of a country's population helps its policy-makers and socio-economic planners to predict for the future what sorts of goods and services might be needed for how many people in which age group. As we have said, a large proportion of children means growing demands for schools; a large number of people entering their child-bearing years signals probable population growth; an increasing number of elderly people may strain pension plans and health services.

The age structure of a population also affects its growth rates. A population that has been growing rapidly will keep growing even when current birth rates slow down to replacement levels. This is because, when the large child population reaches its reproductive years, it will reproduce offsprings, unless that reproduction is kept at replacement level. In fact, reproduction will create an even larger new youth population. Thus, because the same thing is likely to happen again when that new generation reaches its reproductive years, the danger of a rising population spiral becomes higher and higher. For example, in Ethiopia, even if there were slight declines in the average number of children that each woman has, (Total Fertility Rate) in future years, these children will reproduce, producing more individuals than their own generation contains. Thus, the population of the country will continue to expand.

Sex Structure

What is the importance of studying sex structure?

Sex is one of the basic demographic characteristics of a population. **Sex structure** is very important for demographic analysis because it provides useful information about reproductive potential, human resources, and so on. The sex structure of a population is shown by its sex ratio, which is the ratio of male population to female population. Sex ratio is usually expressed as the number of males per 100

females in a population. A ratio greater than 100 shows a greater number – called an excess – of males than females. The formula for calculating a sex ratio is:

$$\text{Sex Ratio} = \frac{\text{Male Population}}{\text{Female Population}} \times 100$$

The sex ratio of Ethiopia’s population increased from 99.4 in 1984 to 101.3 in 1994; and to 101.9, in 2007. The low sex ratio during the 1980s appears to be due to political unrest and civil war. During those years, a large number of the male population was killed or was forced to migrate to other countries to escape being killed or forcibly conscripted into the military.

Table: Population size, by sex and by region, in Ethiopia (2007)

Region	Sex			
	Male		Female	
	Number	%	Number	%
Country – total	37,296,657	50.5	36,621,848	49.5
Tigray	2,124,853	49.2	2,189,603	50.8
Afar	786,338	55.7	624,754	44.3
Amhara	8,636,875	50.2	8,577,181	49.8
Oromiya	13,676,159	50.4	13,482,312	49.6
Somali	2,468,784	55.6	1,970,363	44.4
Benshangul – Gumuz	340,378	50.7	330,469	49.3
SNNP	7,482,051	49.7	7,560,480	50.3
Gambella	159,679	52.0	147,237	48.0
Harari	92,258	50.3	91,086	49.7
Addis Ababa	1,304,518	47.6	1,433,730	52.4
Dire Dawa	171,930	50.2	170,897	49.8
Special Enumerations	52,834	54.7	43,736	45.3

In 2007, Ethiopia had an almost balanced (one-to-one) sex ratio, both in rural and urban areas. The sex ratio for the whole country was 100.8. This meant that in the 2007 estimate, males slightly outnumbered females. Earlier, 1999, estimates indicated slightly different sex ratios. In rural areas, males slightly outnumbered females – the sex ratio was 101.2. In urban areas, females slightly outnumbered males – the sex ratio was 98.5.

Regional sex-ratio data generally supported the preceding data for the nation as a whole, showing slight variations. Male populations exceeded female populations in all regions except in Addis

, Tigray and the SNNPR. In Addis Ababa, whose population is nearly 100% urban and may be considered 'representative' of other urban centres of the country, the number of females is considerably higher than the number of males.

- ✓ Sex ratios in Ethiopia are generally lower for urban areas and higher for rural areas. This is primarily due to large number of females migrating to urban areas.

4.4.4 Impacts of Rapid Population Growth

As we have already discussed, the size of Ethiopia's population has been growing very rapidly. The population growth rate is much higher and is increasing much faster than the economic growth rate and is growing beyond the carrying capacity of the country's natural resources, such as land, water, soil, forest, etc. These negative results of rapid population growth have caused many environmental and socio-economic problems. For example, population growth causes serious environmental degradation in the area where it occurs, including deforestation, pollution, soil erosion, depletion of resources, etc.

Deforestation

What is deforestation? Can you mention some of the problems caused by deforestation? is deforestation a problem in your locality?

- ✓ Deforestation refers to the removal of forest cover of an area without adequate replacement. In other words, it is the process of the indiscriminate destruction of the natural vegetation cover of a forest area.

Forests are the lungs of the earth. They absorb carbon dioxide from the atmosphere and exhale oxygen. They also store energy from the sun, bind topsoil to land, and aid in climate control by capturing and releasing water. They also provide a habitat for innumerable species of plants and animals, serving as a global storehouse of genetic diversity.

The forest cover resource of Ethiopia has been declining significantly over time.

This is caused mainly by rapid population growth and the increasing population's needs for forest resources such as wood and land. Many people in the country wood for cooking, heating, and lightning, as well as for houses and furniture. Similarly, the increasing demand for agricultural and settlement lands is a major cause of forest destruction in the country. In Ethiopia, there is a direct correlation between population density and deforestation – the more people there are in an area, the more trees they cut down. New trees do not spring up to replace the old ones.

- In Ethiopia, rapid population growth leads to deforestation mainly because of peoples' increasing needs for more:
 - ✓ Wood
 - ✓ agricultural land
 - ✓ settlement land
 - ✓ grazing land

Such deforestation has various negative consequences, both directly on the natural environment, and indirectly on the socio-economic conditions of the people. Its direct consequences include the following.

- It accelerates soil erosion.
- It destroys biodiversity.
- It affects rainfall by decreasing evapotranspiration.
- It results in shortages of wood supply.
- It affects the natural beauty of the affected areas.

Pollution

Pollution refers to any undesirable change in natural conditions of water, air, and other components of the natural environment that has negative effects on the health and activities of human beings and other living creatures.

- ❖ In Ethiopia, rapid population growth leads to environmental pollution by increasing emission of the amounts of pollutants such as:
 - Sewage, solid wastes, and pollutant gases generated by households.
 - Pollutant gases, liquids, and solid chemicals generated by expanded industries.
 - Pollutant gases generated by the increasing number of automobiles.
 - Agricultural pollutants, such as fertilizers, pesticides, animal wastes, etc.

4.4.5. Population Policy of Ethiopia

A population policy is a policy that is formulated and implemented by a government in order to plan and control population growth, based on the economic, social, cultural, political, and demographic conditions of the country. It is needed mainly to address population related problems in a country.

- ✓ The population policies of countries can be broadly categorized into two groups as: **anti-natalist and pro-natalist policies.**

- ✓ Anti-natalist population policy seeks to lower fertility rates, in particular, and population growth rates, in general.
- ✓ Pro-natalist population policy seeks to increase fertility rates, in particular, and population growth rates, in general.
- ✓ Similar to most of the developing countries, the type of population policy used in Ethiopia is basically anti-natalist. Such policies promote lowered fertility rates, in particular, and lowered population growth rates, in general.

History of Population Policy in Ethiopia: In Ethiopia, population policies were given low priority before the early 1990s. After the Derg regime, the Transitional Government of Ethiopia (TGE) adopted a national population policy in 1993. The policy was based on the awareness that large population size and continued rapid population growth in Ethiopia can be an enemy of development and can cause economic, social and environmental problems in the country.

Goals and Strategies of Ethiopia's Population Policy

Ethiopia's anti-natal population policy formulates several goals and strategies to regulate population for the overall good of the country.

General objectives of Ethiopia's population policy include:

- Closing the gap between high population growth and low economic productivity, through planned reduction of population growth and increasing economic returns;
- Expediting socio-economic development processes through holistically integrated development programs;
- Reducing the rate of rural-to-urban migration;
- Ensuring environmental protections;
- Reducing morbidity and mortality;
- Raising the economic and social status of women; and
- Improving the social and economic status of vulnerable groups, such as adolescents, children, and the elderly.

Specific objectives of Ethiopia's population policy include:

- Reducing the total fertility rate (TFR) of 7.7 children per woman in 1990 to 4.0 by the year 2015;
- Increasing the prevalence of contraceptive use from 4.0 percent in 1990 to 44 percent in 2015;

- Reducing maternal, infant, and child morbidity and mortality rates, as well as promoting the level of general welfare of the population;
- Significantly increasing female participation at all levels of the educational system;
- Removing all legal and customary practices that prevent women from the full enjoyment of economic and social rights, including the full enjoyment of property rights and access to gainful employment;
- Ensuring spatially balanced population distribution patterns, with a view to maintaining environmental security and extending the scope of development activities;
- Improving productivity in agricultural activities and introducing off-farm and non-agricultural activities for the purpose of employment diversification; and
- Mounting effective country-wide population information and programs that address issues pertaining to small family size and its relationship to human welfare and environmental security.

Ethiopia's Population-Policy Strategies

The strategies by which the goals and objectives of the population policy are to be attained include the following.

- Expanding contraceptive distribution;
- Diversifying available contraceptive methods;
- Raising the minimum age of marriage for girls from 15 years to at least 18 years;
- Promoting breast-feeding as a means of birth-spacing;
- Implementing career counseling services in schools;
- Integrating women into the modern sector of the economy;
- Amending all laws “impeding, in any way, the access of women to all social, economic, and cultural resources”;
- Amending relevant articles and sections of the civil code to remove unnecessary restrictions to “advertisement, propagation and popularization of diverse contraceptive methods”;
- Establishing teenage and youth reproductive health counseling centers;
- Increasing research in reproductive health; and
- Promoting the involvement of males in family planning.

The population policy of Ethiopia acknowledges that existing reproductive health service delivery systems are limited in scope and that choice of family planning methods is limited. To correct these problems, it calls for an expansion of reproductive health service delivery, currently available only through the limited formal health structure, to clinical and community-based outreach services. It also

recommends the involvement of non-governmental organizations (NGOs) in providing reproductive health services, including the widest possible choice of contraceptives.

The policy also acknowledges a need to expand capacity for performing population research and training family planning advisors. To implement these, it calls for family planning to be integrated into the curricula of medical schools, nursing and health assistants' schools, junior colleges, and technical-vocational schools.

The population policy also calls for the expansion of Information, Education, and Communication (IEC), and community involvement in achieving the goals and objectives of the policy. In general, the population policy covers all major ground that needs to be covered in providing directives on the management of population growth in the interest of sustainable development.

4.4.6. Urbanization

Urbanization is the process of population shifts from rural areas to cities, and the resulting growth of urban areas. It is the process whereby large numbers of people leave countryside/rural places and small towns in order to settle in cities and surrounding metropolitan/urban areas. A nation is said to have become more urbanized as its cities grow in number, its urban populations increase in size, and the proportion of its population living in urban areas rises.

The degree of urbanization varies throughout the world but generally reflects the wealth of individual countries. The rich, industrialized countries tend to be the most highly urbanized, while the poor countries tend to be the least urbanized. In the Netherlands, for example, 89 percent of the population is urban, compared to only 16 percent in Ethiopia. In most developing countries, including Ethiopia, most rural migrants to the cities have bettered themselves in comparison to their former standard of living in rural areas. However, the rapid growth of population in urban centers of such countries has been causing serious problems such as overcrowding, substandard housing, homelessness, inadequate municipal services, crime, poverty, and pollution. Today, these characteristics mark the lives of many people in most urban centers of the developing countries. Dealing with these conditions, especially in very large cities like Addis Ababa, presents massive difficulties for governments.

Urban settlements

Urban settlement: refers to town or city settlements. People in urban areas depend on non-agricultural activities. In most cases, it is difficult to give an exact definition for urban centres. Some of the criteria

used to identify urban centres include the official status of urban settlement, size of population, and occupation of the population.

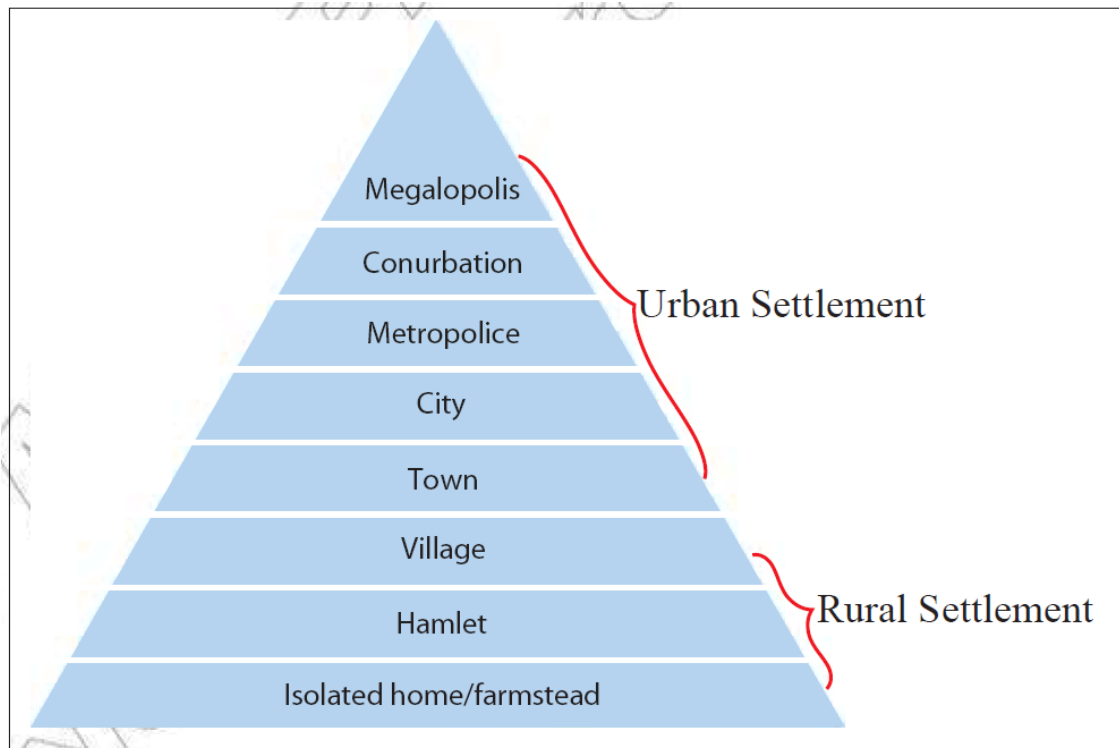


Figure: Hierarchies of settlement

Table: Percentage of Urban Population in Africa by Major Regions (2009)

No	Region	Percentage Urban
1	Sub-Saharan Africa	35
2	Northern Africa	50
3	Western Africa	42
4	Eastern Africa	22
5	Central Africa	41
6	Southern Africa	56
7	Africa	38

Current Urban Challenges

The outcome of rapid urbanization in Africa is that they pose challenges to the growing cities. Cities serve as major growth engines by providing opportunities for employment, education, technology, etc. These conditions, in turn, transfer markets for industrial and agricultural products; consequently, urbanization leads to further expansion of the cities where by their peripheral areas end up in changes

of land use. The conversion of farm lands and watersheds for residential purposes has negative results in food security, water supply and the health of the people. The adverse effects of urbanization in Africa are absence of employment opportunity, unreliable food security, shortage of water, scarcity of shelter and inefficient waste disposal, etc. Urbanization is increasing in both developed and developing countries. Cities in most developing countries are encircled by a number of problems.

Factors affecting urbanization in Ethiopia

Factors Contributing to Urbanization in the nation include: firstly, peoples' need for better living conditions this has been the major cause for the origin, growth and development of many of the urban centers; secondly, people's desire to come close to clustered settlements, many rural Ethiopians have been motivated to move and settle in urban places where there are relatively higher concentration of social services, industries, and employment opportunities.

In the history of Ethiopia, major factors contributed to the origin and development of most urban centers

- ✓ The interconnection of the different parts of the country by all-weather roads, which radiate from Addis Ababa.
- ✓ The five-year Italian occupation, which intensified the construction of roads, and the development of small-scale industries and service giving institutions.

Most urban centers of Ethiopia have developed along major transport routes, which have attracted people to these areas. Consequently, areas with greater transport route network have larger numbers of urban settlements, and higher population densities.

Two main areas have relatively large concentrations of urban centers and urban population: the Shewan and the Harerghe plateaus. Together, these regions account for more than 50 percent of the urban population of the country. The major reasons for this situation are:

- Concentration of industries that results in relatively higher opportunities for employment.
- Concentrations of social services and facilities, such as schools, health institutions, water supplies, electricity, transportation, etc that result in, relatively, better living conditions.

Urbanization in Ethiopia

Urbanization is a recent phenomenon in Ethiopia. It was introduced mainly after the late 19th and early 20th centuries.

Levels and Distribution of Urbanization in Ethiopia

In spite of the high rate of rural-urban migration in Ethiopia, the level of urbanization has been very low in the country. Less than 3 percent of Ethiopia's population lived in urban areas in the 1940s. This figure increased to about 8.5 percent in 1967 and then to 9.7 percent in 1970. The three censuses of the country indicate that the percentage of urban population was 10.6 percent in 1984, 13.7 percent in 1994, and 16.1 percent in 2007.

Although these figures show that urbanization is increasing in Ethiopia, their absolute values also indicate that the country's urbanization is low, and that its urban population is one of the smallest in the world. This indicator reflects the country's status as a very undeveloped nation.

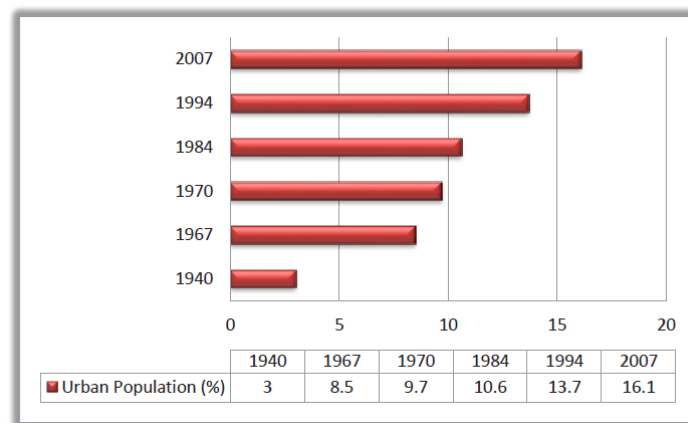


Figure: Trends of Urbanization (Urban Population Growth) in Ethiopia

The other aspect of urbanization in Ethiopia is great variation in the distribution of urban populations illustrated in the following Figure. According to the 2007 Census, the urban population rate ranges from 10 percent in SNNP to 100 percent in Addis Ababa. Despite the fact that Addis Ababa City Administration is entirely urban, its percentage share of Ethiopia's total urban population is only 23.1 percent. The largest share, about 28 percent, is living in Oromiya Region. In contrast, Gambela region has the smallest share of the country's urban population, although more than one-fourth of the region's population lives in urban areas.

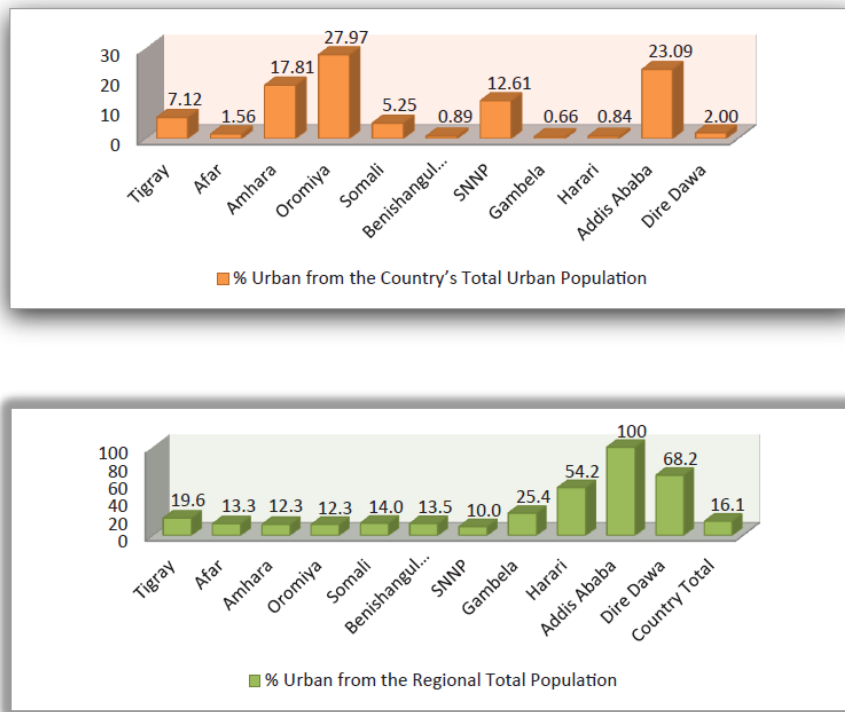


Figure: Regional Distribution of Urban Population in Ethiopia

UNIT SUMMARY

- ✚ Malthusian population theory is very pessimistic about population growth and its effects, while the anti-Malthusian theories are optimistic.
- ✚ Ethiopia is the third most populous country in Africa, with a population of 73.8 million. With a growth rate of 2.23, the country's population is among the fastest growing in the world.
- ✚ In Ethiopia a high percentage of the national population is the young age group. This is the result of high birth rate.
- ✚ In Ethiopia, the population is unevenly distributed; this is due to various physical and human-related factors.
- ✚ Settlements in Ethiopia are broadly categorized as rural and urban. The primary bases for this distinction are the dominant economic activities and population density.
- ✚ Population change in any country is affected by three major factors. These factors are: fertility (births), mortality (deaths), and international migration (movement into or out of country).
- ✚ Ethiopia is characterized by rapid population growth, resulting from a high level of fertility. However, fertility has been declining, moderately especially since the late 1980s.
- ✚ Although, in general, mortality rates have been declining in Ethiopia, as in most countries in sub-Saharan Africa, the country still has a high mortality rate.
- ✚ In Ethiopia, the number of international emigrants is higher than the number of immigrants. Within the country, there have been substantial internal movements of people. This has been due to various reasons, including economic, social, political and natural factors.
- ✚ In Ethiopia, rapid population growth has brought about serious negative impacts on the country's socio-economic development and on its environmental protection.
- ✚ Similar to most of the developing countries, the type of population policy used in Ethiopia is basically anti-natalist, promoting lowered fertility rates, in particular, and lower population growth rates, in general.
- ✚ In Ethiopia, urbanization is a recent phenomenon introduced mainly after the late 19th and early 20th centuries. The country is categorized as one of the least urbanized countries of the world. In Ethiopia, the proportion of urban population varies greatly by region.

UNIT FIVE

ECONOMIC GROWTH AND SUSTAINABLE DEVELOPMENT (21 Hours)

5.1. Economic Activities

Economic activity is the production, distribution and exchange of goods and services. Some examples of these activities are hunting, fishing, farming, grazing, mining, manufacturing, transportation, trade and others. In the world, there are numerous activities through which human beings earn their livings. In order to live, a person must provide for his/her basic necessities such as food, clothing and shelter. For this, people use elements of the natural environment, i.e., natural resources.

5.1.1 Classification of Economic Activities

The economic activities practiced in the world are grouped into five, namely primary, secondary, tertiary, quaternary and quinary. Each type of economic activity is important to a society. The distribution of jobs in a particular economic activity in a country may indicate the level of development of the country.

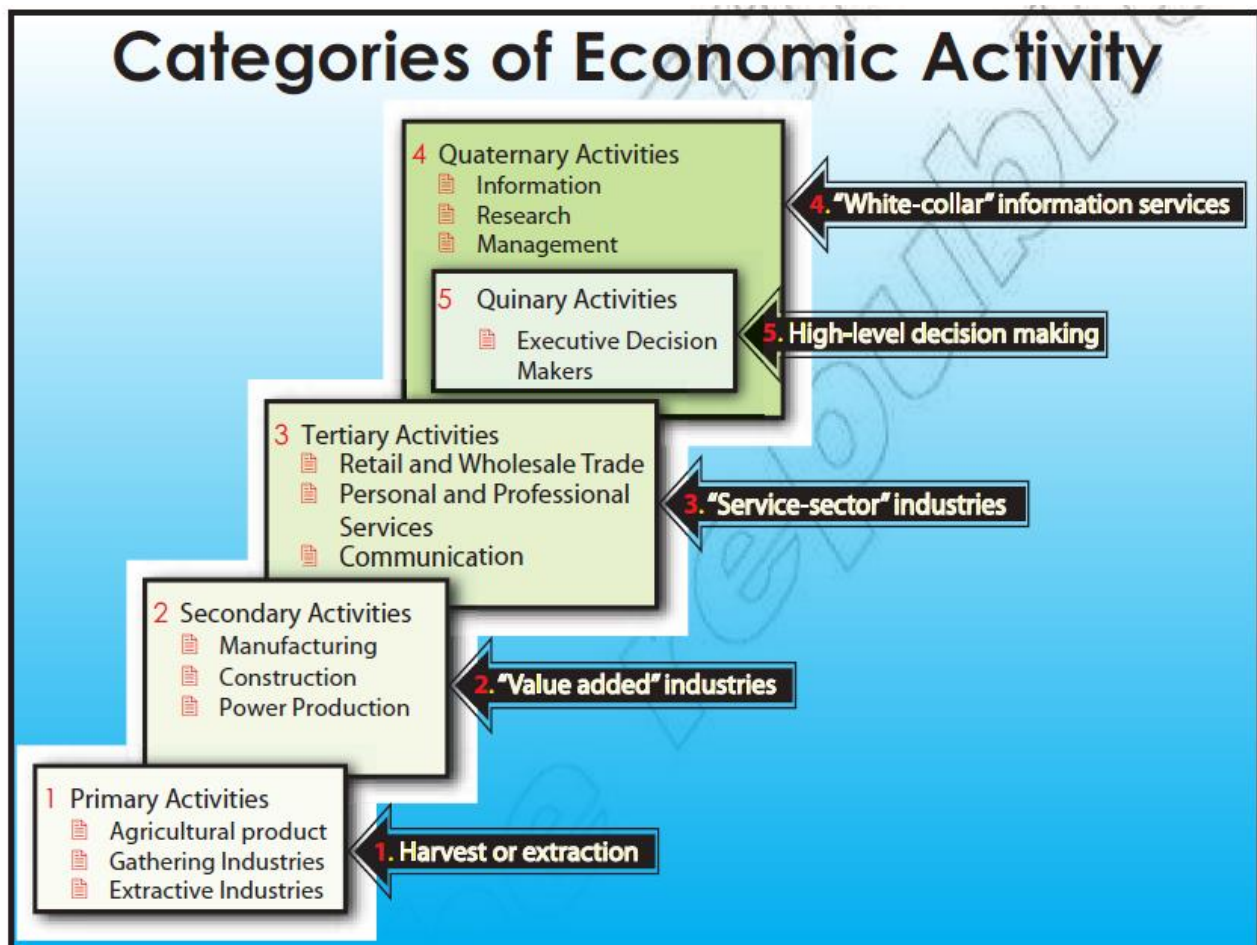


Figure: Categories of economic activity

A. Primary Economic Activities

What are the major primary economic activities practiced in your locality?

What are the main purposes of producing primary goods in your area?

Primary economic activities focus directly on the extraction of resources from the environment. They involve the production of food stuffs and raw materials. These economic activities occur at the beginning of the production cycle, where people live in close contact with the resources of the earth. A few examples of primary economic activities include agriculture, fishing, forestry, and mining. All of these jobs are dependent upon the natural resources of the earth.

Primary economic activities are characterized by the following. They are:

- dependent on the natural environment;
- related to the production of foodstuffs and raw materials through the exploitation of the resources of the earth;
- influenced by the condition of the physical environment in one way or another.

The main types of primary activities include:

- ✓ **Agriculture**
- ✓ **Forestry**
- ✓ **Fishing**
- ✓ **Mining**

Agriculture

What is agriculture? Why is agriculture an important primary economic activity?

Agriculture is the science and art of cultivation of the soil and the rearing of livestock for either local consumption or commercial purposes. It is one of the most important activities of human beings, because it provides them with the most basic necessity; food. Agriculture, as a primary economic activity, aims at solving the basic problems of any society. It provides foodstuffs for the population and raw materials for industries. Above all, it forms the basic livelihood for the majority of the peoples of the world. For example, in Ethiopia about 85 percent of the population is engaged in agricultural activities.

In agricultural activity, land is one of the basic means for production. The characteristic feature of land as a means of production is that when properly used its productivity increases. Another peculiarity of agriculture is that plants and animals are simultaneously means and objects of labor. Moreover agricultural production is seasonal in character, because of the fact that crops ripen at varying stages and times of the year.

Forestry

What is forestry?

A forest is a mass of plants or a wooded area in which trees are the most common features. The extraction of forest products for different purposes by people is called forestry. Forestry is important for the production of wood, timber, gums, nuts, barks, etc.

Today forest products are being used for various domestic and industrial purposes. The following are the most important ones. Forest products are being used:

- for construction purposes;
- as a source of industrial raw materials;
- as a source of foodstuffs such as roots, leaves, fruits, barks, gums, stems, flowers, etc.

Also, indirectly forests are important because they check rapid runoff and consequently control soil erosion. In addition, moisture from forest vegetation has a moderating effect on the climate of the local environment. They also regulate the concentration of greenhouse gases, such as carbon dioxide, thereby regulating the temperature condition of the local environment. A forest is also a habitat for wild animals, and the preservation of a forest ensures the preservation of wild animals.

Fishery

What is fishing?

Fishing is a primary economic activity concerned with the catching and harvesting of fish, other marine creatures, such as whales, seals, pearls, lobsters, crabs, prawns, molluscs, sponges and seaweed. Fishing is one of the oldest occupations of humankind. At present a number of people in some nations depend on fishing as the mainstay of their livelihoods, for example, Norway, Iceland and Japan. But compared to other primary economic activities, fishing is not a very basic economic activity. Currently fishing as an activity provides employment opportunities for only a very small percentage of people. Marine life provides a very minor source of food and non-food products. Ethiopia has a number of lakes, reservoirs and rivers rich in fish resources. However, fishing as an activity is at a low level of development.

Mining

What is mining?

Mining is a primary economic activity concerned with the extraction of mineral bearing substances from the earth's crust. It is closely linked to manufacturing, which is a secondary type of activity. The earth's crust is composed of rock containing minerals. A mineral is an inorganic chemical element or compound found naturally in the crust of the earth. Even though, most minerals are inorganic; there are also some organic minerals, such as coal, petroleum and natural gas. Minerals are not evenly distributed on the earth's surface. The different types of minerals that people use for construction, jewelry making, and in industry come from the mining sector.

B. Secondary Economic activities

What is manufacturing? Is there any difference between manufacturing and industry?

Secondary economic activities include manufacturing, construction and power production. Manufacturing activities take place in factories. It is the activity of making articles. Manufacturing is the activity which turns raw materials into products by using labor, energy and equipment, while industry refers to the place where manufacturing takes place. Also, sometimes the word industry is used to many types of economic activities, for example, the hotel industry, the tourist industry, etc. Therefore, to distinguish the activity of processing raw materials from the other economic activities, we specifically use the term manufacturing.

Manufacturing uses machines, tools and labor to make things for use or sale. The term may refer to a range of human activities, from handicraft to the use of high technology. However, it is most commonly applied to the creation of industrial products in which raw materials are transformed into finished goods on a large scale.

Manufacturing activities are characterized by the following features:

- i. Dependency on raw materials that are obtained from the primary economic sector;
- ii. Transformation of raw materials into finished and/or semi-finished goods;
- iii. Most modern manufacturing industries need power, skilled manpower, huge capital and modern technology;
- iv. Most modern manufacturing industries are characterized by a high level of division of labor;
- v. The production of end products that are ready for consumption or semi-finished goods that serve as an input for other industries.

The main types of Secondary activities include:

- Manufacturing
- Construction
- Power Production

C. Tertiary Economic Activities

What is a tertiary economic activity?

The tertiary sector of the economy, (the service sector or the service industry) is the next type of economic activity. The basic characteristic of this sector is the provision of services.

The tertiary sector involves the provision of services to other businesses as well as to the final consumers. Examples of tertiary economic activities include education, legal services, medical services, trade, transportation services, tourism, etc.

The main types of Tertiary activities include:

- ✓ Trade
- ✓ Tourism

- ✓ Transportation
- ✓ Communication

Trade

What is trade? Can you explain the significance of trade in a society? How many forms of trade exist?

Trade is a tertiary economic activity. It is the process of buying, selling, or exchanging of commodities. The earliest form of trade was bartering, which is the exchange of one article for another of equal value. The emergence of trade is related to the unequal distribution of resources that are essential to satisfying human needs. It also resulted from specialization of skills, such as weaving, metalwork, tannery, pottery, etc. Due to the variety of their occupations, people began to exchange goods in the form of bartering. Later on, they started using metal, salt and paper money as the medium of exchange. Differentiation of products, surplus of production, demand for commodities, differences in culture, adequate transport facilities and suitable world conditions are the main bases of trade.

Transportation

Transportation facilitates the movement of people, commodities, and mail. Nowadays, the types and quality of transportation have made much progress. This has enabled people to travel longer distances in shorter times than ever before. Also, the flow of ideas, beliefs and innovations has become faster and wider. Over the year, faster means of transportation have been introduced, and now distances which used to take days or months to cover can be covered in a matter of minutes or hours. The rapid progress in transportation and other forms of communications have made places all over the earth functionally closer to each other.

Good transportation systems are needed to move goods within a country and abroad. They are also essential to spreading the benefits of health care and education. There are many ways in which goods and people can be transported. In many areas of the world, human porters and animals like horses, donkeys, camels, etc. are the main means of transportation. These are, however, limited to local business. In international business interaction, fast and efficient means of transportation are necessary because of the huge quantities of commodities involved. These include land, water and air transport. The main transportation systems are road, railway, inland waterways, oceans and airways. Each of these has its own advantages and the ideal situation for a country is to have a network of systematically coordinated transport links.

Communication

The word communication involves the transmission of words and messages from one place to another. The following are the major types of communications:

Mail: This includes the distribution of letters, packages as well as money. Example: Postal services

Telephone: is a telecommunications device that transmits and receives sound, most commonly the human voice. Telephones are a point-to-point communication system whose most basic function is to

allow two people separated by large distances to talk to each other. **Example:** Mobile phone, land line telephone.

Radio: It is one of the most important means of communication. All over the world, information is transmitted using radios, with the help of electromagnetic waves.

Television: It is a means of communication used to transmit messages using both images and sounds. Nowadays, it is possible to receive information and entertainment via TV from any corner of the world with the help of satellites.

Internet and other modern communications tools: by using satellites and other electronic equipment like the computer and telephone, new methods of communication have been introduced to the world. Some of them include fax, e-mail and internet.

Tourism

Tourism is a collection of activities, services and industries that delivers a travel experience, and they include transportation, accommodations, eating and drinking establishments, retail shops, entertainment businesses, activity facilities and other hospitality services provided for individuals or groups travelling away from home.

Tourism is a source of both job opportunities and income. It is known as a ‘smokeless industry’. The main tourist attractions include natural and human-made features. The following are some of the reasons for tourism:

- ✚ site-seeing;
- ✚ conferences;
- ✚ sport activities, such as skiing, mountaineering, boating, yachting, fishing, hunting, swimming, etc;
- ✚ health purposes – for example to secure fresh air and sunshine and sometimes to bathe in hot springs etc;
- ✚ research;
- ✚ religious ceremonies.

Features that encourage the development of a tourist industry include:

- good weather conditions;
- beaches, scenery (scenic attractions) or physical landscape features;
- services, such as bathing, boating, recreation, hotels, restaurants, etc.
- accessibility (transport facilities);
- interesting features (historical sites, parks, etc).

D. Quaternary Economic Activities

The quaternary sector may realistically be seen as an advanced form of service activity involving specialized knowledge, technical skills, communication ability, or administrative competence. These activities include education, research, development, financial services and government activities. These

are the activities performed in office buildings, elementary and university classrooms, hospitals and doctors' offices, theatres and television stations. They are activities primarily concentrated in large urban places and require higher levels of education than the other sectors. This section also includes other pure services, such as the entertainment industry.

E. Quinary Economic Activities

What are quinary economic activities? Is there any difference between quaternary and quinary economic activities?

Quinary economic activities are generally considered to be a sub-set of quaternary activities and are those that involve high-level decision making and scientific research skills. It is also a sub-division of the tertiary sector representing the special and highly paid skills of top business executives, government officials, research scientists, financial and legal consultants, and the like. These people find their places of business in major metropolitan centres, in and near major universities and research centres.

5.2 Natural Resources

Natural resources occur naturally within environments when those environments exist in their natural forms, relatively undisturbed by people. A natural resource is often characterized by the amount of biodiversity that exists in various ecosystems. Natural resources are derived from the environment. Many of them are essential for human life, while others are used for satisfying our wants. Natural resources include fertile soil, clean water, minerals, wildlife, vegetation, and energy sources. People use all these resources to improve their lives.

5.2.1 Classification of Natural Resources

In the content of renewability, natural resources can be categorized as follows:

1. **Renewable resources:** are types of resources that can be replaced as they are used. They can be replaced naturally or grown fairly quickly. Forests, water, soil, plant and animal life all can be renewable resources if people manage them carefully. The renewable resources are replaceable in character since they are able to reproduce themselves. For example, forest is a renewable resource that can produce timber and other forest products year after year if it is carefully and scientifically managed.
2. **Non-renewable resources:** As their name suggests, non-renewable resources cannot be replaced once they have been used. The most important example of non-renewable resources are the fossil fuels like, coal, oil and natural gas and minerals. These resources are generally non-replaceable in character.

Resources are not evenly distributed throughout the world. There is a great disparity in the distribution of natural resources. This unevenness in the distribution of natural resources has led to the notion of the "have and have nots" nations, although the total resource quantities of the world have not yet been adequately determined. Through out the world, people cut more and more trees and mine more minerals.

This has led to the occurrence of environmental problems. Sometimes, resources are misused, as a result their quantities become smaller and smaller. This process is called **environmental degradation**. To

overcome this problem, renewable and non-renewable resources should be wisely or sustainably used. Nowadays a number of people believe that the earth's resources are being over used and sometimes misused. If this situation continues uncontrolled, many of the resources will be lost or will deteriorate. Unless something is done quickly, we will face disasters.

In order to overcome this problem, people in many parts of the world practice different resource-conservation measures. Such practices help to maintain the volumes of resources and make them last longer.

5.2.2. NATURAL RESOURCES OF AFRICA AND ITS POLITICS

At the end of this section, you will be able to:

-  *relate major resources of Africa to its exploitation and development of Africa;*
-  *state the actual mineral extraction methods of Africa;*
-  *relate the paradox between the potential and actual resources exploitation in Africa; and*
-  *defend for the advantages of peaceful mechanisms of conflict management around resource utilization in Africa against aggressive mechanisms.*

As some sources indicate, Africa is a naturally endowed continent. Its landmass possesses huge reserve of different kinds of minerals including precious gem stones such as diamond and gold, and petroleum. Its varied topography makes the continent rich in terms of climatic resources and biodiversity. The geology of the continent is also a cause for the presence of various soil types with varying degree of fertility. Water resources are abundant in forms of rivers, lakes and swamps, though there are still many areas with serious shortage of water. The world's second largest population of the continent is also a source of huge human power. Africa's tropical location also makes it potentially conducive for the continent to possess tremendous solar energy. However, all these resources do not seem to serve the cause of the continent's development. Instead, they are becoming causes of conflicts and war.

Major Resources of Africa

Africa is believed to be rich in terms of human and material resources. The physical environment coupled with the continent's varied geology makes Africa prosperous in terms of resources of various nature. Among others, the mineral, agricultural, water and human resources of the continent are of paramount importance for its future development.

Mineral Resources of Africa

Africa's geology consists of various types of rocks and fuel oil. This makes the continent rich in mineral resources. Geographically, Northern Africa is rich in petroleum resources. West and Central Africa are also major regions with huge petroleum reserves. Metallic minerals and other gem stones are abundant in Southern Africa.

As some estimates indicate, Africa has 90% of the world's cobalt and platinum, 50% of the world's gold, 98% of the world's chromium, 70% of the world's tantalite, 64% of the world's manganese and one-third of the world's uranium. The DRC alone has 70% of the world's coltan (ore of Tantalum) and more than 30% of the world's diamond reserve. Guinea is the world's largest exporter of bauxite (ore of aluminum).

North African countries constitute one of the world's major centres of oil production. Libya, Algeria and Nigeria are among Africa's leading producers of crude petroleum. Algeria has a huge reserve of natural gas, as well. North Africa is also rich in phosphate deposits and production, Morocco being the world's leader in its output. Coal, iron ore, uranium, platinum, lead, zinc, and cobalt are also available in Northern Africa, though to a lesser extent. Significant amount of oil reserve is also found in Western and Central Africa. With Nigeria being Africa's top petroleum producer. Other important oil-producing countries of the region include Angola, Gabon, and the Republic of the Congo.

These regions also possess some of the world's most significant sources of cobalt, manganese, potash, bauxite, and copper. Guinea alone has about one-third of the world's reserves of bauxite. Other minerals of significant economic importance in the regions are iron ore, gold, diamond, tin, uranium, phosphate, columbite, and titanium. Likewise, Southern Africa is one of the world's richest source of gold, diamond, and several other rare metals. Among the countries of the region, the republic of

South Africa alone has the largest and most diverse mineral economy. As a result, the country is a leading producer of gold and uncut (raw) diamonds. Zimbabwe is also an important producer of gold. Botswana and Namibia produce significant amount of diamond. Chromium, cobalt, antimony, uranium, lithium, nickel, manganese, asbestos, platinum, titanium, and vanadium are also important minerals produced in southern Africa.

- ❖ Minerals account for about half of the export earnings of 12 African countries. Nearly 90 percent of the exports in Angola, Nigeria, Algeria, Libya, and Zambia come from minerals. As a result, Africa plays a significant role in the world's mineral economy. It produces about:
 - three-quarters of the world's cobalt;
 - half of the global supply of platinum, chromium, and diamonds;
 - one-third of all gold, manganese, and uranium;
 - one-fifth of all bauxite; and
 - one-tenth of the world's petroleum.

Table: Major minerals produced in Africa and the leading producers of the continent

Mineral	Africa's Global share	Leading producers in Africa
Diamonds	77%	Zaire, Botswana, RSA,
Gold	55%	RSA, Ghana, Zimbabwe
Copper	20%	Zambia, DRC, Zimbabwe, RSA,
Bauxite	19 %	Guinea, Ghana and Sierra Leone
Iron ore	4%	Liberia, Mauritania, Morocco, Tunisia, RSA, Zimbabwe and Swaziland
Chromites	41%	RSA and Zimbabwe
Manganese	27%	RSA, Gabon
Petroleum	10%	Nigeria, Libya, Algeria, Egypt and Angola
Tin	13%	DRC(Katanga area) and Nigeria
Cobalt	76%	DRC and Zambia
Coal	6%	RSA , Zimbabwe, Nigeria, Zaire, Zambia, Mozambique, Malawi and Tanzania

Agricultural Resources

Agriculture is the backbone of many African countries. Apart from being the main source of foodstuffs and employment, the sector is a dominant source of agricultural outputs that enter into world market. Africa's physical diversity has made the continent an ideal environment for the production of certain agricultural products that are vitally demanded in the world market. The continent is rich in terms of certain tropical crops such as coffee, cocoa, tea, sugarcane, rubber, palm oil, sisal, cotton and ground nuts. The economy of many African countries is primarily agricultural. They are dependent on the exportation of these tropical crops. Crop production absorbs much of the continent's labour force; and the exportation of crops contributes a lot to the GDP of the exporting countries.

Table: Leading producers of Tropical crops in Africa and Africa's share to the World's production

Crop	Leading producers in Africa	Africa's Global share
Cocoa	Ghana, Cote D'Ivoire, Nigeria, Liberia, Togo, Cameroon	53%
Coffee	Cote D'Ivoire, Ethiopia, Uganda, Kenya, Cameroon, Tanzania, DRC, Angola	20%
Tea	Tanzania, Kenya, Ethiopia, Mozambique, Zimbabwe, Congo, Mauritius	13%
Palm oil	Nigeria, Ivory Coast, and Zaire	16%
Sugar cane	Sudan, Kenya, Uganda, Ethiopia	7%
Rubber	DRC, Ghana, Cote D'Ivoire, Cameroon, Gabon, Sierra Leon, Kenya and Tanzania	6%
Sisal	Tanzania, Kenya and Madagascar	29%
Cotton	Sudan, Nigeria, Ethiopia, Mali, Congo and Egypt	7%
Ground nuts	Nigeria, Senegal and Zaire	29%

Water and Associated Resources

Though the provision of clean water is a critical problem in many countries of Africa, the continent is still rich in terms of water resources. Thousands of rivers that altogether comprise the four major drainage systems of Africa flow over the vast landmass of the continent. The rivers that drain many areas of the continent include some of the world's greatest rivers namely Nile, Congo, Niger and Zambezi. There are also many lakes, a lot of which are concentrated in the Great East African Rift Valley, a site also containing varied biodiversity. The rivers and lakes have tremendous potential in terms of fishing, hydroelectric power generation, irrigation, and mineral extraction.

The rivers and lakes of Africa, as well as the adjacent seas and oceans make the continent rich in terms of fish resources. Major grounds for marine fish such as tuna, sardines, and hake are found some distance off from West African coasts stretching from Morocco to Senegal, and Angola to Namibia. The Nile, Niger, Congo, and Senegal rivers and Lakes Victoria, Tanganyika, Malawi, and Chad are also major sources of freshwater fish. Nile perch is the most common freshwater fish targeted in angling. Morocco, Egypt, South Africa, Ghana, and Nigeria were the top African countries in total fish catch in 1999. Morocco, Namibia, South Africa, Senegal, and Libya exported the most fish in the same period. Morocco is also leading in fish processing industries, producing more canned fish, fish oil, and fish meal than any other African country.

Africa's many large rivers provide the continent with a vast hydro-electric power potential, Which, in fact, has scarcely been exploited. Several major dams including the Aswan High Dam on the Nile River,

the Akosombo Dam on the Volta River, and the Kariba Dam and Cabora Bassa Dam on the Zambezi River have been constructed since 1960. In Ethiopia, the major rivers of the country mainly rivers Tekeze and Ghibe have already been dammed for this purpose. The country has also inaugurated a new dam construction project on the Abay river recently. Irrigation is another potential to be exploited from Africa's rivers and lakes. As these water bodies contain fresh water, they can potentially be exploited for agricultural purposes. As many areas in Africa are rainfall deficient, the rivers and lakes that form the drainage systems of the continent can be used for the production of agricultural products. For instance, the Nile River supports the lives of millions of people in Sudan and Egypt by providing irrigable water to these countries.

, the Awash River in Ethiopia makes such a benefit available to the people of this country. There are many irrigation schemes along the Awash River basin including the Upper and Lower Awash agro-industries and the different sugar cane plantations in the region. However, the huge irrigation of many rivers and lakes in Africa is not yet fully exploited. Despite such available potential, decline in agricultural productivity is a common phenomenon in many African states. As a result, the continent often suffers from recurrent famine. Another most important stockpile associated with water resources in Africa is mineral. The shores and beds of lakes and banks of rivers are of substantial source of minerals. In many countries of the continent, construction materials such as sand, gravel and boulders are being exploited from rivers and lakes.

Gold and other precious metals are also being extracted from rivers in traditional ways in many countries. For example, many people in Ethiopia extract gold from the Baro River in the Gambella Regional State. Similar traditional gold mining practices are also being performed in many African countries.

Human Resources

Africa is the world's second most populous continent. According to the 2009 World Population Data Sheet, the continent is a home for about 1 billion people. Of this number 41% is under 15 and 3% above 65 years of age. The rest 56% is in the productive age. This makes the continent rich in terms of human resource potential. This indicates, firstly, the fact that about 56% of the total population is currently actively productive: and, secondly, the future potential of labour is relatively high since the 41% population of under 15 would soon join the labour force. Yet, we can say that the human resource potential of the continent has never been exploited to its full extent. In simple words, it fails to contribute to the development of the continent since much of the labour force, including the economically active, are unemployed or idle and even underemployed.

Natural Resources Exploitation and Mineral Extraction Methods in Africa

As we have seen so far, Africa has tremendous potential of natural resources. However, the degree of exploitation of these resources of the continent appears being very low. This is partly due to the prevalent lack of good governance, corruption, maladministration and lack of citizens' participation in resource development programs. In addition to this, lack of capital, skilled human power and technology have also contributed for the limited development of the resource exploitation sector in the continent. As a result, most resource extraction techniques are traditional in their nature. This in turn affects the level of

productivity. Forest resources extraction, fishing, cultivation, and other sectors of the economy that are linked with natural resources are dominantly traditional in their nature.

Resource Utilization and Conflict Management

As has been elaborated in preceding discussions, Africa is naturally endowed with varied resources that could potentially contribute a lot to the continent's development. However, the degree of exploitation of these resources is practically insignificant. The huge mineral potential of the continent is untouched. The water resources are untapped. Even in areas where the extraction of minerals is better developed, the resource is becoming more of a cause of conflict than of a socio-economic development. As you might have heard from different sources of information, Africa is the only continent that is well characterized by political unrest, conflicts, civil war and related humanitarian crisis including huge displacement of people. As a matter of fact, the continent is considered as the world's largest refugee camp since millions of Africans live in refugee camps as a result of displacement for reasons of conflicts and civil war.

The major source of conflict in many resource exploitation areas of Africa is related to inappropriate use and allocation of resources. If we consider many of the conflicts and civil wars in Africa, their causes are deep rooted in resource allocation and utilization. In most of the cases, governments fail to provide better opportunities for the residents of the areas where resources, especially minerals, are being exploited. To add fuel to the problem, many governments do not even make the inhabitants of resource extraction areas beneficiaries of the exploited resources. Thus, while many Africans live in extreme poverty expensive resources get tapped and exported to rich countries. This causes conflict. In demand of having their opportunities fulfilled, peoples of such areas confront their governments.

When the government fail to properly address the issue, the conflict intensifies and people die ultimately, the case worsens and civil war occurs. The conflicts that have made Africa an extensive battle field of civil wars are in one way or another associated with unfair utilization and allocation of resources. The Civil wars in Southern Sudan, Liberia, Democratic republic of Congo, Angola, and Rwanda are fought because of inappropriate resource utilization and allocation, besides seeking political power.

Civil war is the exception rather than the rule across much of the developing world. The combination of weak governance structures and resources that offer the promise of windfall gains to those who control production and export is the major cause of violent conflict. In the post-cold-war era, revenues from natural resources have replaced superpower funding as the fuel of war. Between 1990 and 2002, the world saw at least 17 such conflicts in which wealth from natural resource was a primary factor. Diamonds in Angola and Sierra Leone, timber and diamonds in Liberia, gems in Afghanistan, and copper, gold, cobalt and timber in the Democratic Republic of the Congo have all been at the centre of civil conflict.

In the case of the Democratic Republic of the Congo, such a cause of conflict has been accompanied by incursions supported by neighboring states. In this manner, financial flows that could have been used to support human development have frequently been diverted into funding civil wars, with governments, rebels and assorted warlords seeking control over oil, metals, minerals and timber. The wealth from the second largest oil reserves in Africa (Angola) and the fourth largest diamond reserves in the world was

used to fuel a civil war that killed and wounded nearly 1 million people and left other 4 million internally displaced between 1975 and 2002. Today, Angola ranks 160 among 177 countries on the Human Development Index (HDI), with a life expectancy of about 40 years only. The following table gives you a highlight on the duration and causes of civil conflicts in selected countries of Africa.

5.3 Economic Systems

An economic system is the means or structures in a society within which decisions about what to produce, how and when to produce goods and services and where to allocate them are made and implemented. The way a society makes economic decisions may closely reflect the pattern of its culture. It also tells about the society's politics, religion, and even its approach to family.

To achieve economic growth and development, different countries use different approaches or systems. That means economic problems can be solved in various ways. Here are the major economic systems practiced in the world:

- Traditional economic system
- Free market economic system
- Command economic system
- Mixed economic system

Traditional Economy

What are traditional economies? Can you cite a primitive people who still live in traditional economies (especially in Africa)? Do we have such communities in Ethiopia? What are the basic features of a traditional economy?

It is an economic system in which primitive people produce just to feed their households with very little goods or services left over for sale or exchange in the market. Production is geared towards subsistence and basic survival. The system employs simple, backward and traditional means (techniques) in the process of food production.

Market and money are of little importance, and trade is mainly by a barter system. Thus, exchange of goods is limited. Answers to the “for whom” and “how much” questions are fixed by custom, habit, religion, or law in traditional cultures. *Example*, the Bushmen of the Kalahari Desert have rules for sharing their kills among the hunters and their families. The largest share usually goes to the best hunter. In this way, the best hunters survive when games are secured.

- ❖ Traditional economic systems are practiced in extremely backward areas or primitive societies like hunters and gatherers. Generally, it is characterized by:
 - lack of knowledge of science and technology,
 - low productivity.
 - a low level of exchange of goods.

Market Economy

Market economy is an economic system in which individuals, rather than government, make the main decisions regarding economic activities and transactions. Individuals are free to make economic decisions concerning their employment, how to use or accumulate capital, what expenditures to make, and whether to use their resources now or to save them for later consumption. It is important to know that there is no pure free-market economy in the world.

Governments interfere in organizing market economy in many ways and for many reasons. Most governments play an important role in dealing with problems of and employment. They are also concerned with providing education for the society, strengthening national defense, controlling prices of power supply and food, and imposing taxes. The main proponents of free-market economies believe and forward a number of advantages of free market economies, such as encouraging individual responsibility for decisions. They also believe that economic freedom is essential to political freedom.

Free-market economies are also criticized. Opponents believe that a free-market economy cannot ensure basic social values such as alleviating poverty or ensuring equitable income distribution. It may also permit the accumulation of wealth and strong vested interests that could threaten the survival of political freedom.

Generally, market economy is characterized by:

- private property.
- economic freedom.
- prices that are determined by the law of supply and demand.
- decentralized decision making.

Command Economy

In a command economy, the government makes decisions about production and consumption. This implies that the government decides what to produce, how to produce and for whom to produce. Also the governments own all factories, land, housing, etc. The command economic system still exists in some countries, such as North Korea, Cuba and China.

It is characterized by:

- a master plan for supply and price.
- the government develops and enforces plans.
- goods and services are distributed through government agencies.

State ownership of resources actually reduces personal incentive, effort and initiative. At best, in a command economic system, the basic economic questions are answered by a group of planners. They have the power to make economic decisions for the society as a whole. However, this type of economy is fading away in the world.

Mixed Economy

In this type of economic system, both government and private sectors play an important role in answering the “what”, “how” and “for whom” questions concerning society as a whole.

In the mixed system, the government may not only own and run key industries, means of transport, power, and water supply, etc., but it also intervenes to prevent monopolies and ensure free competition. In addition, it might influence prices of agricultural products, rather than allowing them to be influenced by market forces. It might also offer incentives (tax relief, grants, exemptions or penalties) to encourage particular activities (e.g., tree planting).

Most countries of the world have mixed economies (for example, France and Sweden). Some are close to command economy while others rely more on market economy. In a mixed economy, there is competition between the government and the private sectors. Sometimes, those that are unable to compete soon die away.

- ❖ Traditional economic systems make decisions the way they have always been made in the past. Command systems permit a group of planners to answer the basic economic questions. Market systems rely on exchange and a network of prices to determine issues of what, how, for whom, and how much, while mixed economy is characterized partly by free-market.

5.4 Concept and Indicators of Economic Development

Development is a difficult concept to define. The term development means different things to different people. Therefore, possible definitions include a wide range of elements (see the following Figure).

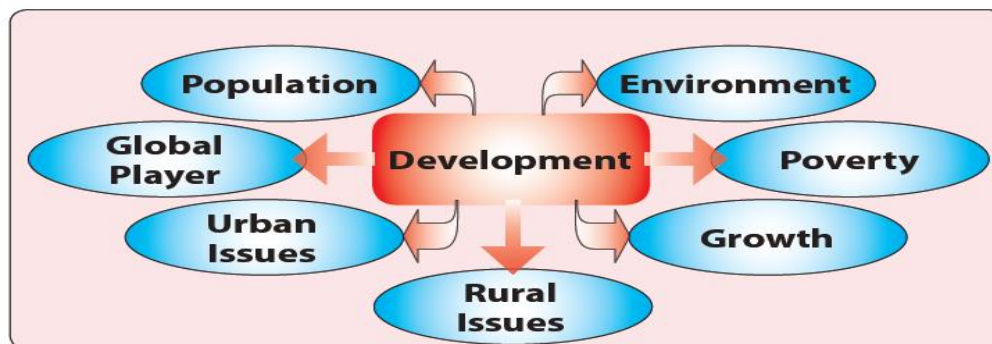


Figure: The Environment of development

Economic development refers to growth in structural and technological change. Because of this, development is both quantitative and qualitative in that it involves increment in production, service provision and income implies, improvements in the overall living conditions of the population. That is why most agree to the idea that a growing economy can definitely promote development. On the other hand, economic growth is usually quantitative as it is related to increasing production of goods and services, thereby allowing a state conceive how much income it has generated.

Economic development is characterized by the following:

- *The increased utilization of the natural resources of a state in order to raise people’s standard of living and to improve the quality of life.*

- *The production of more and more goods and services that enter into markets for sale. For example, we know that people always eat, but as they have meals away from home and pay for restaurant services, the restaurant sector grows up.*
- *The growth in GDP (for instance, the sector listed earlier on in turn, is measured as part of the gross domestic product (GDP) i.e., the total of all goods and services produced within a country.*
- *Division of Labor: the process of development includes specialization in specific area of labour, which consequently leads to “division of labour.” As people take on specialized economic functions, the scale of production increases and the output of each person rises. This type of organizational change is an important part of development as technological progress and mechanical invention or scientific discovery.*
- *Significant changes in the technical and institutional arrangements by means of which output is produced and distributed.*

Indicators of Development

Major Indicators of Development

The level at which a certain economy is said to be developed is directly linked with three most important conditions that are considered as major indicators of development. These are the Gross Domestic Product (**GDP**), the Per capita income (**GNP per capita**) and the **Standard of living** of a certain population.

a) Gross domestic product (GDP): GDP refers to the total value of goods and services produced in a country over a period of time. It can be calculated by either adding up the value of all goods and services produced, or the expenditure on goods and services at the time of sale, or producers’ incomes from the sale of goods or services. However, measuring GDP precisely is quite cumbersome. This is partly because every country has an unofficial economy that is often called a black economy that includes businesses that are not reported to government. Furthermore, GDP measures a country’s economic activity regardless of who owns the productive assets in that country.

For example, the output of Holland-owned companies based in Ethiopia is considered part of Ethiopia’s GDP rather than part of the Holland GDP. The amount of GDP an economy has as well as the major sectors of contributors to the GDP of a country are important indicators of development. Another term, Gross National Product (GNP) is also used to measure a nation’s wealth. GNP refers to the total annual flow of goods and services in the economy of a nation in monetary value. It includes income from within and outside of the state.

b) Per capita income: This one refers to the average amount of money that an individual is expected to have as a result of the state’s GNP. It is computed by dividing the GNP of a country by the total number of its people. As it is a crude measure, per capita income does not tell who exactly gets how much.

However, it indicates how much each individual in a state can potentially get if the total GNP of the state is evenly distributed among all people.

Though not an actual income of an individual, we can generally deduce that the higher the per capita income of a nation is, the higher standard of living and vice-versa.

You can calculate per-capita income, using the following formula:

$$\begin{aligned} \text{Per-capita income} &= \frac{\text{GDP}}{\text{Total Population}} \\ &= \frac{\text{GDP} \pm \left(\begin{array}{l} \text{Income secured from abroad} \\ - \text{income taken away by foreigners} \end{array} \right)}{\text{Total population}} \end{aligned}$$

Example

Suppose gross domestic product (GDP) of a country for a given year is \$50 billion, total population is 25 million, income earned by foreigners from the domestic economy is \$100 million, and income earned by nationals from abroad is \$250 million. What is the percapita income for the country?

$$\begin{aligned} \text{Per capita Income} &= \frac{50 \text{ bill} \pm (250 \text{ mill} - 100 \text{ mill})}{25 \text{ mill}} \\ &= \frac{50 \text{ bill} \pm 150 \text{ mill}}{25 \text{ mill}} \\ &= \frac{50 \text{ bill} + 150 \text{ mill}}{25 \text{ mill}} \quad \left(\begin{array}{l} \text{you have to add them, because} \\ \text{income secured from abroad is} \\ \text{greater than income taken away} \\ \text{by foreigners} \end{array} \right) \\ &= \frac{50,150,000,000}{25,000,000} \\ \text{Per capita income} &= \$2006 \end{aligned}$$

c) Standard of living: This is the threshold of material security measured by the availability of resources for an individual, family, or society. It is perhaps the best measurement of the quality of life of people in a given society. The standard of living that people have in a state is directly related to both GDP and per capita income.

Hence, in economies we say that where we have higher GDP and per capita income, people have better standard of living. Contrary to this, people in countries with low GDP and per capita income have low standard of living. These three factors are strictly interrelated to one another. Thus, a higher GDP means higher per capita income. Likewise, a higher per capita income means better living condition. In well developed economies, there is high GDP, per capita income and standard of living. Consequently, the

things that are essential to satisfy the needs of the population are readily available. The following chart shows you how each of these concepts influence the other.

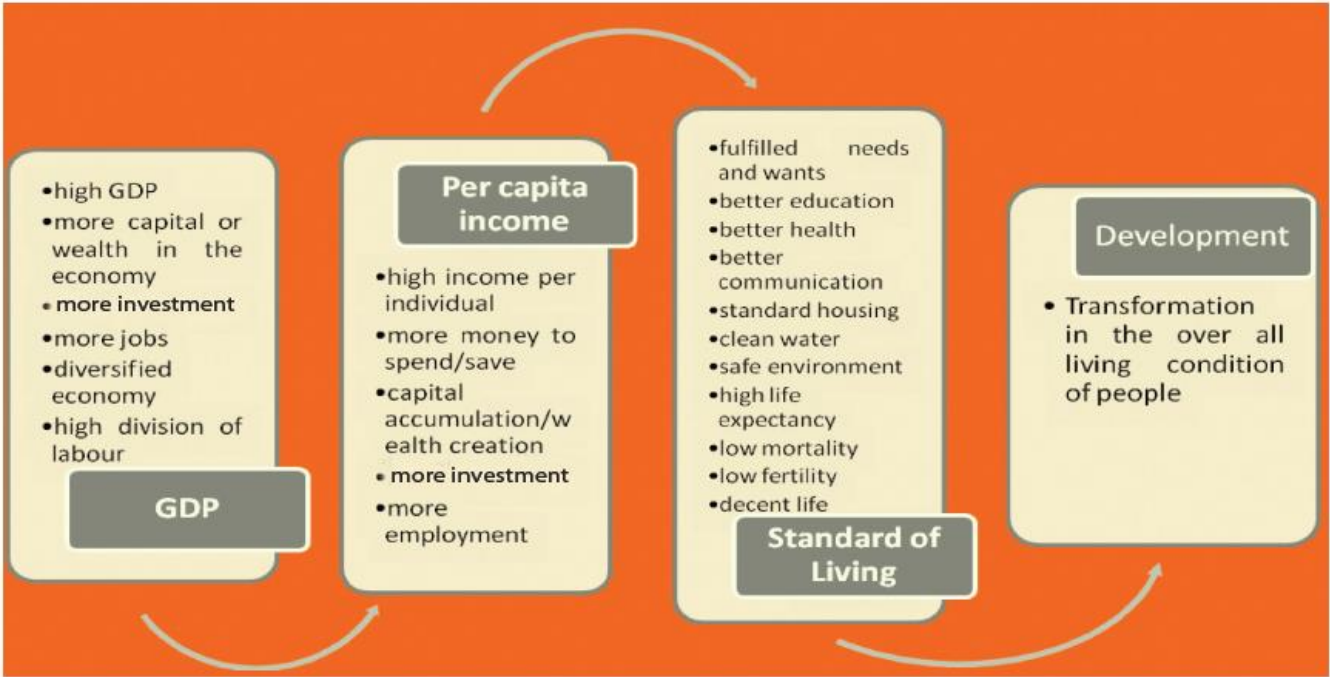


Figure: The major indicators of Development

The following table gives you a highlight of the basic features that characterize developing and developed countries.

Table: Characteristics of Developing and Developed Countries

Developing	Developed
Per capita incomes are low, and capital is scarce.	Per capita incomes are high and capital is readily available.
Wealth is unevenly distributed within individual countries, Example, Colombia, 2.6% of population owns 40% of the national wealth.	Wealth is comparatively evenly distributed, Example, Canada, 10% of population owns 24% of national wealth.
Primary industries dominate national economies.	Manufacturing and service industries dominate national economies.
High proportion of population engaged in subsistence agriculture.	Farming is commercial, efficient, and mechanized.
Populations are rural; but cities are growing rapidly.	Populations are urban, cities growing slowly.
Birth and death rates are high and life expectancy is low. There tends to be a high proportion of children.	Birth and death rates are low and life expectancy is high. High proportion of people over 60 years old.
Inadequate or unbalanced diets resulting from a low consumption of protein; hunger and malnutrition is common.	Adequate supplies of food and balanced diets; overeating sometimes a problem.
Diseases, especially infectious and parasitic diseases, are common. Health care is poor.	Low incidence of disease; good medical services available.
Overcrowding, poor housing, few public services, bad sanitation and poor social conditions.	Social conditions generally good.
Poor educational facilities, high levels of illiteracy and low levels of scientific and technological development.	Education opportunities excellent, high level of literacy, advanced science and technology.
Women may be held in an inferior position in society.	Women are increasingly treated on equal terms with men.

5.5 Sustainable Economic Development

According to the World Commission of Environment and Development (1987), ***“Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs”***.

In other words, sustainability implies that future growth and overall quality of life are critically dependent on the quality of the environment. To destroy the environment in the pursuit of short-term economic goals, jeopardizes present and especially future generations. This is the level of extraction that could be maintained without lessening future levels.

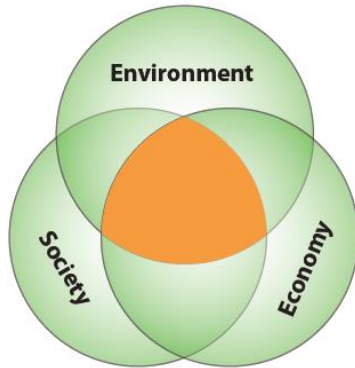


Figure: Components of sustainable development

That is why sustainability is a current paradigm for thinking about a future in which environmental, societal and economic considerations are balanced in the pursuit of development and improved quality of life. Sustainable development has three components, which are shown in the above Figure. The area of overlap (i.e., the intersecting and shaded part of the three circles at the center) represents human wellbeing. As the environment, society and economy become more aligned, the area of overlap increases, and so does human wellbeing. Generally, in order to achieve sustainable development, environmental protection must constitute an integral part of the development process, and cannot be considered in isolation from it. Eradicating poverty and reducing disparities in living standards in different parts of the world are essential goals for achieving sustainable development and meeting the needs of the majority of people.

5.6. Globalization

Globalization is the process by which people their ideas and their activities (economic, cultural, political) in previously relatively separated parts of the world become interconnected, and are drawn to the same social space at the same historical time. It does not imply homogenization, or the elimination of regional differences or uneven development. It leads to increased ecological inter-dependence; and the global world economy is based particularly on the transnational movement of information and the mobile factors of production (capital, labour, technology).

It is said that globalization is a powerful force for poverty reduction, but billions of people in the world are still left out of this process. However, with the development of the present trend, it seems evident that the world will be truly a global village. It must also be noted that there are continuing controversies

and disputes over the value of globalization for the common good in society. Some argue that globalization works to the advantages of the leading industrialized countries of Europe and North America. They argue that a good deal of economic exchange takes place between regions rather than worldwide. For example, the European Union countries mostly trade among themselves, while those of Asia or North America do the same thing. Therefore, there is no international trade that benefits the world community as a whole. And so, globalization adds more to the misery and despair of the already poor and underdeveloped nations, while increasing the wealth of the rich countries.

There are also others who have an opposite and positive view of globalization.

These contenders argue that globalization is not only an economic phenomenon, as it affects all aspect of life. Therefore, they say, it is unavoidable. Because, according to them, the dramatic expansion of science and technology is interwoven with international relations among people of the world and no nation, whether poor or rich, can do away with globalization, but must adjust themselves to this new trend.

The Advantages and Disadvantages of Globalization

Advantages

What advantages do you think globalization has for third-world countries?

1. It creates new opportunities and benefits. For instance, currently some of the Western industrialized nations removed tariff restrictions and opened their markets for the products of third-world countries.
2. There are rules and principles by which globalization is governed. For example, some of the provisions require a country to open up its financial markets and eliminate controls on capital flow (both in and out). This will increase foreign direct investment, promote economic growth and lift the from poverty.
3. Payments, known as “remittances” that developing countries international migrants send back to their countries (home) have become an important measure of the performance of a globalized world economy. For example, in 2009, remittances exceeded \$300 billion, almost three times the total of foreign aid disbursements.

Disadvantages

1. Globalization is unable to avoid the divergent interest clashes between the rich Western industrialized countries and the poor Developing countries.
2. So far, globalization has not brought equality among nations. Although it can minimize the risk of one country harming another, it is difficult to think it will bring equality between the countries. This is because the nature and characteristics of globalization do not allow it to do so.
3. The strong and wealthy nations have the power to promote certain of their advantages.
4. Governments of the Western industrialized nations are ready to accept the principles of free market and make them practical in their relationships with developing countries. But they do this only when they are quite sure that their transcontinental corporations are the winners in the market competition.

5. When the industrialized countries import products from the developing countries, they impose high tariffs because they want to protect and safeguard the interests of their local industries. This clearly shows that they do not exercise free-market principles. Rather they stick to a non-free market economic system.
6. Another problem is that governments of Western industrialized nations subsidize their farmers because they want them to sell their agricultural products at lower prices than those of the imported ones from the developing countries. This throws doubt on the sincerity of the Western industrialized nations when they talk about the importance of globalization for the reduction of poverty in developing countries.
7. Local markets in the developing countries may be dominated by the goods (particularly industrial) imported from industrialized countries the moment they open their doors to globalization.
8. Globalization disregards the protection of the environment and hence developing countries suffer greatly from environmental degradation.

5.7 Economic Growth and Development in Ethiopia

5.7.1 An Overview of Economic Growth and Development Trend in Ethiopia

Economic growth and development are not the same. They have two different meanings. Growth is the quantitative accumulation of the national capital. It is simply a change in output regardless of cultural, political, economic and other improvements. Development is far more complex than growth. Development is a process involving economic as well as social and environmental changes. It involves higher per capita income, equitable education, and job opportunities, greater gender equality, better health and nutrition, cleaner, more sustainable natural environment, a more impartial judicial and legal system, broader civil and political freedoms, and richer cultural life.

Development is also the qualitative transformation of a whole society, a shift to new ways of thinking, and corresponding new relations and new methods of production. Moreover, transformation only qualifies as development if it benefits most people – improves their quality of life and gives them more control over their destinies. This comprehensive process of change has to involve most of the and cannot be limited to modernization at the top or in the capital city.

According to official sources, the economy of Ethiopia shows a forward tendency with a growing rate of 11% per annum and is projected has been growing at a rate of 11% and is projected to increase in the coming 5 years.

When we refer to our topic, Economic Growth and Development Trend in Ethiopia, there has been an encouraging trend since 1996. This fact has been released by Welfare Monitoring surveys as follows.

- In rural areas, the literacy rate increased from 18% in 1995/96 to 31% in 2004/05.
- At country level, gross enrollment ratio at primary grades level has increased from 37.4% in 1996 to 79.8 in 2004/05. Although still relatively high, primary dropouts in rural areas declined from 18.5% in 1996 to 13.6% in 2004 at secondary level from 29.3% to 16.5%. The proportion

of households that are very far away from schools. i.e., 10km in the case of primary school) has fallen substantially.

- The proportion of rural population living more than 20 km from a health facility has fallen from 20% of household, in 1996 to 13% in 2004.
- A consistent decline in malnutrition over time is being shown in both rural and urban area. The rate in urban areas fell from 58% in 1996 to 30% in 2004; and in rural areas fell from 67% to 48%.
- About 92% of rural households are less than 5 kms away from the closest source of drinking water while it is much better in urban area.
More than 82% could get access to drinking water in less than one kilometer distance.
- About 42% of rural households are now less than five kilometers away from the closest all-weather road. The trend over time also indicates that availability of all-weather roads within five kilometer radius has been increasing etc.

In general, Ethiopia has started to have an economic growth at an average 11%, which is appreciable. PASDEP has contributed much to enhance all the achievements made for the last five years, the new Growth and Transformation Program (2010- 15) will promote and lead to increased success within the coming five years.

5.7.2 Major Features of Ethiopian Economy

The nature, performance, and growth of each sector were discussed in detail. Here in this unit, only their basic features are given as we examine the main characteristics of Ethiopia's economy. Try to integrate the information between two units.

This section highlights the major sectors of the Ethiopian economy:

- Agricultural
- Industrial
- distributed (service)

The agricultural sector, which is the dominant sector in many aspects, has performed unsatisfactorily for the last forty or fifty years due to multifaceted determining factors. Although various developmental strategies have been adopted since the early 1960s, their performance has been low. Even in the fiscal years of and 2002/03, 2003/04, 2004/05, 2005/06, 2006/07, 2007/08, the GDP share of agriculture was 44.9%, 47%, 47.4%, 47.2%, 46.3%, 44.6%, respectively.

This apparent decline was due to problems of weather changes, is the main factor on which Ethiopian agriculture depends. However, reforms instituted after 2002 have promoted the sector significantly.

Table: Sector share in GDP

Fiscal year Sector	Share in GDP (%)				
	2005/06	2006/07	2007/08	2008/09	2009/10
Agriculture and allied activities	47.0	46.0	44.6	43	42
Industry	13.0	13.0	13.0	13	13
Service/distributive sector	40	41.0	43.4	45	46
GDP	100	100	100	101	101

Agriculture's main products are food crops, cash crops, industrial crops, fruits and vegetables, flowers, and animal products. Agriculture had been the leading sector followed by the service sector until 2007/08. But, after 2008/09, the service sector emerged as the dominate/leading sector mainly due to the natural factors and economic factors. One of the lower capital and space required at the initial period.

The industrial sector ranked third in its contribution to Ethiopia's GDP. Its growth was small (13.0%) in comparison to that of the agriculture and service sectors. This could be attributed to the short history of industrialization in the country. Its has been modest, not exceeding 7 percent for the last four decades, except the years 1997/98 to 2009/10. The industrial sector consists mainly of large and medium-size industries, small scale and handicraft manufacturing, and the construction sub-sector.

The main products include textiles, food stuffs, beverages, cement, leather and leather products, metallic and non-metallic products, wood, and paper and pulp. Among the industrial sub-sectors, mining and quarrying has shown a tremendous growth performance accounting 44.2 percent of the industrial sector's growth performance. It is followed by large and medium-size manufacturing's and construction. Regarding the links between the various economic sectors, the forward and backward linkages between agriculture and industry have not been very strong. However, recent development programs launched in 2002 and later have practiced the country's industrialization process should rely. This is why the current government adopted the developmental strategy called Agricultural Development-Led Industrialization (ADLI).

ADLI's top-level goals are to ensure:

- economic growth through a rural-center development strategy
- improved living standards for rural and urban dwellers
- effective education services
- expansion of preventive and primary health care facilities

PRESENT FEATURES OF ETHIOPIAN SOCIO-ECONOMIC DEVELOPMENT

Ethiopia's present socio-economic features are treated as follows.

1. GDP per capita: It is believed that Ethiopia's GDP per capita is one of the lowest in the world. PASDEP's target has achieved a growth rate of Ethiopia's GDP per capita status will join the middle in

group countries with in the coming five years (2010-2015). The current status of Ethiopia's GDP per capita is estimated to be US \$344 per annum (source: wikipedia, the free encyclopedia).

2. Literacy rate: Like many of the less developed countries, the illiteracy rate in Ethiopia has been high. The literacy campaign that had under taken before 1991 contributed much to the ongoing task to eradicate illiteracy. Since 1996, the illiteracy rate has decreased by 50%. Welfare Monitoring Surveys Undertaken so far exhibits a sharp rise in gross school enrollment that considerably confirm the decline of illiteracy rate in the country. The current literacy rate of Ethiopia is said to be 42.7% in the year 2009 (Source: wikipedia, the free encyclopedia).

3. Infant Mortality Rate: Like in many developing countries, IMR in Ethiopia has been high. The UNDP 1997 census reveal that Ethiopia's infant mortality rate was 111/000 which even above the average of subsharan countries (97/1000). The recent development strategies adopted to achieve growth and sustained development, has target to lower down the IMR to 47/1000 with in the coming five years.

4. Expected life expectancy: the life expectancy of any country is the reflection of varied social, economic, psychological and other factors. In most developing countries life expectancy does not exceed 55 years. Likewise, in Ethiopia, it was 43 about ten years ago due to the prevalence of HIV impacts. The recent strategies adopted both in the primary health care services and supply of HIV/AIDS medicines extensively, the expected life expectancy in Ethiopia has reached to 55.8 years in 2009/10.

5. Population growth rate: Rapid population growth with an average rate per annum has been one of the many threats to Ethiopia's development pace. Estimates or surveys reveal that 38,000 people are added to the population each week incurring high costs of providing additional services. Addressing this issue GTP the continuation of PASDEP has designed to slow the rapid population growth by applying spacing of births through health service institutions and extension package services

6. Percentage of population with access to clean water: for many decades lack of access to clean water has been our peoples serious problem. Most illness caused both in urban and rural areas are related to unsafe water.

7. Based on the surveys made by a welfare monitoring organization, a greater proportion of rural population had been very far away from sources of drinkable water. Mothers and daughters were travelling long distances to fetch for water. By this; 32% of rural people get their water from unclean rivers and lakes and about 42% from unprotected well or spring. Only 14% used a protected water/well and 10% had access to a public tap water. In the last ten years, the establishment and rehabilitation of water supplies in rural and urban areas have been changing the picture.

8. Level of infrastructural development: This indicator of development includes a vaguely of elements that promote the living standard of people. In our context; we here by examine the road networks, electric power supply, rehabilitation/ resettlement of pastoralists, access to health services and strengthen the urban-rural link.

5.7.3. Challenges and Prospects of Socio-Economic Development for Ethiopia

- **Challenges Of Socio-Economic Development For Ethiopia**

The different factors hindering Ethiopia's progress (challenges) are interconnected and in combination constitute a "poverty trap" that prevents the country from breaking out of poverty. These factors are:

- **Population growth** and the concomitant addition of 2 million persons per year is a major barrier to poverty reduction. For instance, this growth puts tremendous strains on Ethiopia's resource base and the government's ability to deliver services.
- **Land-fragmentation:** more than 50% of Ethiopia's small land holders have less than 2 hectares, which deters the use of modern agricultural input and large-scale farming techniques.
- Environmental degradation has caused frequent droughts.
- Very low productivity
- Low income
- Low levels of investment
- Dependence on unreliable rainfall
- Structural bottlenecks
- Lack of good governance and commitment to accomplishing tasks
- Low infrastructure coverage: although there have been significant improvements during the past 10 years, the level of infrastructure coverage remains remarkably low when compared to elsewhere in the world

Prospects for Ethiopia's Socio- Economic Development

There has been encouraging progress in recent years in improving some basic aspects of life in Ethiopia. Since 1996, the literacy rate has increased by 50%, the rate of malnutrition has fallen by 20%, the share of the population with access to clean water has risen to 38% and there has been a steady decline in the reported incidence of illness. Nonetheless, human development indicators in Ethiopia still remain at low levels compared with global indicators. For example, Ethiopia's maternal mortality, is 673 per 100,000, in 2005/06, which is the highest in the world.

The way to overcome these challenges is to sustain the economic growth achieved in the last few years. Maintaining the priority of such growth is essential to finally having a lasting impact on poverty. This growth process is also essential for financing the necessary social investment for human development. Estimates show that a growth rate of about 8% per annum would have to be sustained to reach the Millennium Development Goal (MDG). This rate would require a great deal of revenue. Existing revenues are insufficient for financing the proposed MDG growth in essential health, education, infrastructure and other services. Therefore, the government requires a massive increase in tax revenue (particularly from the unproductive consumption sector) to achieve the required economic growth of about 6 to 8% per year that is required for meeting the goals of the Growth and Transformation Programme (GTP) period (2010-2015).

Furthermore, Ethiopia's strategy must be based on its most abundant resources: labor and the country's favorable climate (for instance for flowers and other crops).

In sum, progress is needed on every possible front including:

- proper utilization of agricultural potential;

- much more rapid development of the modern sector;
- exploitation of niche markets and opportunities wherever they present themselves;
- expansion of exports: in particular, diversifying exportable items to widen the economic base, reduce susceptibility to shocks and, in the long term, to reduce the prevailing dependency on foreign aid;
- promotion of better links between markets and producers to enable business to take place and to allow people easier access to essential services;
- promotion of capacity building regarding the work force's education and skills.
- slowing the existing rapid population growth; and
- proper utilization of the potential of Ethiopia's women, who constitute about 48.9 percent of the population.

Mode of Delivery

- ✓ Active and interactive methods of teaching need to be employed.

Assessment Modalities

Quizzes , Tests, Model Exam , National Exam

References

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